N° d'ordre NNT :

THESE de DOCTORAT DE L'INSA LYON,
membre de l'Université de Lyon

École Doctorale N° 512
InfoMaths

Spécialité / discipline de doctorat :
Informatique

Soutenue publiquement le 17/07/2026, par :
Johan Erbani

Understanding Machine Learning Vectors: From Confusion Matrices to Byzantine-Robust Federated Learning

---

Devant le jury composé de :

GIANINI Gabriele	Professeur des Universités, Université de Milan-Bicocca, Milan	Rapporteur
TOMMASI Marc	Professeur des Universités, Université de Lille, France	Rapporteur
GOUY-PAILLER Cédric	Directeur de Recherche, CEA, France	Examinateur
RÉVEILLÈRE Laurent	Professeur des Universités, Université de Bordeaux, France	Examinateur
EGYED-ZSIGMOND Elöd	Maître de Conférences HDR, INSA Lyon, France	Directeur de thèse
NURBAKOVA Diana	Maître de Conférences, INSA Lyon, France	Co-directrice de thèse
BEN MOKHTAR Sonia	Directrice de recherche, CNRS, Insa-Lyon, France	Invitée (co-encadrante)

Département FEDORA – INSA Lyon - Ecoles Doctorales

SIGLE	ECOLE DOCTORALE	NOM ET COORDONNEES DU RESPONSABLE
ED 206 CHIMIE	CHIMIE DE LYON https://edchimie.universite-lyon.fr Sec. : Renée EL MELHEM Bâtiment Blaise Pascal, UCB Lyon 1 Tél : 04.72.43.80.46 secretariat@edchimie-lyon.fr	M. Bruno ANDRIOLETTI Université Claude Bernard Lyon 1 Bâtiment F308, BP 2077 43 Boulevard du 11 novembre 1918 69616 Villeurbanne directeur@edchimie-lyon.fr
ED 341 E2M2	ÉVOLUTION, ÉCOSYSTÈME, MICROBIOLOGIE, MODÉLISATION https://e2m2.universite-lyon.fr Sec. : Bénédicte LANZA Bâtiment Atrium, UCB Lyon 1 Tél : 04.72.44.83.62 secretariat.e2m2@univ-lyon1.fr	M. Marc LEMAIRE Université Claude Bernard Lyon 1 Laboratoire MAP – Bâtiment LWOFF 10 rue Dubois 69622 Villeurbanne CEDEX e2m2.codir@listes.univ-lyon1.fr
ED 205 EDISS	INTERDISCIPLINAIRE SCIENCES-SANTÉ https://ediss.universite-lyon.fr Sec. : Bénédicte LANZA Bâtiment Atrium, UCB Lyon 1 Tél : 04.72.44.83.62 secretariat.ediss@univ-lyon1.fr	M. Jérôme LAMARTINE LBTI - UMR5305 CNRS – Université Claude Bernard Lyon 1 Equipe Intégrité Fonctionnelle du Tissu Cutané IBCP - 7 passage du Vercors 69367 Lyon Cedex 07 Tél : 04.72.72.26.66 jerome.lamartine@univ-lyon1.fr
ED 34 EDML	MATÉRIAUX DE LYON https://ed34.universite-lyon.fr Sec. : Yann DE ORDENANA Tél : 04.72.18.62.51 yann.de-ordenana@ec-lyon.fr	M. Stéphane BENAYOUN Ecole Centrale de Lyon Laboratoire LTDS – 36 avenue Guy de Collongue 69134 Ecully CEDEX Tél : 04.72.18.64.37 stephane.benayoun@ec-lyon.fr
ED 160 EEA	ÉLECTRONIQUE, ÉLECTROTECHNIQUE, AUTOMATIQUE https://edeea.universite-lyon.fr Sec. : Philomène TRECOURT Bâtiment Charlotte Perriand, INSA Lyon Tél : 04.72.43.71.70 secretariat.edeea@insa-lyon.fr	M. Philippe DELACHARTRE INSA LYON Laboratoire CREATIS Bâtiment Blaise Pascal, 7 avenue Jean Capelle 69621 Villeurbanne CEDEX Tél : 04.72.43.88.63 philippe.delachartre@insa-lyon.fr
ED 512 INFOMATHS	INFORMATIQUE ET MATHÉMATIQUES http://edinfomaths.universite-lyon.fr Sec. : Renée EL MELHEM Bâtiment Blaise Pascal, UCB Lyon 1 Tél : 04.72.43.80.46 infomaths@univ-lyon1.fr	M. Dragos IFTIMIE Université Claude Bernard Lyon 1 Bâtiment Braconnier, Bureau 232 21 Avenue Claude Bernard 69622 Villeurbanne Cedex Tél : 04.72.44.79.58 direction.infomaths@listes.univ-lyon1.fr
ED 162 MEGA	MÉCANIQUE, ÉNERGÉTIQUE, GÉNIE CIVIL, ACOUSTIQUE https://edmega.universite-lyon.fr Sec. : Philomène TRECOURT Bâtiment Charlotte Perriand, INSA Lyon Tél : 04.72.43.71.70 mega@insa-lyon.fr	M. Etienne PARIZET INSA Lyon Laboratoire LVA Bâtiment St. Exupéry, 25 bis avenue Jean Capelle 69621 Villeurbanne CEDEX etienne.parizet@insa-lyon.fr
ED 483 ScSo	ScSo[1] – https://edsciencessociales.universite-lyon.fr Sec. : Mélina FAVETON Tél : 04.78.69.77.79 melina.faveton@univ-lyon2.fr	M. Bruno MILLY Université Lyon 2 Campus Berges du Rhône 18, quai Claude Bernard Bureau BEL 204 69365 LYON CEDEX 07 bruno.milly@univ-lyon2.fr

[1] ScSo : Architecture, Ergonomie, Géographie-Aménagement-Urbanisme, Histoire-Histoire de l'art, Mondes anciens, Science politique, Sociologie-Anthropologie

You ask me if an ordinary person—by studying hard—would get to be able to imagine these things like I imagine. Of course. I was an ordinary person who studied hard. There's no miracle people. It just happens they got interested in this thing, and they learned all this stuff. They're just people. There's no talent or special miracle ability to understand quantum mechanics or a miracle ability to imagine electromagnetic fields that comes without practice and reading and learning and study. So if you take an ordinary person who's willing to devote a great deal of time and study and work and thinking and mathematics, then he's become a scientist.

— Richard Feynman, *Fun to Imagine: Ways of Thinking*, BBC, August 1983

La découverte est le privilège de l'enfant. C'est du petit enfant que je veux parler, l'enfant qui n'a pas peur encore de se tromper, d'avoir l'air idiot, de ne pas faire sérieux, de ne pas faire comme tout le monde. Il n'a pas peur non plus que les choses qu'il regarde aient le mauvais goût d'être différentes de ce qu'il attend d'elles, de ce qu'elles devraient être, ou plutôt : de ce qu'il est bien entendu qu'elles sont.

— Alexandre Grothendieck, *Récoltes et Semailles*

La bonne méthode, c'est de partir pour faire un tour à pied et d'essayer de se débrouiller avec ce qu'on a, pour justement, en y réfléchissant, créer dans le cerveau ces images mentales. Et peu importe la complexité du problème. Peu importe le caractère impossible au départ de cette création, c'est en séchant, et justement, en s'appropriant progressivement des objets mentaux qui vont correspondre au problème qu'on va progresser.

— Alain Connes, *L'imagination à l'infini*, France Culture, juillet 2018

Abstract

This work studies classification in complex settings. The first part focuses on confusion matrices and aims to characterize model behavior from its outputs. The second part addresses Byzantine-robust federated learning, examining how to defend a distributed system based on local model updates. In both cases, we rely on machine learning vectors—model outputs and gradients—and investigate how to extract meaningful information from them.

Confusion matrix. The confusion matrix is a key tool for understanding, evaluating, and improving model behavior. It is a square matrix built from label–prediction pairs, using test labels and model predictions. Unlike aggregate metrics such as accuracy, it decomposes errors at the class level and provides a detailed view of model behavior.

In heterogeneous settings, where the label distributions of the training and test sets are imbalanced, the confusion matrix becomes harder to interpret. This imbalance biases the matrix entries and can obscure errors arising from similarities between classes.

Multi-label settings, where observations may belong to several classes, and soft-label settings, where observations are represented by non-negative vectors encoding degrees of membership or confidence, better capture real-world complexity. However, extending the confusion matrix to these settings is not straightforward.

As a result, several competing extensions coexist. Most existing works visualize these matrices on benchmark datasets to interpret model behavior or compare them qualitatively with alternative formulations. However, these comparisons lack a clear quantitative criterion, making it unclear which method should be preferred in practice.

Moreover, no existing approach provides a unified formulation of the confusion matrix for single-label, multi-label, and soft-label settings, limiting both the consistency and the scope of machine learning evaluation.

Contributions to the Confusion Matrix. Our main contributions are:

- Propose a principled normalization of the confusion matrix to better recover true class similarity in heterogeneous settings;
- Establish connections between normalization methods, importance sampling, and latent class representations;
- Introduce an optimal transport-based formulation extending confusion matrices to multi-label and soft-label settings, providing a unified framework and linking existing approaches;
- Introduce an experimental framework for evaluating whether confusion matrix extensions accurately recover true model confusion in complex classification settings.

These results are based on the published papers [34, 31] and the under-review paper [30].

Federated learning. Federated learning enables multiple data holders (workers) to collaboratively train a model without sharing their data. Each worker locally updates the global model and sends its update (typically a gradient) to a central server. The server aggregates these updates, updates the model, and broadcasts it back to the workers; this process is repeated at each training round.

In adversarial settings, faulty or malicious (Byzantine) workers may disrupt the training process. Byzantine-robust federated learning aims to mitigate their impact. A common strategy is to discard outlier gradients, based on the assumption that Byzantine gradients deviate more from honest gradients than honest gradients deviate from each other.

However, data heterogeneity—such as label skew, where workers observe different label distributions—also misaligns local updates. As a result, honest workers may appear as outliers, rendering these defense strategies ineffective.

Contributions to Federated Learning. The main contributions on Byzantine-robust federated learning under label skew are:

- Introduce a lightweight local sampling-based defense against Byzantine attacks that does not require validation data;
- Propose a validation-free Byzantine filter based on the observation that honest updates lie within the convex hull of class-wise gradients.

These results are based on the under-review papers [33, 32].

Contents

I Confusion Matrices **3**

I Preliminaries **5**

1 Overview . . . 5
2 Key Properties . . . 6
2.1 Algorithmic Definition . . . 6
2.2 Scalar Evaluation Measures . . . 6
2.3 Consistency with Label–Prediction Pairs . . . 6
2.4 Classifier Diagnostics . . . 7
3 Normalization & Probabilistic Interpretation . . . 7
4 Conclusion . . . 8

II Normalization of Confusion Matrices: Methods and Geometric Interpretations **9**

1 Introduction . . . 10
2 Related Work . . . 11
2.1 Normalization Using Iterative Proportional Fitting . . . 11
2.2 Causes of Errors . . . 12
2.3 Key Takeaways . . . 12
3 Bi-Normalization . . . 12
3.1 Empirical Definition & Desirable Properties . . . 12
3.2 Theoretical Definition, Properties & Estimation . . . 13
4 Normalization as Importance Sampling . . . 14
4.1 Weigthing Sum . . . 14
4.2 Importance sampling . . . 15
5 Geometric View of Normalized Confusion Matrices . . . 16
5.1 Class Representations . . . 16
5.2 Geometric View of Normalized Confusion Matrices . . . 17
6 Experimental Setup . . . 17
6.1 Datasets & Heterogeneity . . . 18
6.2 Models & Training . . . 18
6.3 Metric . . . 18
6.4 Baselines & Experiments . . . 18
7 Empirical Results . . . 19
8 Conclusion . . . 20

III Unifying Confusion Matrices from Single-Label to Soft-Label **23**

1 Introduction . . . 24
2 Related works . . . 25
2.1 Multi-Label Matrices . . . 25
2.2 Soft-Label on the Simplex Matrices . . . 25
2.3 Key Takeaways . . . 25
3 Background . . . 26
3.1 Single-Label Confusion Matrix . . . 26
3.2 How to Transform Predictions into Labels . . . 26
3.3 Informal Introduction to Measures . . . 27
3.4 Informal Introduction to Optimal Transport . . . 28
4 Transport-based Confusion Matrix . . . 29

4.1 Problem Formalization . . . 29
4.2 Cost Function . . . 30
4.3 Solutions . . . 30
4.4 Key Takeaways on the Elementary Matrix . . . 32
4.5 Confusion Matrix . . . 33
5 Comparison with the State of the Art . . . 34
5.1 Unification . . . 34
5.2 Axiomatic Requirements . . . 35
6 Case Study: EXIST . . . 35
7 Conclusion . . . 36

IV Benchmarking Confusion Matrices Beyond Single-Label Classification 39
1 Introduction . . . 40
2 Related Work . . . 40
2.1 Evaluation Methods . . . 40
2.2 Unavailable Ground-Truth & Synthetic Evaluation . . . 41
2.3 Our Contribution . . . 41
3 ConfusionBench . . . 41
3.1 Classifiers from Confusion Matrices . . . 41
3.2 Latent Confusion Matrices . . . 42
3.3 Synthetic Label-Prediction Pairs . . . 42
3.4 Ground-Truth Justification . . . 44
4 Experimental Setup . . . 45
4.1 Scenarios & Hyperparameters . . . 45
4.2 Baselines . . . 45
4.3 Metric . . . 46
5 Evaluation . . . 46
5.1 Multi-label case . . . 46
5.2 Soft-label case . . . 47
6 Limitations . . . 47
7 Conclusion . . . 48

II Byzantine-Robust Federated Learning under Label Skew 57

V Preliminaries 59
1 Overview . . . 59
2 Background . . . 60
2.1 Federated Learning Setting . . . 60
2.2 Threat Model . . . 60
2.3 Data & Label-Skew Model . . . 60
3 Related Work . . . 60
3.1 Robustness Strategies with Homogeneous Data . . . 60
3.2 Robustness Strategies Under Data Heterogeneity . . . 60
4 Manuscript organization . . . 61

VI A Local Sampling Strategy 63
1 NorthStar . . . 64
1.1 Intuition behind NorthStar . . . 65
1.2 Reference Distribution . . . 65
1.3 Algorithm . . . 66
2 Theoretical Arguments . . . 66
2.1 Standard Gradients . . . 67
2.2 Gradients under NorthStar . . . 68
2.3 NorthStar versus Standard Gradients . . . 68
3 Experimental Setup . . . 69

3.1 Datasets . . . 69
3.2 Attacks & Baselines . . . 69
3.3 FL settings . . . 69
4 Results . . . 70
4.1 Effect of Label Skew . . . 70
4.2 Dataset-Level Trends with RFA Alone . . . 70
4.3 Interaction with NNM . . . 71
4.4 Results Averaged over Aggregators . . . 71
4.5 Negative Gains . . . 71
5 Conclusion . . . 71

VII Inside the Convex Hull of Gradients 81
1 HullGuard Filter . . . 82
1.1 Analogy for HullGuard . . . 82
1.2 HullGuard Algorithm . . . 82
1.3 HullGuard in Detail . . . 83
2 Experiments . . . 86
2.1 Experimental Setup . . . 86
2.2 RQ1: How effective is the HullGuard filter? . . . 88
2.3 RQ2: Does HullGuard improve robustness against Byzantine attacks? . . . 88
3 Conclusion . . . 89

VIII Conclusion 91
1 Confusion Matrices . . . 91
2 Federated Learning . . . 92

Part I

Confusion Matrices

Chapter I
Preliminaries

1	Overview	**5**
2	Key Properties	**6**
2.1	Algorithmic Definition	6
2.2	Scalar Evaluation Measures	6
2.3	Consistency with Label–Prediction Pairs	6
2.4	Classifier Diagnostics	7
3	Normalization & Probabilistic Interpretation	**7**
4	Conclusion	**8**

1 Overview

The confusion matrix is a central tool for understanding, evaluating, and improving classification models [63, 64, 43, 100]. Its origin is commonly traced to a 1904 lecture by Karl Pearson to the Draper's Society [67, 5], where he introduced two-way tables to study categorical variables and their dependence. The confusion matrix is a particular type of contingency table that summarizes the empirical joint distribution of true and predicted labels, providing insight into classifier behavior beyond scalar performance measures [63, 64, 43].

Formally, the confusion matrix is a square matrix of size $C \times C$, where C denotes the number of classes. Each entry (i, j) counts the number of instances with true label i predicted as j, for all $i, j \in [C]$, where $[C] = 1, \ldots, C$. It is typically computed on a test set [95].

We illustrate the confusion matrix on a three-class classification problem with classes `Cat`, `Dog`, and `Frog` (see Figure I.1). Diagonal entries correspond to correct predictions, while off-diagonal entries represent misclassifications. For instance, among samples with true label `Dog`, two are predicted as `Cat` and one as `Frog`. The more diagonal the matrix, the better the overall performance.

True / Pred	`Cat`	`Dog`	`Frog`
`Cat`	8	1	1
`Dog`	2	3	1
`Frog`	4	0	2

Figure I.1: Example of a confusion matrix. Rows correspond to true labels and columns to predicted labels.

2 Key Properties

2.1 Algorithmic Definition

Let $\{(x_k, y_k)\}_{k=1}^N$ be a dataset, where x_k denotes the k-th observation and $y_k \in [C]$ its corresponding class label. Let $\hat{y}_k \in [C]$ denote the predicted label for x_k, defined as

$$\hat{y}_k \in \operatorname{argmax}_{i=1}^{C} \Phi(x_k)_i,$$

where Φ is the model, and argmax returns the set of indices achieving the maximum value, from which $\hat{y}_k$ is selected.

Algorithm 1 describes the construction of the confusion matrix $M \in \mathbb{R}^{C \times C}$ from the dataset $\{(y_k, \hat{y}_k)\}_{k=1}^N$.

Algorithm 1 Construction of the Confusion Matrix

Require: Label–prediction pairs $\{(y_k, \hat{y}_k)\}_{k=1}^N$
Initialize $M \in \mathbb{R}^{C \times C}$ with zeros
for $k = 1$ to N **do**
    $M_{y_k, \hat{y}_k} \leftarrow M_{y_k, \hat{y}_k} + 1$
end for
return M

2.2 Scalar Evaluation Measures

Standard evaluation measures, including accuracy, precision, recall, and F1-score, can all be derived from the confusion matrix. For instance,

$$\text{Accuracy} = \frac{\sum_{i=1}^{C} M_{ii}}{M_{++}}, \quad \text{Precision}_k = \frac{M_{kk}}{M_{+k}}, \quad \text{Recall}_k = \frac{M_{kk}}{M_{k+}},$$

where M denotes the confusion matrix, with row, column, and matrix sums defined as

$$M_{i+} = \sum_j M_{ij}, \quad M_{+j} = \sum_i M_{ij}, \quad M_{++} = \sum_{i,j} M_{ij}.$$

These expressions highlight that the entries of the confusion matrix are the building blocks of widely used evaluation metrics, thereby providing a finer-grained view of classification errors and agreement.

2.3 Consistency with Label–Prediction Pairs

The diagonal entries M_{ii} count correct predictions for class i, while off-diagonal entries M_{ij} count misclassifications. This directly implies the following correspondences between label–prediction pairs and the matrix entries:

- **Diagonal & off-diagonal.** The sum of diagonal entries gives the number of correct predictions, while the sum of off-diagonal entries gives the number of errors. The total mass equals the number of label–prediction pairs: $M_{++} = N$.

- **Marginals.** Row and column sums, also referred to as marginals [100], describe the class distributions: M_{i+} gives the number of samples in true class i, while M_{+j} gives the number of samples predicted as class j. In Figure I.1, the label distribution is `Cat` 10, `Dog` 6, and `Frog` 6, while the prediction distribution is `Cat` 14, `Dog` 4, and `Frog` 4.

These properties ensure interpretability through consistency with the underlying set of label–prediction pairs.

2.4 Classifier Diagnostics

The confusion matrix exhibits the following properties:

- **Diagonality.** Higher diagonal values and lower off-diagonal values indicate better classifier performance. The matrix is diagonal if and only if the classifier correctly classifies all observations.
- **Class similarity.** A relatively large off-diagonal entry (i, j), compared to other errors, indicates strong confusion: class j is frequently predicted when the true class is i. This suggests that, from the classifier's perspective, class i is hard to distinguish from class j.
- **Distribution bias.** A row i with globally large values indicates that class i is overrepresented in the true labels, which follows from marginal consistency. Consequently, when row marginals are imbalanced—for instance, if class k is dominant while class l is underrepresented—entries across rows are not directly comparable. Similarly, a column j with globally large values indicates that class j is overrepresented in the predictions, leading to the same limitation. In Figure I.1, this indicates that class `Cat` is overrepresented in both the labels and the predictions.
- **Under & over-prediction.** If the column marginal is much larger than the row marginal, $M_{k+} \ll M_{+k}$, then class k is overpredicted. Conversely, if the row marginal is much larger than the column marginal, $M_{+k} \ll M_{k+}$, then class k is underpredicted. These errors generally reflect systematic model behavior rather than similarity between classes.

Values in the confusion matrix arise from different factors, and their interpretation requires accounting for marginal distributions to make reliable classifier diagnostics. Normalization techniques can help mitigate distribution biases.

3 Normalization & Probabilistic Interpretation

Figure I.2 illustrates standard normalization methods and their effect on row and column sums:

- **Row normalization.** To mitigate label imbalance, a common approach is to apply row normalization:
$$\mathrm{row}(M)_{ij} = \frac{M_{ij}}{M_{i+}},$$
which ensures that each row sums to one. This normalization provides an estimate of the conditional probability $\mathbb{P}(\hat{Y} = j \mid Y = i)$, where Y and $\hat{Y}$ denote the true and predicted label random variables, respectively [43].
- **Column normalization.** To mitigate prediction imbalance, one can apply column normalization:
$$\mathrm{col}(M)_{ij} = \frac{M_{ij}}{M_{+j}},$$
which ensures that each column sums to one. This normalization provides an estimate of the conditional probability $\mathbb{P}(Y = i \mid \hat{Y} = j)$.
- **All normalization.** Finally, normalization by the total mass yields
$$\mathrm{all}(M)_{ij} = \frac{M_{ij}}{M_{++}},$$
which ensures that the entire matrix sums to one. This normalization provides an estimate of the joint probability $\mathbb{P}(Y = i, \hat{Y} = j)$.

These normalizations reduce the impact of label and prediction imbalance on matrix values, making the matrix easier to interpret. They also provide complementary probabilistic interpretations of the confusion matrix, improving the analysis of classifier behavior.

True / Pred	`Cat`	`Dog`	`Frog`	$\mathbf{M_{i+}}$
`Cat`	8	1	1	10
`Dog`	2	3	1	6
`Frog`	4	0	2	6
$\mathbf{M_{+j}}$	14	4	4	22

(a) Raw

True / Pred	`Cat`	`Dog`	`Frog`	$\mathbf{M_{i+}}$
`Cat`	0.80	0.10	0.10	1.00
`Dog`	0.33	0.50	0.17	1.00
`Frog`	0.67	0.00	0.33	1.00
$\mathbf{M_{+j}}$	1.80	0.60	0.60	3.00

(b) Row-normalized

True / Pred	`Cat`	`Dog`	`Frog`	$\mathbf{M_{i+}}$
`Cat`	0.57	0.25	0.25	1.07
`Dog`	0.14	0.75	0.25	1.14
`Frog`	0.29	0.00	0.50	0.79
$\mathbf{M_{+j}}$	1.00	1.00	1.00	3.00

(c) Column-normalized

True / Pred	`Cat`	`Dog`	`Frog`	$\mathbf{M_{i+}}$
`Cat`	0.36	0.05	0.05	0.46
`Dog`	0.09	0.14	0.05	0.28
`Frog`	0.18	0.00	0.09	0.27
$\mathbf{M_{+j}}$	0.63	0.18	0.18	1.00

(d) Fully normalized

Figure I.2: Confusion matrices and their marginals under different normalizations. Row normalization fixes M_{i+}, column normalization fixes M_{+j}, and full normalization fixes M_{++}.

4 Conclusion

This overview illustrates how the confusion matrix provides insight into classifier behavior. Its simplicity and consistency provide a high degree of interpretability and enable reliable analysis.

Chapter II highlights the limitations of standard normalization methods in imbalanced settings, motivating the use of double normalization. It also provides additional perspectives on matrix normalization. Chapter III presents a generalization of the confusion matrix to broader classification settings based on optimal transport theory and the principle of maximum entropy. Finally, Chapter IV introduces a framework for evaluating confusion matrix extensions by measuring their ability to recover known latent confusion patterns from synthetic classifiers, enabling direct comparison between methods.

Chapter II
Normalization of Confusion Matrices: Methods and Geometric Interpretations

1	Introduction	**10**
2	Related Work	**11**
	2.1 Normalization Using Iterative Proportional Fitting	11
	2.2 Causes of Errors	12
	2.3 Key Takeaways	12
3	Bi-Normalization	**12**
	3.1 Empirical Definition & Desirable Properties	12
	3.2 Theoretical Definition, Properties & Estimation	13
4	Normalization as Importance Sampling	**14**
	4.1 Weigthing Sum	14
	4.2 Importance sampling	15
5	Geometric View of Normalized Confusion Matrices	**16**
	5.1 Class Representations	16
	5.2 Geometric View of Normalized Confusion Matrices	17
6	Experimental Setup	**17**
	6.1 Datasets & Heterogeneity	18
	6.2 Models & Training	18
	6.3 Metric	18
	6.4 Baselines & Experiments	18
7	Empirical Results	**19**
8	Conclusion	**20**

This chapter is based on our publication [34].

In heterogeneous settings, confusion matrix values are shaped by two main factors: class similarity—how easily the model confuses two classes—and distribution bias, arising from skewed distributions in the training and test sets. However, confusion matrix values reflect a mix of both factors, making it difficult to disentangle their individual contributions.

To address this, we propose a generalization of row and column normalization based on Iterative Proportional Fitting, referred to as bi-normalization. Unlike standard normalization, this method recovers the underlying structure of class similarity. By disentangling error sources, it enables more accurate diagnosis of model behavior and supports more targeted improvements.

We also establish a correspondence between confusion-matrix normalizations, importance sampling, and the model's internal class representations, thereby providing a clearer interpretation of normalization schemes and what they reveal about a classifier.

Contributions. We highlight key properties of bi-normalization and provide new perspectives on matrix normalizations:

- **Generalizing normalization.** We show that bi-normalization generalizes both row and column normalizations while satisfying key desirable properties.
- **Recovering class similarities.** We show that bistochastic normalization more reliably captures class similarities in heterogeneous settings compared to existing normalization schemes.
- **Importance sampling and class representation views.** We establish connections between confusion matrix normalizations—row, col, all, and bi—and both importance sampling and latent class representations, providing clearer interpretations of normalized confusion matrices.

Code: `https://github.com/JohanErbani/Bi-Normalization`

1 Introduction

While the confusion matrix is a widely used tool [86, 87, 80, 88], the meaning of its values becomes unclear in imbalanced settings.

Sources of confusion. Two main factors shape confusion matrix values: (i) class similarity and (ii) distribution bias [31], as illustrated in Figure II.1. (i) Class similarity refers to confusion between similar classes—the more alike two classes i and j are, the more frequently they are misclassified as each other, increasing values in entries (i, j) and/or (j, i) (Subfigure II.1c). While this source of error is intrinsic to the classes themselves, it also reflects inherent properties of the model: regardless of training, these similarities tend to persist. (ii) Distribution bias refers to errors caused by class imbalance in the predictions or test set. Models trained on imbalanced data tend to overpredict majority classes and underpredict minority ones [13, 71], inflating the columns of majority classes (Subfigure II.1b). Similarly, imbalance in the test set affects the rows: majority classes have higher counts, while minority classes have lower ones (Subfigure II.1a). This source of confusion is incidental and varies with class distributions.

Understanding the sources of classification errors enables targeted improvements. For example, errors from class similarity can be reduced by enhancing preprocessing to better distinguish similar classes or by augmenting the training set with more examples of those classes [42, 105, 103, 2]. Overprediction errors can be reduced by adjusting the loss function, such as reweighting class contributions to reduce the influence of frequent classes [41, 25, 15, 112]. However, confusion matrix values reflect a mix of both class similarity and distribution bias (Subfigure II.1d), making it difficult to disentangle their individual contributions.

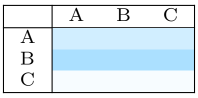
Standard normalizations. In practice, machine learning workflows often apply row, column, or all normalization to confusion matrices:

$$\mathrm{row}(M)_{ij} = \frac{M_{ij}}{M_{i+}}, \quad \mathrm{col}(M)_{ij} = \frac{M_{ij}}{M_{+j}}, \quad \mathrm{all}(M)_{ij} = \frac{M_{ij}}{M_{++}}.$$

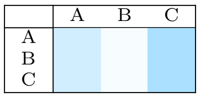
Many papers present several normalized matrices, as each normalization highlights different aspects of the model behavior [43]. However, in imbalanced settings, these normalizations may fail to capture class similarities.

Bi-normalization. Class similarities in the confusion matrix become visible when all rows and columns have the same total, which typically occurs when both the test and training sets are balanced. When both the test set and predictions are imbalanced, a double normalization off rows and columns at the same time is needed to reveal these similarities. This normalization can be achieved using the Iterative Proportional

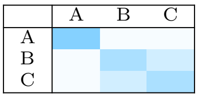
(a) Test set: Imbalance, Predictions: Balance, Class similarity: No



(b) Test set: Balance, Predictions: Imbalance, Class similarity: No



(c) Test set: Balance, Predictions: Balance, Class similarity: Yes



(d) Test set: Imbalance, Predictions: Imbalance, Class similarity: Yes

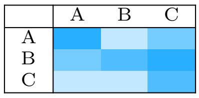


Figure II.1: How class similarity and distribution bias affect confusion matrices. The first three subfigures isolate each factor, while the last combines them: (a) Test set imbalance, with B as the majority class and C as the minority; (b) Prediction imbalance, with C over-predicted and B under-predicted; (c) Class similarities; (d) Combined influence of all factors. Darker colors indicate higher values.

Fitting (IPF) procedure—also known as the Sinkhorn-Knopp algorithm, biproportional fitting, the RAS method, or simply matrix scaling [53]. IPF is widely used in statistics, economics, and computer science [53]. We refer to the normalization produced by IPF as *bi-normalization*, denoted by bi. This normalization yields a confusion matrix in which each row and each column sums to 1: $\mathrm{bi}(M)_{i+} = \mathrm{bi}(M)_{+j} = 1$.

Additionnal perspectives. We also introduce two novel perspectives that provide further insight into matrix normalization:

- Importance sampling: Normalization can be viewed as applying an importance sampling strategy: it reweights each label–prediction pair to match a desired distribution—for instance, balancing labels via row normalization.
- Model representation: Normalization is also related to how the model encodes class representations. We show empirically that the overlap of class clusters in the latent space corresponds—depending on how it is measured—to a particular form of normalized confusion matrix.

Standard and bi normalizations can thus be interpreted through these perspectives, providing additional probabilistic and geometric meaning.

2 Related Work

The first part of this state-of-the-art reviews the application of IPF procedure to normalize contingency tables, while the second focuses on the impact of distribution bias on the confusion matrix.

2.1 Normalization Using Iterative Proportional Fitting

Deming and Stephan [26] proposed the use of the Iterative Proportional Fitting (IPF) procedure to estimate cell probabilities in a contingency table under given marginal constraints.

Ireland and Kullback [54] address the problem of estimating a new contingency table q from a given contingency table p, where each cell q_{ij} represents a probability, subject to known and fixed marginal probabilities $\sum_j q_{ij} = u_i$ and $\sum_i q_{ij} = v_j$. The authors introduce an algorithm minimizing the KL divergence from p to q. This procedure is equivalent to IPF [53].

In the fields of remote sensing and geographic modeling, disagreement between maps and reality is commonly displayed in a confusion matrix [21, 49]. When multiple classification or modeling methods are used, the resulting confusion matrices are typically compared to assess significant differences [49]. Confusion matrix normalization using IPF is a standard analytical technique [20]. In this way, differences in sample sizes used to generate the matrices are eliminated and, therefore, individual cell values within the matrix are directly comparable [20].

For speaker verification system improvements, Nagineni and Hegde [84] use the IPF procedure to normalize confusion matrices and select a cohort set based on similarity modeling for each client speaker.

2.2 Causes of Errors

In the context of neural networks, Erbani et al. [31] examine how imbalanced training and test sets influence confusion matrix entries. They introduce the test–training ranking to distinguish sources of error. When some entries deviate from this criterion, it suggests strong similarity between the corresponding classes. However, class similarities can still impact errors even when the ranking is preserved. Moreover, deviations from the ranking are ambiguous—they do not indicate which entry is too high or too low [31].

2.3 Key Takeaways

Normalizing contingency tables—particularly confusion matrices—using IPF procedure is not a new idea; it is even considered a standard approach in some disciplines. However, to the best of our knowledge, no existing work in machine learning employs this normalization to separate the effects of distribution bias and class similarity within confusion matrixes. This decomposition provides deeper insights into the sources of errors, helping to assess and improve the classifier.

To the best of our knowledge, no prior work has connected confusion matrix normalization to importance sampling or to the structure of class representations in a model's latent space. We introduce two perspectives—probabilistic and geometric—that offer deeper insight into classifier behavior.

3 Bi-Normalization

This section defines the bi-normalization, outline the key properties it should satisfy, and show how the IPF procedure can compute it accordingly.

Theoretical analysis is simpler with positive matrices. Since a non-negative confusion matrix M and its strictly positive counterpart $M + \epsilon$ (for small ϵ) exhibit similar model behavior, we propose using $M + \epsilon$ instead. We now assume that M is a positive confusion matrix.

3.1 Empirical Definition & Desirable Properties

Normalization approximates the confusion matrix that would arise under specific experimental conditions. Row normalization corresponds to the confusion matrix obtained from a balanced test set, whereas column normalization corresponds to that obtained from a model producing balanced predictions.

Empirical definition. Bi-normalization captures class similarities by approximating the confusion matrix under balanced label distributions in both the training and test sets. This is formalized by enforcing both row and column normalization: $\mathrm{bi}(M)_{i+} = \mathrm{bi}(M)_{+j} = 1$.

Standard normalization methods all, row, and col satisfy three properties described below: idempotence, class distribution invariance, and information preservation. By extension, bi-normalization is expected to satisfy these properties as well.

Idempotence. Normalization maps the matrix to specific experimental conditions. For example, the row normalization operator maps the matrix to a setting with balanced label distributions. Once these conditions are reached, further normalization should not modify the matrix, e.g., $\mathrm{row} \circ \mathrm{row}(M) = \mathrm{row}(M)$, where $\circ$ denotes composition.

Class distribution invariance. If two confusion matrices differ only in the experimental conditions targeted by the normalization operator, then normalization yields similar matrices.

For instance, row normalization targets a balanced test set. Thus, regardless of the test set label distribution, a trained model yields similar row-normalized confusion matrices. Formally, let S and T be the confusion matrices obtained from a model Φ on two large test sets with different label distributions (without class extinction). Then, the error patterns across rows remain similar, i.e., $S_{i,:} \underset{\sim}{\propto} T_{i,:}$ for all i, where $\underset{\sim}{\propto}$ denotes equality up to a positive scaling factor[1]. This implies that $\mathrm{row}(S) \approx \mathrm{row}(T)$, showing that the row operator is invariant to variations in the test set label distribution.

This invariance is formalized as invariance under left multiplication by a positive diagonal matrix: $\mathrm{row}(AM) = \mathrm{row}(M)$ for any diagonal matrix $A \in \mathbb{R}_{>0}^{C\times C}$. Similarly, column normalization is invariant under right multiplication by a positive diagonal matrix: $\mathrm{col}(MB) = \mathrm{col}(M)$ for any diagonal matrix $B \in \mathbb{R}_{>0}^{C\times C}$.

Bi-normalization should be invariant under both operations, as it combines row and column normalization, i.e., $\mathrm{bi}(AMB) = \mathrm{bi}(M)$.

Information preservation. Normalization infers, from the original confusion matrix, the one obtained under other experimental conditions. It should therefore preserve as much of the original information as possible, altering it only when necessary.

This requirement can be formalized as an optimization problem that promotes similarity between the original and normalized matrices. We use the Kullback–Leibler (KL) divergence as a dissimilarity measure. Given row and/or column constraints, the normalized matrix is defined as the one closest to the original in terms of KL divergence.

For instance, row normalization satisfies

$$\mathrm{row}(M) \in \underset{\substack{P\in\mathbb{R}_{>0}^{C\times C}:\\ P_{i+}=1,\ P_{+j}=\sum_i \frac{M_{ij}}{M_{i+}}\ \forall i,j}}{\operatorname{argmin}} D_{\mathrm{KL}}(P\|M),$$

where $M \in \mathbb{R}_{>0}^{C\times C}$ is the original confusion matrix, and $D_{\mathrm{KL}}(P\|M)$ denotes the KL divergence from P to M. Proofs and extensions to other normalization methods are provided in the Appendix.

Bi-normalization should minimize this optimization problem under the constraints that each row and each column sums to one.

3.2 Theoretical Definition, Properties & Estimation

This subsection provides the formal definition of the bi-normalization, its properties and how to estimate it.

The bi-normalization is defined as the unique solution of an optimization problem (see Appendix for the proof of existence and uniqueness).

Definition 1. $\mathrm{bi}(M)$ *is the unique minimizer of the following constrained problem:*

$$\mathrm{bi}(M) \in \underset{\substack{P\in\mathbb{R}_{>0}^{C\times C}:\\ P_{i+}=P_{+j}=1\ \forall i,j}}{\operatorname{argmin}} D_{\mathrm{KL}}(P\|M)$$

We now establish key properties:

Proposition 1. $\mathrm{bi}(M)$ *satisfies idempotence, class distribution invariance, and information preservation as described in Subsection Empirical Definition & Desirable Properties.*

[1] For vectors or matrices U and V of the same size, $U \underset{\sim}{\propto} V$ means $U \approx aV$ for some $a \in \mathbb{R}_{>0}$.

The last property is satisfied by definition, while the two others follow directly from it, as shown in the appendix.

Although $\mathrm{bi}(M)$ has no closed form, it can be efficiently estimated by the IPF algorithm (s). In our case, it consists of normalizing the matrix alternately by rows and columns until convergence[2]:

$$\mathrm{col} \circ \mathrm{row} \Big(\ldots \mathrm{col} \circ \mathrm{row} \Big(\mathrm{col} \circ \mathrm{row} \big(M \big) \Big) \ldots \Big) \longrightarrow \mathrm{bi}(M)$$

The IPF procedure always converges when the input matrix M is positive [53]. Let $M_{i+} = u_i$ and $M_{+j} = v_j$ for all i and j, and denote by Q the limit of the IPF procedure[3]. Then, Q solves the following optimization problem [65, 53]:

$$Q \in \underset{\substack{P \in \mathbb{R}_{>0}^{C\times C}: \\ P_{i+} = u_i, P_{+j} = v_j \ \forall\, i,j}}{\operatorname{argmin}} D_{\mathrm{KL}}(P \| M)$$

This directly implies the following:

Proposition 2. $\mathrm{bi}(M)$ *can be approximated using the IPF procedure with row and column constraints set to* 1. *We have* $\mathrm{bi}(M) = Q \approx \hat{Q}$, *where* Q *is the theoretical IPF limit and* $\hat{Q}$ *is its empirical estimate.*

Since $\mathrm{bi}(M)$ is doubly stochastic[4] and M is positive, the IPF procedure converges linearly [53].

4 Normalization as Importance Sampling

This section shows that normalization can be interpreted as an importance sampling strategy used to match a target distribution. It first shows that normalization can be expressed as a weighted sum, then interprets this weighted sum as an importance sampling process.

Let $\{(x_k, y_k)\}_{k=1}^N$ be a single-label dataset, and let $\hat{y}_k \in [C]$ be the model prediction for x_k, defined as

$$\hat{y}_k \in \operatorname{argmax}_{i=1}^C \Phi(x_k)_i,$$

where Φ denotes the model. Let M be the confusion matrix constructed from the label-prediction pairs $\{(y_k, \hat{y}_k)\}_{k=1}^N$.

4.1 Weigthing Sum

By definition, the confusion matrix is $M = \sum_{k=1}^N E_{y_k \hat{y}_k}$, where E_{ij} denotes the $C \times C$ matrix with a 1 at entry (i,j) and 0 elsewhere. Each matrix $E_{y_k \hat{y}_k}$ is a confusion matrix constructed from the single pair $(y_k, \hat{y}_k)$, capturing its individual contribution to M.

Standard and bi normalizations can be obtained by multiplying M on the left and right by diagonal matrices. Specifically, let D^l and D^r denote diagonal matrices (with l for left and r for right), and let M be the original confusion matrix. Then, $D^l M D^r$ yields the normalized version of M. For instance, setting

$$D^l = \mathrm{diag}(\frac{1}{M_{1+}}, \ldots, \frac{1}{M_{C+}}) \text{ and } D^r = I$$

[2]Let $Q^{(t)}$ be the matrix produced by IPF at iteration t. Convergence is reached when $Q^{(t)}_{i+}$ and $Q^{(t)}_{+j}$ are sufficiently close to 1 under a fixed tolerance.

[3]Let $Q^{(t)}$ be the matrix produced by IPF at iteration t. Then $Q^{(t)} \to Q$ as $t \to \infty$.

[4]A doubly stochastic matrix is a square matrix with nonnegative entries such that each row and each column sums to 1.

yields the row-normalized matrix, where $\mathrm{diag}(d_1, \ldots, d_C)$ denotes a diagonal matrix with entries $d_1, \ldots, d_C$ on its diagonal, and I the identity. Other normalizations are presented in the Appendix.

As a result, normalization can be expressed as a weighted sum of the individual contributions $E_{y_k \hat{y}_k}$:

$$D^l M D^r = \sum_{k=1}^{N} D^l_{y_k, y_k} D^r_{\hat{y}_k, \hat{y}_k} E_{y_k \hat{y}_k},$$

where each contribution $E_{y_k \hat{y}_k}$ is weighted by the product $D^l_{y_k, y_k} D^r_{\hat{y}_k, \hat{y}_k}$. For clarity, the weight can be expressed through a weight function ω. For instance, the weight function ω_r (with r for row) for row-normalization is

$$\omega_r : (y, \hat{y}) \mapsto \frac{1}{M_{y+}}, \quad \mathrm{row}(M) = \sum_{k=1}^{N} \omega_r(y_k, \hat{y}_k) E_{y_k \hat{y}_k}.$$

When the dataset is viewed as a realization of random variables, this process corresponds to importance sampling.

4.2 Importance sampling

Let $\{(Y_k, \hat{Y}_k)\}_{k=1}^N$ be independent and identically distributed (i.i.d.) random pairs, representing label–prediction pairs, and f be their joint distribution $f(i,j) = \mathbb{P}(Y = i, \hat{Y} = j)$. The random counterpart of $\mathrm{all}(M)$ converges almost surely (a.s.) to

$$\frac{1}{N} \sum_{k=1}^{N} E_{Y_k \hat{Y}_k} \xrightarrow[N \to \infty]{\text{a.s.}} \mathbb{E}[E_{Y\hat{Y}}],$$

where

$$\mathbb{E}[E_{Y\hat{Y}}]_{ij} = \mathbb{E}[\mathbb{1}_{Y=i, \hat{Y}=j}] = \mathbb{P}(Y = i, \hat{Y} = j) = f(i,j),$$

with $\mathbb{1}$ the indicator function.

Importance sampling refers to Monte Carlo methods that approximate expectations under a target distribution g by reweighting samples drawn from a proposal distribution f [106]. This approach is, for instance, useful when direct samples from g are unavailable. To approximate the expectation of $E_{Y,\hat{Y}}$ when $(Y, \hat{Y})$ is distributed according to g rather than f, we use the estimator $\hat{\mu}_g$, defined as

$$\hat{\mu}_g = \frac{1}{N} \sum_{k=1}^{N} \tfrac{g(Y_k, \hat{Y}_k)}{f(Y_k, \hat{Y}_k)} E_{Y_k \hat{Y}_k} \xrightarrow[N \to \infty]{\text{a.s.}} \mathbb{E}[\tfrac{g(Y, \hat{Y})}{f(Y, \hat{Y})} E_{Y\hat{Y}}] = \mathbb{E}_g[E_{Y\hat{Y}}]$$

with

$$\mathbb{E}_g[E_{Y\hat{Y}}]_{ij} = \mathbb{E}_g[\mathbb{1}_{Y=i, \hat{Y}=j}] = \mathbb{P}_g(Y = i, \hat{Y} = j) = g(i,j),$$

where $\mathbb{E}_g$ and $\mathbb{P}_g$ denote expectation and probability assuming the variables follow the law g. For importance sampling to remain accurate, the variance of each entry of the estimator $\hat{\mu}_g$ must remain small, which occurs when f is approximately proportional to g [106]. By selecting different choices of g, standard normalizations arise naturally.

As an example, consider row normalization (see Appendix for other normalizations). Setting $g(i,j) = \mathbb{P}(\hat{Y} = j \mid Y = i)$[5] recovers the row-normalized confusion matrix. In this case, the importance sampling ratio becomes

$$\frac{g(i,j)}{f(i,j)} = \frac{\mathbb{P}(\hat{Y} = j \mid Y = i)}{\mathbb{P}(Y = i, \hat{Y} = j)} = \frac{1}{\mathbb{P}(Y = i)} \approx \frac{N}{M_{i+}}.$$

Accordingly, the empirical estimator is

$$\tfrac{1}{N} \sum_{k=1}^{N} \tfrac{N}{M_{y_k+}} E_{y_k \hat{y}_k} = \sum_{k=1}^{N} \omega_r(y_k, \hat{y}_k) E_{y_k \hat{y}_k} = \mathrm{row}(M).$$

[5] Note that g is not a normalized distribution, since $\sum_{i,j} g(i,j) = C$ rather than 1.

For row normalization to be reliable, the importance sampling ratio should remain controlled. In particular, this requires that each class i is sufficiently represented in the test set; otherwise, the estimator exhibits high variance and becomes unstable.

Key Takeaways. Normalization is an importance sampling procedure: each pair is reweighted to shift the empirical distribution toward a target one. Reliability requires that the two distributions are not too dissimilar.

5 Geometric View of Normalized Confusion Matrices

This section proposes to model class representations as volumes, and finally link the overlaps of these volumes to normalized confusion matrices.

5.1 Class Representations

This subsection introduces a geometric perspective on class representations in the model's latent space. The key idea is to characterize class clusters as multidimensional histograms.

Representation space & dimensionality reduction. We use the output of the final pre-logit layer as the class representation space, a procedure already employed in previous work [9, 61, 123]. Let ε be the embedding function, so that $\Phi(x) = \text{Softmax} \circ \text{Logit} \circ \varepsilon(x)$, where $\varepsilon(x) \in \mathbb{R}^n$ for each (x, y) in the training set $\mathcal{D}$.

To facilitate histogram construction, we project the embeddings into a lower-dimensional space using Principal Component Analysis (PCA). Let P be the projection matrix with $m < n$, so that $P\varepsilon(x) \in \mathbb{R}^m$.

Class clusters & histograms. For each label i, we define the cluster of projected embedded points as $\mathcal{C}_i = \{P\varepsilon(x) : (x, y) \in \mathcal{D},\ y = i\}$. Similarly, the cluster of points predicted as class j is $\hat{\mathcal{C}}_j = \{P\varepsilon(x) : (x, y) \in \mathcal{D},\ \hat{y} = j\}$.

For each label and prediction clusters, we construct a multidimensional, non-overlapping histogram [104] (see Appendix for details). Each histogram is interpreted as a volume in $\mathbb{R}^{m+1}$, providing a geometric view of the clusters. Figure II.2 shows a toy example illustrating histograms of class clusters.

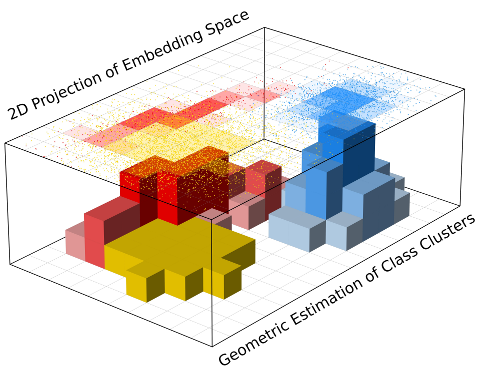


Figure II.2: Toy example showing a 2D projection of embedded observations forming class clusters (top map). Each cluster is associated with a histogram indicating the spatial distribution of points (bottom map).

Weighted histograms. Following the importance sampling perspective (Subsection 4), weighted histograms are constructed using the weight functions ω_a, ω_r, ω_c, and ω_b, corresponding respectively to all, row, col, and bi normalization. For a given bin, its height is defined not by the number of points it contains, but by the sum of their weights.

For instance, consider cluster $\mathcal{C}_i$ and the weighting ω_a. The height of a given bin b is $\sum_{(x,y)\in\mathcal{D}} \omega_a(y,\hat{y})\mathbb{1}_{y=i}\mathbb{1}_{P\varepsilon(x)\in b}$, whereas in the unweighted histogram it is $\sum_{(x,y)\in\mathcal{D}} \mathbb{1}_{y=i}\mathbb{1}_{P\varepsilon(x)\in b}$.

We denote by $V(\mathcal{C},\omega)$ the hypervolume in $\mathbb{R}^{m+1}$ of the weighted histogram built from cluster $\mathcal{C}$ using the weighting ω.

5.2 Geometric View of Normalized Confusion Matrices

This subsection introduces a new matrix, referred to as Geometric Confusion Matrix (GCM), which quantifies the overlap between the weighted histogram of label and prediction clusters:

$$\Big(\mathrm{GCM}_\omega\Big)_{ij} = \lambda\Big(V(\mathcal{C}_i,\omega)\cap V(\hat{\mathcal{C}}_j,\omega)\Big),$$

for $i,j = 1,\dots,C$, where λ denotes the Lebesgue measure. The Lebesgue measure of $V(\mathcal{C}_i,\omega)\cap V(\hat{\mathcal{C}}_j,\omega)$ corresponds to the volume of the intersection between the ω-weighted histograms of $\mathcal{C}_i$ and the one of $\hat{\mathcal{C}}_j$. Unlike the standard confusion matrix, the GCM takes into account the spatial organization of embedded points (see Appendix for details).

We now provide geometric views of confusion matrix normalizations, all claims are supported by experimental results (see Section 7).

All-normalization. All-normalized matrix approximates the overlap between the unweighted histograms of label and prediction clusters:

$$\mathrm{all}(M) \underset{\sim}{\propto} \mathrm{GCM}_{\omega_a},$$

where $\underset{\sim}{\propto}$ means approximately equal up to a scaling factor[6].

Row-normalization. Row-normalized matrix approximates the overlap between normalized label histograms and the resulting weighted prediction histograms:

$$\mathrm{row}(M) \underset{\sim}{\propto} \mathrm{GCM}_{\omega_r}.$$

More precisely, up to a known scaling factor, the label histograms are normalized in the sense that $\lambda\big(V(\mathcal{C}_i,\omega_r)\big) = 1$, whereas the volume of the prediction histograms, $\lambda\big(V(\hat{\mathcal{C}}_j,\omega_r)\big)$ are not fixed a priori (see Appendix for details). Accordingly, label histograms are normalized, while prediction histograms are scaled according to ω_r.

Col-normalization. Similarly, column-normalized matrix approximates the overlap between normalized prediction histograms and resulting weighted label histograms:

$$\mathrm{col}(M) \underset{\sim}{\propto} \mathrm{GCM}_{\omega_c}.$$

Bi-normalization. Finally, bi-normalized matrix approximates the overlap between normalized histograms of both label and prediction clusters:

$$\mathrm{bi}(M) \underset{\sim}{\propto} \mathrm{GCM}_{\omega_b}.$$

These correspondences provide a geometric interpretation of normalized confusion matrices and illustrate how confusion matrices capture the organization of classes in the model's latent space.

6 Experimental Setup

This section describes our experimental setup.

[6] Let M and N be matrices of the same size. We write $M \underset{\sim}{\propto} N$ to indicate that $M \approx \alpha N$ for some scalar $\alpha \in \mathbb{R}_{>0}$.

6.1 Datasets & Heterogeneity

We conduct experiments on four datasets: MNIST [70], Fashion-MNIST [114], CIFAR-10 [60], and STL-10 [18].

Normalization methods differ mainly on non-diagonal matrices. For example, row, column, and bi normalizations yield the same result on diagonal matrices. To encourage non-diagonal confusion matrices, we make MNIST and Fashion-MNIST harder by applying random rotations of 0°, 90°, 180°, or 270° during preprocessing, applied to both the training and test images.

Bi-normalization is particularly useful under heterogeneous settings. To simulate data heterogeneity, we sample from the original datasets with a Dirichlet distribution of concentration α, following [4, 52, 35]. We consider five levels—very low, low, medium, high, and extreme—corresponding to $\alpha \in \{10, 3, 1, 0.3, 0.1\}$.

To avoid class extinction, we enforce a minimum representation of 20% of the original class size. For example, in a dataset with 1000 samples per class, at least 200 samples are retained for each class, while the remaining samples are allocated according to a Dirichlet distribution.

6.2 Models & Training

CNN models (see Appendix for details) are trained using stochastic gradient descent with cross-entropy loss, a batch size of 32, a learning rate of 10^{-3}, a momentum of 0.9, and a weight decay of 10^{-4}.

As mentioned above, the main differences between normalizations appear in the off-diagonal confusion matrices. Therefore, training is limited to a maximum of 10 epochs, and stops earlier if the classifier reaches 60% balanced test accuracy.

6.3 Metric

Let S and T be two confusion matrices in $\mathbb{R}_{\geq 0}^{C\times C}$, possibly already normalized.

To measure the similarity between confusion matrices, we use their overlap:

$$\mathrm{Overlap}(S, T) = \sum_{ij} \min\left(\mathrm{all}(S)_{ij}, \mathrm{all}(T)_{ij}\right),$$

where min denotes the element-wise minimum (see Appendix for the rationale behind reusing all normalization). This score ranges from 0 to 1, with higher values indicating greater similarity. It reaches 1 if and only if $T = S$.

This metric is strictly equivalent to the L^1 distance, since $\|\,\mathrm{all}(S) - \mathrm{all}(T)\|_1 = 2 - 2\,\mathrm{Overlap}(S, T)$ (see Appendix for proof).

6.4 Baselines & Experiments

We compare bi-normalization with standard normalizations: row, col, and all. Each experiment is repeated over MNIST, Fashion-MNIST, CIFAR-10, and STL-10, with five heterogeneity levels and 30 random seeds.

Experiment 1. This experiment compares the confusion matrix obtained from balanced datasets (denoted as M_1) with normalized versions of the confusion matrix obtained from imbalanced datasets (denoted as M_2).

1. Initialize a model with a fixed random seed, then create two deep copies: model 1 and model 2.
2. Train model 1 on a balanced training set and compute its confusion matrix M_1 using a balanced test set.
3. Sample an imbalanced training set and test set, train model 2 on the imbalanced training set, and compute its confusion matrix M_2 on the imbalanced test set.
4. Apply normalization techniques to M_2 and evaluate their similarity to M_1.

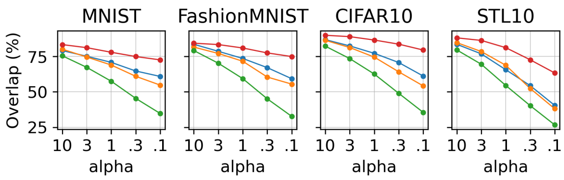
This experiment tests whether bi-normalization reveals class similarities under imbalanced data more effectively than other approaches.

Experiment 2. This experiment, conducted in an imbalanced setting, compares different versions of the GCM—GCMω_a, GCMω_r, GCMω_c, GCMω_b—with normalized confusion matrices derived from M—row(M), col(M), all(M), and bi(M).

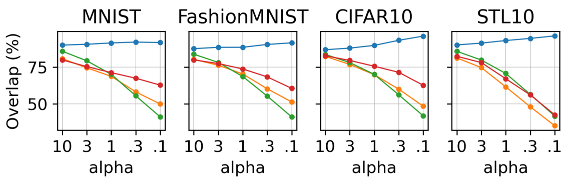
1. Initialize the model with a fixed seed.
2. Sample imbalanced training and test sets. Train the model. Compute both the GCM variants and the normalized confusion matrices using the test set.
3. Measure pairwise similarities.

This experiment evaluates how each normalization reveals the structure of class representations in the model's latent space under a specific weighted histogram.

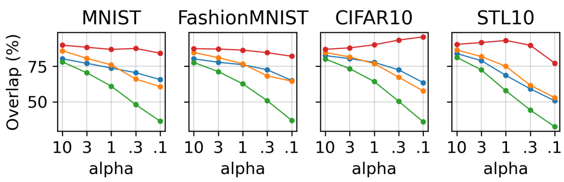
Details on the construction of multivariate histograms required for GCM are provided in the Appendix.



(a) Target: Matrix from balanced setting



(b) Target: GCMω_r



(c) Target: GCMω_b

Figure II.3: Overlap (%) between target matrices and normalized matrices. Legend: ● bi ● row ● col ● all

7 Empirical Results

Figure II.3 shows the results of Experiment 1 (Subfigure II.3a) and Experiment 2 (Subfigures II.3b and II.3c), described in Section 6.4.

Experiment 1. This experiment evaluates how well different normalizations recover class similarities. Starting from a confusion matrix obtained under an imbalanced setting, normalization aims to approximate the one obtained under a balanced setting.

Across datasets and heterogeneity levels, Subfigure II.3a shows that bi-normalization consistently achieves the highest overlap, outperforming other methods. As heterogeneity increases, (i) the gap between bi-normalization and the other methods tends to widen, and (ii) the ability to recover class similarity diminishes regardless of the normalization method.

This behavior is expected: the greater the imbalance between the training and test sets, the more distribution bias is introduced into the matrix. This bias tends to obscure class similarities, making row, col, and all normalizations unsuitable. Moreover,

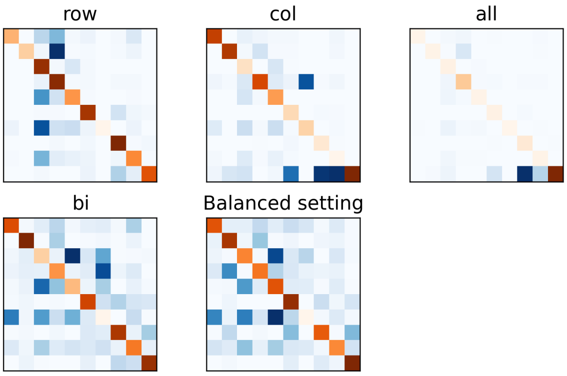


Figure II.4: One instance (seed 0) on Fashion-MNIST with $\alpha = 0.3$. Bi-normalization best recovers class relationships under distribution shifts. Diagonal variations are shown in orange; off-diagonal in blue, with darker shades indicating higher values.

higher heterogeneity leads to prediction and test class distributions that deviate further from a balanced setting, making normalization procedures less reliable, as discussed in Section 4.

Figure II.4 further illustrates that bi-normalization captures the class similarity patterns present in the balanced confusion matrix, whereas other methods often produce vertical or horizontal stripes.

By definition, bi-normalization is expected to achieve the lowest KL divergence. Figure A.2 in the Appendix confirms this, providing further evidence of its ability to recover class similarities.

Experiment 2. This experiment evaluates the correspondence between the organization of class representations in the model's latent space and normalized confusion matrices.

Subfigure II.3b demonstrates a clear correspondence between GCMω_r and row normalization. Overlap increases as heterogeneity grows. We observe similar results for GCMω_a with all normalizations, and for GCMω_c with column normalization (see Appendix).

Subfigure II.3c shows that the correspondence between GCMω_b and bi-normalization is weaker. This makes sense, as the granularity differs from the other approaches. For instance, in GCMω_r, all embedded points of the same label receive the same weight, while in GCMω_b, embedded points are weighted by both label and prediction. This indicates that spatial separation is finer under GCMω_b weighting than with other weightings. Nonetheless, bi-normalization still exhibits higher correspondence than other methods.

This experiment confirms the validity of the geometric interpretation of normalizations described in Section 5.2.

8 Conclusion

We revisited confusion matrix normalization and showed that bi-normalization provides a principled way to isolate class similarity from distribution bias. While the use of IPF for contingency tables is well established in other fields, its value for interpreting classifier behavior in machine learning has been largely overlooked. We demonstrated that bi-normalization reveals class similarities more clearly than standard methods, especially in heterogeneous settings.

We also connected confusion matrix normalization to importance sampling. From this perspective, normalization reweights label–prediction pairs to estimate classifier behavior under a target distribution rather than the observed one. This interpretation clarifies both the role and the limits of normalization.

Finally, we provided a geometric interpretation of normalized confusion matrices. It showed that confusion matrices reflect intersections between class clusters in the latent space of a model. By applying the corresponding normalization weights to these clusters, one recovers the associated normalized confusion matrix. This connects matrix normalization to the spatial organization of class representations learned by the model.

Chapter III
Unifying Confusion Matrices from Single-Label to Soft-Label

1	Introduction		**24**
2	Related works		**25**
	2.1	Multi-Label Matrices	25
	2.2	Soft-Label on the Simplex Matrices	25
	2.3	Key Takeaways	25
3	Background		**26**
	3.1	Single-Label Confusion Matrix	26
	3.2	How to Transform Predictions into Labels	26
	3.3	Informal Introduction to Measures	27
	3.4	Informal Introduction to Optimal Transport	28
4	Transport-based Confusion Matrix		**29**
	4.1	Problem Formalization	29
	4.2	Cost Function	30
	4.3	Solutions	30
	4.4	Key Takeaways on the Elementary Matrix	32
	4.5	Confusion Matrix	33
5	Comparison with the State of the Art		**34**
	5.1	Unification	34
	5.2	Axiomatic Requirements	35
6	Case Study: EXIST		**35**
7	Conclusion		**36**

This chapter is based on our publication [31] and Section Case Study: EXIST on our collaboration [85].

The confusion matrix is well established in the single-label setting, where each observation belongs to exactly one class. While multi-label and soft-label settings better reflect the complexity of real-world data, extending the confusion matrix to these settings is not straightforward.

We introduce the *Transport-based Confusion Matrix* (TCM), a generalization of the confusion matrix applicable to single, multi, and soft-label settings. TCM preserves the familiar structure of the standard confusion matrix and coincides with it in the single-label case. It captures errors that were previously inaccessible, improving both the consistency and the scope of machine learning evaluation.

Contributions. We introduce the *Transport-based Confusion Matrix* (TCM), which offers the following advantages:

- **Universality.** A unified framework for analyzing model errors across single-label, multi-label, and soft-label settings. To the best of our knowledge, it is also the only method applicable to the soft-label setting.
- **Unification.** In the single-label setting, TCM coincides with the standard confusion matrix. It also preserves partial consistency with existing extensions, thereby bridging previously incompatible approaches.
- **Reliability and Interpretability.** By combining optimal transport theory with the principle of maximum entropy, TCM is both theoretically grounded and interpretable.

Code: `https://github.com/JohanErbani/TCM`

1 Introduction

No existing approach provides a unified formulation of the confusion matrix for single-label, multi-label, and soft-label settings, limiting both the consistency and the scope of machine learning evaluation.

Calssification settings. The standard setting is *single-label*, where each observation belongs to exactly one class. For instance, an image may be labeled as either `Cat`, `Dog`, or `Frog`. In this case, the label space $\mathcal{Y}$ is defined as $\mathcal{Y} = [C]$, where C is the number of classes. Equivalently, it can be represented as the set of one-hot vectors:

$$\mathcal{Y} = \{\mathbf{y} \in \{0, 1\}^C : \|\mathbf{y}\|_1 = 1\}.$$

In *multi-label classification*, each observation may belong to several classes simultaneously. For example, an image can contain both a cat and a dog and thus be associated with both labels. In this case, $\mathcal{Y} = \{0, 1\}^C$. In the *soft-label* setting, each observation is associated with non-negative real values over classes, representing confidence or degree of membership. For example, a scientific paper may be labeled as 0.8 `Mathematics`, 0.4 `Physics`, and 0.2 `Computer Science`, reflecting how much it relates to each field. In this case, $\mathcal{Y} = \mathbb{R}^C_{\geq 0}$. Multi and soft-label frameworks are widely used in applications such as medical diagnosis, emotion recognition, and image or text classification [120, 110, 72, 119, 58, 77, 122, 19, 66, 23, 73]. Soft labels are particularly useful for capturing annotator disagreement in subjective tasks, and have been shown to improve generalization, robustness, and calibration [107, 19].

Problem & motivation. Error decomposition is straightforward in the single-label setting. For example, if the model predicts `Apple` instead of the true label `Pear`, the confusion is unambiguous. In multi and soft-label settings, however, the decomposition becomes unclear. For instance, if the model predicts `Apple, Lemon` while the true label is `Banana, Pear`, it is unclear whether `Banana` is confused with `Apple` and `Pear` with `Lemon`, whether the opposite assignment is more appropriate, or whether the confusion pattern is more complex. The problem becomes even harder when the numbers of predicted and true classes differ, or when labels are soft.

As a result, multiple extensions of the confusion matrix co-exist [63, 51, 64, 43, 10, 100, 31]. These definitions differ, and no formulation is designed to simultaneously handle single-label, multi-label, and soft-label settings, leading to inconsistencies in machine learning evaluation tools.

Transport-based Confusion Matrix. We introduce the Transport-based Confusion Matrix (TCM), a unified framework that extends the confusion matrix to multi and soft-label settings while preserving its structure and usability. Our approach is grounded in optimal transport theory and the principle of maximum entropy.

2 Related works

2.1 Multi-Label Matrices

The Scikit-learn library [89] provides a method for evaluating errors in multi-label classification, yielding one confusion matrix per class. However, this class-wise analysis is insufficient to fully capture a model's behavior [51, 64].

Krstinić et al. [63] introduce four formulas to compute elementary confusion matrices, each tailored to a specific case (e.g., too many or too few predicted classes). Similarly, Heydarian, Doyle, and Samavi [51] propose three formulas that yield comparable experimental results [62]. Krstinić et al. [64] recover standard precision and recall by constructing two confusion matrices, one per metric. These approaches lack theoretical justification for their design choices.

Görtler et al. [43] propose a confusion matrix for hierarchical and multi-output labels. In the multi-label setting without hierarchy, each class is represented by a binary variable (present/absent), yielding a $2C \times 2C$ matrix. Each pair of true class i and predicted class j is associated with a 2×2 submatrix:

$$\begin{array}{c|c} \mathbb{P}(Y^i = 1, \hat{Y}^j = 1) & \mathbb{P}(Y^i = 1, \hat{Y}^j = 0) \\ \hline \mathbb{P}(Y^i = 0, \hat{Y}^j = 1) & \mathbb{P}(Y^i = 0, \hat{Y}^j = 0) \end{array}$$

where Y^i and $\hat{Y}^j$ denote binary variables indicating the presence of true class i and predicted class j, respectively. This formulation departs from the standard $C \times C$ confusion matrix and is less directly interpretable.

2.2 Soft-Label on the Simplex Matrices

Binaghi et al. [10] and Silván-Cárdenas and Wang [100] propose confusion matrices for restricted soft-label scenarios, where both labels and predictions are simplex points (i.e., non-negative vectors summing to 1).

Binaghi et al. [10] introduce a matrix based on fuzzy set theory. However, this approach loses the familiar behavior of the confusion matrix: even perfect predictions may yield a non-diagonal matrix [93].

In geographic information science, Silván-Cárdenas and Wang [100] propose a method to compare terrestrial images. Each pixel is associated with multiple classes; one image serves as the reference and the other as the approximation. Each entry (i, j) is defined as an interval, where the lower (resp. upper) bound corresponds to the minimum (resp. maximum) sub-pixel class overlap.

2.3 Key Takeaways

Existing solutions often lack theoretical justification, deviate from the familiar behavior of the confusion matrix, and appear mutually incompatible. Moreover, no method simultaneously integrates single-label, multi-label, and soft-label frameworks, leading to inconsistencies across evaluation settings.

To address these limitations, we propose a unifying approach: the Transport-based Confusion Matrix. Grounded in optimal transport and the principle of maximum entropy, this framework preserves the behavior of the classical confusion matrix, accommodates all labeling settings simultaneously, and reconciles approaches that previously appeared incompatible. To the best of our knowledge, this perspective has not yet been explored and provides a more comprehensive and interpretable evaluation of model performance.

The main state-of-the-art approaches discussed in this paper are summarized in Table III.1.

Table III.1: Overview of state-of-the-art confusion matrices and TCM in single-label, multi-label, soft-label frameworks restricted to simplex point labels and predictions (Simplex), and unrestricted soft-label frameworks.

Matrix	Single	Multi	Simplex	Soft	Description
Confusion Matrix (CM)	✓	✗	✗	✗	Each entry (i, j) counts how often i is predicted as j.
Multi-Label Confusion Matrix (MLCM) [63]	✓	✓	✗	✗	The confusion matrix relies on four distinct formulas, each corresponding to a specific case.
Multi-Label Confusion Tensor (MLCT) [64]	✓	✓	✗	✗	Two matrices whose diagonals match classical recall and precision scores. Interpretation of off-diagonal entries is unclear.
Sub-pixel Confusion Matrix (SCM) [100]	✓	✗	✓	✗	It computes the minimum and maximum sub-pixel class overlap, yielding interval-valued entries.
Transport-based Confusion Matrix (TCM)	✓	✓	✓	✓	Grounded in optimal transport and maximum entropy.

3 Background

This section introduces the concepts required to define TCM constructions. The first subsection describes how a confusion matrix can be decomposed into elementary matrices. The second subsection presents an interpretation of these elementary matrices that allows us to generalize the confusion matrix. This perspective connects naturally to optimal transport (Subsection 3.4), where the notion of measures is central (Subsection 3.3).

3.1 Single-Label Confusion Matrix

Let $\{(x_k, y_k)\}_{k=1}^N$ be a single-label dataset, and let $\hat{y}_k \in [C]$ be the model prediction for x_k, defined as

$$\hat{y}_k \in \operatorname{argmax}_{i=1}^C \Phi(x_k)_i,$$

where Φ denotes the model. Let M be the confusion matrix constructed from the label-prediction pairs $\{(y_k, \hat{y}_k)\}_{k=1}^N$.

The confusion matrix can be written as a sum of elementary matrices, each corresponding to a label–prediction pair:

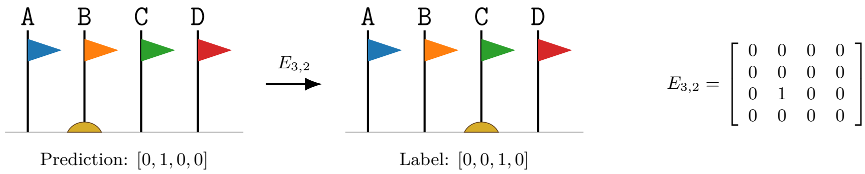
$$M = \sum_{k=1}^{N} E_{y_k \hat{y}_k}, \tag{III.1}$$

where E_{ij} is the $C \times C$ matrix with a 1 at position (i, j) and 0 elsewhere. Each $E_{y_k \hat{y}_k}$ is itself a confusion matrix constructed from a single label–prediction pair $(y_k, \hat{y}_k)$. We refer to such matrices as *elementary matrices*.

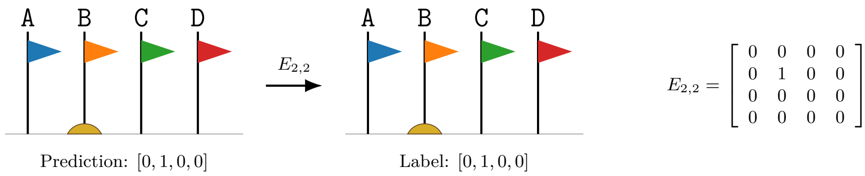
Equation (III.1) shows that extending the confusion matrix to more general settings reduces to extending the definition of the elementary matrix. Summing these generalized elementary matrices over the dataset then yields a generalized confusion matrix.

3.2 How to Transform Predictions into Labels

In the single-label setting, each elementary matrix $E_{y_k \hat{y}_k}$ can be interpreted as a set of instructions that transforms the prediction $\hat{y}_k$ into the label y_k. To illustrate this



(a) Prediction `B`, label `C`



(b) Prediction `B`, label `B`

Figure III.1: Illustration of elementary matrices as transport instructions. For a misclassification where class `B` is predicted instead of class `C`, the mass is transported from class `B` to class `C`. For a correct prediction in class `B`, the mass stays at class B, so no transport is needed.

idea, we introduce the following analogy, depicted in Figure III.1.

Consider an area with marker flags, each representing a class. For a classification task with classes `A`, `B`, `C`, and `D`, there are four flags. At the base of the flag corresponding to the predicted class, we place one kilogram of sand. This represents the value 1, while the other flags, with no sand, represent 0. For example, if the model predicts `B`, the one-hot vector is $[0, 1, 0, 0]$, which corresponds to one kilogram of sand at flag `B`.

The elementary matrix specifies how to move this sand to match the true label. Entry (i, j) indicates the amount of sand to be moved from flag j to flag i. For instance, if the true label is `C`, the associated elementary matrix is $E_{3,2}$, with entry 1 at $(3, 2)$ and 0 elsewhere; this indicates that one kilogram of sand must be transported from flag `B` to flag `C`. After the transfer, all the sand lies under flag `C`, giving the one-hot vector $[0, 0, 1, 0]$, which matches the label. In contrast, if the prediction is correct, the elementary matrix is $E_{2,2}$. This corresponds to moving one kilogram of sand from flag `B` to flag `B`, so no change is needed.

This interpretation naturally connects to optimal transport, where each elementary matrix can be viewed as a transport plan. We first introduce the notion of a measure, which provides the appropriate framework for representing mass distributions in optimal transport.

3.3 Informal Introduction to Measures

We briefly introduce measures; see [101] for a formal treatment.

A measure assigns a non-negative value to certain sets, called *measurable sets*. These sets are subsets of a space denoted by X. This collection of measurable sets is called a σ-algebra on X. Intuitively, a measure acts like a weighing scale: it assigns a weight to each measurable set, while the σ-algebra specifies which sets can be measured. A set not included in the σ-algebra cannot be assigned a weight. Let μ be a measure, and let A and B be two measurable sets, with $\varnothing$ denoting the empty set. A measure satisfies the following natural properties: $\mu(\varnothing) = 0$; for any disjoint sets A and B, $\mu(A \cup B) = \mu(A) + \mu(B)$; and for any sets $A \subseteq B$, then $\mu(A) \leq \mu(B)$.

A probability measure is a measure with total mass equal to one, i.e., $\mu(X) = 1$. We focus on discrete probability measures on a finite space, where the σ-algebra is the power set of X, that is, the collection of all subsets of X, denoted by $\mathcal{P}(X)$.

For instance, consider a coin flip experiment with $X = \{\text{heads}, \text{tails}\}$. Then

$$\mathcal{P}(X) = \{\varnothing, \{\text{heads}\}, \{\text{tails}\}, X\}$$

If the coin always lands on heads, it can be modeled by the Dirac measure δ_{heads}. A Dirac measure at x assigns mass to a set depending solely on whether the set contains

x or not. For any measurable set A,

$$\delta_x(A) = \begin{cases} 0 & \text{if } x \notin A \\ 1 & \text{if } x \in A \end{cases}$$

If the coin lands on heads with probability $p_{\text{heads}} \in [0, 1]$ and on tails with probability $p_{\text{tails}} = 1 - p_{\text{heads}}$, then μ is a Bernoulli measure given by

$$\mu = p_{\text{heads}} \delta_{\text{heads}} + p_{\text{tails}} \delta_{\text{tails}}.$$

More generally, a discrete probability measure [82] takes the form

$$\mu = \sum_{x \in X} p_x \, \delta_x, \quad \text{with } p_x \geq 0 \text{ and } \sum_{x \in X} p_x = 1.$$

For any $A \subseteq X$,

$$\mu(A) = \sum_{x \in A} p_x$$

3.4 Informal Introduction to Optimal Transport

We give a short introduction to optimal transport adapted to our needs; see [108] for a formal treatment.

Optimal transport studies how to transform a measure $\hat{\mu}$ into another μ while minimizing a transportation cost.

We consider discrete probability measures of the form

$$\mu = \sum_{i=1}^{C} p_i \delta_{a_i}, \quad \hat{\mu} = \sum_{j=1}^{C} q_j \delta_{b_j},$$

where $p_i, q_j \geq 0$ and $\sum_i p_i = \sum_j q_j = 1$.

A transport plan is a matrix $\pi \in \mathbb{R}_{\geq 0}^{C \times C}$, where π_{ij} represents the amount of mass transported from b_j to a_i. The set of admissible transport plans is

$$T(\mu, \hat{\mu}) = \left\{ \pi \in \mathbb{R}_{\geq 0}^{C \times C} \ : \ \sum_{j=1}^{C} \pi_{ij} = p_i, \ \sum_{i=1}^{C} \pi_{ij} = q_j \right\}. \tag{III.2}$$

These constraints ensure that all the mass from the source measure is transported to match the target measure, and that transported mass remains non-negative.

To break ties between admissible transport plans, a cost function

$$c : \{a_i\}_{i=1}^{C} \times \{b_j\}_{j=1}^{C} \to \mathbb{R}_{\geq 0}$$

is needed. It assigns a cost to transporting one unit of mass from b_j to a_i [108]. A low cost encourages mass transfer between two locations, whereas a high cost discourages it. For instance, in a physical setting, one typically seeks to move mass over minimal distances, making the Euclidean distance a natural choice for the cost function.

The optimal transport problem, known as the Kantorovich problem, consists in finding a transport plan minimizing the total cost:

$$\underset{\pi \in T(\mu, \hat{\mu})}{\operatorname{argmin}} \sum_{i,j=1}^{C} c(a_i, b_j) \, \pi_{ij}. \tag{III.3}$$

Solutions are called *optimal transport plans*.

Consider again the sand-and-flags analogy from Subsection 3.2. Suppose the model predicts `B` while the true label is `C`. We represent the label and the prediction by the measures μ and $\hat{\mu}$, respectively:

$$\mu = \delta_{\texttt{C}}, \quad \text{and} \quad \hat{\mu} = \delta_{\texttt{B}},$$

Algorithm 2 Transport-Based Confusion Matrix (TCM)

Require: Label-prediction pairs $\{(\mathbf{y}^{(k)}, \hat{\mathbf{y}}^{(k)})\}_{k=1}^{N}$, with $(\mathbf{y}^{(k)}, \hat{\mathbf{y}}^{(k)}) \in \mathbb{R}^C_{\geq 0} \times \mathbb{R}^C_{\geq 0}$, and a weighting function $w : \mathbb{R}^C_{\geq 0} \times \mathbb{R}^C_{\geq 0} \to \mathbb{R}_{\geq 0}$

Initialize $M \leftarrow 0 \in \mathbb{R}^{C \times C}$
for each label-prediction pair $(\mathbf{y}^{(k)}, \hat{\mathbf{y}}^{(k)})$ **do**
  $\mathbf{u} \leftarrow \dfrac{\mathbf{y}^{(k)}}{\|\mathbf{y}^{(k)}\|_1}, \quad \hat{\mathbf{u}} \leftarrow \dfrac{\hat{\mathbf{y}}^{(k)}}{\|\hat{\mathbf{y}}^{(k)}\|_1}$
  $\mathbf{v} \leftarrow \min(\mathbf{u}, \hat{\mathbf{u}})$ {element-wise minimum}
  if $\mathbf{u} = \hat{\mathbf{u}}$ **then**
    $\pi^{(k)} \leftarrow \mathrm{diag}(\mathbf{v})$
  else
    $\pi^{(k)} \leftarrow \mathrm{diag}(\mathbf{v}) + \dfrac{(\mathbf{u} - \mathbf{v}) \otimes (\hat{\mathbf{u}} - \mathbf{v})}{\|\mathbf{u} - \mathbf{v}\|_1}$
  end if
  $M \leftarrow M + w(\mathbf{y}^{(k)}, \hat{\mathbf{y}}^{(k)})\pi^{(k)}$
end for
return M

where the probability space is $X = \{\mathtt{A}, \mathtt{B}, \mathtt{C}, \mathtt{D}\}$, and the σ-algebra $\mathcal{P}(X)$. The goal is to transform $\hat{\mu}$ into μ. In the sand-and-flags analogy (see Subsection 3.2), this corresponds to moving the sand to the appropriate flags. The set of admissible transport plans $T(\mu, \hat{\mu})$ consists of all plans that transfer sand from the prediction configuration to the label configuration. In this setting, there exists a unique admissible plan, namely $T(\mu, \hat{\mu}) = \{E_{3,2}\}$. Consequently, the solution to the Kantorovich problem is $E_{3,2}$, independently of the chosen cost function.

4 Transport-based Confusion Matrix

This section details how to compute TCM, which is summarized in Algorithm 2. We begin with the Kantorovich problem relevant to our setting and define the associated cost function, whose solution yields a closed-form expression for TCM. We then interpret the formulas underlying TCM and explain the role of the weighting factor.

Let $\mathbf{y}$ and $\hat{\mathbf{y}}$ denote two vectors in $\mathbb{R}^C$ representing a ground-truth label and its associated prediction, respectively. For instance, in a three-class setting, a single-label annotation $y = 2$ can be encoded as $\mathbf{y} = [0, 1, 0]$. In a multi-label setting, if the model predicts classes 1 and 2, this corresponds to $\hat{\mathbf{y}} = [1, 1, 0]$. In a soft-label setting, the label may take non-negative real values, e.g., $\mathbf{y} = [0.8, 0.5, 0]$.

4.1 Problem Formalization

Adopting an optimal transport perspective on elementary matrices provides a natural way to extend them to multi-label and soft-label settings. To make optimal transport applicable, we interpret each label–prediction pair as a pair of measures and solve the associated Kantorovich problem:

$$\mathbf{y} \text{ is mapped to } \sum_{k=1}^{C} \mathbf{y}_k \delta_k, \quad \text{and } \hat{\mathbf{y}} \text{ is mapped to } \sum_{k=1}^{C} \hat{\mathbf{y}}_k \delta_k,$$

where the underlying space is $[C]$ equipped with the σ-algebra $\mathcal{P}([C])$.

For the set of admissible transport plans to be non-empty, the total mass of the measures associated with $\mathbf{y}$ and $\hat{\mathbf{y}}$ must be equal, i.e.,

$$\|\mathbf{y}\|_1 = \|\hat{\mathbf{y}}\|_1.$$

This condition is generally not satisfied in multi or soft-label settings, where the prediction may assign too much or too little mass. To address this, we compare normalized vectors, $\mathbf{y}/\|\mathbf{y}\|_1$ and $\hat{\mathbf{y}}/\|\hat{\mathbf{y}}\|_1$, which ensures equal total mass (equal to 1) and yields the following probability measures:

$$\frac{\mathbf{y}}{\|\mathbf{y}\|_1} \text{ is mapped to } \sum_{k=1}^{C} \frac{\mathbf{y}_k}{\|\mathbf{y}\|_1}\delta_k, \quad \text{and } \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1} \text{ is mapped to } \sum_{k=1}^{C} \frac{\hat{\mathbf{y}}_k}{\|\hat{\mathbf{y}}\|_1}\delta_k$$

The Kantorovich problem is then formulated as

$$\underset{\pi \in T(\mathbf{y},\hat{\mathbf{y}})}{\arg\min} \sum_{i,j=1}^{C} c(i,j)\,\pi_{ij}, \tag{III.4}$$

where $T(\mathbf{y},\hat{\mathbf{y}})$ denotes the set of admissible transport plans associated with the normalized label and prediction:

$$T(\mathbf{y},\hat{\mathbf{y}}) = \left\{ \pi \in \mathbb{R}_{\geq 0}^{C\times C} \ : \ \pi_{i+} = \frac{\mathbf{y}_i}{\|\mathbf{y}\|_1},\ \pi_{+j} = \frac{\hat{\mathbf{y}}_j}{\|\hat{\mathbf{y}}\|_1} \right\}. \tag{III.5}$$

To complete the formulation, it remains to specify the cost function.

4.2 Cost Function

The cost function indicates the cost of transporting a unit of mass from one location to another [108].

In our setting, it reflects how difficult it is for the model to confuse class i with class j. For instance, if a classifier systematically confuses `A` and `B`, but never `A` and `C`, a suitable cost function would satisfy $c(\mathtt{A},\mathtt{B}) = 0$ and $c(\mathtt{A},\mathtt{C}) = \infty$. However, such information is not available a priori and is precisely what we aim to infer through the confusion matrix.

We therefore adopt a minimal assumption: a class is identical to itself and distinct from all others. This idea is encoded by the following cost function, which assigns 0 when the classes are identical and 1 otherwise:

$$c(i,j) = \begin{cases} 0 & \text{if } i = j \\ 1 & \text{if } i \neq j \end{cases}$$

This cost function is known as the *discrete metric* [82, 108].

4.3 Solutions

The following proposition characterizes the solutions to the Kantorovich problem (Equation III.4):

Proposition 3. *A matrix $\pi \in T(\mathbf{y},\hat{\mathbf{y}})$ is an optimal transport plan if and only if*

$$\mathrm{diag}(\pi) = \min\left(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}\right), \tag{III.6}$$

where $\mathrm{diag}(\pi)$ *denotes the diagonal of* π*, and* $\min(\mathbf{y},\hat{\mathbf{y}})$ *is understood element-wise.*

According to Proposition 3, the set of optimal transport plans, denoted $T^{\mathrm{opt}}(\mathbf{y},\hat{\mathbf{y}})$, is defined as

$$T^{\mathrm{opt}}(\mathbf{y},\hat{\mathbf{y}}) = \{\pi \in T(\mathbf{y},\hat{\mathbf{y}}) : \mathrm{diag}(\pi) = \min(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1})\} \tag{III.7}$$

The next proposition shows that $T^{\mathrm{opt}}(\mathbf{y},\hat{\mathbf{y}})$ typically contains infinitely many solutions:

Proposition 4. *The set $T^{opt}(\mathbf{y}, \hat{\mathbf{y}})$ contains a unique optimal transport plan if and only if either the prediction is exact (i.e., $\frac{\mathbf{y}}{\|\mathbf{y}\|_1} = \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}$), or there exists at most one overestimated class (i.e., $\frac{\mathbf{y}_k}{\|\mathbf{y}\|_1} < \frac{\hat{\mathbf{y}}_k}{\|\hat{\mathbf{y}}\|_1}$), or there exists at most one underestimated class (i.e., $\frac{\hat{\mathbf{y}}_k}{\|\hat{\mathbf{y}}\|_1} < \frac{\mathbf{y}_k}{\|\mathbf{y}\|_1}$). Otherwise, it contains infinitely many optimal solutions.*

According to Propositions 3 and 4, the following hold for optimal plans:

(i) There is a unique optimal plan when the number of classes satisfies $C \leq 3$;

(ii) Any surplus mass from overestimated classes is transported only to underestimated classes;

(iii) A plan is diagonal if and only if $\mathbf{y} = \hat{\mathbf{y}}$.

These statements are demonstrated in Appendix.

Because the cost function relies on minimal assumptions—namely, that a class is identical to itself and distinct from all others—excess mass can be redistributed to any underestimated class without affecting the cost. In particular, the transport cost from a class i to any other class $j \neq i$ is identical. As a result, there are typically infinitely many optimal transport plans.

To break ties between optimal plans, we adopt the principle of maximum entropy: when only partial information is available, one should select the probability distribution that maximizes entropy while satisfying the known constraints [47]. This yields the least biased estimate consistent with the available information. This principle is widely used in physics and has been applied in fields such as ecology, spectroscopy, imaging, and neuroscience [6]. The following proposition shows that there exists a unique entropy maximizer among optimal plans and provides its formula.

Proposition 5. *Let f be the function defined by*

$$\begin{aligned} f: \quad \Delta^{C-1} \times \Delta^{C-1} \quad &\to \quad \mathbb{R}_{\geq 0}^{C \times C} \\ (u, \hat{u}) \quad &\mapsto \quad \operatorname{diag}\big(\min(u, \hat{u})\big) + \tfrac{\left(u - \min(u,\hat{u})\right) \otimes \left(\hat{u} - \min(u,\hat{u})\right)}{\|u - \min(u,\hat{u})\|_1}, \end{aligned} \tag{III.8}$$

where $\Delta^{C-1} = \{u \in \mathbb{R}_{\geq 0}^{C} : \sum_{i=1}^{C} u_i = 1\}$ denotes the probability simplex, and the minimum is taken elementwise. By continuous extension, f is also well defined at $u = \hat{u}$.

Then, there exists a unique entropy-maximizing transport plan in $T^{\text{opt}}(\mathbf{y}, \hat{\mathbf{y}})$, denoted by $\pi(\mathbf{y}, \hat{\mathbf{y}})$, and it is given by

$$\pi(\mathbf{y}, \hat{\mathbf{y}}) = f\Big(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}\Big)$$

Proposition 5 establishes a unified closed-form expression for the elementary matrix, valid across single, multi, and soft-label settings.

Remark. Considering the formula of f in Proposition 5, when $u = \hat{u}$, the denominator $\|u - \min(u, \hat{u})\|_1$ becomes zero. In the Appendix, we show that as $\|u - \hat{u}\|_1 \to 0$,

$$\frac{\big(u - \min(u, \hat{u})\big) \otimes \big(\hat{u} - \min(u, \hat{u})\big)}{\|u - \min(u, \hat{u})\|_1} \to 0 \in \mathbb{R}^{C \times C}.$$

Thus, we can rewrite

$$f = \begin{cases} \operatorname{diag}\big(\min(u, \hat{u})\big) & \text{if } u = \hat{u}, \\ \operatorname{diag}\big(\min(u, \hat{u})\big) + \frac{\left(u - \min(u,\hat{u})\right) \otimes \left(\hat{u} - \min(u,\hat{u})\right)}{\|u - \min(u,\hat{u})\|_1} & \text{otherwise.} \end{cases} \tag{III.9}$$

These equalities are formalized through the concept of continuous extension.

4.4 Key Takeaways on the Elementary Matrix

To better understand the behavior of elementary matrices, we summarize here the key foundational and behavioral properties that characterize them. An interactive demonstration is also available at:

`https://johanerbani.github.io/Elementary_matrix_of_TCM/`

Within the TCM framework, elementary matrices satisfy the following principles:

(i) Labels and predictions are compared *relatively* rather than absolutely. This is ensured by normalizing both the label and prediction vectors.

(ii) A normalized label can be obtained from the normalized prediction by transporting units of predicted mass toward the appropriate classes. This transformation is formalized using optimal transport theory combined with the principle of maximum entropy, and can be intuitively understood through the sand-and-flag analogy (see Subsection 3.2).

(iii) Elementary matrices define transport plans: entry (i, j) quantifies the mass transported from class j to class i to transform the normalized prediction into the normalized label. In particular, row sums match the normalized label distribution, column sums match the normalized prediction distribution, and the total sum equals one.

(iv) The quantity that remains in place after alignment with the label distribution is considered *correct*. These quantities are recorded on the matrix diagonal: entry (i, i) represents the common mass at component i between the normalized label and prediction vectors.

(v) The quantity that is moved to achieve alignment with the label distribution is considered *incorrect*. Excess mass from overestimated classes (i.e., where the prediction exceeds the label) is transported to underestimated classes (i.e., where the prediction is lower than the label). These quantities are recorded off-diagonal: entry (i, j) is nonzero only when class i is underestimated and class j is overestimated. In this case, it is proportional to both the deficit in class i and the excess mass in class j, i.e.,

$$\left(\frac{\mathbf{y}_i}{\|\mathbf{y}\|_1} - \frac{\hat{\mathbf{y}}_i}{\|\hat{\mathbf{y}}\|_1}\right)\left(\frac{\hat{\mathbf{y}}_j}{\|\hat{\mathbf{y}}\|_1} - \frac{\mathbf{y}_j}{\|\mathbf{y}\|_1}\right).$$

(vi) Elementary matrices follow specific value patterns. They are diagonal if and only if the prediction is exact, in which case

$$\pi(\mathbf{y}, \hat{\mathbf{y}}) = \mathrm{diag}(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}) = \mathrm{diag}(\frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1})$$

When the prediction is not exact, only the rows corresponding to underestimated classes and the columns corresponding to overestimated classes contain nonzero values. The greater the underestimation of a class, the larger the values in the corresponding row tend to be; similarly, the greater the overestimation, the larger the values in the corresponding column. Moreover, if an entry (i, j) is nonzero, then the entry (j, i) is zero.

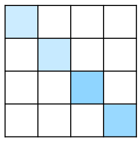
The above points are illustrated below:

$$\pi(\mathbf{y},\hat{\mathbf{y}}) = \quad \text{Correctly positioned mass} \quad + \quad \text{Mispositioned mass}$$

$$\downarrow \qquad\qquad\qquad\qquad \downarrow$$

$$\text{Diag. matrix} \qquad\qquad \text{Matrix with zero diag.}$$

$$\downarrow \qquad\qquad\qquad\qquad \downarrow$$



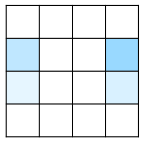
$$\underbrace{\qquad\qquad}_{\pi(\mathbf{y},\hat{\mathbf{y}})_{ii}}$$

$$\shortparallel$$

$$\Big(\text{Common mass in } i\Big)$$

$$\shortparallel$$

$$\min\left(\frac{\mathbf{y}_i}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}_i}{\|\hat{\mathbf{y}}\|_1}\right)$$



$$\underbrace{\qquad\qquad}_{\pi(\mathbf{y},\hat{\mathbf{y}})_{ij}}$$

$$\shortparallel$$

$$\frac{\Big(\text{Deficit in } i\Big)\Big(\text{Excess in } j\Big)}{\Big(\text{Total mispositioned mass}\Big)}$$

$$\shortparallel$$

$$\frac{\left(\frac{\mathbf{y}_i}{\|\mathbf{y}\|_1} - \min\left(\frac{\mathbf{y}_i}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}_i}{\|\hat{\mathbf{y}}\|_1}\right)\right)\left(\frac{\hat{\mathbf{y}}_j}{\|\hat{\mathbf{y}}\|_1} - \min\left(\frac{\mathbf{y}_j}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}_j}{\|\hat{\mathbf{y}}\|_1}\right)\right)}{\left\|\frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1} - \min\left(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}\right)\right\|_1}$$

The elementary matrix interprets prediction errors as mass transport from predictions to labels. By separating correctly positioned mass from redistributed mass, while preserving the marginal distributions, it provides a concise and interpretable representation of discrepancies between predictions and labels.

4.5 Confusion Matrix

Let $\{(\mathbf{y}_k, \hat{\mathbf{y}}_k)\}_{k=1}^N$ denote the label–prediction vectors used to build the confusion matrix. Here, the subscript k indexes the data points, not the vector components.

The Transport-based Confusion Matrix (TCM) is defined as

$$\mathrm{TCM} = \sum_{k=1}^{N} w(\mathbf{y}_k, \hat{\mathbf{y}}_k)\, \pi(\mathbf{y}_k, \hat{\mathbf{y}}_k), \tag{III.10}$$

where $\pi(\mathbf{y}_k, \hat{\mathbf{y}}_k)$ is the optimal transport plan maximizing the entropy (see Proposition 5), and $w : \mathbb{R}^C_{\geq 0} \times \mathbb{R}^C_{\geq 0} \to \mathbb{R}_{\geq 0}$ is a weighting function. The scalar $w(\mathbf{y}_k, \hat{\mathbf{y}}_k)$ weights the contribution of $\pi(\mathbf{y}_k, \hat{\mathbf{y}}_k)$.

In multi or soft-label settings, the total mass (or number of active classes) may vary across label–prediction pairs, motivating the use of weighting schemes. Examples include:

- $w : (\mathbf{y}, \hat{\mathbf{y}}) \mapsto 1$, where all contributions are equally weighted;
- $w : (\mathbf{y}, \hat{\mathbf{y}}) \mapsto \|\mathbf{y}\|_1$, where the weight increases with the number of active classes in the label. For instance, an observation labeled `Apple, Lemon` contributes twice as much as one labeled `Apple`;
- $w : (\mathbf{y}, \hat{\mathbf{y}}) \mapsto \frac{1}{\|\mathbf{y}\|_1}$, where the weight decreases with the number of active classes in the label. This can help break ties between class confusions, as label vectors with many active classes tend to distribute mass more evenly across the elementary matrix.

The choice of weighting is left to the user and may depend on the application. In practice, comparing multiple weightings can provide additional insight into model behavior.

Overall, the TCM aggregates elementary transport plans, providing a global representation of model behavior across the dataset. By summing these plans, it captures both correctly positioned mass and systematic misallocations, while preserving the underlying label and prediction distributions. Through weighting, it allows the matrix to adapt to specific needs. This construction extends the confusion matrix to multi and soft-label settings in a unified manner.

5 Comparison with the State of the Art

This section shows that TCM unifies several state-of-the-art approaches (Subsection 5.1) and is axiomatically more consistent (Subsection 5.2).

5.1 Unification

TCM follows the same underlying principles as existing confusion matrices and unifies several formulations that previously appeared incompatible.

To generalize the single-label confusion matrix, any extension to multi or soft-label settings should coincide with it in the single-label case.

Proposition 6. *TCM and the single-label matrix are identical in the single-label framework.*

Other state-of-the-art matrices, such as the MLCM and SCM, also exhibit this property.

In the MLCM framework, multi-label elementary matrices are defined through four ad hoc formulas, one for each case:

(i) Exact prediction, $\mathbf{y} = \hat{\mathbf{y}}$;

(ii) The label is a subset of the prediction, $\mathbf{y} \leq \hat{\mathbf{y}}$ (elementwise) and $\mathbf{y} \neq \hat{\mathbf{y}}$;

(iii) The prediction is a subset of the label, $\hat{\mathbf{y}} \leq \mathbf{y}$ (elementwise) and $\mathbf{y} \neq \hat{\mathbf{y}}$;

(iv) None of the above configurations, i.e., the general case.

Proposition 7. *In case (i), (ii), and (iii) MLCM and TCM weighted by* $w : (\mathbf{y}, \hat{\mathbf{y}}) \mapsto \|\mathbf{y}\|_1$ *produce identical elementary matrices.*

Our formula for case (iv) follows from a single, theoretically grounded expression. In contrast, the formula in [63] is ad hoc, constructed specifically for case (iv) without theoretical justification.

In the soft-label setting restricted to labels and predictions on the simplex (i.e., non-negative vectors summing to 1), Silván-Cárdenas and Wang [100] introduce SCM, a matrix with interval-valued entries. Although optimal transport is not explicitly used, their elementary matrix is closely related to ours. In particular, our approach recovers the contributions of SCM, as shown in the following proposition:

Proposition 8. *Elementary matrices from SCM framework can be computed as follows: for each entry* (i, j)*, select* $\underline{\pi}$ *and* $\overline{\pi}$ *in* $T^{opt}(\mathbf{y}, \hat{\mathbf{y}})$ *such that* $\underline{\pi}_{ij} \leq \pi_{ij} \leq \overline{\pi}_{ij}$ *for all* $\pi \in T^{opt}(\mathbf{y}, \hat{\mathbf{y}})$*, then define entry* (i, j) *as* $[\underline{\pi}_{ij}, \overline{\pi}_{ij}]$*.*

For a given label–prediction pair on the simplex, each entry of the TCM contribution lies within the corresponding SCM interval. While SCM entries (i, j) capture worst and best-case confusion scenarios, providing a pairwise description of errors, the TCM instead offers a global description that accounts for all classes simultaneously and preserves both marginal distributions and total mass. In particular, row sums match the label distribution, column sums match the prediction distribution, and the total sum equals one.

Overall, TCM provides a unified framework for confusion matrices across single-label, multi-label, and soft-label settings. It recovers key properties of existing approaches while replacing ad hoc constructions with a single formulation based on optimal transport.

5.2 Axiomatic Requirements

To extend the confusion matrix to soft classifications, Silván-Cárdenas and Wang [100] propose that an elementary matrix should satisfy two key properties. First, *diagonalization*: the matrix is diagonal if and only if the prediction is exact. Second, *marginal consistency*: the row sums match the label vector, while the column sums match the prediction vector used to construct the elementary matrix.

Diagonalization allows clear identification of perfect matches. Marginal consistency ensures that performance metrics derived from the confusion matrix, such as accuracy or F1 score, remain consistent with the label and prediction vectors, thereby improving interpretability [100].

Proposition 9. *In the TCM framework, the elementary matrices satisfy both diagonalization and marginal consistency.*

In contrast, several state-of-the-art approaches do not meet these criteria. In particular, SCM [100], due to its interval-valued entries, does not satisfy marginal consistency, whether one considers the lower bound, the upper bound, or the midpoint of the intervals.

Overall, among existing approaches, TCM is the only method that simultaneously satisfies the proposed axiomatic properties while preserving the standard confusion matrix structure, making it the most consistent and interpretable framework.

6 Case Study: EXIST

This section presents a short case study illustrating the use of TCM.

EXIST is a series of scientific events and shared tasks on sexism identification in social networks [92]. One dataset contains several thousand annotated tweets for studying sexism detection and characterization. The benchmark defines three tasks described Table III.2: (i) sexism detection, (ii) intention classification, and (iii) fine-grained sexism categorization. Each tweet is labeled by six annotators with diverse profiles to reduce bias and capture uncertainty. In the soft-label setting, labels are computed as the average of annotator votes. We analyze a model from the EXIST 2025; see [85] for details on the architecture and training.

Figure III.2 presents the TCMs for the three tasks. The main conclusions are:

- **Task 1.** The matrix is strongly diagonal, indicating high classification performance with limited confusion between classes.
- **Task 2.** Diagonal values remain dominant, confirming good overall performance. However, `Direct` and `Non-sexist` show both high diagonals and strong columns, indicating over-prediction. Rows for `Judgemental` and `Reported` have weaker diagonals and are frequently confused with `Direct` and `Non-sexist`.
- **Task 3.** The diagonal structure is preserved, and `Non-sexist` remains over-predicted. `Sexual Violence` has a weak column, indicating it is rarely predicted. Rows for `Misogyny Non-Sexual Violence` and `Sexual Violence` show weak diagonals and higher off-diagonal mass, indicating frequent errors. Notable confusions include `Misogyny Non-Sexual Violence` being predicted as `Non-sexist`, and `Sexual Violence` being confused with `Misogyny Non-Sexual Violence`, `Stereotyping Dominance`, and `Non-sexist`.

Task	Setting	Class	Definition
1	Simplex	Non-sexist	Not sexist
		Sexist	Sexist or about sexism (e.g., denouncing a sexist act)
2	Simplex	Non-sexist	Not sexist
		Direct	Explicitly sexist
		Reported	Reports a sexist situation
		Judgemental	Denounces injustice against women
3	Soft	Non-sexist	Not sexist
		Ideological Inequality	Rejects gender equality or discredits feminism
		Stereotyping Dominance	Promotes stereotypes or male superiority
		Objectification	Reduces women to objects
		Sexual Violence	Sexual harassment or sexual content
		Misogyny Non-Sexual Violence	Expresses hatred or violence toward women

Table III.2: EXIST 2025 tweet classification tasks. Simplex denotes the soft-label setting, where labels lie on the simplex (i.e., non-negative vectors summing to 1).

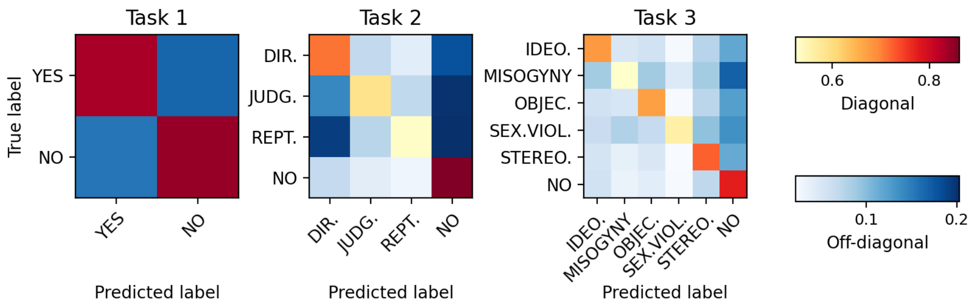


Figure III.2: Row-normalized TCMs with weights $w(\mathbf{y}, \hat{\mathbf{y}}) = 1$, illustrating model performance on the soft-label versions of Tasks 1, 2 and 3 from EXIST 2025. Since diagonal values are significantly larger than off-diagonal ones, two shared color scales are used to better highlight variations within each group.

Overall, all tasks show a clear diagonal structure, indicating good performance. However, the confusion matrices exhibit vertical stripes, reflecting under and over-prediction. Some errors are more pronounced, notably between `Reported` and `Direct`, and between `Misogyny Non-Sexual Violence` and `Non-sexist`.

7 Conclusion

This work addresses key limitations of existing confusion matrix extensions, including limited theoretical grounding, restricted applicability, and reduced interpretability.

We introduced the Transport-based Confusion Matrix (TCM), a unified framework based on optimal transport and the principle of maximum entropy. TCM extends the standard confusion matrix to multi-label and soft-label settings while remaining identical to it in the single-label case. It provides a closed-form solution, preserves the familiar matrix structure, and satisfies key consistency properties.

We also showed that existing approaches share common principles that can be recovered within the TCM framework. In this sense, TCM clarifies the relationships between previous methods while replacing ad hoc constructions with a single theoretically grounded formulation.

Overall, TCM offers a consistent and interpretable tool for analyzing model behavior across a broad range of classification settings.

Chapter IV
Benchmarking Confusion Matrices Beyond Single-Label Classification

1	Introduction		**40**
2	Related Work		**40**
	2.1	Evaluation Methods	40
	2.2	Unavailable Ground-Truth & Synthetic Evaluation	41
	2.3	Our Contribution	41
3	ConfusionBench		**41**
	3.1	Classifiers from Confusion Matrices	41
	3.2	Latent Confusion Matrices	42
	3.3	Synthetic Label-Prediction Pairs	42
	3.4	Ground-Truth Justification	44
4	Experimental Setup		**45**
	4.1	Scenarios & Hyperparameters	45
	4.2	Baselines	45
	4.3	Metric	46
5	Evaluation		**46**
	5.1	Multi-label case	46
	5.2	Soft-label case	47
6	Limitations		**47**
7	Conclusion		**48**

This chapter is based on our preprint [30].

While multi-label and soft-label settings better reflect the complexity of real-world data, they also give rise to multiple competing definitions of confusion matrices, with no principled way to compare them. Existing work typically evaluates these formulations on benchmark datasets, but does not assess whether they faithfully represent the intrinsic behavior of the underlying model.

We introduce ConfusionBench, a benchmark for evaluating confusion matrix extensions in multi-label and soft-label classification settings. The benchmark relies on stochastic classifiers with controlled latent confusion structures, providing a ground-truth confusion matrix for assessing the recovery ability of confusion matrix extensions. Our results reveal substantial differences between existing matrices that prior evaluation practices failed to expose.

Contributions. Our contributions are as follows:

- **ConfusionBench.** We introduce ConfusionBench, a benchmark to evaluate how well confusion matrix extensions recover ground-truth confusions in multi-label and soft-label settings.
- **Empirical comparison.** We provide a systematic empirical comparison of existing confusion matrix extensions across controlled scenarios varying the number of classes, the complexity of confusion patterns, and the type of label representation.

1 Introduction

In multi-label and soft-label settings, several confusion matrix extensions coexist [63, 64, 43, 51, 31]. This leads to inconsistent model behavior interpretations and provides no clear guidance on which method should be used.

Problem & motivation. Multi-label settings, where observations may belong to several classes, and soft-label settings, where observations are represented by non-negative vectors encoding degrees of membership or confidence, better capture real-world complexity [109, 78, 76, 102, 22, 77, 16, 111, 107, 19]. However, extending the confusion matrix to these settings is not straightforward.

As a result, multiple extensions of confusion matrices have been proposed. Most existing works [63, 51, 64, 43, 10, 100, 31] either visualize their matrices on benchmark datasets to interpret model behavior or compare them qualitatively with alternative formulations. However, because no ground-truth confusion matrix characterizing the intrinsic behavior of a model is available, these comparisons lack a clear quantitative criterion, making it unclear which method should be preferred in practice.

ConfusionBench. We introduce ConfusionBench, a framework for evaluating confusion matrices based on their ability to recover known confusion patterns. Our approach reverses the standard pipeline: instead of estimating confusion from a trained model, we construct stochastic classifiers with controlled, known confusion structures, and generate synthetic label–prediction pairs accordingly. This provides a ground-truth reference matrix, enabling direct and quantitative comparison between methods.

2 Related Work

2.1 Evaluation Methods

Existing works evaluate confusion matrix extensions using four main approaches.

Qualitative analysis. Matrices are visualized on benchmark datasets to interpret model behavior [63, 51, 64, 43, 10, 100, 31]. This mainly illustrates how to read the proposed representation.

Relative comparison. Methods are compared against existing formulations to highlight similarities and differences [64, 100, 31, 10, 62]. While informative, this only provides relative comparisons and does not assess absolute quality.

Metric recovery. Some works evaluate whether the proposed matrix can recover scalar performance metrics [64, 100, 31, 10].

Axiomatic validity. Some works define desirable properties that a confusion matrix extension should satisfy in order to remain consistent with the classical single-label case [100, 31]. For example, an extension may be required to reduce to the standard confusion matrix for single-label predictions, or to be diagonal under perfect predictions. Such criteria are useful for ruling out invalid formulations, but they do not quantify how well a valid method captures the underlying confusion behavior.

Algorithm 3 ConfusionBench

Require: Number of classes C, number of samples N, density parameter d, maximum number of attempts T, maximum number of active classes L, dropout probability p, maximum number of components per class S, set of confusion matrix extensions $\mathcal{A}$

Ensure: Evaluation scores for all methods in $\mathcal{A}$

```
Sample Q using Algorithm 4 with inputs (C, d, T)
Generate D = {(y_k, ŷ_k)}_{k=1}^N using Algorithm 5 with inputs (N, C, Q, L, p, S)
for each method A ∈ 𝒜 do
    Compute M_A ← A(D)
    Evaluate the agreement between row(M_A) and Q using quantitative metrics
end for
return  Evaluation scores for all methods in 𝒜
```

While useful, these approaches do not provide a ground-truth confusion matrix describing the true behavior of the model. As a result, they cannot determine which method best captures the underlying confusion structure.

2.2 Unavailable Ground-Truth & Synthetic Evaluation

The absence of ground truth is a common challenge in machine learning. To address this issue, synthetic generative settings are widely used in several areas of machine learning to provide controlled ground-truth structures for evaluation.

In causal inference, synthetic data are commonly used to evaluate the recovery of known causal structures [36, 40, 68, 83]. Similarly, synthetic benchmarks in disentanglement analysis, explainable artificial intelligence, and visual question answering [79, 59, 55, 81, 74] have been introduced to provide controlled, reproducible, and unbiased ground-truth structures, enabling precise evaluation of specific reasoning capabilities.

2.3 Our Contribution

We introduce ConfusionBench, a synthetic benchmark for evaluating confusion matrix extensions in multi-label and soft-label settings. By constructing stochastic classifiers with controlled confusion patterns, ConfusionBench provides a ground-truth reference matrix that enables direct and quantitative comparison between methods.

3 ConfusionBench

ConfusionBench evaluates confusion matrix extensions by comparing the estimated confusion matrix with a known target confusion structure. The closer a method is to the target matrix, the more faithfully it captures the intrinsic behavior of the classifier. The ConfusionBench procedure is detailed below and summarized in Algorithm 3.

3.1 Classifiers from Confusion Matrices

In the single-label setting, the normalized confusion matrix admits a well-known probabilistic interpretation [43] including:

$$\mathrm{all}(P)_{ij} = \frac{P_{ij}}{P_{++}} \approx \mathbb{P}(Y = i, \hat{Y} = j), \qquad \mathrm{row}(P)_{ij} = \frac{P_{ij}}{P_{i+}} \approx \mathbb{P}(\hat{Y} = j \mid Y = i),$$

where $P_{++} = \sum_{i,j} P_{ij}$ and $P_{i+} = \sum_j P_{ij}$. Therefore, the row-normalized confusion matrix characterizes the conditional prediction behavior of the classifier given the true class.

Building on this observation, ConfusionBench constructs stochastic classifiers whose behavior is fully determined by a row-stochastic matrix Q (i.e., $Q_{ij} \geq 0$ and

$\sum_{j=1}^{C} Q_{ij} = 1$ for all i), representing its latent confusion patterns. Given a sample (x, y) with label $y \in [C]$, the classifier produces a prediction $\hat{y} \in [C]$ sampled from the distribution defined by the y-th row of Q, that is, $\hat{y} = j$ with probability Q_{yj}.

In contrast to standard classifiers, which produce predictions based on inputs x and are typically deterministic at inference time, this synthetic classifier is stochastic and conditioned only on the labels. This construction allows us to control and know exactly the intrinsic confusion pattern of the classifier, thereby providing a target confusion matrix to recover.

3.2 Latent Confusion Matrices

To generate model behaviors with controlled confusion patterns, we construct a random row-stochastic matrix $Q \in \mathbb{R}_{\geq 0}^{C \times C}$. Each row i describes the behavior of the classifier conditional on the true class: when the true class is i, the classifier predicts class j with probability Q_{ij}.

The matrix Q is generated as follows. First, we sample $D = \lfloor dC^2 \rfloor$ independent values from $\mathrm{Unif}(0, 1)$. Second, we initialize a zero matrix in $\mathbb{R}_{\geq 0}^{C \times C}$, randomly place the C largest sampled values on the diagonal, and randomly place the remaining $D - C$ values off the diagonal. Finally, we row-normalize the matrix.

The first two steps produce confusion patterns with diagonal dominance, modeling classifiers that perform better than random guessing. The final step ensures row-stochasticity. The density parameter d controls the proportion of nonzero entries: small values produce sparse matrices, while larger values produce denser confusion patterns.

Algorithm 4 summarizes the construction of Q, and Figure IV.1 shows heatmaps of the resulting latent confusion matrices.

3.3 Synthetic Label-Prediction Pairs

A dataset of label–prediction pairs $\{(y_k, \hat{y}_k)\}_{k=1}^{N}$ is required to derive confusion matrix extensions. Algorithm 5 generates such pairs across different classification settings. We detail the procedure below, but first provide an interpretation of the generation process.

Observation-level interpretation. For simplicity, our framework generates classifier predictions $\hat{y}_k$ directly from the labels y_k, without explicitly modeling the observations x_k. Nevertheless, this procedure admits a natural interpretation.

Each observation x_k is composed of n_k components $\{q_l\}_{l=1}^{n_k}$. Each component is associated with at most one class, independently of the labeling setting considered, and may also be associated with no class. Labels y_k and predictions $\hat{y}_k$ are obtained by independently processing these components and aggregating the resulting class information.

The nature of the components and the aggregation rule depend on the classification setting. In single-label classification, each observation contains a single component, i.e., $n_k = 1$ for all k. This component represents an object. For instance, it may correspond to a `Dog`. Aggregation then simply activates the predicted class. In multi-label classification, components also represent objects, but their number may vary across observations and be larger than one, i.e., $n_k \geq 1$. Aggregation activates every class predicted by at least one component. In soft-label classification, observations are decomposed into many components representing class patterns, such as tokens associated with `Physics` topics. Class membership is measured by the number of components assigned to each class.

Regardless of the classification setting, the classifier processes these components independently and assigns each of them either to a class or to no class. Three cases may occur:

- **Labeled-predicted.** The component belongs to class i. The classifier detects it and predicts class j with probability Q_{ij}.

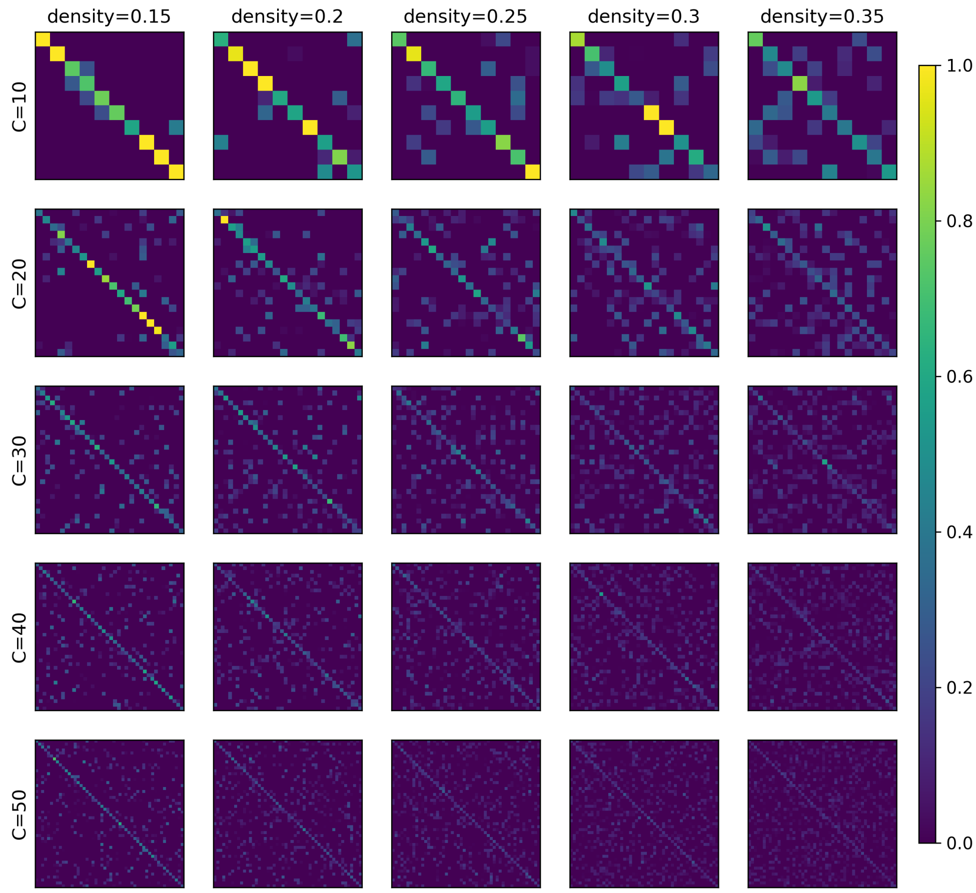


Figure IV.1: Latent confusion matrices (run 0) across different numbers of classes, ranging from simple confusion patterns (Max samples = 1) to more complex ones (Max samples = 4), and from concentrated patterns ($\alpha = 0.1$) to more dispersed ones ($\alpha = 0.8$).

- **Unlabeled-predicted.** The component is assigned to no class, but the classifier interprets it as close to class i. It then predicts a class according to the latent confusion pattern $Q_{i,:}$.
- **Labeled-unpredicted.** The component belongs to a class, but it is not detected and therefore triggers no prediction.

The unlabeled-unpredicted case is ignored, since it has no effect on either the label or the prediction. We also exclude void labels and void predictions: at least one component must belong to a class, and at least one component must be assigned to a class by the classifier. In the single-label setting, there is only one component, so these assumptions ensure that this component is assigned to a class.

This interpretation is formalized in Algorithm 5, even though the observation x_k is not explicitly modeled. Labels are assumed to be directly induced by the observation. Prediction generation only requires the amount of components associated with each class, which is encoded in the label. No-class components are not explicitly represented, but are mimicked through dropout: removing label classes mimics the unlabeled-predicted case, while removing prediction classes mimics the labeled-unpredicted case. Finally, Algorithm 5 keeps only nonzero pairs: the label must contain at least one active class, and the model must predict at least one class.

Single-label dataset. The single-label setting is obtained from Algorithm 5 by setting $L = S = 1$. Each sample therefore contains one active class and one component. We initialize y_k and $\hat{y}_k$ as zero vectors, select a class i, set $[y_k]_i = 1$, sample j from $Q_{i,:}$, and set $[\hat{y}_k]_j = 1$.

Multi-label dataset. The multi-label setting is obtained by setting $L > 1$ and $S = 1$. Each sample may contain several active classes, with one component per active class. After initializing y_k and $\hat{y}_k$, we select l active classes in y_k. For each active class i, we sample a predicted class j from $Q_{i,:}$ and activate j in $\hat{y}_k$.

At this stage, labels and predictions contain the same number of active classes, which is unrealistic in practice. We therefore introduce a dropout mechanism that randomly removes active labels or predictions, modeling missing labels and missing predictions.

Soft-label dataset. The soft-label setting is obtained by setting $L, S > 1$. Each active class may then be supported by several components. For each active class i, we sample an integer $m_i \in \{1, \dots, S\}$, representing the number of components supporting class i, and set

$$[y_k]_i \leftarrow m_i.$$

Each component is then predicted independently according to $Q_{i,:}$. Thus, for $r = 1, \dots, m_i$, we sample $j \sim Q_{i,:}$ and update

$$[\hat{y}_k]_j \leftarrow [\hat{y}_k]_j + 1.$$

Repeating this process over all active classes yields soft-label vectors $(y_k, \hat{y}_k)$, whose entries count class-level evidence.

As in the multi-label case, dropout is applied to create differences in support and total mass between labels and predictions.

3.4 Ground-Truth Justification

This subsection gives two independent arguments for why any row-normalized confusion matrix extension should be consistent with the latent matrix Q.

Consistency across classification settings. The stochastic classifier defined by Q has the same confusion pattern regardless of the classification setting. Therefore, a confusion matrix extension should recover the same matrix across single-label, multi-label, and soft-label settings.

Let $\{(y_k, \hat{y}_k)\}_{k=1}^{N}$ be a dataset generated in the single-label setting by Algorithm 5, and let M be the standard confusion matrix computed from this dataset. Since predictions are sampled from the rows of Q, the row-normalized confusion matrix converges almost surely to Q:

$$\mathrm{row}(M) \underset{N \to \infty}{\longrightarrow} Q \quad \text{almost surely.}$$

Moreover, valid confusion matrix extensions reduce to the standard confusion matrix in the single-label setting. Hence, they are consistent estimators of Q in this regime.

This yields a natural requirement: confusion matrix extensions recover Q in the single-label setting. Since the underlying classifier confusion pattern does not change when moving to other classification settings, they should also recover Q in those settings.

Beyond co-occurrence. The single-label confusion matrix M estimates label–prediction co-occurrence:

$$\mathrm{all}(M)_{ij} \underset{N \to \infty}{\longrightarrow} \mathbb{P}(Y = i, \hat{Y} = j) \quad \text{almost surely.}$$

However, we view a confusion matrix as aiming to capture a more specific relation: class i is present in the observation and is recognized as class j. The target is therefore

$$\mathbb{P}(\text{class } i \text{ is recognized as class } j).$$

This relation is stricter than co-occurrence. For example, suppose that x contains a `Dog` component that produces no prediction, and a no-class component predicted as `Wolf`. Then the label `Dog` and the prediction `Wolf` co-occur, but the model has not confused `Dog` with `Wolf`.

In real data, this component-level relation is not directly observable. In our synthetic framework, it is available because the decision process is known. We can therefore define an ideal matrix $\hat{Q}$: initialize $\hat{Q}$ to zero, and increment $\hat{Q}_{ij}$ each time a component of class i is predicted as class j. Components with no class, or predictions with no class, do not update the matrix, since they do not encode a class-to-class confusion.

After row-normalization, this ideal matrix converges independently of the classification setting:

$$\mathrm{row}(\hat{Q})_{ij} \underset{N\to\infty}{\longrightarrow} Q_{ij} = \mathbb{P}(\text{class } i \text{ is recognized as class } j \mid i \text{ is present}) \quad \text{almost surely.}$$

Thus, any confusion matrix extension that aims to capture this confusion relation should recover Q after row-normalization.

In the single-label setting of our framework, each observation contains one labeled component and produces one predicted class. Hence, component-level recognition coincides with label–prediction co-occurrence. The standard confusion matrix therefore estimates the intended quantity in this setting.

Based on these arguments, consistency with Q is a desirable property for any confusion matrix extension M. The closer $\mathrm{row}(M)$ is to Q, the more faithfully M captures the underlying classifier behavior.

4 Experimental Setup

4.1 Scenarios & Hyperparameters

We evaluate both the multi-label and soft-label settings over a grid of synthetic scenarios. The latent confusion matrix is generated with density $d \in \{0.15, 0.20, 0.25, 0.30\}$, ranging from sparser to denser confusion patterns. We additionally vary the dropout probability, with $p \in \{0, 0.1, 0.2, 0.3, 0.4\}$, to control the amount of missing labels or predictions. Finally, we vary the number of classes, with $C \in \{10, 20, 30, 40, 50\}$.

For each configuration, we generate $N = 15000$ label–prediction pairs and estimate the corresponding confusion matrices. The number of active classes per label vector is limited to at most 10. Each experiment is repeated over 5 random runs.

4.2 Baselines

We consider confusion matrices satisfying the following requirements: being square matrices of size C and being diagonal under perfect predictions.

We compare the following methods. The Multi-Label Confusion Matrix (MLCM) [63], and the two confusion tensors proposed by Krstinić et al. [64]: the Multi-Label Confusion Tensor Recall ($\mathrm{MLCT}^{\mathrm{r}}$) and Precision ($\mathrm{MLCT}^{\mathrm{p}}$).

The Sub-pixel Confusion Matrix (SCM) [100] is designed for label–observation pairs lying on the simplex and produces interval-valued matrices. Following the authors' representations, we consider three variants: $\mathrm{SCM}^{\mathrm{min}}$ (lower bound), $\mathrm{SCM}^{\mathrm{mid}}$ (midpoint), and $\mathrm{SCM}^{\mathrm{max}}$ (upper bound). We additionally evaluate SCM using normalized labels and normalized predictions.

The Transport-based Confusion Matrix (TCM) [31] includes a weighting factor w that must be specified according to the application context. We consider two variants: a constant weighting factor $w : (y, \hat{y}) \mapsto 1$, denoted $\mathrm{TCM}^{\mathrm{one}}$, and an inverse-label weighting factor $w : (y, \hat{y}) \mapsto 1/|y|$, denoted $\mathrm{TCM}^{\text{inv-lab}}$. In $\mathrm{TCM}^{\mathrm{one}}$, all label–prediction pairs contribute equally. In $\mathrm{TCM}^{\text{inv-lab}}$, pairs associated with labels containing more classes contribute less to the final matrix.

In the multi-label setting, we compare all aforementioned methods. In the soft-label setting, we only compare SCM and TCM variants, as the other methods are not designed for soft labels.

4.3 Metric

To quantify the agreement between estimated and latent confusion matrices, we consider the overlap metric [34], with a variant in which M is first row-normalized to match the row-stochastic structure of Q:

$$\text{Overlap}(M) = \sum_{i,j} \min\left(\text{all} \circ \text{row}(M)_{ij}, \text{all}(Q)_{ij}\right) \in [0, 1].$$

The overlap quantifies the amount of shared probability mass between the distributions $\text{all} \circ \text{row}(M)$ and $\text{all}(Q)$.

We additionally report the average row-wise squared ℓ_2 distance between Q and $\text{row}(M)$:

$$\frac{1}{C} \sum_i \sum_j \left(\text{row}(M)_{ij} - Q_{ij}\right)^2$$

It complements the overlap by penalizing large discrepancies in probability mass, rather than only measuring the amount of shared mass between the two distributions. Lower values indicate better agreement between the estimated and latent confusion matrices.

Finally, we report the Jensen–Shannon divergence with logarithm in base 2:

$$\text{JSD}\left(\text{all} \circ \text{row}(M), \text{all}(Q)\right) \in [0, 1].$$

This divergence has an information-theoretic interpretation: it measures the inefficiency incurred when the two distributions are represented by their common mixture [29]. Unlike the Kullback–Leibler divergence, it is symmetric and bounded. Lower values indicate better agreement.

5 Evaluation

5.1 Multi-label case

Figures IV.2, IV.3, and IV.4 report the results in the multi-label setting. We compare the estimated matrices with the latent matrix Q using overlap, Jensen–Shannon divergence, and the average row-wise squared ℓ_2 distance.

Overall, $\text{TCM}^{\text{inv-lab}}$ gives the best recovery of the latent confusion matrix. It obtains the highest overlap and the lowest Jensen–Shannon divergence in almost all scenarios. It also remains among the best methods for the squared ℓ_2 distance. This is consistent with the data generation process: the classifier behavior is defined independently of the number of active classes in the label, while labels with fewer active classes induce simpler confusion patterns that are easier to recover. As a result, weighting each label-prediction pair by the inverse number of active labels provides a more faithful representation of the classifier behavior.

TCM^{one} is also competitive. It is usually worse than $\text{TCM}^{\text{inv-lab}}$, but it remains better than most non-transport methods across the three metrics. This shows that the transport formulation already captures the row-wise structure of Q well, even without the inverse-label correction.

The SCM variants have more mixed behavior. SCM^{mid} is usually the strongest SCM variant according to overlap and Jensen–Shannon divergence. However, the squared ℓ_2 distance reveals a limitation of SCM^{min}: it can have acceptable overlap or divergence while still producing large row-wise errors. This suggests that SCM^{min} may recover part of the shared mass, but distributes this mass less accurately within rows than the other methods.

The multi-label-specific methods, MLCM, MLCT^r, and MLCT^p, are less competitive. They generally obtain lower overlap and higher Jensen–Shannon divergence than the transport-based methods. Thus, although they are designed for multi-label classification, they do not recover the latent row-stochastic confusion behavior as accurately in this benchmark.

The number of classes has a clear effect on recovery quality. As C increases, overlap decreases, while Jensen–Shannon divergence and the squared ℓ_2 distance increase. This is expected: for a fixed number of samples, increasing C spreads the same information over a larger matrix, making the latent confusion structure harder to recover. Increasing the density d and the dropout probability also makes the task harder, but their effect is weaker than that of C. Overall, the ranking of the methods remains stable across scenarios.

5.2 Soft-label case

Figures IV.5, IV.6, and IV.7 report the results in the soft-label setting. Since MLCM and MLCT are not designed for soft labels, we only compare TCM and SCM variants.

The conclusions are similar to the multi-label case, but the differences between methods are clearer. $\text{TCM}^{\text{inv-lab}}$ consistently gives the best recovery of Q. It achieves the highest overlap, the lowest Jensen–Shannon divergence, and one of the lowest squared ℓ_2 distances across densities, dropout levels, and class cardinalities.

TCM^{one} is again the second most reliable method overall. It does not match $\text{TCM}^{\text{inv-lab}}$, but it clearly improves over the SCM variants on Jensen–Shannon divergence and squared ℓ_2 distance. This confirms that the transport-based formulation extends naturally to soft-label data.

The SCM variants are less stable. SCM^{mid} is generally the best SCM variant for overlap, while SCM^{max} often gives the largest Jensen–Shannon divergence. The squared ℓ_2 distance again highlights a strong failure mode of SCM^{min}, which produces much larger row-wise errors than the other methods.

As in the multi-label setting, increasing the number of classes, density, or dropout makes recovery harder. Nevertheless, the ranking of the methods remains stable. These results show that ConfusionBench distinguishes methods that all may otherwise all appear valid according to standard axiomatic criteria.

6 Limitations

ConfusionBench has several limitations, which we discuss below.

Synthetic data. ConfusionBench relies on synthetic data rather than real-world datasets and predictions produced by trained models. This design choice makes the latent confusion structure observable, thereby providing a ground truth for evaluating confusion matrix extensions.

Specific classifier. ConfusionBench classifiers are not standard classifiers. They are stochastic and untrained, and they decompose observations into independent components, ignoring co-occurrence relations between classes. However, they still capture a basic classifier behavior: the true class is more likely to be recognized than confused with other classes.

They are therefore valid classifiers for our purpose. Confusion matrices are computed only from label–prediction pairs and are classifier-agnostic. In principle, they should reflect the behavior of any classifier, including this synthetic one.

Moreover, because ConfusionBench classifiers treat classes independently and ignore complex inter-class dependencies, their ground-truth confusion structures are likely easier to recover than those of real trained models. The benchmark should therefore be seen as a favorable controlled setting.

Component assumptions. Our data generation process implicitly assumes that observations can be decomposed into components (see the observation-level interpretation). The nature of these components depends on the label setting. Components are assumed to belong to at most one class, to have equal importance, and to be unique in the single-label setting. These assumptions are not realistic.

Real-world observations can be discretized, but their underlying components are not determined by the classification setting. A component may be associated with several classes, components may have different importance, and observations usually contain many components regardless of the label setting.

This design choice is nevertheless useful: it provides a common component-level description across single-label, multi-label, and soft-label classification. It allows us to decompose classifier behavior in the same way across settings, which is the key requirement for defining an observable ground-truth confusion structure.

Overall, the results reported in ConfusionBench should be interpreted as a controlled analysis of confusion matrix extensions, not as a universal ranking of these methods. Under different assumptions, the ranking of methods may differ.

7 Conclusion

We introduced ConfusionBench, a synthetic benchmark for evaluating confusion matrix extensions beyond the single-label setting. Instead of estimating confusion from an unknown classifier, ConfusionBench constructs stochastic classifiers with known latent confusion structures, enabling direct quantitative evaluation of how accurately generalized confusion matrices recover the underlying confusion behavior.

Our experiments in both multi-label and soft-label settings show that existing confusion matrix extensions differ substantially in their ability to recover the latent confusion behavior. Across a wide range of controlled scenarios, the transport-based confusion matrices are the most faithful to the underlying structure, with the inverse-label weighting variant usually obtaining the best agreement. SCM-based methods provide a reasonable but less accurate alternative, while multi-label specific methods such as MLCM and MLCT are less effective.

These findings highlight the importance of evaluating confusion matrices not only through qualitative visualizations, metric recovery, or axiomatic consistency, but also through their faithfulness to the intrinsic behavior of the classifier. ConfusionBench provides a principled way to perform this evaluation by making the target confusion structure observable.

Algorithm 4 Sampling a Latent Confusion Matrix

Require: Number of classes C, density parameter $d \in [1/C, 1]$

Ensure: A row-stochastic matrix $Q \in \mathbb{R}_{\geq 0}^{C \times C}$ with diagonal dominance

Set the number of nonzero entries:

$$D \leftarrow \lfloor dC^2 \rfloor$$

Sample D independent values:

$$v_1, \ldots, v_D \sim \mathrm{Unif}(0, 1)$$

Sort the values in decreasing order:

$$v_{(1)} \geq v_{(2)} \geq \cdots \geq v_{(D)}$$

Initialize

$$M \leftarrow \mathbf{0}_{C \times C}$$

//Place the largest values on the diagonal

Sample a random ordering $i_1, \ldots, i_C$ of $\{1, \ldots, C\}$

for $k = 1$ **to** C **do**

  Set

$$M_{i_k i_k} \leftarrow v_{(k)}$$

end for

//Place the remaining values off the diagonal

Sample an ordered list of $D - C$ distinct positions uniformly without replacement from

$$\{(i, j) \in \{1, \ldots, C\}^2 : i \neq j\}$$

Let

$$(i_{C+1}, j_{C+1}), \ldots, (i_D, j_D)$$

be these sampled off-diagonal positions

for $k = C + 1$ **to** D **do**

  Set

$$M_{i_k j_k} \leftarrow v_{(k)}$$

end for

//Normalize rows to obtain a stochastic matrix

Set

$$Q \leftarrow \mathrm{row}(M)$$

return Q

Algorithm 5 Synthetic Data Generation

Require: Number of samples N, number of classes C, latent confusion matrix $Q \in \mathbb{R}_{\geq 0}^{C \times C}$, maximum number of active classes L, dropout probability p, maximum number of components per class S
Ensure: Soft-label count pairs $\{(y_k, \hat{y}_k)\}_{k=1}^{N}$, with $y_k, \hat{y}_k \in \mathbb{N}_0^C$

Initialize $k \leftarrow 1$
while $k \leq N$ **do**
// Initialize a candidate label–prediction pair
Initialize $y \leftarrow \mathbf{0}_C$ and $\hat{y} \leftarrow \mathbf{0}_C$
// Sample the active true classes in the observation
Sample the number of active true classes:

$$\ell \sim \text{Unif}\{1, \dots, L\}$$

Sample a set $A \subseteq \{1, \dots, C\}$ of ℓ distinct classes uniformly without replacement
// Generate the components
for each $i \in A$ **do**
Sample the number of components supporting class i:

$$m_i \sim \text{Unif}\{1, \dots, S\}$$

Set

$$[y]_i \leftarrow m_i$$

end for
// Generate component predictions from the latent confusion matrix
for each $i \in A$ **do**
for $r = 1$ **to** $[y]_i$ **do**
Sample a predicted class:

$$j \sim \text{Categorical}(Q_{i,:})$$

Update the prediction:

$$[\hat{y}]_j \leftarrow \max([\hat{y}]_j + 1, S)$$

end for
end for
// Apply dropout to create missing labels or missing predictions
Sample $u \sim \text{Unif}(0, 1)$
if $u < 0.5$ **then**
// Remove some label components
for each i such that $[y]_i > 0$ **do**
Sample the number of dropped label components:

$$b_i \sim \text{Binomial}([y]_i, p)$$

Remove them from the label:

$$[y]_i \leftarrow [y]_i - b_i$$

end for
else
// Remove some prediction components
for each j such that $[\hat{y}]_j > 0$ **do**
Sample the number of dropped prediction components:

$$b_j \sim \text{Binomial}([\hat{y}]_j, p)$$

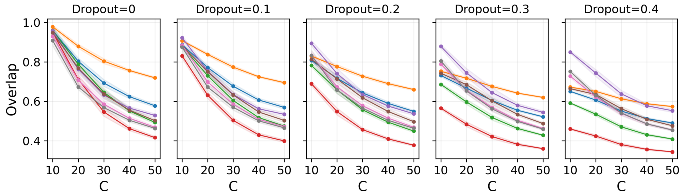
Remove them from the prediction:

$$[\hat{y}]_j \leftarrow [\hat{y}]_j - b_j$$

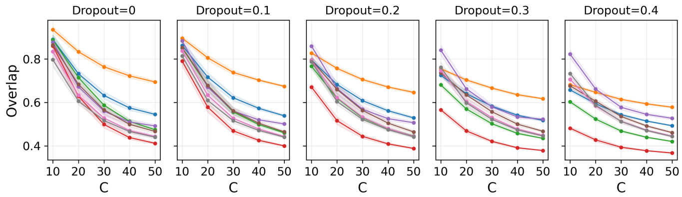
end for
end if
// Keep only valid pairs with at least one label and one prediction
if $y \neq \mathbf{0}_C$ and $\hat{y} \neq \mathbf{0}_C$ **then**
Accept the pair:

$$(y_k, \hat{y}_k) \leftarrow (y, \hat{y})$$

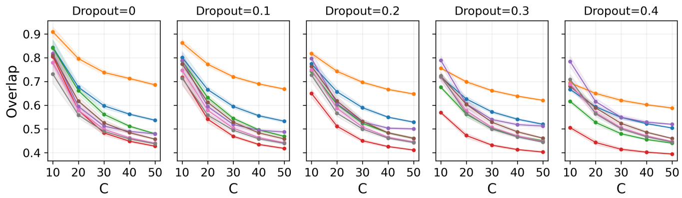
$k \leftarrow k + 1$
end if
end while
return $\{(y_k, \hat{y}_k)\}_{k=1}^{N}$



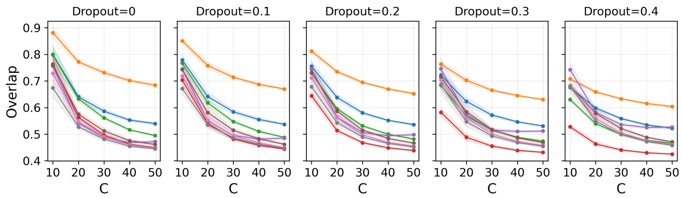
(a) $d = 0.15$



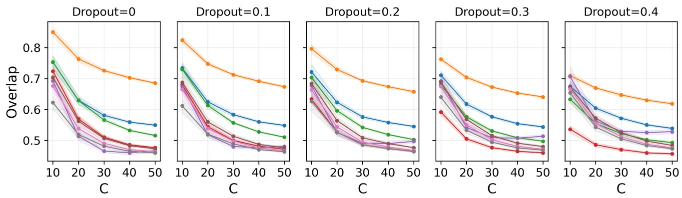
(b) $d = 0.2$



(c) $d = 0.25$

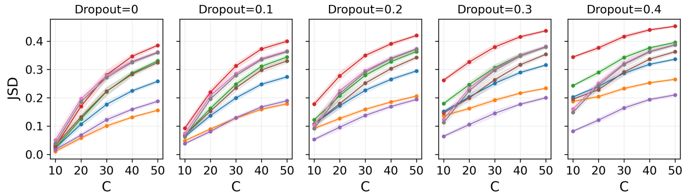


(d) $d = 0.3$

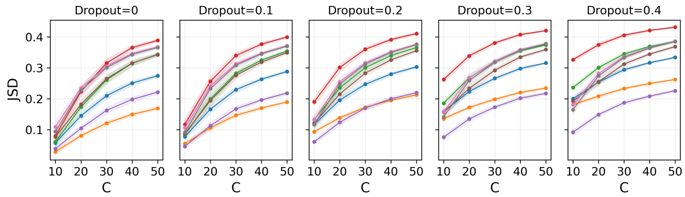


(e) $d = 0.35$

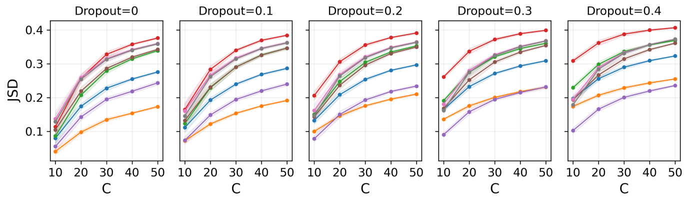
Figure IV.2: **Overlap** in the **multi-label** setting between the latent ground-truth confusion matrix and the empirical confusion matrices. The density parameter varies from sparse ($d = 0.15$) to denser ($d = 0.35$) confusion patterns. The dropout parameter models total-mass discrepancy between labels and predictions, from 0 (same total mass) to 0.4 (high discrepancy). The number of classes ranges from 10 to 50. Results are averaged over 15 seeds, with shaded regions indicating ± one standard deviation. Legend: • TCM^{one} • $\text{TCM}^{\text{inv-lab}}$ • SCM^{mid} • SCM^{max} • SCM^{min} • MLCM • MLCT^{r} • MLCT^{p}



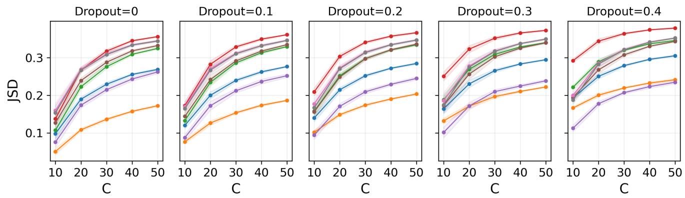
(a) $d = 0.15$



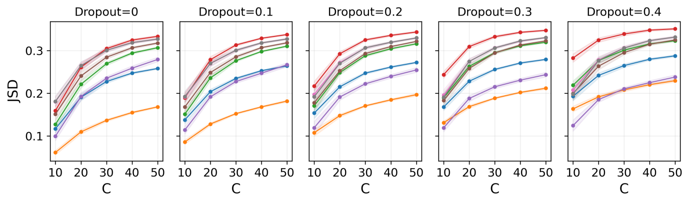
(b) $d = 0.2$



(c) $d = 0.25$

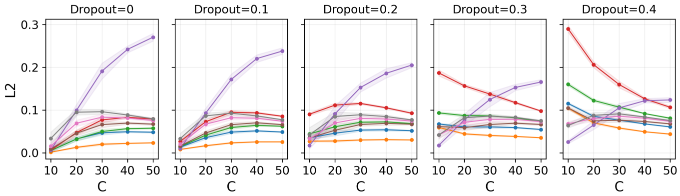


(d) $d = 0.3$

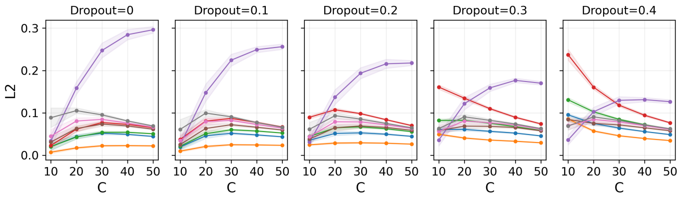


(e) $d = 0.35$

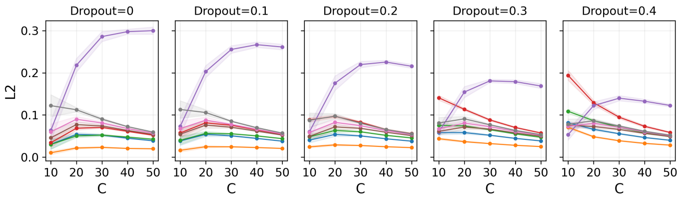
Figure IV.3: **Jensen–Shannon divergence** in the **multi-label** setting between the latent ground-truth confusion matrix and the empirical confusion matrices. The density parameter varies from sparse ($d = 0.15$) to denser ($d = 0.35$) confusion patterns. The dropout parameter models total-mass discrepancy between labels and predictions, from 0 (same total mass) to 0.4 (high discrepancy). The number of classes ranges from 10 to 50. Results are averaged over 15 seeds, with shaded regions indicating ± one standard deviation. Legend: • $\mathrm{TCM}^{\mathrm{one}}$ • $\mathrm{TCM}^{\mathrm{inv\text{-}lab}}$ • $\mathrm{SCM}^{\mathrm{mid}}$ • $\mathrm{SCM}^{\mathrm{max}}$ • $\mathrm{SCM}^{\mathrm{min}}$



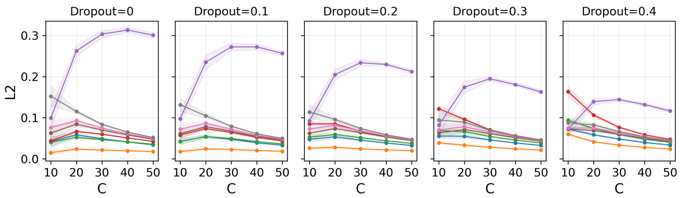
(a) $d = 0.15$



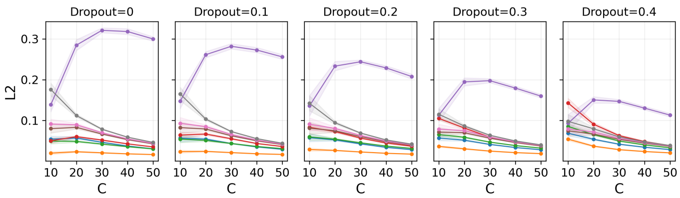
(b) $d = 0.2$



(c) $d = 0.25$

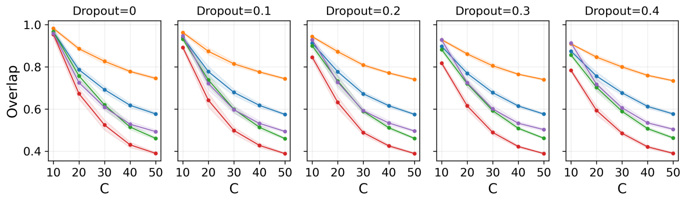


(d) $d = 0.3$

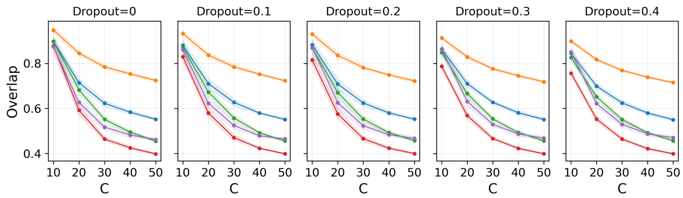


(e) $d = 0.35$

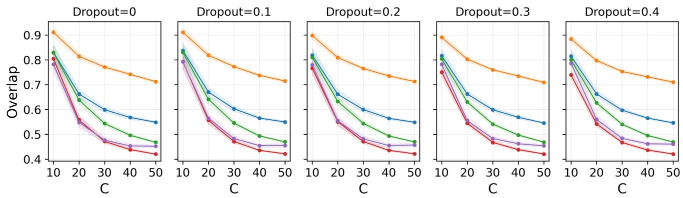
Figure IV.4: **Squared ℓ_2 distance** in the **multi-label** setting between the latent ground-truth confusion matrix and the empirical confusion matrices. The density parameter varies from sparse ($d = 0.15$) to denser ($d = 0.35$) confusion patterns. The dropout parameter models total-mass discrepancy between labels and predictions, from 0 (same total mass) to 0.4 (high discrepancy). The number of classes ranges from 10 to 50. Results are averaged over 15 seeds, with shaded regions indicating $\pm$ one standard deviation. Legend: • TCM$^{\text{one}}$ • TCM$^{\text{inv-lab}}$ • SCM$^{\text{mid}}$ • SCM$^{\text{max}}$ • SCM$^{\text{min}}$ • MLCM • MLCT$^{\text{r}}$ • MLCT$^{\text{p}}$



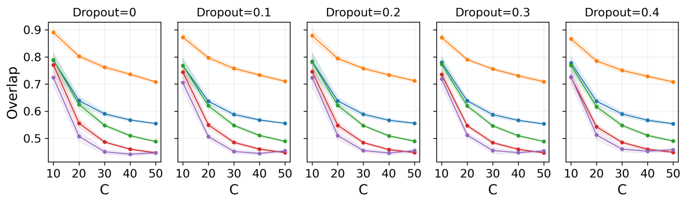
(a) $d = 0.15$



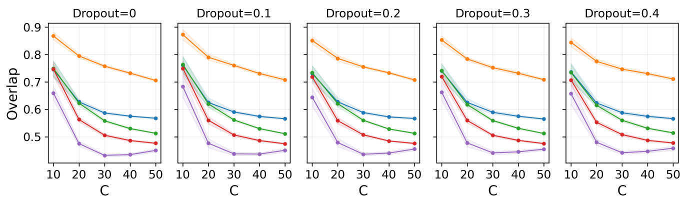
(b) $d = 0.2$



(c) $d = 0.25$

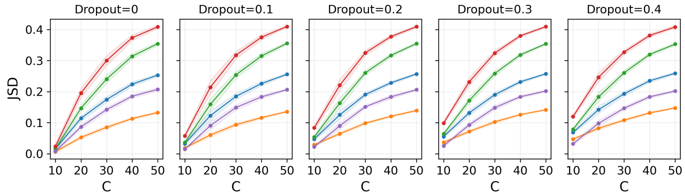


(d) $d = 0.3$

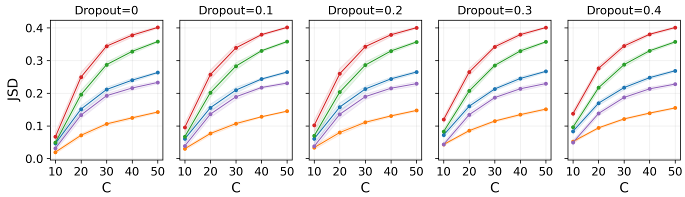


(e) $d = 0.35$

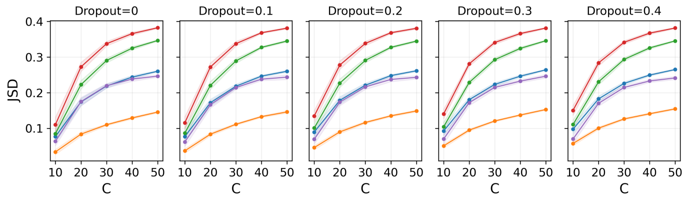
Figure IV.5: **Overlap** in the **soft-label** setting between the latent ground-truth confusion matrix and the empirical confusion matrices. The density parameter varies from sparse ($d = 0.15$) to denser ($d = 0.35$) confusion patterns. The dropout parameter models total-mass discrepancy between labels and predictions, from 0 (same total mass) to 0.4 (high discrepancy). The number of classes ranges from 10 to 50. Results are averaged over 15 seeds, with shaded regions indicating ± one standard deviation. Legend: • $\mathrm{TCM}^{\mathrm{one}}$ • $\mathrm{TCM}^{\mathrm{inv\text{-}lab}}$ • $\mathrm{SCM}^{\mathrm{mid}}$ • $\mathrm{SCM}^{\mathrm{max}}$ • $\mathrm{SCM}^{\mathrm{min}}$



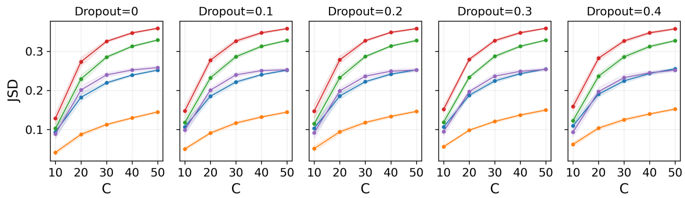
(a) $d = 0.15$



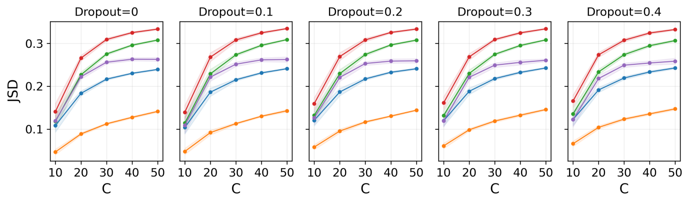
(b) $d = 0.2$



(c) $d = 0.25$

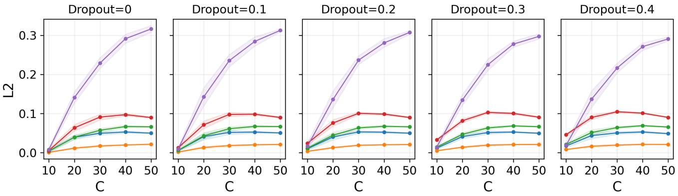


(d) $d = 0.3$

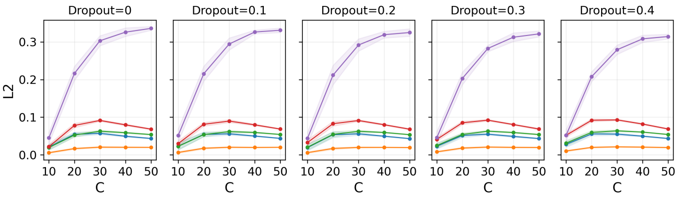


(e) $d = 0.35$

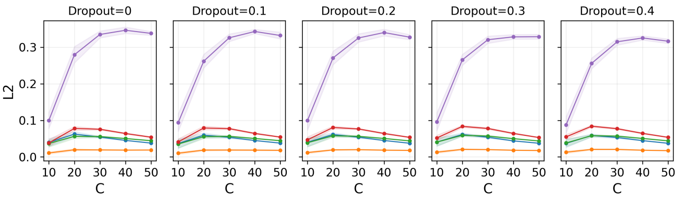
Figure IV.6: **Jensen–Shannon divergence** in the **soft-label** setting between the latent ground-truth confusion matrix and the empirical confusion matrices. The density parameter varies from sparse ($d = 0.15$) to denser ($d = 0.35$) confusion patterns. The dropout parameter models total-mass discrepancy between labels and predictions, from 0 (same total mass) to 0.4 (high discrepancy). The number of classes ranges from 10 to 50. Results are averaged over 15 seeds, with shaded regions indicating ± one standard deviation. Legend: ● $\mathrm{TCM}^{\mathrm{one}}$ ● $\mathrm{TCM}^{\mathrm{inv\text{-}lab}}$ ● $\mathrm{SCM}^{\mathrm{mid}}$ ● $\mathrm{SCM}^{\mathrm{max}}$ ● $\mathrm{SCM}^{\mathrm{min}}$



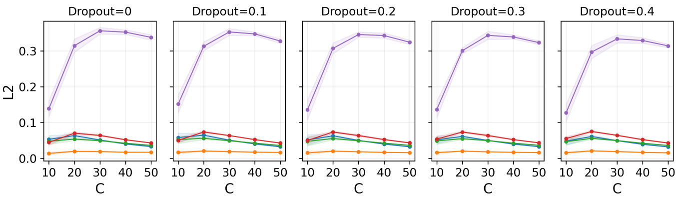
(a) $d = 0.15$



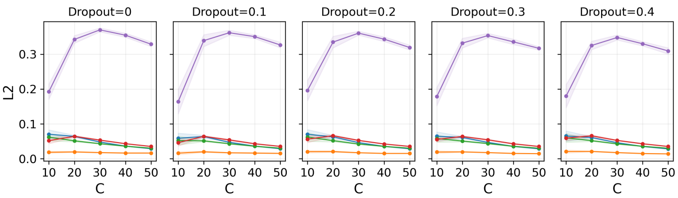
(b) $d = 0.2$



(c) $d = 0.25$



(d) $d = 0.3$



(e) $d = 0.35$

Figure IV.7: **Squared ℓ_2 distance** in the **soft-label** setting between the latent ground-truth confusion matrix and the empirical confusion matrices. The density parameter varies from sparse ($d = 0.15$) to denser ($d = 0.35$) confusion patterns. The dropout parameter models total-mass discrepancy between labels and predictions, from 0 (same total mass) to 0.4 (high discrepancy). The number of classes ranges from 10 to 50. Results are averaged over 15 seeds, with shaded regions indicating ± one standard deviation. Legend: ● TCM$^{\text{one}}$ ● TCM$^{\text{inv-lab}}$ ● SCM$^{\text{mid}}$ ● SCM$^{\text{max}}$ ● SCM$^{\text{min}}$

Part II

Byzantine-Robust Federated Learning under Label Skew

Chapter V
Preliminaries

1	Overview		**59**
2	Background		**60**
	2.1	Federated Learning Setting	60
	2.2	Threat Model	60
	2.3	Data & Label-Skew Model	60
3	Related Work		**60**
	3.1	Robustness Strategies with Homogeneous Data	60
	3.2	Robustness Strategies Under Data Heterogeneity	60
4	Manuscript organization		**61**

1 Overview

Federated learning (FL) enables a set of distributed workers to train a shared model while keeping their raw data local [45, 44]. At each round, a trusted server broadcasts the current global model. Each worker computes an update from its local data and sends it back to the server, which aggregates the received updates to form the next global model. We focus on the synchronous gradient-based setting, where honest worker updates are stochastic gradients computed on local minibatches.

Under ideal conditions, the server aggregates honest gradients by averaging. However, some workers may be faulty or malicious and may send arbitrary corrupted updates, called Byzantine updates. Byzantine-robust training aims to limit the influence of such workers [44, 45]. A standard strategy replaces averaging with a robust aggregation rule [45, 44]. These rules are designed to reduce the influence of abnormal updates, but their guarantees often rely on honest gradients being sufficiently concentrated around a common direction [44, 4, 90, 7]. This assumption often fails under data heterogeneity, a central challenge in federated learning [4, 56, 37].

Data heterogeneity increases the dispersion of honest gradients, making Byzantine updates harder to distinguish from legitimate ones [4, 45, 56]. This work focuses on label skew, a common form of heterogeneity where workers have different label distributions but share the same class-conditional feature distributions. For example, in a federated medical project, hospitals may observe different disease prevalences because of their specialization. One hospital may treat many pneumonia cases, while another rarely does. This corresponds to label skew when $P_i(Y)$ varies across hospitals, while $P_i(X \mid Y = c)$ remains identical across hospitals for every class c. Under label skew, each worker optimizes a local objective biased toward its own label distribution [56, 121, 7]. As a result, honest gradients can differ substantially even in the absence of Byzantine behavior.

2 Background

2.1 Federated Learning Setting

We consider a federated learning setting with n workers and a trusted central server. The goal is to train a parametrized classifier Φ_θ over C classes without centralizing the workers' data. Each honest worker i owns a local dataset $\mathcal{D}_i$ of observation–label pairs $(x, y) \in \mathcal{X} \times [C]$, where $\mathcal{X}$ is the input space and $[C] = \{1, \dots, C\}$.

At each communication round, the server broadcasts the current parameter vector θ_t to the workers. Each worker computes a local update from its data, typically a stochastic gradient estimated on a mini-batch, and sends it back to the server. The server aggregates the received updates to produce the next global model.

2.2 Threat Model

We study Byzantine-robust federated learning with $n = h + f$ workers, where h workers are honest and $f < h$ may be Byzantine. The set of honest workers is denoted by $\mathcal{H} \subset [n]$. The server is assumed to be trusted.

Byzantine workers may send arbitrary updates to the server. Following the standard threat model in [4, 56, 7], we assume that Byzantine workers may have full knowledge of the honest workers' updates. The objective of the server is therefore to compute an aggregate update close to the one that would have been obtained from honest workers only.

2.3 Data & Label-Skew Model

Gathering all local datasets $\mathcal{D}_i$, let $\mathcal{D} = \cup_{i \in \mathcal{H}} \mathcal{D}_i = \{(x_k, y_k)\}_{k=1}^{N}$ denote the global training set of size N. The dataset $\mathcal{D}$ contains only samples from honest workers. Although Byzantine workers may also possess data, these samples are excluded as they are not used in our analysis.

The label-skew setting is defined by two key assumptions. First, the label distributions vary across workers, meaning that the class proportions in the local dataset $\mathcal{D}_i$ may differ from one worker to another. Second, for any class $c \in [C]$, the conditional feature distribution is shared across all workers. In other words, samples belonging to class c are drawn from a common distribution L^c, independently of the worker that holds them.

3 Related Work

3.1 Robustness Strategies with Homogeneous Data

Robust aggregation methods mitigate the impact of Byzantine updates during training [91, 45]. Examples include coordinate-wise median (CWMed) [118], coordinate-wise trimmed mean (CwTM) [118], Krum [11], Minimum Diameter Averaging [46], and Comparative Gradient Elimination [48]. These methods strongly rely on the assumption that honest workers have homogeneous data. Their robustness under data heterogeneity remains unclear [44, 4, 90].

Beyond robust aggregators, additional strategies enhance robustness in homogeneous settings by leveraging a clean dataset held by the server to evaluate and filter worker updates [116, 14, 38, 94].

3.2 Robustness Strategies Under Data Heterogeneity

To reduce gradient variance caused by data heterogeneity, a common approach applies a pre-aggregation step before the main aggregation rule, leading to a two-step process. Examples include Bucketing [56], and Nearest-Neighbor Mixing (NNM) [4]. For instance, when combining Bucketing with CWMed, the server first applies Bucketing to the received gradients to reduce heterogeneity, then applies CWMed to produce the final robust update.

Bucketing [56] randomly groups inputs, averages each group, and feeds these averages to the aggregation rule. This reduces the variance of honest updates in expectation, but in some iterations it may fail to reduce heterogeneity, allowing Byzantine workers to disrupt training [44]. In contrast, NNM [4] deterministically replaces each gradient with the average of its $n - f$ nearest neighbors. Although it provides optimal probabilistic robustness guarantees, its high computational cost limits scalability in large distributed systems [44].

Other strategies adopt filtering mechanisms that remove suspected Byzantine clients prior to aggregation, as in FedREDefense [117] and Gradient Splitting (GAS) [75]. FedREDefense [117] checks whether a client update is consistent with a genuine training process by reconstructing it through training on optimized synthetic data. However, updates obtained from any valid training procedure can bypass this detection, for instance in label-flipping attacks [117]. GAS [75] mitigates the curse of dimensionality by decomposing high-dimensional gradients into low-dimensional subvectors and applying a robust aggregation rule to compute reliability scores for each client. Gradients with high scores are identified as honest and aggregated to form the global update, which also alleviates gradient heterogeneity across clients.

Finally, some strategies rely on access to a server-side dataset, such as BOBA [7] and RAGE [24], an assumption that may be unrealistic in many federated learning scenarios.

4 Manuscript organization

Chapter VI proposes a lightweight worker-side defense mechanism, in contrast to standard server-side approaches. It aligns honest gradients from the start through a sampling mechanism. Finally, Chapter VII introduces a server-side mechanism designed to filter Byzantine updates.

Chapter VI
A Local Sampling Strategy

1 NorthStar **64**
  1.1 Intuition behind NorthStar 65
  1.2 Reference Distribution 65
  1.3 Algorithm 66
2 Theoretical Arguments **66**
  2.1 Standard Gradients 67
  2.2 Gradients under NorthStar 68
  2.3 NorthStar versus Standard Gradients 68
3 Experimental Setup **69**
  3.1 Datasets 69
  3.2 Attacks & Baselines 69
  3.3 FL settings 69
4 Results **70**
  4.1 Effect of Label Skew 70
  4.2 Dataset-Level Trends with RFA Alone 70
  4.3 Interaction with NNM 71
  4.4 Results Averaged over Aggregators 71
  4.5 Negative Gains 71
5 Conclusion **71**

This chapter is based on our publication [33].

Under label skew, honest updates can differ substantially across workers, making Byzantine updates harder to distinguish. Most existing Byzantine defenses operate at the server level: they modify the aggregation rule, apply a pre-aggregation transformation, filter suspicious updates, or rely on a server-side validation dataset. In all cases, they attempt to identify Byzantine behavior among already heterogeneous honest updates.

To address this challenge, we introduce *NorthStar*, a worker-side mechanism. It changes the construction of local mini-batches so that honest workers compute gradients using a shared reference label distribution. This directly targets one source of honest-gradient dissimilarity under label skew: the mismatch between local class proportions.

NorthStar is therefore complementary to existing robust aggregation methods. It can be applied before server-side defenses, reducing the heterogeneity of honest updates at its source. This makes aggregation easier without requiring additional communication during training or access to a clean server-side dataset.

Contributions. We propose a lightweight, principled defense mechanism against Byzantine workers under label skew:

- **Lightweight Defense Mechanism.** We propose *NorthStar*, a strategy that improves Byzantine robustness by reducing honest-gradient dissimilarity;
- **Theoretical Results.** We provide asymptotic arguments explaining the effectiveness of NorthStar. In particular, we show that standard sampling induces a label-distribution mismatch term in honest-gradient dissimilarity, whereas NorthStar removes this term;
- **Experimental Results.** We empirically show that NorthStar consistently improves robustness across multiple models, datasets, aggregation rules, and Byzantine attack scenarios.

Code: `https://github.com/JohanErbani/NorthStar`

1 NorthStar

NorthStar is a worker-side sampling procedure in which each worker locally constructs its mini-batches according to a reference label distribution shared across workers. This reference distribution is broadcast once by the server before training begins.

This section first presents the intuition behind the design of NorthStar, then introduces the reference distribution and provides its algorithm.

Algorithm 6 NorthStar Local Training at Worker k

Require: Local dataset $\mathcal{D}_k = \{\mathcal{D}_k^{(c)}\}_{c=1}^{C}$ partitioned into label-wise subsets, positions $\{s_k^c\}_{c=1}^{C}$ initialized to 1 at the first training round, reference distribution $\mathbf{p} = (p_1, \ldots, p_C)$, mini-batch size B, model Φ_θ with parameter θ_t, loss function ℓ

Ensure: Local stochastic gradient g_k

$\mathcal{B} \leftarrow \emptyset$

for $c = 1$ **to** C **do**

  $B_c \leftarrow \lceil B p_c \rceil$

  $n_c \leftarrow |\mathcal{D}_k^{(c)}|$

  if $n_c > 0$ **then**

    Construct $\mathcal{B}_c$ by taking the next B_c samples from $\mathcal{D}_k^{(c)}$ in cyclic order, starting from index s_k^c

$$\mathcal{B}_c \leftarrow \Big\{(x, y) \in \mathcal{D}_k^{(c)} : \text{indexed by } s_k^c,\ s_k^c + 1 \ (\mathrm{mod}\, n_c),\ \ldots,\ s_k^c + B_c - 1 \ (\mathrm{mod}\, n_c)\Big\}$$

    $\mathcal{B} \leftarrow \mathcal{B} \cup \mathcal{B}_c$

    Update s_k^c to the next position after the selected block

$$s_k^c \leftarrow s_k^c + B_c - 1 \ (\mathrm{mod}\, n_c) + 1$$

  end if

end for

$g_k \leftarrow \nabla_\theta \left(\frac{1}{|\mathcal{B}|} \sum_{(x,y)\in\mathcal{B}} \ell\big(y, \Phi_\theta(x)\big) \right) \Big|_{\theta=\theta_t}$

return g_k

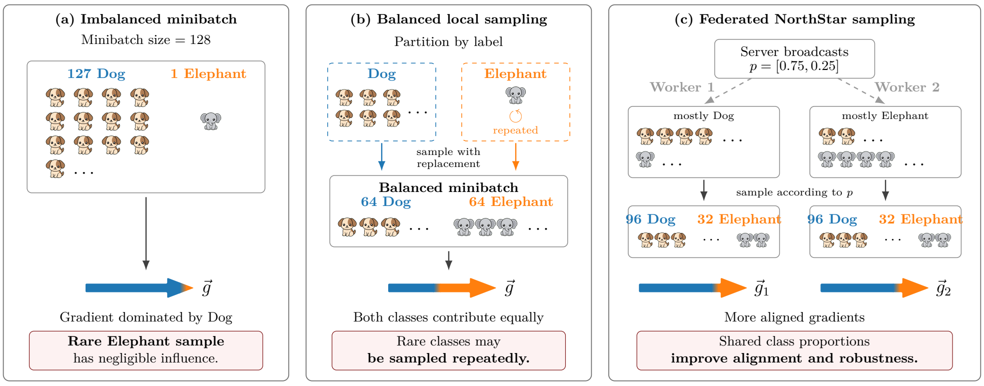


Figure VI.1: Intuition behind NorthStar; (a) An imbalanced minibatch yields a gradient dominated by the majority class; (b) Balanced sampling with replacement reduces majority-class domination; (c) In federated learning, NorthStar sampling encourages workers to align their local gradients with a common reference distribution that is broadcast once by the server.

1.1 Intuition behind NorthStar

Figure VI.1 illustrates the intuition behind NorthStar sampling.

Centralized learning. Consider a binary classification problem with classes `Dog` and `Elephant`. Let a mini-batch of size 128 be composed almost entirely of `Dog` images, with only a single `Elephant` example. The resulting gradient is then dominated by the `Dog` class, as it mainly reflects the direction that minimizes the loss over the majority of samples. Consequently, the `Elephant` example has negligible influence on the update.

A natural way to mitigate this imbalance is to partition the local dataset into label-wise subsets and perform balanced sampling. For instance, one may sample with replacement 64 images from the `Dog` subset and 64 from the `Elephant` subset. This leads to two immediate consequences: (i) the gradient is no longer dominated by `Dog`, as both classes contribute equally; and (ii) if the local dataset contains only a single `Elephant` example, it is repeated 64 times. While such repetition may introduce bias and overfitting in centralized settings, in federated learning it helps each worker account for classes in the appropriate proportion.

Federated learning. Now consider a federated learning setting with two honest workers. Before training, the server broadcasts to workers a reference class distribution, e.g., $[0.75, 0.25]$, indicating that mini-batches should contain 75% `Dog` and 25% `Elephant`. This imbalance can reflect intrinsic class difficulty: `Dog` exhibits high intra-class variability (e.g., many breeds), whereas `Elephant` is more homogeneous.

Assume the first worker predominantly observes `Dog` images, while the second mostly observes `Elephant` images. Each worker partitions its local dataset by label and samples with replacement according to the reference distribution. In practice, each mini-batch contains $96 = 128 \times 0.75$ `Dog` images and $32 = 128 \times 0.25$ `Elephant` images. As a result, both workers assign comparable importance to each class, leading to more aligned gradients and improved robustness at the system level.

Notably, this procedure induces repeated sampling: the first worker repeatedly reuses its few `Elephant` examples, while the second oversamples `Dog` images throughout training. Despite this, our experiments suggest that the alignment effect across clients outweighs the local redundancy.

1.2 Reference Distribution

NorthStar relies on a *reference distribution* shared across workers to guide the local sampling process. This reference distribution is represented by a simplex vector $\mathbf{p} \in \mathbb{R}^C_{\geq 0}$, i.e., a non-negative vector of size C such that $\sum_{i=1}^{C} p_i = 1$. The vector $\mathbf{p}$ defines the target class proportions enforced during local training.

We describe three possible choices for the reference distribution $\mathbf{p}$.

Shared local distributions. In a Byzantine-free setting, each worker sends its local label distribution to the server once before training. The server aggregates these distributions and broadcasts the resulting global distribution $\mathbf{p}$, where

$$p_i = \frac{|\mathcal{D}^{(i)}|}{|\mathcal{D}|},$$

$|\mathcal{D}^{(i)}|$ denotes the total number of samples belonging to class i across all workers, and $|\mathcal{D}|$ is the total number of samples in the system. This approach has two limitations: (i) local label distributions may reveal private information about the workers' data, and (ii) in the presence of Byzantine workers, malicious participants can bias the reference distribution by reporting false distributions.

To mitigate these issues, we compute the coordinate-wise median of the local label distributions using a secure multi-party computation (MPC) protocol [1]. This preserves privacy and limits the ability of Byzantine workers to exploit label statistics. In our setting, the protocol introduces negligible overhead: the sharing step is performed only once before training and involves only C coordinates, where C is the number of classes and is typically small.

Prior knowledge. The global label distribution is assumed to be known in advance from expert knowledge or historical data. For instance, in tumor detection from MRI scans or crop type classification from satellite imagery, national statistics can provide a reliable estimate of the label distribution. In this case, we expect

$$p_i \approx \frac{|\mathcal{D}^{(i)}|}{|\mathcal{D}|}.$$

A priori. All workers may agree on a predefined reference distribution, independent of the true label distribution. For example, in fair hiring systems, classes can be weighted equally to mitigate algorithmic bias, i.e.,

$$p_i = \frac{1}{C}.$$

1.3 Algorithm

Algorithm 6 summarizes the local training procedure used by NorthStar at worker k. The method assumes that, before training begins, the server broadcasts a reference label distribution $\mathbf{p} = (p_1, \ldots, p_C)$ to all workers. Each worker then partitions its local dataset $\mathcal{D}_k$ into label-wise subsets $\mathcal{D}_k^{(c)}$, one for each class $c \in \{1, \ldots, C\}$.

At each local training step, worker k constructs a mini-batch by allocating approximately Bp_c samples to class c, where B is the mini-batch size. More precisely, the worker sets $B_c = \lceil Bp_c \rceil$ and selects B_c samples from the corresponding subset $\mathcal{D}_k^{(c)}$. For each class c, samples are selected cyclically from $\mathcal{D}_k^{(c)}$. This makes the worker traverse the available examples of class c before reusing them, thereby minimizing unnecessary repetitions.

If a class is absent from the local dataset, it is skipped. The final mini-batch is obtained by aggregating the selected samples across all available classes. Worker k then computes the stochastic gradient on this NorthStar mini-batch and sends it to the server, as in standard federated learning. Thus, NorthStar only modifies the local sampling procedure and does not require additional communication after the initial broadcast of $\mathbf{p}$.

2 Theoretical Arguments

This section provides theoretical arguments supporting the effectiveness of NorthStar. All proofs are provided in the appendix.

We analyze a single local training step and do not model the cumulative effect of repeated sampling over the full training trajectory. The results are asymptotic with respect to the number of honest workers, i.e., they describe the limit as $|\mathcal{H}| \to \infty$.

We consider a label-skew setting where, for each class c, observations follow a client-independent conditional distribution $\mathcal{L}_c$. Worker i owns $M_i^{(c)}$ samples of class c, denoted by $x_{i,1}^{(c)}, \dots, x_{i,M_i^{(c)}}^{(c)}$, which form its class-specific dataset $\mathcal{D}_i^{(c)}$. These samples are i.i.d. from $\mathcal{L}_c$. The quantities $M_i^{(c)}$ may vary across workers. For each fixed class c, they are assumed to be i.i.d. across workers, independent of the class-c observations, and strictly positive, i.e., $M_i^{(c)} > 0$, for analytical simplicity.

Robust aggregation relies on the similarity of honest gradients. We therefore study the honest-gradient dissimilarity

$$\frac{1}{|\mathcal{H}|} \sum_{i \in \mathcal{H}} \|g_i - g^\star\|^2,$$

where g_i is the gradient of honest worker i and $g^\star$ is the expected honest gradient, i.e., $\mathbb{E}[\frac{1}{|\mathcal{H}|} \sum_{i \in \mathcal{H}} g_i]$.

2.1 Standard Gradients

We first consider standard local gradients, without NorthStar sampling.

For analytical simplicity, we assume stratified mini-batches: worker i samples classes according to its local label distribution $\mathbf{q}_i = (q_{i,1}, \dots, q_{i,C})$ where

$$q_{i,c} = \frac{M_i^{(c)}}{\sum_{d=1}^C M_i^{(d)}}.$$

Let $\mathbf{q} = (q_1, \dots, q_C)$ denote the expected honest label distribution, with $q_c = \mathbb{E}[q_{i,c}]$. The standard local stochastic gradient is

$$g_i = \sum_{c=1}^C q_{i,c} g_i^{(c)}, \quad g_i^{(c)} = \frac{1}{B_i^{(c)}} \sum_{\ell=1}^{B_i^{(c)}} \nabla_\theta \ell(\Phi_\theta(X_{i,\ell}^{(c)}), c), \quad B_i^{(c)} = q_{i,c} b.$$

Here, $X_{i,\ell}^{(c)}$ is the ℓ-th sampled class-c example in the mini-batch. For simplicity, we ignore integer-rounding effects and assume that $B_i^{(c)} = q_{i,c} b$ is an integer. The expected honest gradient is

$$g^\star = \sum_{c=1}^C q_c g^{(c)}, \qquad g^{(c)} = \mathbb{E}_{x \sim \mathcal{L}_c} \left[\nabla_\theta \ell(\Phi_\theta(x), c) \right].$$

Asymptotically, gradient dissimilarity splits into two terms: (i) a class-wise estimation error; and (ii) a label-distribution mismatch term.

Proposition 10. *Assume that honest workers are i.i.d. and that, for each class c, $\mathbb{E}[\|g_i^{(c)}\|^2] < \infty$.*

Then

$$\frac{1}{|\mathcal{H}|} \sum_{i \in \mathcal{H}} \|g_i - g^\star\|^2$$

$$\xrightarrow[|\mathcal{H}| \to \infty]{\text{a.s.}} \underbrace{\mathbb{E}\left[\left\| \sum_{c=1}^C q_{i,c} \left(g_i^{(c)} - g^{(c)} \right) \right\|^2 \right]}_{\textit{class-wise estimation error}} + \underbrace{\mathbb{E}\left[\left\| \sum_{c=1}^C (q_{i,c} - q_c) g^{(c)} \right\|^2 \right]}_{\textit{label-distribution mismatch}}$$

Thus, under label skew, standard gradients can remain dissimilar even when class-wise gradients are accurately estimated, because workers combine classes with different weights.

2.2 Gradients under NorthStar

We now analyze a simplified NorthStar step based on random sampling with replacement. In practice, NorthStar uses cyclic sampling, which reduces repetitions and is therefore expected to be less detrimental to convergence.

At each local step, worker i samples B_c examples from $\mathcal{D}_i^{(c)}$, where $B_c = p_c b > 0$ is shared across workers and determined by the broadcasted reference distribution $\mathbf{p} = (p_1, \ldots, p_C)$. We assume $p_c > 0$ for every class c and ignore integer-rounding effects, so that B_c is treated as an integer. Let $u_{i,k}^{(c)}$ be the number of times sample $x_{i,k}^{(c)}$ is selected. Conditionally on $M_i^{(c)}$,

$$(u_{i,1}^{(c)}, \ldots, u_{i,M_i^{(c)}}^{(c)}) \sim \text{Multinomial}\left(B_c; \frac{1}{M_i^{(c)}}, \ldots, \frac{1}{M_i^{(c)}}\right).$$

The NorthStar gradient is

$$g_{\mathrm{NS}\,i} = \sum_{c=1}^{C} p_c g_{\mathrm{NS}\,i}^{(c)}, \quad g_{\mathrm{NS}\,i}^{(c)} = \frac{1}{B_c} \sum_{k=1}^{M_i^{(c)}} u_{i,k}^{(c)} \nabla_\theta \ell(\Phi_\theta(x_{i,k}^{(c)}), c).$$

The expected honest gradient is

$$g_{\mathrm{NS}}^{\star} = \sum_{c=1}^{C} p_c g_{\mathrm{NS}}^{(c)}, \quad g_{\mathrm{NS}}^{(c)} = g^{(c)} = \mathbb{E}_{x \sim \mathcal{L}_c}\left[\nabla_\theta \ell(\Phi_\theta(x), c)\right].$$

Asymptotically, gradient dissimilarity contains only one term: a class-wise estimation error.

Proposition 11. *Assume that honest workers are i.i.d. and that, for each class c, $\mathbb{E}[\|g_{\mathrm{NS}\,i}^{(c)}\|^2] < \infty$.*
Then

$$\frac{1}{|\mathcal{H}|} \sum_{i \in \mathcal{H}} \|g_{\mathrm{NS}\,i} - g_{\mathrm{NS}}^{\star}\|^2 \xrightarrow[|\mathcal{H}| \to \infty]{\text{a.s.}} \underbrace{\mathbb{E}\left[\left\|\sum_{c=1}^{C} p_c \left(g_{\mathrm{NS}\,i}^{(c)} - g^{(c)}\right)\right\|^2\right]}_{\textit{class-wise estimation error under NorthStar}}$$

NorthStar gradient dissimilarity remains nonzero due to class-wise stochastic estimation error, but not because of worker-specific class proportions.

2.3 NorthStar versus Standard Gradients

Gradient dissimilarity. Under standard sampling, asymptotic honest-gradient dissimilarity decomposes as

$$\Big(\text{gradient dissimilarity without NorthStar}\Big) = \Big(\text{class-wise estimation error}\Big) + \Big(\text{label-distribution mismatch}\Big)$$

With NorthStar, the corresponding decomposition becomes

$$\Big(\text{NorthStar gradient dissimilarity}\Big) = \Big(\text{class-wise estimation error}\Big)$$

Thus, standard sampling contains an additional term induced by worker-specific label proportions. NorthStar removes this source of dissimilarity by replacing local class weights with a shared reference distribution.

Since the remaining class-wise estimation terms have the same nature in both decompositions, they are expected to have comparable magnitude. In contrast, the label-distribution mismatch term can dominate under strong label skew and appears only in the standard procedure. This explains why NorthStar reduces honest-gradient dissimilarity, as observed empirically.

Target model. The next proposition shows that, when the reference distribution matches the global distribution, i.e., $\mathbf{p} = \mathbf{q}$, NorthStar preserves the expected target of the standard procedure.

Proposition 12. *If the broadcasted reference distribution satisfies* $\mathbf{p} = \mathbf{q}$*, then NorthStar preserves the expected honest gradient:*

$$g^\star_{\mathrm{NS}} = g^\star.$$

This provides insight into the effectiveness of NorthStar: it aligns honest gradients by replacing worker-specific class weights with a shared reference distribution, while preserving the target model when $\mathbf{p} = \mathbf{q}$.

3 Experimental Setup

3.1 Datasets

Our experiments are conducted on the following datasets : CIFAR-10 [60], MNIST [69], Fashion-MNIST [113], and Kuzushiji-MNIST [17]. We divide training data among the workers using a Dirichlet distribution to generate non-IID local datasets, following the approach of [4, 52], with concentration parameters $\alpha \in \{0.1, 1, 10\}$. For each dataset, the test set follows the same class distribution as its corresponding training set.

3.2 Attacks & Baselines

We consider the following attacks : A Little Is Enough (ALIE) [8], Fall Of Empire (FOE) [115], Mimic [56], Min-Sum [99] and Label Flipping (LF) [3]. We use the following robust aggregators : Coordinate-Wise Median (CWMed) [118], Coordinate-Wise Trimmed Mean (CwTM) [11] and RFA [91]. We also compare performance with and without pre-aggregation, using Nearest Neighbor Mixing (NNM) [4]. We use the CrossEntropy loss as a baseline to evaluate our approach.

We exclude robustness methods that require additional data or heavy computation, since NorthStar is intended as a lightweight worker-side complement to existing defenses.

3.3 FL settings

We consider 60 workers, with $f \in \{2, 8, 14, 20, 26\}$ Byzantine workers. We train a four-layer CNN for CIFAR-10, and a two-layer CNN for MNIST, Fashion-MNIST and Kuzushiji-MNIST (models in Appendix). The models are trained with batch size 128 and momentum $\beta = 0.9$. Momentum has shown strong robustness against Byzantine behavior in both heterogeneous [4] and homogeneous settings [28, 57, 39]. Model performance is evaluated by computing the accuracy averaged over the last 200 training steps.

4 Results

This section evaluates the impact of NorthStar on Byzantine-robust federated learning under label skew. We first analyze the results obtained with RFA, then discuss the aggregated results across robust aggregators.

4.1 Effect of Label Skew

Tables VI.1, VI.2, VI.3, and VI.4 report the results obtained with RFA on CIFAR-10, MNIST, Fashion-MNIST, and Kuzushiji-MNIST. Gains are computed as the difference between NorthStar and the baseline. Negative gains are shown in red, positive gains in green, and gains above five percentage points in bold green.

Across datasets, NorthStar becomes more useful as label skew increases. When $\alpha = 10$, local label distributions are close to homogeneous, so CE and NS usually obtain similar performance. Gains are therefore small, and sometimes slightly negative.

The effect becomes clearer for $\alpha = 1$ and $\alpha = 0.1$. In these regimes, the baseline often degrades as the number of Byzantine workers increases, while NorthStar mitigates this degradation in many settings. This is especially visible on CIFAR-10 and Fashion-MNIST, and in collapse cases on MNIST and Kuzushiji-MNIST.

Overall, these results support the intuition of Section 1: under label skew, honest gradients differ partly because workers use different class proportions. By enforcing a shared reference distribution in local mini-batches, NorthStar reduces this source of dissimilarity and makes robust aggregation more effective.

4.2 Dataset-Level Trends with RFA Alone

MNIST. The effect of NorthStar depends on dataset difficulty and on how close the baseline is to saturation. On MNIST, baseline accuracies are often close to 99% when heterogeneity is mild or moderate. In this regime, there is little room for improvement, and NorthStar mostly leads to negligible variations, sometimes slightly negative. This is expected: when the baseline already reaches near-optimal accuracy, enforcing a shared sampling distribution cannot improve much and may introduce small fluctuations due to repeated examples. However, when the baseline collapses under strong heterogeneity or large Byzantine counts, NorthStar can produce large recovery effects. For instance, with RFA under ALIE at $\alpha = 0.1$ and $f = 20$, accuracy increases from 18.16% to 87.05%, corresponding to a gain of 68.89 percentage points.

Kuzushiji-MNIST. On Kuzushiji-MNIST, NorthStar has limited effect under mild heterogeneity, $\alpha = 10$, where gains are small and sometimes negative. The benefits become clearer for $\alpha = 1$, especially when the number of Byzantine workers increases. In this regime, NorthStar improves accuracy under MinSum, FOE, and LF, with gains reaching 19.80, 32.98, and 17.84 points, respectively. Under strong heterogeneity, $\alpha = 0.1$, the improvement is larger and more consistent. NorthStar mitigates several collapse cases, with gains of 28.76 points under Mimic with $f = 14$, 58.51 points under MinSum with $f = 20$, and 42.26 points under FOE with $f = 14$.

CIFAR-10 & Fashion-MNIST. On harder datasets CIFAR-10 and Fashion-MNIST, the gains are more gradual and more consistently visible across attacks. Under moderate heterogeneity, $\alpha = 1$, NorthStar often improves accuracy by several percentage points, especially when the number of Byzantine workers increases. On CIFAR-10, for example, RFA gains 9.05 points under ALIE with $f = 14$, 7.85 points under ALIE with $f = 20$, and 29.86 points under MinSum with $f = 14$. On Fashion-MNIST, the recovery can be even stronger: with RFA under ALIE at $\alpha = 1$ and $f = 26$, NorthStar improves accuracy by 43.67 points.

The trend becomes clearer under strong heterogeneity, $\alpha = 0.1$. In this setting, standard training often suffers from severe performance degradation, while NorthStar substantially improves robustness in most configurations. On CIFAR-10, gains with RFA are positive for almost all attacks and Byzantine counts, often exceeding 5 points and reaching more than 20 points in some cases. On Fashion-MNIST, the gains are also large and systematic, with improvements such as 40.91 points under ALIE with $f = 20$, 28.41 points under MinSum with $f = 26$, and 44.66 points under FOE with $f = 26$.

4.3 Interaction with NNM

NNM generally makes the baseline more robust, leaving less room for NorthStar to improve. When $\alpha = 10$, gains with NNM+RFA are therefore usually small and close to zero, since both label skew and baseline instability are limited.

Under stronger heterogeneity, NorthStar remains complementary to NNM in many cases. With NNM+RFA, gains reach 31.67 points on CIFAR-10 under ALIE with $f = 26$ at $\alpha = 0.1$, 24.68 points under MinSum with $f = 14$, and 17.45 points on Fashion-MNIST under MinSum with $f = 26$ at $\alpha = 1$. On Kuzushiji-MNIST, NorthStar also yields strong gains, such as 54.00 points under MinSum with $f = 26$ at $\alpha = 1$.

Overall, NNM and NorthStar are often complementary and largely beneficial, but their combination becomes less stable when heterogeneity is extreme and local class-wise samples are scarce.

4.4 Results Averaged over Aggregators

Tables VI.5, VI.7, and VI.6 report results averaged over CWMed, CwTM, and RFA, both with and without NNM. These aggregated results confirm the trends observed with RFA alone. NorthStar provides the largest gains under moderate and severe label skew, while its effect is smaller under mild heterogeneity.

This shows that the method is not tied to a specific aggregation rule. Instead, it acts as a complementary worker-side mechanism that improves the quality and alignment of honest updates before they reach the server.

4.5 Negative Gains

The largest negative gains remain isolated.

With RFA alone, the worst drop without NNM occurs on CIFAR-10 under FOE at $\alpha = 10$ and $f = 26$, where NorthStar loses 30.60 points; with NNM+RFA, the worst drop occurs on Fashion-MNIST under MinSum at $\alpha = 0.1$ and $f = 20$, with a loss of 29.98 points.

However, these cases are not representative of the global trend. When results are averaged over aggregators, the worst drops are much smaller: -9.15 points without NNM and -14.10 points with NNM.

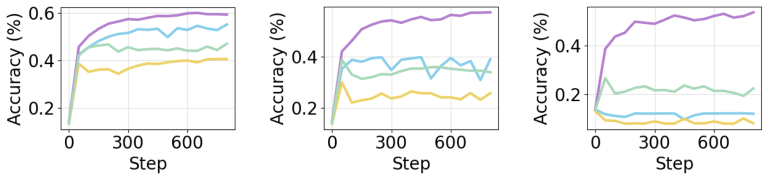
Compared with the many positive gains observed across datasets, attacks, and heterogeneity levels, these failures appear as epiphenomena rather than a systematic limitation.

5 Conclusion

We introduced *NorthStar*, a worker-side sampling strategy for Byzantine-robust federated learning under label skew. Rather than changing the server-side aggregation rule, NorthStar acts locally: honest workers build mini-batches according to a shared reference label distribution, reducing the gradient dissimilarity induced by heterogeneous class proportions.

Our analysis shows that standard sampling introduces a label-distribution mismatch term in honest-gradient dissimilarity. NorthStar removes this term by enforcing a common sampling distribution, while preserving the expected training target when the reference distribution matches the expected honest label distribution.

Empirically, NorthStar improves robustness across datasets, attacks, aggregation rules, and Byzantine counts, especially under strong label skew. These results show



(a) $f = 8$ (b) $f = 14$ (c) $f = 26$

Figure VI.2: CIFAR-10 test accuracy under ALIE with RFA and label-skew parameter $\alpha = 0.1$. Legend: • NorthStar, • NNM+NorthStar, • RFA, • NNM+RFA.

that Byzantine robustness can be improved not only through stronger aggregation, but also by reducing honest-gradient heterogeneity at its source.

Table VI.1: CIFAR10, RFA. Mean accuracy (%) over the last 200 steps. We compare cross-entropy only (CE) and cross-entropy with NorthStar (NS) across $\alpha \in \{10, 1, 0.1\}$, attacks, and Byzantine counts $f \in \{2, 8, 14, 20, 26\}$.

			ALIE					Mimic					MinSum					FOE					LF				
			2	8	14	20	26	2	8	14	20	26	2	8	14	20	26	2	8	14	20	26	2	8	14	20	26
10	RFA	CE	73.85	69.37	60.89	49.42	30.96	74.20	72.83	67.88	62.27	56.74	72.60	61.28	53.75	39.65	8.29	71.50	61.81	55.32	47.17	36.63	74.26	73.78	72.95	71.80	67.36
		NS	74.38	70.12	60.77	50.59	38.16	74.84	72.97	68.20	62.64	57.84	72.46	62.81	55.60	34.76	11.23	71.89	62.15	55.87	39.86	6.03	74.82	73.84	73.23	71.66	67.82
		gain	0.53	0.75	-0.12	1.17	**7.20**	0.64	0.14	0.32	0.37	1.10	-0.14	1.53	1.85	-4.89	2.94	0.39	0.34	0.55	-7.31	-30.60	0.56	0.06	0.28	-0.14	0.46
	NNM+RFA	CE	73.85	72.99	69.66	72.22	73.91	74.29	72.89	71.17	69.29	63.63	73.23	67.17	61.58	56.11	49.71	71.24	67.66	62.18	57.59	50.20	74.35	74.34	74.41	74.60	74.58
		NS	75.12	72.88	69.80	68.31	73.97	74.91	73.60	72.00	69.55	63.89	73.50	69.75	62.78	56.85	49.45	72.16	68.46	63.41	57.85	48.07	75.01	74.60	74.92	74.43	74.30
		gain	1.27	-0.11	0.14	-3.91	0.06	0.62	0.71	0.83	0.26	0.26	0.27	2.58	1.20	0.74	-0.26	0.92	0.80	1.23	0.26	-2.13	0.66	0.26	0.51	-0.17	-0.28
1	RFA	CE	72.29	64.60	53.24	34.91	20.47	72.80	70.93	66.49	57.57	47.49	69.05	60.91	54.12	43.08	4.96	68.65	62.09	56.74	48.11	37.32	72.79	71.44	70.22	67.89	60.64
		NS	72.33	67.12	57.85	47.33	34.38	73.08	70.54	66.19	57.60	52.51	69.80	62.18	59.01	47.44	5.72	69.88	62.16	59.70	50.46	8.01	73.22	71.76	71.31	70.47	66.35
		gain	0.04	2.52	4.61	**12.42**	**13.91**	0.28	-0.39	-0.30	0.03	**5.02**	0.75	1.27	4.89	4.36	0.76	1.23	0.07	2.96	2.35	-29.31	0.43	0.32	1.09	2.58	**5.71**
	NNM+RFA	CE	72.00	66.49	62.49	57.24	66.44	72.99	71.42	69.17	65.43	57.74	70.03	55.14	57.42	50.75	38.13	68.12	63.41	57.71	50.86	42.44	72.94	72.11	71.55	71.34	69.83
		NS	72.18	71.76	71.55	65.19	72.25	72.89	71.72	70.97	68.28	61.37	70.47	70.12	69.48	67.55	66.07	69.17	67.69	66.29	62.91	49.78	73.36	72.63	73.14	73.12	73.31
		gain	0.18	**5.27**	**9.06**	**7.95**	**5.81**	-0.10	0.30	1.80	2.85	3.63	0.44	**14.98**	**12.06**	**16.80**	**27.94**	1.05	4.28	**8.58**	**12.05**	**7.34**	0.42	0.52	1.59	1.78	3.48
0.1	RFA	CE	56.58	40.10	24.44	22.22	8.48	59.82	60.45	51.34	41.50	28.09	56.58	46.82	38.26	26.70	4.56	57.05	47.76	41.27	31.26	19.04	58.49	56.90	56.76	49.01	48.60
		NS	63.85	53.66	36.93	29.92	12.05	63.16	61.42	53.90	38.90	36.77	60.89	51.76	43.99	26.10	8.36	60.95	52.43	46.17	27.90	4.30	62.99	58.91	59.51	55.72	49.76
		gain	**7.27**	**13.56**	**12.49**	**7.70**	3.57	3.34	0.97	2.56	-2.60	**8.68**	4.31	4.94	**5.73**	-0.60	3.80	3.90	4.67	4.90	-3.36	-14.74	4.50	2.01	2.75	**6.71**	1.16
	NNM+RFA	CE	55.14	45.01	34.69	31.09	21.01	59.18	61.36	59.57	52.69	31.90	55.99	35.72	11.23	29.89	8.65	56.17	51.79	41.56	34.09	26.90	57.77	57.11	57.63	52.17	51.71
		NS	63.91	59.49	56.96	50.46	52.68	63.37	63.57	62.87	57.30	49.50	61.53	49.56	35.91	39.34	9.35	61.00	55.75	51.41	38.19	17.82	62.96	59.58	61.08	58.34	55.31
		gain	**8.77**	**14.48**	**22.27**	**19.37**	**31.67**	4.19	2.21	3.30	4.61	**17.60**	**5.54**	**13.84**	**24.68**	**9.45**	0.70	4.83	3.96	**9.85**	4.10	-9.08	**5.19**	2.47	3.45	**6.17**	3.60

Table VI.2: MNIST, RFA. Mean accuracy (%) over the last 200 steps. We compare cross-entropy only (CE) and cross-entropy with NorthStar (NS) across $\alpha \in \{10, 1, 0.1\}$, attacks, and Byzantine counts $f \in \{2, 8, 14, 20, 26\}$.

			ALIE					Mimic					MinSum					FOE					LF				
			2	8	14	20	26	2	8	14	20	26	2	8	14	20	26	2	8	14	20	26	2	8	14	20	26
10	RFA	CE	98.73	98.76	98.45	98.21	95.49	98.72	98.61	98.47	98.25	98.06	98.55	98.04	97.08	97.03	69.86	98.50	97.94	97.57	97.51	95.80	98.74	98.82	98.73	98.78	98.71
		NS	98.69	98.59	98.36	98.04	95.01	98.68	98.62	98.45	98.22	98.14	98.66	98.02	96.97	95.00	92.28	98.59	97.92	96.95	95.88	94.70	98.73	98.83	98.76	98.72	98.55
		gain	-0.04	-0.17	-0.09	-0.17	-0.48	-0.04	0.01	-0.02	-0.03	0.08	0.11	-0.02	-0.11	-2.03	**22.42**	0.09	-0.02	-0.62	-1.63	-1.10	-0.01	0.01	0.03	-0.06	-0.16
	NNM+RFA	CE	98.71	98.74	98.65	98.57	98.46	98.74	98.74	98.50	98.32	97.76	98.68	98.35	97.91	97.23	96.59	98.50	98.41	97.98	97.76	96.43	98.69	98.74	98.67	98.67	98.80
		NS	98.70	98.77	98.72	98.52	98.34	98.73	98.72	98.58	98.27	97.89	98.68	98.37	98.10	97.61	96.15	98.59	98.46	98.28	97.76	96.07	98.70	98.80	98.82	98.86	98.66
		gain	-0.01	0.03	0.07	-0.05	-0.12	-0.01	-0.02	0.08	-0.05	0.13	0.00	0.02	0.19	0.38	-0.44	0.09	0.05	0.30	0.00	-0.36	0.01	0.06	0.15	0.19	-0.14
1	RFA	CE	98.65	98.67	98.28	97.41	67.35	98.64	98.50	98.21	97.87	97.08	98.49	98.21	97.70	96.76	69.17	98.50	98.26	97.92	97.28	87.77	98.69	98.75	98.63	98.59	98.59
		NS	98.61	98.52	98.37	97.54	92.00	98.56	98.38	98.19	98.05	97.58	98.63	98.23	97.51	96.39	93.15	98.48	98.12	97.41	95.99	94.65	98.64	98.66	98.58	98.57	98.60
		gain	-0.04	-0.15	0.09	0.13	**24.65**	-0.08	-0.12	-0.02	0.18	0.50	0.14	0.02	-0.19	-0.37	**23.98**	-0.02	-0.14	-0.51	-1.29	**6.88**	-0.05	-0.09	-0.05	-0.02	0.01
	NNM+RFA	CE	98.73	98.72	98.63	98.51	98.50	98.70	98.63	98.50	98.30	97.58	98.61	98.26	98.09	88.81	87.81	98.37	97.75	97.49	93.60	90.86	98.72	98.77	98.74	98.71	98.79
		NS	98.55	98.68	98.59	98.42	98.36	98.60	98.51	98.40	98.17	97.52	98.27	98.28	97.85	97.80	97.78	98.18	98.22	97.90	97.85	96.92	98.62	98.73	98.74	98.67	98.66
		gain	-0.18	-0.04	-0.04	-0.09	-0.14	-0.10	-0.12	-0.10	-0.13	-0.06	-0.34	0.02	-0.24	**8.99**	**9.97**	-0.19	0.47	0.41	4.25	**6.06**	-0.10	-0.04	0.00	-0.04	-0.13
0.1	RFA	CE	98.58	98.21	96.22	18.16	8.65	98.65	97.90	96.75	93.46	82.56	98.35	95.17	92.71	55.54	21.97	98.35	96.36	94.74	76.42	19.66	98.65	98.70	98.63	98.30	97.16
		NS	98.46	98.16	97.48	87.05	18.01	98.30	98.02	97.60	96.93	91.87	98.18	97.34	94.46	86.47	47.41	98.20	97.17	94.80	88.13	67.85	98.39	98.37	98.24	98.33	98.15
		gain	-0.12	-0.05	1.26	**68.89**	**9.36**	-0.35	0.12	0.85	3.47	**9.31**	-0.17	2.17	1.75	**30.93**	**25.44**	-0.15	0.81	0.06	**11.71**	**48.19**	-0.26	-0.33	-0.39	0.03	0.99
	NNM+RFA	CE	98.66	98.40	94.15	84.11	74.89	98.72	98.66	97.73	96.12	91.79	98.37	88.67	85.81	66.13	35.71	98.09	88.63	83.17	69.17	39.30	98.77	98.72	98.77	98.73	98.63
		NS	98.52	98.25	98.13	98.14	98.29	98.52	98.33	98.10	97.16	96.33	98.08	97.54	96.83	88.65	14.69	97.92	97.43	61.17	45.26	41.23	98.51	98.49	98.48	98.36	98.39
		gain	-0.14	-0.15	3.98	**14.03**	**23.40**	-0.20	-0.33	0.37	1.04	4.54	-0.29	**8.87**	**11.02**	**22.52**	-21.02	-0.17	**8.80**	-22.00	-23.91	1.93	-0.26	-0.23	-0.29	-0.37	-0.24

Table VI.3: Fashion-MNIST, RFA. Mean accuracy (%) over the last 200 steps. We compare cross-entropy only (CE) and cross-entropy with NorthStar (NS) across $\alpha \in \{10, 1, 0.1\}$, attacks, and Byzantine counts $f \in \{2, 8, 14, 20, 26\}$.

			ALIE					Mimic					MinSum					FOE					LF				
			2	8	14	20	26	2	8	14	20	26	2	8	14	20	26	2	8	14	20	26	2	8	14	20	26
10	RFA	CE	86.86	86.96	84.40	81.16	74.62	86.98	87.17	86.32	85.42	81.83	86.22	83.17	82.18	78.63	57.67	86.24	84.29	83.14	79.80	71.52	86.91	87.15	86.98	86.50	86.28
		NS	87.17	86.84	85.81	82.83	75.65	87.11	87.18	85.97	85.22	83.05	86.49	84.59	81.73	76.10	74.16	86.59	83.67	81.39	78.19	74.81	87.02	87.08	86.86	86.72	86.65
		gain	0.31	-0.12	1.41	1.67	1.03	0.13	0.01	-0.35	-0.20	1.22	0.27	1.42	-0.45	-2.53	**16.49**	0.35	-0.62	-1.75	-1.61	3.29	0.11	-0.07	-0.12	0.22	0.37
	NNM+RFA	CE	86.87	86.71	86.63	87.04	87.11	87.05	87.08	86.44	85.55	83.00	85.75	85.41	82.19	82.21	77.09	85.88	84.05	82.23	79.64	77.07	86.95	87.23	87.13	86.92	87.07
		NS	87.16	87.05	87.16	87.00	86.74	87.00	87.06	86.40	85.85	83.48	86.92	85.64	84.58	83.12	80.05	86.30	85.38	84.73	81.94	78.29	87.13	87.13	86.83	86.84	87.14
		gain	0.29	0.34	0.53	-0.04	-0.37	-0.05	-0.02	-0.04	0.30	0.48	1.17	0.23	2.39	0.91	2.96	0.42	1.33	2.50	2.30	1.22	0.18	-0.10	-0.30	-0.08	0.07
1	RFA	CE	86.57	85.15	80.64	75.05	26.12	86.52	86.40	85.67	82.36	78.92	85.60	82.12	78.36	72.99	51.98	85.57	82.59	79.55	78.35	67.00	86.59	86.60	86.10	85.87	83.33
		NS	87.15	86.18	84.60	80.24	69.79	86.78	86.38	86.18	85.40	82.56	85.86	84.02	82.41	78.15	71.88	86.58	83.97	82.63	79.16	74.69	87.03	86.95	86.86	86.38	86.17
		gain	0.58	1.03	3.96	**5.19**	**43.67**	0.26	-0.02	0.51	3.04	3.64	0.26	1.90	4.05	**5.16**	**19.90**	1.01	1.38	3.08	0.81	**7.69**	0.44	0.35	0.76	0.51	2.84
	NNM+RFA	CE	86.64	85.97	85.39	84.88	84.73	86.91	86.54	85.56	83.74	80.83	85.05	81.91	78.65	69.15	65.05	83.69	80.96	79.77	77.73	64.74	87.17	87.02	86.51	86.34	85.06
		NS	87.02	86.71	86.58	86.29	85.67	86.91	86.77	86.22	85.67	83.96	86.11	85.77	84.82	84.82	82.50	85.76	83.85	84.62	83.84	79.10	87.10	86.77	86.84	86.57	86.55
		gain	0.38	0.74	1.19	1.41	0.94	0.00	0.23	0.66	1.93	3.13	1.06	3.86	**6.17**	**15.67**	**17.45**	2.07	2.89	4.85	**6.11**	**14.36**	-0.07	-0.25	0.33	0.23	1.49
0.1	RFA	CE	85.57	81.24	73.42	29.06	8.65	85.52	85.58	82.44	71.60	55.93	82.16	69.54	65.35	53.13	10.50	81.82	70.75	67.16	64.74	17.31	86.02	86.20	86.68	84.60	77.06
		NS	86.52	83.31	78.85	69.97	18.01	86.71	86.37	85.82	82.25	69.87	85.53	81.83	80.48	74.95	38.91	85.33	79.77	79.41	76.36	61.97	86.33	86.32	87.12	85.79	83.98
		gain	0.95	2.07	**5.43**	**40.91**	**9.36**	1.19	0.79	3.38	**10.65**	**13.94**	3.37	**12.29**	**15.13**	**21.82**	**28.41**	3.51	**9.02**	**12.25**	**11.62**	**44.66**	0.31	0.12	0.44	1.19	**6.92**
	NNM+RFA	CE	86.16	81.67	80.17	77.77	70.56	85.87	85.96	81.43	74.40	67.65	79.37	74.26	65.08	57.15	24.61	78.41	75.81	69.99	53.34	35.96	86.22	86.25	86.56	84.60	77.16
		NS	86.08	84.58	81.64	81.38	86.12	86.35	86.53	86.35	76.91	73.38	82.76	79.40	75.46	27.17	16.92	82.47	81.05	79.04	32.56	34.17	86.15	86.69	87.22	86.24	86.90
		gain	-0.08	2.91	1.47	3.61	**15.56**	0.48	0.57	4.92	2.51	**5.73**	3.39	**5.14**	**10.38**	-29.98	-7.69	4.06	**5.24**	**9.05**	-20.78	-1.79	-0.07	0.44	0.66	1.64	**9.74**

Table VI.4: Kuzushiji-MNIST, RFA. Mean accuracy (%) over the last 200 steps. We compare cross-entropy only (CE) and cross-entropy with NorthStar (NS) across $\alpha \in \{10, 1, 0.1\}$, attacks, and Byzantine counts $f \in \{2, 8, 14, 20, 26\}$.

			ALIE					Mimic					MinSum					FOE					LF				
			2	8	14	20	26	2	8	14	20	26	2	8	14	20	26	2	8	14	20	26	2	8	14	20	26
10	RFA	CE	89.91	88.64	84.95	81.04	67.21	88.81	89.47	85.82	85.94	81.75	88.14	84.10	80.92	77.79	72.42	86.68	85.64	81.77	80.61	70.38	90.87	89.45	83.11	81.01	71.10
		NS	90.00	88.29	85.22	78.20	70.20	89.33	90.16	86.72	85.26	80.39	89.11	85.58	85.50	81.31	76.01	87.29	82.09	78.05	76.39	73.98	90.74	87.34	85.22	80.76	71.76
		gain	0.09	-0.35	0.27	-2.84	2.99	0.52	0.69	0.90	-0.68	-1.36	0.97	1.48	4.58	3.52	3.59	0.61	-3.55	-3.72	-4.22	3.60	-0.13	-2.11	2.11	-0.25	0.66
	NNM+RFA	CE	89.86	89.75	89.44	90.35	89.99	89.93	89.54	88.37	88.03	85.08	88.75	83.88	83.45	83.32	79.91	87.84	85.78	85.12	82.75	79.70	89.51	87.43	89.65	88.81	88.33
		NS	90.07	88.57	89.94	89.09	88.95	89.46	90.00	86.17	85.49	80.39	91.08	87.13	86.65	81.67	77.76	89.67	85.83	78.63	78.61	66.31	91.04	89.85	88.72	89.24	88.56
		gain	0.21	-1.18	0.50	-1.26	-1.04	-0.47	0.46	-2.20	-2.54	-4.69	2.33	3.25	3.20	-1.65	-2.15	1.83	0.05	-6.49	-4.14	-13.39	1.53	2.42	-0.93	0.43	0.23
1	RFA	CE	87.77	87.34	80.93	66.25	54.60	86.92	89.69	88.42	86.57	80.02	84.62	82.78	64.71	64.31	58.66	85.70	83.04	61.09	55.57	37.77	88.14	83.83	80.09	69.02	48.39
		NS	87.72	86.90	83.28	70.13	56.13	87.90	89.03	85.98	85.63	82.30	88.29	86.97	84.51	80.65	67.01	84.14	80.47	77.55	73.61	70.75	87.87	87.43	82.37	77.40	66.23
		gain	-0.05	-0.44	2.35	3.88	1.53	0.98	-0.66	-2.44	-0.94	2.28	3.67	4.19	**19.80**	**16.34**	**8.35**	-1.56	-2.57	**16.46**	**18.04**	**32.98**	-0.27	3.60	2.28	**8.38**	**17.84**
	NNM+RFA	CE	87.15	91.53	90.34	88.35	89.92	87.98	89.81	89.97	88.56	84.66	84.75	78.39	83.03	75.09	53.33	82.37	86.24	84.48	81.40	66.78	89.80	87.75	89.05	87.05	87.53
		NS	90.03	89.03	90.69	89.37	89.23	88.90	89.01	86.84	87.07	84.55	87.27	88.87	86.18	86.88	83.95	83.47	87.30	84.67	78.61	77.01	89.56	90.27	90.09	89.57	89.20
		gain	2.88	-2.50	0.35	1.02	-0.69	0.92	-0.80	-3.13	-1.49	-0.11	2.52	**10.48**	3.15	**11.79**	**30.62**	1.10	1.06	0.19	-2.79	**10.23**	-0.24	2.52	1.04	2.52	1.67
0.1	RFA	CE	86.81	62.22	44.29	31.85	15.89	86.77	71.25	49.91	38.58	31.81	82.14	54.06	15.40	0.00	21.38	79.24	52.51	17.85	0.00	0.00	85.84	51.70	42.58	29.89	22.97
		NS	87.80	76.99	58.29	28.13	24.45	89.06	85.22	78.67	53.08	41.43	83.42	68.11	56.72	58.51	41.28	83.93	72.96	60.11	34.00	0.00	88.78	82.99	71.95	44.21	29.91
		gain	0.99	**14.77**	**14.00**	-3.72	**8.56**	2.29	**13.97**	**28.76**	**14.50**	**9.62**	1.28	**14.05**	**41.32**	**58.51**	**19.90**	4.69	**20.45**	**42.26**	**34.00**	0.00	2.94	**31.29**	**29.37**	**14.32**	**6.94**
	NNM+RFA	CE	90.54	76.85	84.98	85.83	68.09	90.33	89.85	87.46	83.42	60.69	79.21	70.74	11.39	20.77	24.46	84.01	71.14	72.86	62.57	13.96	90.45	90.34	88.38	82.45	67.75
		NS	89.49	87.94	82.05	88.62	85.15	89.36	87.26	88.14	84.84	62.48	87.28	71.79	73.97	27.75	25.90	85.42	81.25	78.53	73.45	18.03	88.18	87.41	87.41	86.44	83.25
		gain	-1.05	**11.09**	-2.93	2.79	**17.06**	-0.97	-2.59	0.68	1.42	1.79	**8.07**	1.05	**62.58**	**6.98**	1.44	1.41	**10.11**	**5.67**	**10.88**	4.07	-2.27	-2.93	-0.97	3.99	**15.50**

Table VI.5: CIFAR-10. Mean accuracy (%) over the last 200 steps for $\overline{\text{agg}}$ (average over CWMed, CwTM, and RFA without NNM) and $\overline{\text{agg} + \text{NNM}}$ (same with NNM). We compare cross-entropy only (CE) and cross-entropy with NorthStar (NS) across $\alpha \in \{10, 1, 0.1\}$, attacks, and Byzantine counts $f \in \{2, 8, 14, 20, 26\}$.

			ALIE					Mimic					MinSum					FOE					LF				
			2	8	14	20	26	2	8	14	20	26	2	8	14	20	26	2	8	14	20	26	2	8	14	20	26
10	$\overline{\text{agg}}$	CE	73.45	66.58	56.51	45.32	31.93	73.86	72.48	68.97	64.54	57.70	72.47	62.90	53.72	30.60	15.58	70.76	61.30	54.30	46.06	32.23	73.86	72.97	71.25	68.98	63.64
		NS	74.02	67.08	56.62	44.11	36.13	74.55	73.04	69.64	64.88	59.18	72.90	64.64	56.32	43.14	21.91	71.21	62.02	56.27	44.18	23.08	74.55	73.46	72.16	69.11	64.34
		gain	0.58	0.50	0.11	-1.22	4.21	0.69	0.57	0.67	0.34	1.48	0.43	1.75	2.60	**12.55**	**6.32**	0.45	0.71	1.97	-1.88	-9.15	0.70	0.49	0.91	0.13	0.70
	$\overline{\text{agg} + \text{NNM}}$	CE	73.77	72.38	68.82	68.11	67.68	74.14	72.89	71.16	69.31	63.58	73.25	67.21	60.92	55.70	49.12	71.19	67.45	62.50	57.29	49.23	74.39	74.28	74.39	74.70	74.64
		NS	74.86	72.21	68.58	65.19	65.33	74.89	73.78	72.05	69.82	63.77	73.38	69.57	62.93	56.39	48.00	72.07	67.98	63.47	57.42	47.27	74.96	74.61	74.84	74.61	74.58
		gain	1.09	-0.17	-0.24	-2.92	-2.35	0.75	0.90	0.89	0.52	0.20	0.13	2.36	2.01	0.69	-1.11	0.89	0.53	0.97	0.14	-1.96	0.58	0.34	0.45	-0.09	-0.07
1	$\overline{\text{agg}}$	CE	71.19	60.80	46.68	30.76	22.20	71.61	69.72	64.90	56.46	48.75	67.27	57.06	39.55	28.08	14.30	66.59	57.99	47.78	36.45	24.48	71.32	69.25	67.40	64.08	55.89
		NS	71.97	63.41	53.42	41.72	29.87	72.90	70.99	67.58	61.45	53.58	71.09	61.38	54.52	39.86	16.82	69.11	60.96	54.81	44.02	19.37	72.98	71.59	70.55	68.32	63.63
		gain	0.78	2.61	**6.74**	**10.96**	**7.67**	1.29	1.27	2.68	4.99	4.84	3.82	4.33	**14.97**	**11.78**	2.52	2.51	2.97	**7.03**	**7.57**	-5.11	1.66	2.34	3.14	4.24	**7.74**
	$\overline{\text{agg} + \text{NNM}}$	CE	71.93	66.66	61.36	57.60	66.13	73.02	71.52	69.01	65.33	56.60	69.79	55.12	56.53	46.85	34.87	68.10	63.07	57.90	51.42	40.02	72.72	72.03	71.38	71.19	68.81
		NS	72.30	71.71	69.93	65.03	68.45	72.90	71.61	71.01	68.31	61.17	70.51	70.20	69.91	68.25	66.99	69.05	67.79	66.77	61.56	52.05	73.34	72.72	73.08	73.20	73.39
		gain	0.37	**5.05**	**8.57**	**7.43**	2.32	-0.12	0.09	2.00	2.98	4.56	0.72	**15.08**	**13.38**	**21.40**	**32.11**	0.95	4.72	**8.87**	**10.14**	**12.03**	0.63	0.70	1.70	2.01	4.58
0.1	$\overline{\text{agg}}$	CE	50.61	31.26	22.13	19.48	10.68	55.76	53.83	43.98	33.20	28.39	51.74	31.36	17.78	16.45	13.91	51.46	35.21	20.61	17.32	14.65	54.72	52.55	48.42	43.68	41.69
		NS	60.53	42.99	30.16	24.37	12.38	60.38	59.02	53.66	42.36	36.75	56.37	39.70	32.13	23.08	20.16	55.59	41.58	29.15	18.40	7.38	58.30	54.84	54.00	49.64	46.44
		gain	**9.92**	**11.73**	**8.03**	4.89	1.69	4.62	**5.19**	**9.68**	**9.16**	**8.36**	4.63	**8.34**	**14.35**	**6.63**	**6.25**	4.13	**6.37**	**8.54**	1.08	-7.27	3.58	2.29	**5.57**	**5.97**	4.75
	$\overline{\text{agg} + \text{NNM}}$	CE	55.51	45.33	34.44	31.14	23.08	59.17	61.23	59.15	53.25	30.45	56.08	34.49	16.22	16.82	13.45	55.97	49.11	40.24	34.27	20.85	57.65	57.30	57.67	51.08	49.59
		NS	63.91	59.26	55.36	47.88	47.89	63.36	63.32	63.11	57.68	48.94	61.42	47.49	37.01	36.99	16.43	60.99	55.67	49.56	39.46	15.53	63.00	59.56	61.00	57.46	53.98
		gain	**8.40**	**13.93**	**20.92**	**16.74**	**24.82**	4.19	2.09	3.96	4.43	**18.50**	**5.33**	**12.99**	**20.78**	**20.16**	2.98	**5.02**	**6.56**	**9.32**	**5.19**	-5.32	**5.34**	2.26	3.32	**6.38**	4.40

Table VI.6: MNIST. Mean accuracy (%) over the last 200 steps for $\overline{\text{agg}}$ (average over CWMed, CwTM, and RFA without NNM) and $\overline{\text{agg} + \text{NNM}}$ (same with NNM). We compare cross-entropy only (CE) and cross-entropy with NorthStar (NS) across $\alpha \in \{10, 1, 0.1\}$, attacks, and Byzantine counts $f \in \{2, 8, 14, 20, 26\}$.

			ALIE					Mimic					MinSum					FOE					LF				
			2	8	14	20	26	2	8	14	20	26	2	8	14	20	26	2	8	14	20	26	2	8	14	20	26
10	$\overline{\text{agg}}$	CE	98.66	98.64	98.30	97.89	95.79	98.66	98.59	98.33	97.93	97.65	98.55	98.16	97.19	95.98	86.48	98.41	97.84	97.58	97.04	95.26	98.67	98.76	98.53	98.20	97.10
		NS	98.67	98.47	98.27	97.91	95.79	98.66	98.61	98.34	97.94	97.50	98.66	98.04	97.00	95.39	93.20	98.49	97.71	97.17	96.16	95.18	98.71	98.69	98.56	97.72	96.08
		gain	0.01	-0.18	-0.04	0.01	0.00	0.01	0.02	0.01	0.00	-0.15	0.11	-0.12	-0.18	-0.59	**6.72**	0.08	-0.13	-0.41	-0.87	-0.08	0.04	-0.07	0.03	-0.49	-1.02
	$\overline{\text{agg} + \text{NNM}}$	CE	98.70	98.69	98.67	98.67	98.62	98.71	98.73	98.53	98.35	97.66	98.63	98.46	98.09	97.41	96.39	98.53	98.27	97.90	97.65	96.34	98.73	98.76	98.70	98.73	98.72
		NS	98.70	98.69	98.74	98.51	98.56	98.73	98.71	98.59	98.28	97.78	98.68	98.43	98.29	97.82	96.40	98.60	98.30	98.01	97.40	96.26	98.70	98.73	98.77	98.80	98.77
		gain	-0.00	0.00	0.07	-0.15	-0.06	0.02	-0.02	0.06	-0.07	0.12	0.05	-0.03	0.20	0.41	0.01	0.08	0.04	0.11	-0.25	-0.08	-0.03	-0.03	0.07	0.07	0.04
1	$\overline{\text{agg}}$	CE	98.47	98.32	97.63	96.00	80.90	98.56	98.40	98.17	97.65	96.08	98.20	97.81	96.93	93.79	69.93	98.24	97.70	96.38	91.36	50.72	98.46	98.47	98.07	97.31	94.44
		NS	98.53	98.29	98.20	97.44	93.68	98.57	98.35	98.14	97.81	96.97	98.49	98.00	97.13	96.15	94.10	98.38	97.72	96.95	95.65	92.06	98.56	98.57	98.14	97.33	92.88
		gain	0.06	-0.03	0.56	1.44	**12.78**	0.01	-0.05	-0.02	0.17	0.89	0.29	0.19	0.20	2.36	**24.17**	0.14	0.02	0.57	4.30	**41.34**	0.11	0.10	0.08	0.02	-1.56
	$\overline{\text{agg} + \text{NNM}}$	CE	98.71	98.72	98.54	98.44	98.65	98.70	98.65	98.45	98.31	97.28	98.53	98.24	98.10	90.94	87.77	98.36	97.92	97.46	94.94	89.03	98.73	98.78	98.70	98.72	98.77
		NS	98.55	98.60	98.57	98.48	98.44	98.60	98.49	98.40	98.15	97.43	98.21	98.25	97.96	97.87	98.00	98.17	98.23	97.97	97.48	96.34	98.67	98.66	98.70	98.60	98.64
		gain	-0.16	-0.12	0.03	0.04	-0.21	-0.10	-0.15	-0.05	-0.16	0.15	-0.33	0.01	-0.14	**6.93**	**10.23**	-0.19	0.31	0.51	2.55	**7.32**	-0.06	-0.12	-0.01	-0.11	-0.13
0.1	$\overline{\text{agg}}$	CE	86.24	93.09	83.42	37.10	9.84	90.94	89.63	76.00	70.69	53.72	88.78	50.94	38.76	32.30	22.02	86.07	67.25	48.19	40.08	22.04	93.38	91.15	85.41	66.58	67.11
		NS	97.56	97.09	93.13	67.78	28.82	97.50	95.47	94.21	89.70	67.83	96.73	93.52	86.42	75.60	51.27	96.84	92.99	81.68	69.17	37.76	97.66	97.34	95.64	91.81	85.37
		gain	**11.32**	4.00	**9.71**	**30.69**	**18.98**	**6.56**	**5.84**	**18.22**	**19.01**	**14.10**	**7.95**	**42.58**	**47.66**	**43.30**	**29.25**	**10.77**	**25.74**	**33.49**	**29.09**	**15.73**	4.28	**6.19**	**10.23**	**25.23**	**18.26**
	$\overline{\text{agg} + \text{NNM}}$	CE	98.67	98.35	95.96	84.61	86.18	98.73	98.54	97.62	92.45	88.27	98.40	88.85	59.54	32.30	33.91	98.02	92.02	66.73	54.56	23.57	98.76	98.74	98.75	98.73	98.55
		NS	98.51	98.29	98.11	98.09	98.23	98.51	98.31	98.07	97.23	89.73	98.10	97.47	90.45	48.80	19.81	97.77	97.37	72.28	44.95	26.93	98.52	98.44	98.45	98.35	98.43
		gain	-0.16	-0.06	2.15	**13.48**	**12.06**	-0.22	-0.23	0.46	4.78	1.46	-0.30	**8.62**	**30.92**	**16.50**	-14.10	-0.24	**5.35**	**5.56**	-9.62	3.36	-0.24	-0.30	-0.30	-0.37	-0.12

Table VI.7: Fashion-MNIST. Mean accuracy (%) over the last 200 steps for $\overline{\text{agg}}$ (average over CWMed, CwTM, and RFA without NNM) and $\overline{\text{agg} + \text{NNM}}$ (same with NNM). We compare cross-entropy only (CE) and cross-entropy with NorthStar (NS) across $\alpha \in \{10, 1, 0.1\}$, attacks, and Byzantine counts $f \in \{2, 8, 14, 20, 26\}$.

			ALIE					Mimic					MinSum					FOE					LF				
			2	8	14	20	26	2	8	14	20	26	2	8	14	20	26	2	8	14	20	26	2	8	14	20	26
10	$\overline{\text{agg}}$	CE	86.75	86.07	82.69	80.03	73.39	86.71	86.58	85.58	84.74	82.27	85.99	83.68	82.12	77.97	69.08	85.62	83.86	82.22	79.47	73.98	86.61	86.17	84.48	82.76	78.70
		NS	87.04	86.54	84.82	81.21	75.27	87.12	87.10	86.06	85.21	82.62	86.83	84.93	83.46	79.53	77.72	86.31	83.95	82.61	80.41	76.52	87.05	86.83	86.04	85.04	82.33
		gain	0.29	0.48	2.13	1.18	1.88	0.41	0.52	0.48	0.48	0.35	0.84	1.26	1.35	1.56	**8.63**	0.69	0.08	0.39	0.95	2.54	0.44	0.67	1.55	2.28	3.63
	$\overline{\text{agg} + \text{NNM}}$	CE	86.82	86.57	86.69	86.84	86.73	87.04	87.08	86.48	85.54	82.59	85.81	85.00	82.79	82.06	76.84	85.84	84.03	81.73	79.44	77.51	86.93	87.11	87.10	86.94	87.23
		NS	87.11	86.93	86.93	86.82	86.68	87.06	87.08	86.31	85.75	83.15	86.87	85.58	84.53	82.59	79.97	86.28	85.13	84.00	81.90	79.30	87.12	87.24	87.14	87.04	87.27
		gain	0.28	0.36	0.24	-0.02	-0.05	0.02	0.01	-0.17	0.21	0.56	1.06	0.58	1.73	0.53	3.13	0.43	1.10	2.27	2.46	1.79	0.19	0.13	0.04	0.10	0.04
1	$\overline{\text{agg}}$	CE	85.05	82.38	77.32	70.25	49.68	85.28	84.48	83.58	80.83	77.49	83.68	79.49	75.50	69.19	52.15	83.69	79.74	75.13	71.82	52.03	85.13	83.88	81.40	74.05	68.31
		NS	86.92	85.48	82.87	78.90	69.26	86.71	86.31	85.80	84.46	82.37	86.12	84.35	81.88	79.47	74.69	86.00	84.01	81.70	79.18	74.80	86.83	86.46	85.57	83.45	79.22
		gain	1.87	3.10	**5.55**	**8.64**	**19.57**	1.43	1.83	2.22	3.63	4.87	2.44	4.86	**6.38**	**10.28**	**22.54**	2.32	4.26	**6.57**	**7.35**	**22.77**	1.70	2.58	4.16	**9.40**	**10.91**
	$\overline{\text{agg} + \text{NNM}}$	CE	86.68	85.92	85.14	84.65	84.57	86.78	86.48	85.57	83.49	80.69	84.84	81.96	79.14	74.47	65.77	83.67	81.18	79.20	77.28	59.35	87.11	86.98	86.28	86.01	82.41
		NS	87.04	86.72	86.27	86.25	85.88	86.93	86.77	86.20	85.62	84.03	85.96	85.87	85.14	85.03	83.23	85.70	84.24	84.43	83.56	78.50	87.08	86.76	86.88	86.63	86.60
		gain	0.35	0.80	1.13	1.60	1.31	0.15	0.29	0.63	2.13	3.34	1.12	3.90	**6.00**	**10.56**	**17.47**	2.04	3.06	**5.23**	**6.28**	**19.15**	-0.03	-0.22	0.60	0.62	4.18
0.1	$\overline{\text{agg}}$	CE	77.44	70.69	59.25	24.75	9.93	78.21	73.99	67.50	56.45	42.78	71.13	42.24	27.94	33.44	19.14	70.96	50.10	31.46	33.88	17.01	78.68	76.61	72.33	66.12	56.55
		NS	80.07	76.56	70.56	55.66	30.83	81.17	78.11	76.32	67.92	51.93	79.31	71.44	69.59	61.78	36.70	78.72	73.31	67.99	51.88	28.93	79.80	82.92	79.21	75.23	72.21
		gain	2.63	**5.87**	**11.31**	**30.91**	**20.90**	2.96	4.12	**8.82**	**11.47**	**9.15**	**8.18**	**29.20**	**41.65**	**28.34**	**17.56**	**7.76**	**23.20**	**36.53**	**18.00**	**11.93**	1.12	**6.31**	**6.88**	**9.11**	**15.66**
	$\overline{\text{agg} + \text{NNM}}$	CE	86.14	81.53	80.11	78.01	63.55	85.86	85.32	81.58	74.20	65.68	81.13	73.37	60.94	38.51	27.48	77.29	73.11	62.04	53.89	30.88	86.13	85.87	86.10	82.99	73.76
		NS	86.09	84.98	83.56	81.63	86.25	86.42	86.42	86.12	77.29	72.27	83.60	77.04	73.77	29.30	22.07	82.24	80.49	77.12	59.62	29.31	86.16	86.41	87.15	85.77	84.62
		gain	-0.05	3.45	3.45	3.61	**22.70**	0.56	1.10	4.53	3.09	**6.58**	2.47	3.67	**12.83**	-9.21	-5.41	4.96	**7.38**	**15.08**	**5.73**	-1.57	0.03	0.53	1.06	2.79	**10.86**

Table VI.8: Kuzushiji-MNIST. Mean accuracy (%) over the last 200 steps for $\overline{\text{agg}}$ (average over CWMed, CwTM, and RFA without NNM) and $\overline{\text{agg} + \text{NNM}}$ (same with NNM). We compare cross-entropy only (CE) and cross-entropy with NorthStar (NS) across $\alpha \in \{10, 1, 0.1\}$, attacks, and Byzantine counts $f \in \{2, 8, 14, 20, 26\}$.

			ALIE					Mimic					MinSum					FOE					LF				
			2	8	14	20	26	2	8	14	20	26	2	8	14	20	26	2	8	14	20	26	2	8	14	20	26
10	$\overline{\text{agg}}$	CE	89.78	89.35	86.02	84.51	70.51	87.99	88.83	85.39	83.92	82.03	87.87	83.47	81.13	78.02	63.74	87.38	84.83	81.74	79.72	72.06	89.74	88.87	84.33	83.12	76.99
		NS	90.56	88.16	87.34	82.56	69.37	89.09	88.85	86.90	84.20	79.64	88.10	84.29	82.86	76.39	70.12	88.50	83.13	77.45	75.21	71.36	90.45	87.77	86.60	84.13	77.78
		gain	0.78	-1.19	1.31	-1.95	-1.14	1.09	0.02	1.51	0.28	-2.39	0.23	0.82	1.73	-1.63	**6.39**	1.12	-1.70	-4.29	-4.51	-0.70	0.71	-1.10	2.26	1.01	0.79
	$\overline{\text{agg} + \text{NNM}}$	CE	89.32	90.11	89.92	90.22	89.78	89.99	89.17	88.35	88.12	85.10	88.69	82.99	83.49	79.62	76.92	87.86	85.77	84.51	81.66	77.55	89.58	87.87	89.26	89.32	88.95
		NS	90.02	89.12	89.69	89.36	89.65	89.44	90.06	86.43	85.61	80.73	90.58	87.47	85.91	81.56	77.14	89.54	85.99	79.66	79.24	69.04	91.06	89.68	89.16	88.61	88.44
		gain	0.70	-0.99	-0.23	-0.86	-0.13	-0.55	0.89	-1.92	-2.51	-4.37	1.89	4.47	2.42	1.95	0.22	1.68	0.22	-4.85	-2.41	-8.50	1.47	1.81	-0.10	-0.71	-0.51
1	$\overline{\text{agg}}$	CE	85.74	87.46	78.39	73.58	44.50	84.55	87.84	86.55	83.61	80.02	83.56	78.69	67.77	63.41	46.38	84.34	82.59	67.62	61.51	41.23	85.77	83.48	80.96	71.18	58.39
		NS	89.31	88.05	86.23	77.02	58.75	88.80	88.20	85.79	83.97	80.80	87.45	86.29	84.26	79.89	68.50	87.16	83.12	78.98	76.99	68.91	89.08	88.11	84.30	80.12	72.67
		gain	3.58	0.59	**7.85**	3.44	**14.25**	4.25	0.37	-0.77	0.36	0.78	3.90	**7.60**	**16.49**	**16.48**	**22.12**	2.82	0.52	**11.36**	**15.48**	**27.68**	3.32	4.63	3.34	**8.95**	**14.28**
	$\overline{\text{agg} + \text{NNM}}$	CE	88.01	90.97	90.37	89.32	89.97	88.12	89.83	89.80	88.49	84.65	85.09	80.30	82.99	70.44	43.67	82.20	85.16	83.91	81.10	53.67	89.42	87.53	88.23	87.44	86.76
		NS	89.89	89.53	89.87	89.44	89.31	88.55	88.97	86.97	86.69	84.48	86.94	88.10	86.11	85.45	83.87	83.17	87.03	85.30	80.93	79.14	89.41	90.44	89.20	89.50	89.09
		gain	1.88	-1.44	-0.50	0.12	-0.66	0.43	-0.86	-2.83	-1.80	-0.18	1.85	**7.80**	3.12	**15.01**	**40.20**	0.97	1.87	1.39	-0.17	**25.48**	-0.01	2.90	0.97	2.06	2.33
0.1	$\overline{\text{agg}}$	CE	68.23	61.01	47.14	22.52	11.76	70.67	63.72	48.57	47.70	33.58	67.33	42.33	25.62	20.36	13.88	66.81	41.96	27.90	19.49	0.00	67.86	59.02	54.49	48.83	40.66
		NS	69.91	68.52	56.60	38.22	22.14	77.72	75.80	71.15	58.62	44.24	68.72	63.51	50.93	56.54	24.35	71.33	62.89	53.50	39.38	13.21	73.74	72.97	62.83	52.40	44.55
		gain	1.68	**7.51**	**9.45**	**15.69**	**10.38**	**7.05**	**12.09**	**22.58**	**10.92**	**10.66**	1.38	**21.17**	**25.31**	**36.18**	**10.47**	4.52	**20.94**	**25.59**	**19.89**	**13.21**	**5.88**	**13.95**	**8.34**	3.58	3.89
	$\overline{\text{agg} + \text{NNM}}$	CE	90.19	84.07	82.75	73.97	55.69	90.44	89.92	87.56	83.05	61.23	81.79	66.34	28.97	27.41	22.06	83.25	72.29	53.94	42.05	17.21	90.45	90.46	88.73	84.29	70.40
		NS	89.60	84.67	80.00	88.61	85.44	89.50	87.77	88.02	84.34	66.77	86.72	71.56	65.80	30.38	25.74	85.26	81.29	75.04	51.38	22.24	88.45	88.08	87.99	86.51	84.54
		gain	-0.59	0.60	-2.74	**14.64**	**29.75**	-0.94	-2.15	0.46	1.29	**5.53**	4.93	**5.21**	**36.83**	2.97	3.68	2.01	**8.99**	**21.10**	**9.33**	**5.03**	-2.00	-2.37	-0.75	2.22	**14.14**

Chapter VII

Inside the Convex Hull of Gradients

1 HullGuard Filter **82**

1.1 Analogy for HullGuard 82

1.2 HullGuard Algorithm 82

1.3 HullGuard in Detail 83

2 Experiments **86**

2.1 Experimental Setup 86

2.2 RQ1: How effective is the HullGuard filter? 88

2.3 RQ2: Does HullGuard improve robustness against Byzantine attacks? 88

3 Conclusion **89**

This chapter is based on our publication [32].

Many robust protocols assess update honesty through a proximity criterion: honest gradients are expected to be close to one another, while Byzantine gradients are treated as outliers. Other methods rely on a server-side validation set. Under label skew, honest gradients can naturally differ, so gradient proximity is not a reliable proxy for honesty. In this setting, a more suitable criterion is consistency at the level of class-wise gradients.

We introduce *HullGuard*, a server-side filter based on a geometric property of honest updates under label skew: they are noisy convex combinations of global class-wise gradients. HullGuard exploits this structure by asking workers to share their mini-batch label distributions at each training step. It then filters workers whose updates are inconsistent with the induced class-wise geometry. In our experiments, HullGuard removes most Byzantine workers during training, outperforms existing strategies, and can be combined with standard robust aggregation methods. We complement these empirical results with a theoretical analysis that helps explain its effectiveness.

Contributions. We propose a principled defense mechanism against Byzantine workers under label skew:

- **Structural property.** We show that honest updates satisfy a structural property under label skew: each honest gradient can be written as a noisy convex combination of global class-wise gradients. This property helps distinguish heterogeneity caused by label skew from deviations induced by Byzantine behavior.
- **Effective Byzantine filtering.** We introduce HullGuard, a Byzantine filter that exploits this structure by selecting workers whose updates are most consistent with the global class-wise gradients.
- **Theoretical analysis.** We provide theoretical arguments that help explain the effectiveness of HullGuard, with full proofs given in the appendix.
- **Experimental evaluation.** We evaluate HullGuard under increasingly challenging label-skew and Byzantine settings. The results show that HullGuard filters out most Byzantine workers during training, improves robustness over competing methods in difficult settings, and maintains stable performance even when existing approaches degrade.

Code: `https://github.com/JohanErbani/HullGuard`

1 HullGuard Filter

This section introduces HullGuard. The first part presents an analogy to help understand the mechanism behind HullGuard. The second part provides an overview of the algorithm. The final part presents the algorithmic details and theoretical arguments.

1.1 Analogy for HullGuard

Imagine a university recruiter selecting rowers for the university rowing team. Among n candidates, the recruiter must select the best h athletes. To compare them, K time trials are organized. In each trial, a team of h rowers is randomly sampled and attempts to achieve the best possible time. Let $A_1, \ldots, A_K$ denote the sampled teams and $t(A_k)$ the time obtained by team A_k.

The recruiter would like to assess the contribution of each candidate to the team's performance. For each candidate i, the recruiter estimates the probability that a team performs better when candidate i is present than when the candidate is absent, yielding a score $s(i)$:

$$\begin{aligned} s(i) &= \frac{|\{(k,l) : t(A_k) < t(A_l) \text{ with } i \in A_k, i \notin A_l\}|}{|\{k : i \in A_k\}|\,|\{l : i \notin A_l\}|} \\ &\approx \mathbb{P}\big(t(A) < t(A') \mid i \in A,\ i \notin A'\big) \end{aligned}$$

The recruiter then selects the h candidates with the highest scores. This strategy can be theoretically grounded under simple assumptions (see Proposition 19 in Appendix).

1.2 HullGuard Algorithm

We now present an overview of HullGuard. The description is intentionally informal; italicized terms will be formally defined in the following subsections. Algorithm 7 summarizes the procedure. At each round, workers send their update to the server along with the label distribution of the data used to compute it.

Sampling & evaluating worker groups. The goal of HullGuard is to identify honest workers (analogous to good rowers) and distinguish them from Byzantine workers (bad rowers). The algorithm begins by sampling K groups (analogous to teams) $A_1, \ldots, A_K$ of size h.

Each group is evaluated using a performance score. In the recruiter story this score was the time $t(A)$. In HullGuard, the score is the *empirical Frobenius variance of the global class-wise gradient estimator* (i.e., the variance of an estimator), denoted $\hat{\sigma}^2(A)$.

Variance of class-wise gradient estimates. To compute $\hat{\sigma}^2(A)$, we proceed as follows. For each group A, we randomly sample subgroups $B_1, \ldots, B_{K'}$ such that $B_k \subset A$ for $k = 1, \ldots, K'$. For each subgroup, we compute an estimate of the *global class-wise gradients* (i.e., a matrix per subgroup), producing $\hat{\mathcal{U}}_1, \ldots, \hat{\mathcal{U}}_{K'} \in \mathbb{R}^{C \times D}$. We then compute the *empirical Frobenius variance* based on these matrices, denoted $\hat{\sigma}^2(A)$. This quantity measures how much the *global class-wise gradients* vary within A.

The key observation is that, under label skew, honest updates are a noisy convex combinations[1] of vectors called *global class-wise gradients*. These vectors are identical across workers. As a result, a group A composed mostly of honest workers produces similar gradient estimates across the subgroups B_k, leading to a small group score $\hat{\sigma}^2(A)$ (analogous to a small time: the lower, the better).

In contrast, Byzantine updates tend to exhibit *class-wise gradients* that differ from those of honest workers in order to be disruptive. Consequently, when a group A contains Byzantine workers, their presence increases the variability of the estimates, i.e., the matrices $\hat{\mathcal{U}}_1, \ldots, \hat{\mathcal{U}}_{K'}$ differ more. In summary, the more honest workers a group A contains, the smaller its *empirical Frobenius variance* $\hat{\sigma}^2(A)$.

Worker scoring & selection. Following the same process as in the recruiter, HullGuard assigns a score to each worker based on pairwise comparisons between group scores $\hat{\sigma}^2(A_1)$,..., $\hat{\sigma}^2(A_K)$ and selects the h workers with the highest scores, thereby filtering out Byzantine workers.

Robust aggregation. Finally, a user-chosen robust aggregation step is applied. It mitigates two issues: some Byzantine workers may remain after HullGuard filtering, and honest gradients may still be heterogeneous due to data distribution differences across workers, causing the update to deviate from the honest mean gradient. For instance, the coordinate-wise median can be used.

1.3 HullGuard in Detail

This subsection introduces the notion of class-wise gradients, explains why honest updates are noisy convex combinations of global class-wise gradients, describes how class-wise gradients are estimated, and shows why the variance of these estimators within a group can act as a Byzantine detector.

Local gradient as a convex combination of local class-wise gradients. For each worker i, the local gradient can be written as a convex combination of local class-wise gradients:

$$\underbrace{\nabla \mathcal{L}_i = \sum_{c=1}^{C} \frac{|\mathcal{D}_i^c|}{|\mathcal{D}_i|} \nabla \mathcal{L}_i^c,}_{\text{local update}} \quad \text{with} \quad \underbrace{\nabla \mathcal{L}_i^c = \frac{1}{|\mathcal{D}_i^c|} \sum_{(x,y) \in \mathcal{D}_i^c} \nabla \ell\big(c, \Phi(x)\big),}_{\text{local class-wise gradient}} \tag{VII.1}$$

where $\mathcal{D}_i^c \subset \mathcal{D}_i$ is the subset of samples belonging to class c. The class-wise gradient $\nabla \mathcal{L}_i^c$ is defined as the zero vector if class c is not present in $\mathcal{D}_i$.

Equation (VII.1) shows that the local gradient is a convex combination of class-wise gradients, with weights given by the local class proportions.

Local gradient as a noisy combination of global class-wise gradients. Local gradient $\nabla \mathcal{L}_i$ can be written with global (not local) class-wise gradients $\nabla \mathcal{L}^c$ and an

[1]A vector u is a convex combination of vectors $v_1, \ldots, v_C$ if there exist coefficients $\alpha_1, \ldots, \alpha_C$ such that $\alpha_c \geq 0$, $\sum_{c=1}^{C} \alpha_c = 1$, and $u = \sum_{c=1}^{C} \alpha_c v_c$.

error term ε_i:

$$\nabla \mathcal{L}_i = \sum_{c=1}^{C} \frac{|\mathcal{D}_i^c|}{|\mathcal{D}_i|} \nabla \mathcal{L}^c + \varepsilon_i \tag{VII.2}$$

$$\text{with} \quad \underbrace{\nabla \mathcal{L}^c = \frac{1}{|\mathcal{D}^c|} \sum_{(x,y)\in\mathcal{D}^c} \nabla \ell\big(c, \Phi(x)\big)}_{\text{global class-wise gradient (not worker-specific)}} \quad , \quad \underbrace{\varepsilon_i = \sum_{c=1}^{C} \frac{|\mathcal{D}_i^c|}{|\mathcal{D}_i|} (\nabla \mathcal{L}_i^c - \nabla \mathcal{L}^c)}_{\text{noise}}$$

Under label skew, the error term has zero mean, i.e., $\mathbb{E}[\varepsilon_i] = 0$. Furthermore, global class-wise gradients are not worker-specific; they depend only on the global dataset $\mathcal{D}$.

Equation (VII.2) shows that each honest update is a noisy convex combination of global class-wise gradients.

Class-wise gradient estimator. For any selected subgroup $B_1, \dots, B_{K'}$ of workers, the gradients and label proportions used to produce their updates define a linear system from which the global class-wise gradients can be estimated.

Consider a subgroup B of s honest workers indexed by $\{1, \dots, s\}$. By row stacking Equation (VII.2), we obtain

$$\underbrace{\begin{pmatrix} \nabla\mathcal{L}_1 \\ \vdots \\ \nabla\mathcal{L}_s \end{pmatrix}}_{\text{updates } K\times D} = \underbrace{\begin{pmatrix} |\mathcal{D}_1^1|/|\mathcal{D}_1| & \dots & |\mathcal{D}_1^C|/|\mathcal{D}_1| \\ & \vdots & \\ |\mathcal{D}_s^1|/|\mathcal{D}_s| & \dots & |\mathcal{D}_s^C|/|\mathcal{D}_s| \end{pmatrix}}_{\text{label distributions } K\times C} \underbrace{\begin{pmatrix} \nabla\mathcal{L}^1 \\ \vdots \\ \nabla\mathcal{L}^C \end{pmatrix}}_{\text{global class gradients } C\times D} + \underbrace{\begin{pmatrix} \varepsilon_1 \\ \vdots \\ \varepsilon_s \end{pmatrix}}_{\text{noise } K\times D}$$

$$\begin{array}{ccccccc} \downarrow & & \downarrow & & \downarrow & & \downarrow \\ U & = & P & & \mathcal{U} & + & E \end{array}$$

where U denotes the update matrix, P the label distribution matrix, $\mathcal{U}$ the global class gradient matrix, and E the noise matrix.

To estimate the global class-wise gradients, we consider the following minimization problem

$$\min_{X\in\mathbb{R}^{C\times D}} \|PX - U\|_F, \tag{VII.3}$$

which seeks the C vectors that best explain the updates according to P. Indeed, we have

$$\|PX - U\|_F^2 = \sum_{i=1}^{s} \left\| \sum_{c=1}^{C} \frac{|\mathcal{D}_i^c|}{|\mathcal{D}_i|} (X_{c:} - \nabla\mathcal{L}_i^c) \right\|_2^2,$$

where $X_{1:}, \dots, X_{C:}$ denote the rows of X. When sufficient data are available (i.e., when s is sufficiently large), a natural candidate for X is the matrix of global class-wise gradients $\mathcal{U}$. The solution to Problem (VII.3) is given by the least-squares estimator [27]

$$P^{\dagger}U \in \operatorname*{argmin}_{X\in\mathbb{R}^{C\times D}} \|PX - U\|_F, \tag{VII.4}$$

where $P^{\dagger}$ denotes the Moore–Penrose pseudoinverse[2] of P.

If workers have sufficiently different label distributions (i.e., the label distribution matrix P has full column rank), this estimator is unbiased and, under an additional assumption, optimal among linear estimators (see Proposition 20 in Appendix). Specifically, if the variance of the loss gradient is class-independent, it achieves the minimum variance among all linear unbiased estimators.

In HullGuard, for each group A, we sample subgroups $B_1, B_2, \dots, B_{K'}$. For each subgroup B_i, we estimate the global class-wise gradient using the least-squares estimator $\hat{\mathcal{U}}_i = P_i^{\dagger} U_i$, where P_i and U_i denote here the label distribution and the updates matrices corresponding to subgroup B_i.

Variance as a Byzantine detector. The Frobenius variance [96] is defined as

[2] The Moore–Penrose pseudoinverse $A^{\dagger}$ of a matrix A extends the notion of matrix inversion to non-square or rank-deficient matrices. When A is invertible, the pseudoinverse coincides with the inverse, i.e., $A^{\dagger} = A^{-1}$.

Definition 2 (Frobenius variance)**.** *The variance of a random matrix X with respect to the Frobenius norm is defined as* $\sigma^2(X) = \mathbb{E}\big[\|X - \mathbb{E}[X]\|_F^2\big]$.

Once the matrices $\hat{\mathcal{U}}_1, \dots, \hat{\mathcal{U}}_{K'}$ are computed, we estimate the Frobenius variance of the class-wise gradient estimator as

$$\underbrace{\hat{\sigma}^2(A) = \frac{1}{K'-1}\sum_{k=1}^{K'} \left\|\hat{\mathcal{U}}_k - \hat{\mu}\right\|_F^2,}_{\text{empirical Frobenius variance of class gradients}} \qquad \text{where } \hat{\mu} = \frac{1}{K'}\sum_{k=1}^{K'}\hat{\mathcal{U}}_k$$

The more similar the workers in group A are—where similarity is measured through their class-wise gradients across subgroups—the lower the empirical variance $\hat{\sigma}^2(A)$.

For theoretical purposes, we make the following assumptions: (i) the number of Byzantine workers satisfies $f < N/3$, which is equivalent to $f < h/2$; (ii) as with honest workers, Byzantine updates are also convex combinations of C vectors, and their label distributions are consistent (i.e., Byzantine updates are also derived from their class-wise gradients); (iii) for any subgroup B, the associated label distribution matrix has full column rank. All details are provided in the appendix.

The intuition is the following. The variance measures worker similarity. A group A composed solely of Byzantine workers could exhibit high internal similarity and therefore a low variance. However, since we assume $f < h/2$, Byzantine workers are always in the minority, and no such group can exist. Under the constraint $f < h/2$, the variance does not only capture similarity across workers, but similarity to the majority, i.e., similarity to honest workers. Consequently, as soon as Byzantine workers behave differently from honest ones (from a class-wise gradient perspective), their presence increases the variance.

We formalize this observation by studying the Frobenius variance of the class-wise gradient estimator for a fixed group A, denoted by $\sigma^2(A)$, which we estimate by $\hat{\sigma}^2(A)$. Let h' and f' denote the numbers of honest and Byzantine workers in A. Before proceeding, we introduce the random variables involved in the analysis. To avoid heavy notation, we present realizations of these random variables.

Consider a sampled subgroup $B \subset A$. Let P and Q denote the label distribution matrices associated with the honest and Byzantine workers in B, respectively. Let U and V denote the corresponding updates, defined in the same way as previously. Define

$$\hat{u} = P^{\dagger}U, \quad \hat{v} = Q^{\dagger}V, \quad w = (X+Y)^{-1}X, \quad X = P^{\top}P, \quad \text{and } Y = Q^{\top}Q.$$

Then $\hat{u}$ is a realization of the random variable $\hat{\mathcal{U}}$, which represents the honest class-wise gradient estimator computed from the honest elements of A. Similarly, $\hat{v}$ is a realization of the random variable $\hat{\mathcal{V}}$, the corresponding estimator obtained from Byzantine workers.

Regarding w, the matrix $X = P^{\top}P$ captures the structure of honest workers' label distributions. When these distributions are diverse and cover different classes, X spans the class space well; when they are similar or concentrated on a few classes, X becomes nearly low rank and carries less information. The same interpretation holds for Y for Byzantine workers. Thus, w measures the share of total information $X + Y$ contributed by honest workers, while $I - w = (X+Y)^{-1}Y$ measures the contribution of Byzantine workers. The matrix w is a realization of the random matrix W.

We can now explain how the variance associated with a fixed group A decomposes:

$$\sigma^2(A) = \underbrace{\mathbb{E}\left[\left\|W(\hat{\mathcal{U}} - \mathcal{U})\right\|_F^2\right]}_{\substack{\text{variance of honest}\\ \text{estimator weighted by } W}} + \underbrace{\mathbb{E}\left[\left\|(I - W)(\hat{\mathcal{V}} - \mathcal{V})\right\|_F^2\right]}_{\substack{\text{variance of Byzantine}\\ \text{estimator weighted by } (I-W)}} + \underbrace{\mathbb{E}\left[\|(W - \mathbb{E}[W])(\mathcal{V} - \mathcal{U})\|_F^2\right]}_{\substack{\text{variance of } W \text{ weighted by}(\mathcal{V}-\mathcal{U}),\\ \text{the difference between honest and}\\ \text{Byzantine global class-wise gradients}}}, \tag{VII.5}$$

where $\mathcal{U}$ and $\mathcal{V}$ denote the class-wise gradient matrices of honest and Byzantine workers, respectively. The first term corresponds to the variance of the honest estimator in A, weighted by the information contained in the honest label distributions through W. The second term is the analogous contribution for Byzantine workers. The last term captures the variability of W, scaled by the difference between the honest and Byzantine global class-wise gradients.

In HullGuard, the score of each worker depends on its membership in groups with low variance. Therefore, to bypass the HullGuard filter, Byzantine workers are incentivized to adopt strategies that reduce the group variance $\sigma^2(A)$ while remaining disruptive. The first term mainly depends on honest workers. The second term can be driven to zero by an appropriate Byzantine strategy. However, reducing the last term is more challenging. Byzantine workers would need either to make the variance of W vanish—which does not directly depend on them—or to match the honest class-wise gradients, which would no longer hinder model convergence.

In summary, obtaining a high worker score requires Byzantine workers to reduce $\sigma^2(A)$. However, achieving this effectively forces their class-wise gradients $\mathcal{V}$ to align with the honest ones $\mathcal{U}$.

Worker score. Once the empirical Frobenius variances $\sigma^2(A_1), \dots, \sigma^2(A_K)$ have been computed, HullGuard assigns each worker i a score $s(i)$ defined as

$$\begin{aligned} s(i) &= \frac{|\left\{(k,l) : \hat{\sigma}^2\left(A_k\right) < \hat{\sigma}^2\left(A_l\right) \text{ with } i \in A_k, i \notin A_l\right\}|}{|\left\{k : i \in A_k\right\}| \, |\left\{l : i \notin A_l\right\}|} \\ &\approx \mathbb{P}\big(\sigma^2(A) < \sigma^2(A') \mid i \in A,\ i \notin A'\big) \end{aligned}$$

A sufficient condition for this score to be theoretically grounded is that the variance $\sigma^2(A)$ increases with the number of Byzantine workers in A. Under some simplifying assumptions—Byzantine label distributions mimic the honest ones, and Byzantine class-wise gradients are sufficiently disruptive compared to the honest ones—we show that this property holds in Appendix.

2 Experiments

To guide the analysis, we address the following research questions:

RQ1. How effective is the HullGuard filter?

RQ2. Does HullGuard improve robustness against Byzantine attacks?

2.1 Experimental Setup

Datasets. We conduct our experiments on the following datasets : CIFAR-10 [60], MNIST [70], Fashion MNIST (FMNIST) [114] and EuroSAT [50]. Training data is divided among the workers using a Dirichlet distribution to generate non-IID local datasets, following the approach of [4, 52], with concentration parameters $\alpha \in \{0.1, 1, 10\}$. For each dataset, the test set follows the same class distribution as its corresponding training set.

Algorithm 7 HullGuard

Require: $\{(p_i, g_i)\}_{i=1}^{n}$ where $p_i \in \mathbb{R}_+^C$ is the class-distribution vector of worker i and $g_i \in \mathbb{R}^D$ is its gradient
Ensure: Robust aggregated gradient $\tilde{g} \in \mathbb{R}^D$
Class attributes: n (number of workers), h (number of honest workers), K (number of groups), K' (number of sub-subsets for variance estimation), s (sub-subset size), RobustAgg (robust aggregation rule)
Sample K groups $A_1, \dots, A_K \subset [n]$ uniformly at random, with $|A_k| = h$ for all $k \in [K]$ {Groups may overlap; elements are unique within each group}
for $k = 1$ to K **do**
  Sample K' subgroups $B_{k,1}, \dots, B_{k,K'} \subset A_k$ of size s {Subgroups may overlap; elements are unique within each group}
  for $j = 1$ to K' **do**
    $\hat{\mathcal{U}}_{k,j} \leftarrow$ EstimateClassWiseGradients$(B_{k,j}) \in \mathbb{R}^{C \times D}$
  end for
  $\hat{\sigma}^2(A_k) \leftarrow$ EmpiricalFrobeniusVariance$(\hat{\mathcal{U}}_{k,1}, \dots, \hat{\mathcal{U}}_{k,K'})$
end for
for $i = 1$ to n **do**
  $s(i) \leftarrow \dfrac{\left|\left\{(k,l) : \hat{\sigma}^2(A_k) < \hat{\sigma}^2(A_l) \text{ such that } i \in A_k,\ i \notin A_l\right\}\right|}{|\{k : i \in A_k\}|\ |\{l : i \notin A_l\}|}$
end for
$H \leftarrow \mathrm{Top}_h(s(1), \dots, s(n))$ {Select the h workers with the highest scores}
$\tilde{g} \leftarrow$ RobustAgg$(\{g_i\}_{i \in H})$ {Apply robust aggregation on the selected workers}
return $\tilde{g}$

Attacks & baselines. We consider the following attacks : A Little Is Enough (ALIE) [8], Fall Of Empire (FOE) [115], Mimic [56], Min-Sum [99] and Label Flipping (LF) [3]. We compare our solution to state-of-the-art robust aggregators : Coordinate-Wise Median (CWMed) [118], Coordinate-Wise Trimmed Mean (CwTM) [11], RFA [91] and GAS [75]. We also compare performance with and without pre-aggregation, including Bucketing (BKT) [56] and Nearest Neighbor Mixing (NNM) [4]. Due to limited computational resources, NNM is evaluated only on the CIFAR-10 dataset, where it exhibits trends similar to those observed with the other methods. Results can be found in appendix in Table D.6.

HullGuard requires the label distribution used to compute each update. We consider three Byzantine distributions: uniform, Dirac, and random. The main text reports results for the random case. The others show very similar behavior, indicating that HullGuard is largely invariant to these label distribution attacks. Results are provided in the Appendix.

FL settings. We consider 60 workers, with $f \in \{8, 14, 20, 26\}$ Byzantine workers. We train a four-layer CNN for CIFAR-10 and EuroSAT, and a two-layer CNN for MNIST and FMNIST (model in Appendix).The models are trained with batch size 32 and momentum $\beta = 0.9$. Momentum has shown strong robustness against Byzantine behavior in both heterogeneous [4] and homogeneous settings [28, 57, 39]. Model performance is evaluated by computing the accuracy averaged over the last 200 training steps.

HullGuard parameters. We set $K = 1000$ sampled groups and $K' = 5$ sampled subgroups per group, with subgroup size $s = 2C$. In practice, the user chooses an aggregation method applied after the filter. We use Bucketing—with bucket size $s = 5$, corresponding to an estimated 10% fraction of Byzantine workers among the h selected workers—combined with a coordinate-wise median aggregator. In the Appendix, we also evaluate HullGuard with GAS (Table D.1), which exhibits a similar overall trend.

Table VII.1: Mean number of Byzantine workers bypassing the HullGuard filter during training, averaged across all attacks (ALIE, FOE, Mimic, MinSum, LF).

		8	14	20	26
CIFAR10	10	1.34	0.80	0.50	0.59
	1	1.37	0.79	0.41	0.58
	0.1	4.10	2.92	1.90	1.79
MNIST	10	1.64	1.02	0.46	0.69
	1	1.64	1.01	0.76	0.93
	0.1	3.13	2.30	1.59	2.01
FMNIST	10	1.18	0.76	0.48	0.64
	1	1.22	0.99	0.82	1.14
	0.1	3.14	2.55	1.75	2.42
EuroSAT	10	1.25	0.65	0.30	0.41
	1	1.35	0.80	0.46	0.59
	0.1	5.46	3.17	1.78	1.62

2.2 RQ1: How effective is the HullGuard filter?

Table VII.1 evaluates the effectiveness of the HullGuard filtering stage by reporting the mean number of Byzantine workers that pass the filter during training, averaged across all attacks. Lower values indicate a more effective filter.

Overall, the results show that HullGuard successfully blocks the majority of Byzantine updates across all datasets and configurations. For low and moderate heterogeneity levels ($\alpha = 10$ and $\alpha = 1$), the number of Byzantine workers that bypass the filter remains close to one or below in most settings, indicating that the filter almost completely removes adversarial contributions.

We also observe is that the filtering performance improves as the number of Byzantine workers increases. For instance, on CIFAR10 with $\alpha = 10$, the average number of Byzantine workers passing the filter decreases from 1.34 (8 workers) to 0.59 (26 workers). Similar trends appear across other settings, suggesting that the Frobenius variance of class-wise gradients increases with the proportion of Byzantine workers, making worker scores more discriminative.

When $\alpha = 0.1$, more Byzantine workers occasionally pass the filter. However, even in this highly heterogeneous regime, their number remains well below the total number of Byzantine participants, meaning that the filter still removes most adversarial updates before aggregation.

These results demonstrate that HullGuard's filter is highly effective at eliminating Byzantine workers prior to aggregation.

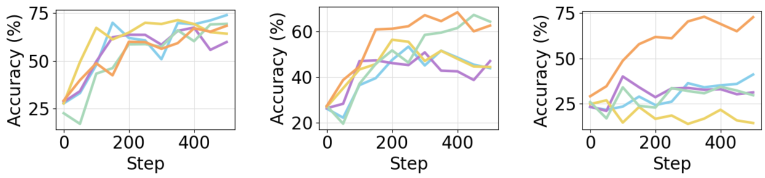
2.3 RQ2: Does HullGuard improve robustness against Byzantine attacks?

Table VII.2 compares the final accuracy of state of the art methods face to HullGuard on CIFAR-10 under different attacks, heterogenity.

Overall, the results show that HullGuard substantially improves robustness in the most challenging scenarios, particularly when the number of workers increases or when attacks strongly degrade the performance of existing aggregators.

First, under the ALIE attack, most baseline aggregators suffer severe performance degradation as the number of workers grows. For example, with $\alpha = 10$, CWMed drops from 80.01 (8 workers) to 17.48 (26 workers). In contrast, HullGuard maintains stable performance around $78 - -80\%$ accuracy regardless the number of Byzantine workers. This leads to very large improvements over the best baseline, reaching gains of 50.66 and 60.39 points for 20 and 26 workers, respectively. Similar patterns appear for $\alpha = 1$ and $\alpha = 0.1$, where HullGuard again prevents the dramatic accuracy collapse observed for other methods.

Second, under FOE and Mimic attacks, HullGuard also provides significant robustness improvements when attacks become more harmful. For instance, with $\alpha = 10$ and 26 workers, RFA collapses to 3.60% accuracy under FOE, whereas HullGuard maintains 80.40%, producing a gain of 13.35 points over the best baseline. Under



(a) $f = 8$ (b) $f = 14$ (c) $f = 26$

Figure VII.1: Training Accuracy for dataset EuroSAT, under the attack ALIE, with Dirichlet concentration parameter 0.1. Legend : • BKT + CWMed • BKT + CwTM • BKT + RFA • BKT + GAS • HullGuard

Mimic with the same configuration, HullGuard improves accuracy by 23.90 points. Similar gains appear for $\alpha = 1$ and $\alpha = 0.1$, especially for large numbers of workers.

Third, for weaker or less disruptive attacks such as MinSum and LF, existing robust aggregators already maintain strong performance (around 80% accuracy). In these settings HullGuard performs comparably to state-of-the-art methods, with small positive or negative gains typically within a few percentage points.

Finally, HullGuard shows remarkable consistency, never collapsing in contrast to its competitors. Its robustness advantage becomes more pronounced as the difficulty of the setting increases. In particular, the largest improvements consistently appear when the number of Byzantine workers is high (20 and 26) or under stronger attacks, where competing methods often fail while HullGuard maintains stable performance.

Similar trends are observed across the other datasets, confirming that HullGuard consistently improves robustness against strong Byzantine attacks while remaining competitive with state-of-the-art methods when attacks are weaker. This behavior is illustrated in Figure VII.1, which shows training on EuroSAT under high data heterogeneity ($\alpha = 0.1$). Additional results for CIFAR-10, MNIST, Fashion-MNIST, and EuroSAT are reported in the Appendix, in Tables D.2, D.3, D.4, and D.5, respectively.

3 Conclusion

In this chapter, we introduced HullGuard, a Byzantine filtering method for federated learning under label-skew heterogeneity. The method leverages a key property of this setting: despite update heterogeneity, honest updates remain noisy convex combinations of global class-wise gradients. By exploiting this structure, HullGuard detects and filters workers whose updates are inconsistent with those of honest participants before aggregation.

Our experiments show that HullGuard effectively removes Byzantine workers during training and significantly improves robustness under strong attacks, including regimes where existing defenses often fail. HullGuard does not require server-side validation data.

We also provide theoretical insight explaining why the filter is difficult to evade: Byzantine workers that attempt to bypass it must align their class gradients with those of honest workers, which limits their capacity to disrupt training.

Table VII.2: Mean Accuracy (%) comparison over last 200 training steps between robust aggregators (CWMed, CwTM, RFA, GAS with BKT, and HullGuard) on CIFAR-10 for different concentration parameter $\alpha \in \{10, 1, 0.1\}$ under several attacks (ALIE, FOE, Mimic, MinSum, LF) and different numbers of workers $f \in \{8, 14, 20, 26\}$. The Gain row reports the accuracy improvement of HullGuard over the best competing method for each configuration.

		ALIE				FOE				Mimic				MinSum				LF			
		8	14	20	26	8	14	20	26	8	14	20	26	8	14	20	26	8	14	20	26
$\alpha = 10$	CWMed	80.01	64.81	24.43	17.48	78.77	77.08	48.94	29.59	78.37	76.74	62.04	52.41	81.03	81.13	81.04	80.20	77.09	75.72	69.98	61.98
	CwTM	77.69	58.69	22.06	17.71	80.21	78.66	64.38	30.33	80.25	78.15	66.16	52.81	81.29	81.29	81.07	80.67	79.49	77.37	71.26	66.63
	RFA	77.68	61.67	23.32	18.97	80.03	78.16	64.14	3.60	80.03	77.95	55.90	48.17	81.51	81.35	81.39	80.40	79.71	78.71	78.34	75.36
	GAS	75.47	45.12	27.31	18.70	80.27	78.21	76.82	67.05	78.26	74.03	71.61	56.73	80.53	81.28	80.92	80.56	80.58	80.25	80.30	80.36
	HullGuard	80.35	78.58	77.97	79.36	80.26	77.96	77.52	80.40	80.49	78.06	77.24	80.63	80.65	78.71	77.65	80.82	80.39	77.98	77.38	80.05
	Gain	0.34	**13.77**	**50.66**	**60.39**	-0.01	-0.70	0.70	**13.35**	0.24	-0.09	**5.63**	**23.90**	-0.86	-2.64	-3.74	0.15	-0.19	-2.27	-2.92	-0.31
$\alpha = 1$	CWMed	79.24	52.87	18.10	13.03	76.41	73.93	34.44	16.95	75.84	74.20	53.87	42.03	79.18	79.84	79.78	78.80	75.16	72.99	65.10	56.11
	CwTM	76.65	42.06	23.58	12.98	79.17	77.58	57.85	25.86	78.17	76.18	60.23	43.96	79.81	79.42	79.20	79.19	78.04	75.92	69.05	56.77
	RFA	76.08	50.00	25.38	14.34	78.68	77.17	53.68	16.89	77.51	76.30	52.16	38.91	79.89	79.91	79.19	79.19	77.70	76.40	75.66	66.06
	GAS	74.73	31.25	16.49	12.58	78.14	77.06	73.94	56.96	76.73	71.39	67.48	50.09	79.73	80.05	79.24	78.74	78.81	78.40	78.79	77.58
	HullGuard	79.15	76.25	74.94	77.61	78.79	75.36	74.34	79.02	78.91	75.61	74.16	78.54	79.47	76.52	75.11	78.45	79.16	75.53	74.68	79.03
	Gain	-0.09	**23.38**	**49.56**	**63.27**	-0.38	-2.22	0.40	**22.06**	0.74	-0.69	6.68	**28.45**	-0.42	-3.53	-4.67	-0.74	0.35	-2.87	-4.11	1.45
$\alpha = 0.1$	CWMed	64.38	29.47	11.08	7.87	60.18	50.06	28.52	10.75	61.97	53.87	29.50	25.47	67.39	68.01	65.66	67.22	57.58	52.71	36.75	38.78
	CwTM	60.29	26.04	12.71	16.68	63.86	53.55	32.05	28.49	63.93	58.12	42.36	27.10	66.82	67.84	64.82	66.33	60.01	55.33	38.01	35.98
	RFA	61.19	34.11	18.72	4.30	64.93	62.47	25.98	7.20	64.46	62.64	37.96	26.92	66.97	67.54	64.97	64.94	61.28	62.18	55.71	51.46
	GAS	58.88	15.70	15.04	4.30	65.01	64.03	50.63	22.53	64.94	58.40	50.12	28.00	66.83	68.34	65.54	66.74	64.09	62.88	57.31	59.89
	HullGuard	64.20	65.80	62.04	55.36	65.07	61.03	56.21	65.36	63.82	64.89	57.22	64.86	66.50	67.51	63.41	66.89	60.85	59.74	54.18	62.35
	Gain	-0.18	**31.69**	**43.32**	**38.68**	0.06	-3.00	**5.58**	**36.87**	-1.12	2.25	**7.10**	**36.86**	-0.89	-0.83	-2.25	-0.33	-3.24	-3.14	-3.13	2.46

Chapter VIII
Conclusion

1	Confusion Matrices	**91**
2	Federated Learning	**92**

This thesis studied machine learning vectors to better understand and improve classification systems. The first part focused on model outputs through confusion matrices. The second part focused on local gradients in Byzantine-robust federated learning.

1 Confusion Matrices

The first part of this thesis focuses on confusion matrices. Although simple and widely used, confusion matrices become harder to interpret, define, and assess in more complex settings.

Chapter II studied normalization methods for confusion matrices in heterogeneous settings. In these settings, class imbalance in the test set and prediction imbalance can distort matrix values and hide similarities between classes. To address this issue, we introduced *bi-normalization*, based on Iterative Proportional Fitting, which normalizes rows and columns simultaneously. This reduces the influence of distribution bias and better reveals class-similarity structure.

We also connected confusion matrix normalization to importance sampling, interpreting it as a reweighting of label–prediction pairs toward a target distribution. This clarifies both the role and the limits of normalization. Finally, we gave a geometric interpretation of confusion matrices in the model latent space. Matrix entries can be understood as overlaps between class clusters, and normalizing these clusters recovers the corresponding normalized confusion matrix.

Chapter III extended the confusion matrix beyond the single-label setting. We introduced the *Transport-based Confusion Matrix*, a formulation based on optimal transport and the principle of maximum entropy. This approach applies to single-label, multi-label, and soft-label classification while preserving the familiar square-matrix structure of the standard confusion matrix. It also unifies several existing formulations and provides a consistent way to analyze errors in more complex classification settings.

Chapter IV addressed the problem of evaluating confusion matrix extensions. Since real datasets do not provide a ground-truth confusion matrix in multi-label and soft-label settings, we introduced ConfusionBench, a synthetic benchmark based on stochastic classifiers with controlled latent confusion structures. This framework makes it possible to quantitatively evaluate whether a confusion matrix extension recovers the true behavior of a classifier. The results show that transport-based approaches are particularly effective for recovering latent confusion patterns in complex classification settings.

Overall, these contributions provide a more general and principled view of confusion matrices. They clarify and extend the normalization process [34], generalize confusion matrices beyond the single-label setting [31], and introduce a benchmark for comparing competing extensions [30].

Future work could develop more systematic ways to use confusion matrices to improve classifiers, for example through new loss functions, active learning strategies, or model-selection criteria based on error structure rather than aggregate accuracy. Another direction would be to build confusion matrices from a causal perspective, in order to distinguish observed label–prediction associations from more structural relations between classes.

2 Federated Learning

The second part of this thesis focused on Byzantine-robust federated learning under label skew. In this setting, honest workers have different label distributions, making their gradients naturally dissimilar and reducing the effectiveness of many Byzantine-robust aggregation methods.

Chapter VI introduced *NorthStar*, a worker-side sampling strategy designed to reduce label-induced gradient dissimilarity. Instead of acting at the server level or requiring a validation set, NorthStar aligns local gradients from the start by using a shared reference label distribution. By reducing dissimilarity among honest gradients, it makes Byzantine updates easier to distinguish and improves system robustness. The theoretical analysis shows that, asymptotically, NorthStar removes a label-distribution mismatch term from honest-gradient dissimilarity. The experiments show that NorthStar improves Byzantine robustness, especially under moderate to strong label skew, and can be combined with other robustness strategies such as NNM.

Chapter VII introduced *HullGuard*, a server-side filtering method based on the geometry of honest gradients under label skew. The key observation is that honest updates can be written as noisy convex combinations of shared class-wise gradients. HullGuard exploits this structure to identify and remove workers whose updates are inconsistent with this class-wise geometry before aggregation. Experiments show that HullGuard filters most Byzantine workers and remains effective in regimes where existing defenses may collapse.

Overall, NorthStar [33] and HullGuard [32] address the same difficulty from two complementary perspectives. NorthStar reduces honest-gradient heterogeneity at its source, before updates are sent to the server. HullGuard acts later, at the server level, by filtering Byzantine workers through a class-wise notion of gradient consistency rather than raw gradient closeness. NorthStar is therefore lighter and local, while HullGuard is more computationally demanding but provides a stronger structural filter in more adversarial settings.

Future work could strengthen the theoretical analysis with non-asymptotic guarantees and results that hold beyond expectation. Another direction is to study NorthStar and HullGuard under broader forms of heterogeneity, such as feature skew or mixed heterogeneity.

Chapter A
Normalization of Confusion Matrices: Methods and Geometric Interpretations

1 Additional Results **94**
2 IPF & RAS Algorithms **94**
3 Overlap Metric **95**
3.1 Why Normalize? 95
3.2 Equivalence with L_1 Distance 95
4 Experiments Details **95**
4.1 Histogram in Experiments 96
4.2 Model Architectures 96
5 Information Preservation, Idempotence & Class Distribution Invariance **96**
5.1 Information Preservation for Standards Normalizations . 96
5.2 Idempotence & Class Distribution Invariance of Bi Normalization 97
6 Importance Sampling **97**
7 Geometric Confusion Matrix **98**
7.1 Normalized Histograms Under Weighting Schemes . . . 98
7.2 Geometric Confusion Matrix 99
8 Equivalence Scaling Lemma **99**
8.1 Problem Reformulation 100
8.2 Karush-Kuhn-Tucker Conditions 100
8.3 Solution 101
8.4 Convexity of f 101

The supplementary materials are organized as follows: **Section 1** provides additional results that further support our propositions. **Section 2** describes the Iterative Proportional Fitting (IPF) algorithm and an equivalent formulation known as the RAS algorithm, whose variations with respect to IPF are later exploited. **Section 3** details the overlap metric used in our experiments. **Section 4** presents experimental details, including the multidimensional histogram setup and the model architectures. **Section 5** provides proofs of key properties such as information preservation, idempotence, and class distribution invariance. **Section 6** describes additional weighting schemes mentioned in Subsection Normalization as Importance Sampling. **Section 7** derives and proves the explicit formula for the Geometric Confusion Matrix (GCM). **Section 8** demonstrates an auxiliary lemma used in the preceding sections.

Algorithm 8 Iterative Proportional Fitting (IPF)

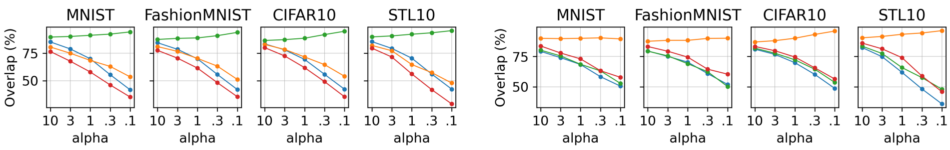
Require: Initial matrix $M \in \mathbb{R}_{>0}^{C\times C}$, target row sums $u \in \mathbb{R}_{>0}^{C}$, target column sums $v \in \mathbb{R}_{>0}^{C}$ such that $u_+ = v_+$, tolerance $\epsilon > 0$, maximum number of steps T

Initialize $Q^{(0)} \leftarrow M$, $t \leftarrow 0$
repeat
  for each row $i = 1$ to C **do**
    $Q_{i,:}^{(t+1)} \leftarrow Q_{i,:}^{(t)} \cdot \frac{u_i}{Q_{i+}^{(t)}}$
  end for
  for each column $j = 1$ to C **do**
    $Q_{:,j}^{(t+2)} \leftarrow Q_{:,j}^{(t+1)} \cdot \frac{v_j}{Q_{+j}^{(t+1)}}$
  end for
  $t \leftarrow t + 2$
until $\|Q_{:+}^{(t)} - u\|_1 + \|Q_{+:}^{(t)} - v\|_1 \leq \epsilon$ **or** $t \geq T$
return $\widehat{Q} \leftarrow Q^{(t)}$

1 Additional Results

Figure A.1 presents the remaining results of Experiment 2 (see Subsection Baselines & Experiments) with target matrices GCM_a and GCM_c. We observe trends consistent with those obtained for GCM_r: GCM_a aligns closely with the all normalization (Subfigure A.1a), while GCM_c aligns closely with the col normalization (Subfigure A.1b). These results further support the geometric interpretation of normalizations.

Figure A.2 reports the results for Experiment 1 (see Subsection Baselines & Experiments) using the KL divergence instead of the overlap metric. It shows that bi-normalization still achieves the highest similarity with the confusion matrix under a balanced setting, as expected.



(a) Target: $\mathrm{GCM}\omega_a$ (b) Target: $\mathrm{GCM}\omega_c$

Figure A.1: Overlap (%) between target matrices and normalized matrices. Legend: • bi • row • col • all

2 IPF & RAS Algorithms

This section presents the IPF (Algorithm 8) and RAS (Algorithm 9) procedures.

According to Lemma 2 (Section 8), there exist positive diagonal matrices D^l and D^r such that $\mathrm{bi}(M) = D^l M D^r$. The RAS method estimates D^l and D^r. This algorithm will be used in the following section and is equivalent to Algorithm 8; see [53] for further details.

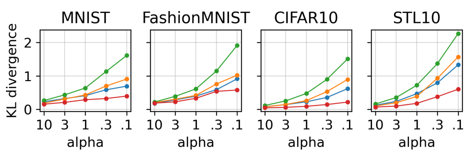


Figure A.2: KL divergence (%) between matrix from balanced setting and normalized matrices. Legend: • bi • row • col • all

Algorithm 9 RAS method

Require: Positive matrix $M \in \mathbb{R}_{>0}^{C \times C}$, target row sums $r \in \mathbb{R}_{>0}^{C}$, target column sums $c \in \mathbb{R}_{>0}^{C}$ such that $r_+ = c_+$, tolerance $\epsilon > 0$, maximum number of steps T

Initialize $D^{r\,(0)} \leftarrow I$, $t \leftarrow 0$

repeat

  for each $i = 1$ to C **do**

    $D_{ii}^{l\,(t+1)} \leftarrow \dfrac{r_i}{\sum_j M_{ij} D_{jj}^{r\,(t)}}$

  end for

  for each $j = 1$ to C **do**

    $D_{jj}^{r\,(t+2)} \leftarrow \dfrac{c_j}{\sum_i M_{ij} D_{ii}^{l\,(t+1)}}$

  end for

  $t \leftarrow t + 2$

until $\|(D^{l\,(t)} M D^{r\,(t)})_{:+} - r\|_1 + \|(D^{l\,(t)} M D^{r\,(t)})_{+:} - c\|_1 \leq \epsilon$ **or** $t \geq T$

return $\widehat{D}^l \leftarrow D^{l\,(t)}$, $\widehat{D}^r \leftarrow D^{r\,(t)}$

3 Overlap Metric

This section explains the rationale for normalizing confusion matrices when computing the Overlap metric, and shows its equivalence with the L_1 distance.

3.1 Why Normalize?

The entries of a confusion matrix are only meaningful relative to each other; absolute values lack interpretability. For example, $P_{ii} = 3$ indicates perfect classification if $P_{i+} = 3$, but poor performance if $P_{i+} = 30$. More generally, for any $\lambda > 0$, P and λP represent the same model behavior.

Comparisons between two matrices are meaningful when they share the same total sum, i.e., $P_{++} = Q_{++}$. For instance, $P_{ii} = 1$ may seem worse than $Q_{ii} = 10$, but if $P_{++} = 3$ and $Q_{++} = 30$, the relative accuracies are identical: $P_{ii}/P_{++} = Q_{ii}/Q_{++}$.

For consistency, we compare normalized confusion matrices: $\mathrm{all}(P) = P/P_{++}$ and $\mathrm{all}(Q) = Q/Q_{++}$. This ensures that behaviors are preserved and enables consistent comparisons.

3.2 Equivalence with L_1 Distance

Let P and Q be two matrices such that $P_{++} = Q_{++} = 1$. Then,

$$
\begin{aligned}
\|P - Q\|_1 &= \sum_{ij} |P_{ij} - Q_{ij}| \\
&= \sum_{ij} \max(P_{ij}, Q_{ij}) - \min(P_{ij}, Q_{ij}) \\
&= \sum_{ij : P_{ij} \geq Q_{ij}} P_{ij} - \min(P_{ij}, Q_{ij}) \qquad + \sum_{ij : Q_{ij} > P_{ij}} Q_{ij} - \min(P_{ij}, Q_{ij}) \\
&= \sum_{ij} P_{ij} - \min(P_{ij}, Q_{ij}) + \sum_{ij} Q_{ij} - \min(P_{ij}, Q_{ij}) \\
&= 2 - 2\,\mathrm{Overlap}(P, Q)
\end{aligned}
$$

Thus, the Overlap metric is directly related to the L_1 distance.

4 Experiments Details

This section presents details on the multidimensional histogram setup and the model architectures.

4.1 Histogram in Experiments

First, the projection space is divided into a regular grid of m-dimensional hyperrectangles,[1] which define the histogram bins. We then construct non-overlapping histograms [104] associated with the clusters $\mathcal{C}_k$ and $\widehat{\mathcal{C}}_k$ for $k = 1, \ldots, C$.

Considering a cluster $\mathcal{C}$ and its unweighted histogram, each bin height represents the number of points from $\mathcal{C}$ that fall within the bin (i.e., hyperrectangle) under consideration. In the weighted case, the bin height instead corresponds to the sum of the point weights, using one of these weighting schemes ω_a, ω_r, ω_c, or ω_b.

In the experiments, we fix the projected space dimensionality to $m = 10$ and specify the bin width accordingly. Following the recommendation of [98] (see Subsection 3.4, Eq. (3.66)), we adopt the optimal bin width $3.5\ \sigma_k\ n^{-1/(2+m)}$, where σ_k denotes the empirical standard deviation along the k-th dimension, n is the sample size, and m is the data dimensionality.

4.2 Model Architectures

We use the following abbreviations to describe our architectures: $L(n)$ denotes a fully connected linear layer with n outputs; R is a ReLU activation; $C(c)$ represents a 2D convolutional layer with c output channels, kernel size 5 for CIFAR datasets and 3 otherwise, padding 0 for CIFAR and 1 otherwise, and stride 1; M denotes 2D max pooling with kernel size 2; B stands for batch normalization; and D represents dropout with a fixed probability of 0.25.

In line with Allouah et al. [4], the model architectures are defined as follows. For MNIST and Fashion-MNIST, the architecture is $(1, 28 \times 28) - C(8) - R - M - C(16) - R - M - L(64) - R - L(10)$. For CIFAR-10 and STL-10, the architecture is $(3, 32 \times 32) - C(64) - R - B - C(64) - R - B - M - D - C(128) - R - B - C(128) - R - B - M - D - L(128) - R - D - L(10)$.

5 Information Preservation, Idempotence & Class Distribution Invariance

This section demonstrates that normalization methods satisfy the properties of information preservation, idempotence, and class distribution invariance.

5.1 Information Preservation for Standards Normalizations

We state that, for a positive confusion matrix M, the normalizations minimize the KL divergence under specific constraints:

$$\mathrm{all}(M) \in \underset{\substack{P \in \mathbb{R}_{>0}^{C\times C}: \\ P_{i+} = \frac{M_{i+}}{M_{++}}, P_{+j} = \frac{M_{+j}}{M_{++}}\ \forall i,j}}{\operatorname{argmin}} D_{\mathrm{KL}}(P \| M), \qquad \mathrm{row}(M) \in \underset{\substack{P \in \mathbb{R}_{>0}^{C\times C}: \\ P_{i+} = 1, P_{+j} = \sum_i \frac{M_{ij}}{M_{i+}}\ \forall i,j}}{\operatorname{argmin}} D_{\mathrm{KL}}(P \| M),$$

$$\mathrm{col}(M) \in \underset{\substack{P \in \mathbb{R}_{>0}^{C\times C}: \\ P_{i+} = \sum_j \frac{M_{ij}}{M_{+j}}, P_{+j} = 1\ \forall i,j}}{\operatorname{argmin}} D_{\mathrm{KL}}(P \| M)$$

We observe that these normalizations can be expressed as the product of diagonal matrices and the original confusion matrix:

$$\mathrm{row}(M) = \mathrm{diag}(\tfrac{1}{M_{1+}}, \ldots, \tfrac{1}{M_{C+}}) M I, \quad \mathrm{col}(M) = I M\, \mathrm{diag}(\tfrac{1}{M_{+1}}, \ldots, \tfrac{1}{M_{+C}}), \tag{A.1}$$

[1] In geometry, a hyperrectangle generalizes a rectangle (in 2D) and a rectangular cuboid (in 3D) to higher dimensions [97].

$$\text{and } \operatorname{all}(M) = IMI/M_{++}.$$

According to Lemma 2 (Section 8), these normalization procedures indeed minimize the KL divergence, which concludes the proof.

5.2 Idempotence & Class Distribution Invariance of Bi Normalization

In Proposition 1, we state that the bi-normalization satisfies idempotence,

$$\operatorname{bi} \circ \operatorname{bi}(M) = \operatorname{bi}(M),$$

and class distribution invariance, that is, for any diagonal matrices $A, B \in \mathbb{R}_{>0}^{C\times C}$,

$$\operatorname{bi}(AMB) = \operatorname{bi}(M).$$

We prove these properties in what follows.

Idempotence. By definition, $\operatorname{bi} \circ \operatorname{bi}(M)$ is the solution to the problem

$$\underset{\substack{P\in\mathbb{R}_{>0}^{C\times C}:\\ P_{i+} = P_{+j} = 1\ \forall\, i,j}}{\operatorname{argmin}} \quad D_{\mathrm{KL}}(P \| \operatorname{bi}(M))$$

Moreover, we observe that $I \operatorname{bi}(M) I$ satisfies the marginal constraints. By Lemma 2, $I \operatorname{bi}(M) I$ solves the above problem, and since the solution is unique, we have

$$\operatorname{bi} \circ \operatorname{bi}(M) = I \ \operatorname{bi}(M)\, I = \operatorname{bi}(M),$$

which establishes the idempotence property.

Class Distribution Invariance. By Lemma 2, there exist diagonal matrices D^l and D^r such that

$$\operatorname{bi}(AMB) = D^l AMBD^r.$$

By definition of the bi-normalization, the row and column sums satisfy

$$(D^l AMBD^r)_{i+} = (D^l AMBD^r)_{+j} = 1 \quad \forall i, j.$$

Additionally, $D^l A$ and BD^r are diagonal. Therefore, by Lemma 2, $\operatorname{bi}(AMB)$ is the unique solution to

$$\underset{\substack{P\in\mathbb{R}_{>0}^{C\times C}:\\ P_{i+} = 1, P_{+j} = 1\ \forall\, i,j}}{\operatorname{argmin}} \quad D_{\mathrm{KL}}(P \| M)$$

This implies $\operatorname{bi}(AMB) = \operatorname{bi}(M)$, completing the proof.

6 Importance Sampling

This section details the weighting schemes mentioned in Subsection Normalization as Importance Sampling.

From the equalities presented in Equation (A.1) (Subsection 5.1) and by applying the same procedure described in the main part of the paper, it directly follows that

$$\omega_a : (y, \hat{y}) \mapsto \frac{1}{M_{++}} \quad \text{and} \quad \operatorname{all}(M) = \sum_{k=1}^{N} \omega_a(y_k, \hat{y}_k) E_{y_k \hat{y}_k},$$

and

$$\omega_c : (y, \hat{y}) \mapsto \frac{1}{M_{+\hat{y}}} \quad \text{and} \quad \operatorname{col}(M) = \sum_{k=1}^{N} \omega_c(y_k, \hat{y}_k) E_{y_k \hat{y}_k}.$$

According to Lemma 2 (Section 8), there exist positive diagonal matrices D^l and D^r such that $\operatorname{bi}(M) = D^l M D^r$, using Algorithm 9 (Section 2), we estimate these matrices by matrices $\widehat{D}^l$ and $\widehat{D}^r$. It follows

$$\omega_b : (y, \hat{y}) \mapsto \widehat{D}^l_y \widehat{D}^r_{\hat{y}} \quad \text{and} \quad \operatorname{bi}(M) = \sum_{k=1}^{N} \omega_b(y_k, \hat{y}_k) E_{y_k \hat{y}_k}.$$

7 Geometric Confusion Matrix

This section establishes the volumes of the various scaled histograms introduced previously, and compares the analytical form of the classical confusion matrix with that of the geometric confusion matrix.

7.1 Normalized Histograms Under Weighting Schemes

In Section Geometric View of Normalized Confusion Matrices, we state that, under specific weighting schemes, some histograms are normalized up to a scaling factor. We prove this statement below.

We first introduce the following lemma:

Lemma 1. *Let $\mathcal{C}$ be a cluster of points in the model's latent space, based on dataset $\mathcal{D}$. Let P be a projection matrix, and let ω be the weighting function. Then,*

$$\lambda(V(\mathcal{C},\omega)) = r \sum_{(x,y)\in\mathcal{D}} \omega(y,\hat{y}) \mathbb{1}_{P\varepsilon(x)\in\mathcal{C}},$$

where λ denotes the Lebesgue measure, $\mathbb{1}$ is the indicator function, and r the volume of hyperrectangles which split the projection space.

Proof. We begin by decomposing the histogram into a union of hyperrectangles which grid the projection space, denoted by $\mathcal{V}$. Since $\mathcal{D}$ contains a finite number of points, the covering set $\mathcal{V}$ is also finite. By construction of the weighted histogram, we have

$$\begin{aligned} V(\mathcal{C},\omega) &= \bigcup_{v\in\mathcal{V}} (V(\mathcal{C},\omega)\cap v) \\ &= \bigcup_{v\in\mathcal{V}} \underbrace{r_1^v \times \cdots \times r_m^v}_{\text{bin/hyperrectangle splitting the projection space}} \times \underbrace{\left[0, \sum_{(x,y)\in\mathcal{D}} \omega(y,\hat{y})\mathbb{1}_{P\varepsilon(x)\in\mathcal{C}\cap v}\right]}_{\text{bin height}} \end{aligned}$$

where $r_1^v, \ldots, r_m^v$ denote the intervals describing the sides of each hyperrectangle.

Its Lebesgue measure equals the sum of the measures of the individual hyperrectangles:

$$\lambda(V(\mathcal{C},\omega)) = \sum_{v\in\mathcal{V}} \lambda(r_1^v \times \cdots \times r_m^v \times \left[0, \sum_{(x,y)\in\mathcal{D}} \omega(y,\hat{y})\mathbb{1}_{P\varepsilon(x)\in\mathcal{C}\cap v}\right])$$

The Lebesgue measure of a hyperrectangle is the product of its side lengths. Only the heights vary, while the other side lengths are bin-independent. Let $r = \prod_{i=1}^m \lambda(r_i^v)$. Then,

$$\lambda(V(\mathcal{C},\omega)) = \sum_{v\in\mathcal{V}} r \sum_{(x,y)\in\mathcal{D}} \omega(y,\hat{y})\mathbb{1}_{P\varepsilon(x)\in\mathcal{C}\cap v} = r \sum_{(x,y)\in\mathcal{D}} \omega(y,\hat{y})\mathbb{1}_{P\varepsilon(x)\in\mathcal{C}}$$

which concludes the proof. □

According to Lemma 1, we have

$$\begin{aligned} \lambda(V(\mathcal{C}_i,\omega_r)) &= r \sum_{(x,y)\in\mathcal{D}} \omega_r(y,\hat{y})\mathbb{1}_{P\varepsilon(x)\in\mathcal{C}_i} = r \sum_{(x,y)\in\mathcal{D}} \frac{1}{M_{i+}}\mathbb{1}_{P\varepsilon(x)\in\mathcal{C}_i} = r, \\ \lambda(V(\widehat{\mathcal{C}}_j,\omega_c)) &= r \sum_{(x,y)\in\mathcal{D}} \omega_c(y,\hat{y})\mathbb{1}_{P\varepsilon(x)\in\widehat{\mathcal{C}}_j} = r \sum_{(x,y)\in\mathcal{D}} \frac{1}{M_{+j}}\mathbb{1}_{P\varepsilon(x)\in\widehat{\mathcal{C}}_j} = r, \\ \lambda(V(\mathcal{C}_i,\omega_b)) &= r \sum_{(x,y)\in\mathcal{D}} \omega_b(y,\hat{y})\mathbb{1}_{P\varepsilon(x)\in\mathcal{C}_i} = r \sum_{(x,y)\in\mathcal{D}} \widehat{D}_i^l \widehat{D}_{\hat{y}}^r \mathbb{1}_{P\varepsilon(x)\in\mathcal{C}_i} = r\,\mathrm{bi}(M)_{i+} = r, \\ \lambda(V(\widehat{\mathcal{C}}_j,\omega_b)) &= r \sum_{(x,y)\in\mathcal{D}} \omega_b(y,\hat{y})\mathbb{1}_{P\varepsilon(x)\in\widehat{\mathcal{C}}_j} = r \sum_{(x,y)\in\mathcal{D}} \widehat{D}_y^l \widehat{D}_j^r \mathbb{1}_{P\varepsilon(x)\in\widehat{\mathcal{C}}_j} = r\,\mathrm{bi}(M)_{+j} = r. \end{aligned}$$

In that sense, these histograms are normalized using weighting schemes up to a scaling factor r, corresponding to the fixed volume of the hyperrectangles partitioning the projection space.

7.2 Geometric Confusion Matrix

The Geometric Confusion Matrix is defined as

$$\Big(\mathrm{GCM}_\omega\Big)_{ij} = \lambda\Big(V(\mathcal{C}_i,\omega) \cap V(\widehat{\mathcal{C}}_j,\omega)\Big)$$

We begin by deriving an explicit formula for GCM_ω. Following the same approach as in proof of Lemma 1, and noting that the grid of hyperrectangles partitioning the projection space is shared across all histograms, the intersection of two histograms is given by:

$$\begin{aligned}
&V(\mathcal{C}_i,\omega) \cap V(\widehat{\mathcal{C}}_j,\omega)\\
&= \bigcup_{v\in\mathcal{V}} r_1^v \times \cdots \times r_m^v \times \left[0, \sum_{(x,y)\in\mathcal{D}} \omega(y,\hat{y})\mathbb{1}_{P\varepsilon(x)\in\mathcal{C}_i\cap v}\right] \quad \bigcap \\
&\qquad\qquad \bigcup_{v\in\mathcal{V}} r_1^v \times \cdots \times r_m^v \times \left[0, \sum_{(x,y)\in\mathcal{D}} \omega(y,\hat{y})\mathbb{1}_{P\varepsilon(x)\in\widehat{\mathcal{C}}_j\cap v}\right]\\
&= \bigcup_{v\in\mathcal{V}} r_1^v \times \cdots \times r_m^v \times \left[0, \min\left(\sum_{(x,y)\in\mathcal{D}} \omega(y,\hat{y})\mathbb{1}_{P\varepsilon(x)\in\mathcal{C}_i\cap v}, \sum_{(x,y)\in\mathcal{D}} \omega(y,\hat{y})\mathbb{1}_{P\varepsilon(x)\in\widehat{\mathcal{C}}_j\cap v}\right)\right]
\end{aligned}$$

As a result, the (i,j) entry of the geometric confusion matrix becomes:

$$\Big(\mathrm{GCM}_\omega\Big)_{ij} = r\sum_{v\in\mathcal{V}} \min\left(\sum_{(x,y)\in\mathcal{D}} \omega(y,\hat{y})\mathbb{1}_{P\varepsilon(x)\in\mathcal{C}_i\cap v}, \sum_{(x,y)\in\mathcal{D}} \omega(y,\hat{y})\mathbb{1}_{P\varepsilon(x)\in\widehat{\mathcal{C}}_j\cap v}\right)$$

The normalized confusion matrix can also be expressed as a sum over grid cells, analogously to GCM_ω. Let norm denote a normalization operator among all, row, col, and bi, and let ω be its associated weighting scheme. It is straightforward to obtain:

$$\begin{aligned}
\mathrm{norm}(M)_{ij} &= \sum_{(x,y)\in\mathcal{D}} \omega(i,j)\mathbb{1}_{y=i}\mathbb{1}_{\hat{y}=j}\\
&= \sum_{(x,y)\in\mathcal{D}} \omega(i,j)\mathbb{1}_{P\varepsilon(x)\in\mathcal{C}_i\cap\widehat{\mathcal{C}}_j}\\
&= \sum_{v\in\mathcal{V}}\sum_{(x,y)\in\mathcal{D}} \omega(i,j)\mathbb{1}_{P\varepsilon(x)\in\mathcal{C}_i\cap\widehat{\mathcal{C}}_j\cap v}
\end{aligned}$$

This explicit formula for GCM_ω highlights how the geometric confusion matrix incorporates spatial information. The geometric confusion matrix compares in each grid cell the number of points labeled i (regardless of prediction) with the number of points predicted as j (regardless of true label). As a result, in a grid cell containing only points with label i and prediction j, the contribution to GCM_ω matches that of the standard confusion matrix. In contrast, in a cell containing a mix of labels and predictions, the minimum operation produces a contribution that differs from the simple count of points labeled i and predicted as j—that is, from the contribution in the standard confusion matrix.

8 Equivalence Scaling Lemma

This section demonstrates the following lemma:

Lemma 2. *Let M be a positive matrix in $\mathbb{R}_{>0}^{C\times C}$, and r and c, be two positive vectors in $\mathbb{R}_{>0}^{C}$ such that $r_+ = c_+$. Then, any matrix M^* of the form $M^* = D^l M D^r$, where D^l and D^r are positive diagonal matrices such that*

$$(D^l M D^r)_{i+} = r_i \quad \text{and} \quad (D^l M D^r)_{+j} = c_j \quad \text{for all } i, j,$$

is the unique solution to the following problem:

$$M^* \in \underset{\substack{P\in\mathbb{R}_{>0}^{C\times C}:\\ P_{i+}=r_i,\, P_{+j}=c_j\ \forall i,j}}{\operatorname{argmin}} D_{\mathrm{KL}}(P\|M)$$

Note that this result is already known in a different form, referred to as equivalence scaling (see Section 3. Different approaches to equivalence scaling in [53]). We do not claim to introduce a new result with this lemma; rather, we were unable to find this particular formulation in the literature.

8.1 Problem Reformulation

We use the method of Lagrange multipliers. We define the row constraints for $i = 1, \ldots, C$ as:

$$h_i : \begin{array}{ccc} \mathbb{R}_{>0}^{C\times C} & \to & \mathbb{R} \\ P & \mapsto & P_{i+} - r_i \end{array}$$

and the column constraints for $j = 1, \ldots, C$ as:

$$g_j : \begin{array}{ccc} \mathbb{R}_{>0}^{C\times C} & \to & \mathbb{R} \\ P & \mapsto & P_{+j} - c_j \end{array}$$

Since adding a constant to the objective function does not affect the arguments of the minima, we define $f(P) := D_{\mathrm{KL}}(P\|M) - 1$ to simplify the derivative calculations.

The optimization problem becomes:

$$M^* \in \underset{\substack{P\in\mathbb{R}_{>0}^{C\times C}:\\ h_i(P)=g_j(P)=0\ \forall i,j}}{\operatorname{argmin}} f(P)$$

The functions f, h_i, and g_j are continuously differentiable.

8.2 Karush-Kuhn-Tucker Conditions

To find a solution, we apply the Karush-Kuhn-Tucker (KKT) conditions (see *Section 5.5.3: KKT Optimality Conditions* in [12]). These conditions are necessary for a point M^* to be a local minimum when a constraint qualification is satisfied. In our case, the Linearity Constraint Qualification (LCQ) holds, since the constraints h_i and g_j are affine.

Moreover, since f is convex (as shown in a subsection below), and the constraint set is convex, the KKT conditions are also sufficient for optimality [12]. Thus, any point satisfying the KKT conditions is a global minimum.

In other words, any feasible matrix P^* and multipliers μ^* and ν^* satisfying:

$$\nabla_P L(P^*, \mu^*, \nu^*) = 0$$

is a global minimum, where the Lagrangian is given by:

$$L(P, \mu, \nu) = f(P) + \sum_{i=1}^{C} \mu_i h_i(P) + \sum_{j=1}^{C} \nu_j g_j(P)$$

Since f is strictly convex (Subsection *Convexity of f*), this minimum is also unique.

8.3 Solution

Let P be an admissible point. We compute the gradient of the Lagrangian[2]:

$$\nabla_P L(P, \mu, \nu)_{ij} = \left[\ln\left(\frac{P_{ij}}{M_{ij}}\right) + \mu_i + \nu_j \right]_{ij}$$

Setting the gradient to zero gives:

$$\ln\left(\frac{P_{ij}}{M_{ij}}\right) + \mu_i + \nu_j = 0 \quad \Leftrightarrow \quad P_{ij} = \exp(-\mu_i) M_{ij} \exp(-\nu_j)$$

We define diagonal matrices D^l and D^r with entries:

$$(D^l)_{ii} = \exp(-\mu_i), \qquad (D^r)_{jj} = \exp(-\nu_j)$$

Then the solution takes the form:

$$P = D^l M D^r$$

In conclusion, any matrix of the form $D^l M D^r$ that satisfies the given row and column constraints is the unique solution to the minimization problem.

8.4 Convexity of f

Let $P, P' \in \mathbb{R}_{>0}^{C\times C}$ be two distinct matrices, and let $\gamma \in (0, 1)$. The function f is strictly convex if and only if

$$f(\gamma P + (1-\gamma)P') < \gamma f(P) + (1-\gamma) f(P').$$

We now prove this inequality. We first show that the KL divergence is strictly convex in P when M is fixed.

$$\begin{aligned} D_{\mathrm{KL}}(\gamma P + (1-\gamma)P' \| M) &= \sum_{ij} \left(\gamma P_{ij} + (1-\gamma)P'_{ij}\right) \ln\left(\frac{\gamma P_{ij} + (1-\gamma)P'_{ij}}{\gamma M_{ij} + (1-\gamma)M_{ij}}\right) \\ &< \sum_{ij} \gamma P_{ij} \ln\left(\frac{P_{ij}}{M_{ij}}\right) + (1-\gamma)P'_{ij} \ln\left(\frac{P'_{ij}}{M_{ij}}\right) \\ &= \gamma D_{\mathrm{KL}}(P\|M) + (1-\gamma) D_{\mathrm{KL}}(P'\|M) \end{aligned}$$

where the strict inequality follows from the log-sum inequality, which becomes strict when $P \neq P'$. This implies that f is strictly convex, which completes the proof.

[2]Though the gradient is a vector, we index it here by (i, j) for clarity.

Chapter B
Unifying Confusion Matrices from Single-Label to Soft-Label

1 Proof of Proposition 3 . . . . **103**
2 Proof of Proposition 4 . . . . **104**
3 Corollaries of Propositions 3 & 4 . . . . **106**
4 Proof of proposition 5 . . . . **106**
4.1 Case $\frac{\mathbf{y}}{\|\mathbf{y}\|_1} = \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}$ and continuous extension 107
4.2 Case $\frac{\mathbf{y}}{\|\mathbf{y}\|_1} \neq \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}$ 108
5 Proof of Proposition 6 . . . . **110**
6 Proof of proposition 7 . . . . **110**
6.1 Case $Y = \hat{Y}$ 111
6.2 Case $Y \subsetneq \hat{Y}$ and $\hat{Y} \subsetneq Y$ 111
6.3 $Y \not\subset \hat{Y}$ and $\hat{Y} \not\subset Y$ 111
7 Proof of Proposition 8 . . . . **112**
7.1 Interval Bounds for Optimal Plan Entries 112
7.2 Existence of Elements Attaining the Bounds 112

1 Proof of Proposition 3

We first observe that

$$\sum_{i,j} c(i,j)\pi_{ij} = \sum_{i \neq j} \pi_{ij} = 1 - \mathrm{Tr}(\pi),$$

since π has total mass 1. Therefore, minimizing the transport cost is equivalent to maximizing $\mathrm{Tr}(\pi)$.

For any admissible plan $\pi \in T(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1})$, the marginal constraints imply

$$\pi_{ii} \leq \frac{\mathbf{y}_i}{\|\mathbf{y}\|_1} \quad \text{and} \quad \pi_{ii} \leq \frac{\hat{\mathbf{y}}_i}{\|\hat{\mathbf{y}}\|_1},$$

hence

$$\pi_{ii} \leq \min(\frac{\mathbf{y}_i}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}_i}{\|\hat{\mathbf{y}}\|_1}).$$

Summing over i yields

$$\mathrm{Tr}(\pi) = \sum_{i=1}^{C} \pi_{ii} \leq \sum_{i=1}^{C} \min(\frac{\mathbf{y}_i}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}_i}{\|\hat{\mathbf{y}}\|_1}).$$

It remains to show that this upper bound is achievable. Define $d := \min(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1})$ and set

$$\pi_{ii} = d_i \quad \text{for all } i.$$

Let the residual vectors be

$$r := \frac{\mathbf{y}}{\|\mathbf{y}\|_1} - d, \qquad s := \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1} - d.$$

By construction, $r, s \geq 0$ and

$$\sum_{i=1}^{C} r_i = \sum_{i=1}^{C} s_i.$$

Therefore, there exists a nonnegative matrix $(\pi_{ij})_{i \neq j}$ with row sums r and column sums s, which completes π into a feasible transport plan in $T(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1})$.

Thus, the upper bound is attained, and we conclude that

$$\max_{\pi \in T(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1})} \operatorname{Tr}(\pi) = \sum_{i=1}^{C} \min(\frac{\mathbf{y}_i}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}_i}{\|\hat{\mathbf{y}}\|_1}).$$

This proves optimality.

2 Proof of Proposition 4

Let $(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1})$ be a label–prediction pair, and let π be an optimal plan. By Proposition 3, we have

$$\operatorname{diag}(\pi) = \min(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}).$$

Therefore, the row sums of the off-diagonal terms are

$$\mathbf{u} := \frac{\mathbf{y}}{\|\mathbf{y}\|_1} - \min(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}),$$

and the column sums are

$$\hat{\mathbf{u}} := \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1} - \min(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}).$$

We first note that:

1. $\mathbf{u}$ and $\hat{\mathbf{u}}$ have the same total mass, that is,

$$\mathbf{u}_+ = \hat{\mathbf{u}}_+,$$

2. any non-uniqueness of optimal plans can only arise from the off-diagonal terms, since the diagonal is fixed by optimality.

We now distinguish several cases.

If $\frac{\mathbf{y}}{\|\mathbf{y}\|_1} = \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}$, then $\mathbf{u} = \hat{\mathbf{u}} = 0$. Hence, all off-diagonal terms vanish, and the optimal plan is unique.

Assume next that only one class is underestimated, that is, there exists some i such that

$$\frac{\hat{\mathbf{y}}_i}{\|\hat{\mathbf{y}}\|_1} < \frac{\mathbf{y}_i}{\|\mathbf{y}\|_1}, \qquad \text{and} \qquad \frac{\mathbf{y}_k}{\|\mathbf{y}\|_1} \leq \frac{\hat{\mathbf{y}}_k}{\|\hat{\mathbf{y}}\|_1} \quad \text{for all } k \neq i.$$

Then $\mathbf{u}$ vanishes everywhere except at coordinate i. Equivalently, all off-diagonal mass is received by row i. Since $\pi_{jj} = \min(\frac{\mathbf{y}_j}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}_j}{\|\hat{\mathbf{y}}\|_1})$ for every j, admissibility implies that, for every $j \neq i$,

$$\frac{\hat{\mathbf{y}}_j}{\|\hat{\mathbf{y}}\|_1} = \pi_{+j} = \pi_{jj} + \pi_{ij} = \min(\frac{\mathbf{y}_j}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}_j}{\|\hat{\mathbf{y}}\|_1}) + \pi_{ij},$$

and therefore
$$\pi_{ij} = \frac{\hat{\mathbf{y}}_j}{\|\hat{\mathbf{y}}\|_1} - \min(\frac{\mathbf{y}_j}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}_j}{\|\hat{\mathbf{y}}\|_1}).$$
Thus, every off-diagonal coefficient is uniquely determined, and the optimal plan is unique. The same argument applies when only one class is overestimated.

We now assume that none of the previous cases holds. Then there exist at least two underestimated classes, indexed by $i_1, \dots, i_K$, and at least two overestimated classes, indexed by $j_1, \dots, j_L$, with $K \geq 2$ and $L \geq 2$. Equivalently,

$$\mathbf{u}_{i_k} > 0 \quad \text{for all } k = 1, \dots, K, \qquad \hat{\mathbf{u}}_{j_\ell} > 0 \quad \text{for all } \ell = 1, \dots, L.$$

We show that in this case there exist at least two distinct optimal plans.

Since the diagonal part is fixed, the off-diagonal part defines a transport problem with marginals $\mathbf{u}$ and $\hat{\mathbf{u}}$. It is therefore sufficient to construct two distinct admissible couplings between these marginals.

Define a first plan π^a by transporting the mass greedily from the overestimated classes $j_1, j_2, \dots, j_L$ to the underestimated classes $i_1, i_2, \dots, i_K$, in this order. More precisely, one starts with column j_1 and fills rows $i_1, i_2, \dots, i_K$ successively until the mass $\hat{\mathbf{u}}_{j_1}$ is exhausted; one then proceeds similarly with column j_2, and so on. Define a second plan π^b in the same way, except that the underestimated classes are filled in the reverse order $i_K, i_{K-1}, \dots, i_1$.

Both constructions yield admissible transport plans for the off-diagonal mass, and therefore optimal plans once the diagonal $\min(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1})$ is added back. Moreover, $\pi^a \neq \pi^b$. We now show that
$$\pi^a_{i_1 j_1} > \pi^b_{i_1 j_1}.$$

By construction of π^a, the mass sent from column j_1 is first assigned to row i_1, so that
$$\pi^a_{i_1 j_1} = \min(\mathbf{u}_{i_1}, \hat{\mathbf{u}}_{j_1}) > 0.$$
On the other hand, for any $\pi \in T^{\text{opt}}(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1})$, one has

$$\pi_{ij} \leq \min(\mathbf{u}_i, \hat{\mathbf{u}}_j),$$

since $\mathbf{u}_i$ is the total off-diagonal mass required in row i, and $\hat{\mathbf{u}}_j$ is the total off-diagonal mass available in column j.

We now show that this upper bound is not attained at (i_1, j_1) for π^b.

First, assume by contradiction that

$$\pi^b_{i_1 j_1} = \mathbf{u}_{i_1}.$$

Since π^b fills column j_1 starting from rows $i_K, i_{K-1}, \dots, i_1$, this implies that column j_1 still has available mass when reaching row i_1. Therefore, all previous rows in the construction order must have sent their entire mass to j_1, that is,

$$\pi^b_{i_k j_1} = \mathbf{u}_{i_k} \qquad \text{for all } k = 1, \dots, K.$$

Hence,
$$\hat{\mathbf{u}}_{j_1} \geq \sum_{k=1}^{K} \mathbf{u}_{i_k} = \mathbf{u}_+.$$
Since $\mathbf{u}_+ = \hat{\mathbf{u}}_+$, this implies

$$\hat{\mathbf{u}}_{j_1} = \hat{\mathbf{u}}_+ \quad \text{and} \quad \hat{\mathbf{u}}_{j_\ell} = 0 \text{ for all } \ell \neq 1,$$

which contradicts the assumption that at least two classes are overestimated. Therefore,

$$\pi^b_{i_1 j_1} < \mathbf{u}_{i_1}.$$

Next, assume by contradiction that

$$\pi^b_{i_1 j_1} = \hat{\mathbf{u}}_{j_1}.$$

During the construction of π^b, column j_1 is filled starting from rows $i_K, i_{K-1}, \ldots$. Since j_1 is the first column treated, any row with positive mass encountered before i_1 would contribute strictly positively to column j_1. Thus, this equality implies

$$\mathbf{u}_{i_k} = 0 \qquad \text{for } k = 2, \ldots, K,$$

which contradicts the assumption that at least two classes are underestimated. Hence,

$$\pi^b_{i_1 j_1} < \hat{\mathbf{u}}_{j_1}.$$

Combining the two strict inequalities, we obtain

$$\pi^b_{i_1 j_1} < \min(\mathbf{u}_{i_1}, \hat{\mathbf{u}}_{j_1}) = \pi^a_{i_1 j_1}.$$

Thus, $\pi^a \neq \pi^b$, and there exist at least two distinct optimal plans.

Finally, since $T^{\text{opt}}(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1})$ is convex, every convex combination

$$\lambda \pi^a + (1-\lambda)\pi^b, \qquad \lambda \in [0,1],$$

also belongs to $T^{\text{opt}}(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1})$. Therefore, there are infinitely many optimal plans.

3 Corollaries of Propositions 3 & 4

As soon as $C \leq 3$, there is either equality or at most one overestimated or one underestimated class. As a result, there is a unique optimal plan by Proposition 4.

By optimality, the row sums of the off-diagonal terms are

$$\mathbf{u} := \frac{\mathbf{y}}{\|\mathbf{y}\|_1} - \min(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}),$$

and the column sums are

$$\hat{\mathbf{u}} := \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1} - \min(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}).$$

By admissibility, any surplus mass from overestimated classes is transported only to underestimated classes.

An optimal plan is diagonal if and only if $\min(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1})_+ = \frac{\mathbf{y}}{\|\mathbf{y}\|_1}_+ = \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}_+$, that is, if and only if $\frac{\mathbf{y}}{\|\mathbf{y}\|_1} = \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}$.

4 Proof of proposition 5

Let's show that the matrix

$$\pi(\mathbf{y}, \hat{\mathbf{y}}) =$$
$$\operatorname{diag}\Big(\min(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1})\Big) + \frac{\Big(\frac{\mathbf{y}}{\|\mathbf{y}\|_1} - \min(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1})\Big) \otimes \Big(\frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1} - \min(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1})\Big)}{\|\frac{\mathbf{y}}{\|\mathbf{y}\|_1} - \min(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1})\|_1}$$

is in $T^{\text{opt}}(y, \hat{y})$ and maximises the entropy. Moreover, let's demonstrate that $\pi(\mathbf{y}, \hat{\mathbf{y}})$ becomes

$$\pi(\mathbf{y}, \hat{\mathbf{y}}) = \operatorname{diag}\Big(\min(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1})\Big),$$

when $y = \hat{y}$ by continuous extension.

4.1 Case $\frac{\mathbf{y}}{\|\mathbf{y}\|_1} = \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}$ and continuous extension

When $\frac{\mathbf{y}}{\|\mathbf{y}\|_1} = \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}$, there is a unique optimal plan (see Section 3), namely

$$\operatorname{diag}(\min(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1})),$$

which is therefore the unique entropy maximizer.

Now let $(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1})$ be such that $\frac{\mathbf{y}}{\|\mathbf{y}\|_1} \neq \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}$. We show that, as

$$\|\frac{\mathbf{y}}{\|\mathbf{y}\|_1} - \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}\|_1 \to 0,$$

the formula in Proposition 5 converges to

$$\pi(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}) = \operatorname{diag}(\min(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1})),$$

thereby proving that the formula remains valid by continuous extension at $\frac{\mathbf{y}}{\|\mathbf{y}\|_1} = \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}$.

Set

$$E := \frac{\left(\frac{\mathbf{y}}{\|\mathbf{y}\|_1} - \min(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1})\right) \otimes \left(\frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1} - \min(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1})\right)}{\|\frac{\mathbf{y}}{\|\mathbf{y}\|_1} - \min(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1})\|_1}.$$

It is enough to prove that

$$E \to 0 \in \mathbb{R}^{C\times C} \qquad \text{as} \qquad \|\frac{\mathbf{y}}{\|\mathbf{y}\|_1} - \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}\|_1 \to 0.$$

For every $i, j \in \{1, \ldots, C\}$, the entry E_{ij} is

$$E_{ij} = \frac{\left(\frac{\mathbf{y}_i}{\|\mathbf{y}\|_1} - \min(\frac{\mathbf{y}_i}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}_i}{\|\hat{\mathbf{y}}\|_1})\right)\left(\frac{\hat{\mathbf{y}}_j}{\|\hat{\mathbf{y}}\|_1} - \min(\frac{\mathbf{y}_j}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}_j}{\|\hat{\mathbf{y}}\|_1})\right)}{\|\frac{\mathbf{y}}{\|\mathbf{y}\|_1} - \min(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1})\|_1}.$$

We prove that $E_{ij} \to 0$ by deriving an upper bound depending on $\|\frac{\mathbf{y}}{\|\mathbf{y}\|_1} - \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}\|_1$.

First, observe that

$$\begin{aligned}
\frac{\mathbf{y}_i}{\|\mathbf{y}\|_1} - \min(\frac{\mathbf{y}_i}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}_i}{\|\hat{\mathbf{y}}\|_1}) &= \max(\frac{\mathbf{y}_i}{\|\mathbf{y}\|_1} - \frac{\hat{\mathbf{y}}_i}{\|\hat{\mathbf{y}}\|_1}, 0) \leq \|\frac{\mathbf{y}}{\|\mathbf{y}\|_1} - \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}\|_1, \\
\frac{\hat{\mathbf{y}}_j}{\|\hat{\mathbf{y}}\|_1} - \min(\frac{\mathbf{y}_j}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}_j}{\|\hat{\mathbf{y}}\|_1}) &= \max(\frac{\hat{\mathbf{y}}_j}{\|\hat{\mathbf{y}}\|_1} - \frac{\mathbf{y}_j}{\|\mathbf{y}\|_1}, 0) \leq \|\frac{\mathbf{y}}{\|\mathbf{y}\|_1} - \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}\|_1.
\end{aligned}$$

Hence,

$$0 \leq E_{ij} \leq \frac{\|\frac{\mathbf{y}}{\|\mathbf{y}\|_1} - \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}\|_1^2}{\|\frac{\mathbf{y}}{\|\mathbf{y}\|_1} - \min(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1})\|_1}.$$

Next, we use the identity

$$\begin{aligned}
\|\frac{\mathbf{y}}{\|\mathbf{y}\|_1} - \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}\|_1 &= \sum_{k=1}^{C} |\frac{\mathbf{y}_k}{\|\mathbf{y}\|_1} - \frac{\hat{\mathbf{y}}_k}{\|\hat{\mathbf{y}}\|_1}| \\
&= \sum_{k:\, \frac{\hat{\mathbf{y}}_k}{\|\hat{\mathbf{y}}\|_1} < \frac{\mathbf{y}_k}{\|\mathbf{y}\|_1}} (\frac{\mathbf{y}_k}{\|\mathbf{y}\|_1} - \frac{\hat{\mathbf{y}}_k}{\|\hat{\mathbf{y}}\|_1}) + \sum_{k:\, \frac{\mathbf{y}_k}{\|\mathbf{y}\|_1} < \frac{\hat{\mathbf{y}}_k}{\|\hat{\mathbf{y}}\|_1}} (\frac{\hat{\mathbf{y}}_k}{\|\hat{\mathbf{y}}\|_1} - \frac{\mathbf{y}_k}{\|\mathbf{y}\|_1}) \\
&= \sum_{k=1}^{C} (\frac{\mathbf{y}_k}{\|\mathbf{y}\|_1} - \min(\frac{\mathbf{y}_k}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}_k}{\|\hat{\mathbf{y}}\|_1})) + \sum_{k=1}^{C} (\frac{\hat{\mathbf{y}}_k}{\|\hat{\mathbf{y}}\|_1} - \min(\frac{\mathbf{y}_k}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}_k}{\|\hat{\mathbf{y}}\|_1})) \\
&= \|\frac{\mathbf{y}}{\|\mathbf{y}\|_1} - \min(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1})\|_1 + \|\frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1} - \min(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1})\|_1.
\end{aligned}$$

Since
$$\sum_{k=1}^{C}(\frac{\mathbf{y}_k}{\|\mathbf{y}\|_1}-\min(\frac{\mathbf{y}_k}{\|\mathbf{y}\|_1},\frac{\hat{\mathbf{y}}_k}{\|\hat{\mathbf{y}}\|_1}))=\sum_{k=1}^{C}(\frac{\hat{\mathbf{y}}_k}{\|\hat{\mathbf{y}}\|_1}-\min(\frac{\mathbf{y}_k}{\|\mathbf{y}\|_1},\frac{\hat{\mathbf{y}}_k}{\|\hat{\mathbf{y}}\|_1})),$$
we obtain
$$\|\frac{\mathbf{y}}{\|\mathbf{y}\|_1}-\frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}\|_1=2\|\frac{\mathbf{y}}{\|\mathbf{y}\|_1}-\min(\frac{\mathbf{y}}{\|\mathbf{y}\|_1},\frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1})\|_1.$$
Substituting this into the previous bound gives
$$0\leq E_{ij}\leq 2\|\frac{\mathbf{y}}{\|\mathbf{y}\|_1}-\frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}\|_1.$$
Therefore,
$$E_{ij}\to 0 \qquad \text{as} \qquad \|\frac{\mathbf{y}}{\|\mathbf{y}\|_1}-\frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}\|_1\to 0,$$
for all $i,j\in\{1,\dots,C\}$. Hence,
$$E\to 0\in\mathbb{R}^{C\times C}.$$

We have thus proved that, by continuous extension at $\frac{\mathbf{y}}{\|\mathbf{y}\|_1}=\frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}$, one has
$$\pi(\frac{\mathbf{y}}{\|\mathbf{y}\|_1},\frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1})=\operatorname{diag}\Big(\min(\frac{\mathbf{y}}{\|\mathbf{y}\|_1},\frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1})\Big).$$

4.2 Case $\frac{\mathbf{y}}{\|\mathbf{y}\|_1}\neq\frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}$

Define, for each $k\in\{1,\dots,C\}$,
$$d_k:=\min(\frac{\mathbf{y}_k}{\|\mathbf{y}\|_1},\frac{\hat{\mathbf{y}}_k}{\|\hat{\mathbf{y}}\|_1}),\qquad r_k:=\frac{\mathbf{y}_k}{\|\mathbf{y}\|_1}-\min(\frac{\mathbf{y}_k}{\|\mathbf{y}\|_1},\frac{\hat{\mathbf{y}}_k}{\|\hat{\mathbf{y}}\|_1}),$$
$$c_k:=\frac{\hat{\mathbf{y}}_k}{\|\hat{\mathbf{y}}\|_1}-\min(\frac{\mathbf{y}_k}{\|\mathbf{y}\|_1},\frac{\hat{\mathbf{y}}_k}{\|\hat{\mathbf{y}}\|_1}).$$
By construction,
$$\frac{\mathbf{y}_k}{\|\mathbf{y}\|_1}=d_k+r_k,\qquad \frac{\hat{\mathbf{y}}_k}{\|\hat{\mathbf{y}}\|_1}=d_k+c_k,\qquad r_kc_k=0.$$
Moreover,
$$\sum_{k=1}^{C}r_k=\sum_{k=1}^{C}c_k=:m>0,$$
since $\frac{\mathbf{y}}{\|\mathbf{y}\|_1}\neq\frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}$.

From admissibility and optimality constraints, any feasible matrix M must satisfy
$$M_{kk}=d_k,\qquad \sum_{j:j\neq k}M_{kj}=r_k,\qquad \sum_{i:i\neq k}M_{ik}=c_k.$$
Therefore, the problem reduces to maximizing
$$H(M):=-\sum_{i=1}^{C}\sum_{j=1,j\neq i}^{C}M_{ij}\ln(M_{ij})$$
over the off-diagonal entries, under
$$M_{ij}\geq 0,\qquad \sum_{j\neq i}M_{ij}=r_i,\qquad \sum_{i\neq j}M_{ij}=c_j.$$

Define the index sets of underestimated and overestimated classes as
$$A:=\{i:r_i>0\}=\{i:\frac{\mathbf{y}_i}{\|\mathbf{y}\|_1}>\frac{\hat{\mathbf{y}}_i}{\|\hat{\mathbf{y}}\|_1}\},\qquad B:=\{j:c_j>0\}=\{j:\frac{\hat{\mathbf{y}}_j}{\|\hat{\mathbf{y}}\|_1}>\frac{\mathbf{y}_j}{\|\mathbf{y}\|_1}\}.$$

Since $r_k c_k = 0$ for every k, the sets A and B are disjoint. Also, if $i \notin A$, then $r_i = 0$, hence $M_{ij} = 0$ for all $j \neq i$. Likewise, if $j \notin B$, then $c_j = 0$, hence $M_{ij} = 0$ for all $i \neq j$. Thus the only potentially nonzero off-diagonal entries are those with $(i, j) \in A \times B$.

We thus solve

$$\max_{M_{ij} \geq 0} \; - \sum_{(i,j) \in A \times B} M_{ij} \ln(M_{ij})$$

subject to

$$\sum_{j \in B} M_{ij} = r_i \quad (i \in A), \qquad \sum_{i \in A} M_{ij} = c_j \quad (j \in B).$$

The objective is strictly concave on the convex feasible set, so any critical point is the unique maximizer. Introduce Lagrange multipliers $(\alpha_i)_{i \in A}$ and $(\beta_j)_{j \in B}$ and consider

$$\mathcal{L}(M, \alpha, \beta) = - \sum_{(i,j) \in A \times B} M_{ij} \ln(M_{ij}) + \sum_{i \in A} \alpha_i \Big(\sum_{j \in B} M_{ij} - r_i \Big) + \sum_{j \in B} \beta_j \Big(\sum_{i \in A} M_{ij} - c_j \Big).$$

For any $(i, j) \in A \times B$, stationarity gives

$$\frac{\partial \mathcal{L}}{\partial M_{ij}} = -(\ln M_{ij} + 1) + \alpha_i + \beta_j = 0,$$

hence

$$M_{ij} = e^{\alpha_i - 1} e^{\beta_j}.$$

Thus M_{ij} factorizes as

$$M_{ij} = a_i b_j.$$

Using the row and column constraints,

$$r_i = \sum_{j \in B} M_{ij} = a_i \sum_{j \in B} b_j, \qquad c_j = \sum_{i \in A} M_{ij} = b_j \sum_{i \in A} a_i,$$

leading to

$$M_{ij} = \frac{r_i c_j}{\big(\sum_{k \in A} a_k\big)\big(\sum_{l \in B} b_l\big)} = \frac{r_i c_j}{\lambda}$$

To identify λ, we use the normalization condition $M_{++} = 1$. First, observe that

$$m = \sum_{k \in A} r_k = \sum_{k \in A} \frac{\mathbf{y}_k}{\|\mathbf{y}\|_1} - \min\Big(\frac{\mathbf{y}_k}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}_k}{\|\hat{\mathbf{y}}\|_1}\Big) = \Big\| \frac{\mathbf{y}}{\|\mathbf{y}\|_1} - \min\Big(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}\Big) \Big\|_1,$$

$$m = \sum_{l \in B} c_l = \sum_{l \in B} \frac{\hat{\mathbf{y}}_l}{\|\hat{\mathbf{y}}\|_1} - \min\Big(\frac{\mathbf{y}_l}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}_l}{\|\hat{\mathbf{y}}\|_1}\Big) = \Big\| \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1} - \min\Big(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}\Big) \Big\|_1,$$

and therefore $\| \frac{\mathbf{y}}{\|\mathbf{y}\|_1} - \min(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}) \|_1 = \| \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1} - \min(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}) \|_1$.

It follows that

$$\begin{aligned}
1 = M_{++} &= \sum_{k=1}^{C} M_{kk} + \sum_{k \neq l} M_{kl} \\
&= \min\Big(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}\Big)_+ + \sum_{(k,l) \in A \times B} \frac{r_k c_l}{\lambda} \\
&= \min\Big(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}\Big)_+ + \frac{\big(\sum_{k \in A} r_k\big)\big(\sum_{l \in B} c_l\big)}{\lambda} \\
&= \min\Big(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}\Big)_+ + \frac{m^2}{\lambda}.
\end{aligned}$$

Since $\| \frac{\mathbf{y}}{\|\mathbf{y}\|_1} \|_1 = 1$ and

$$\Big\| \frac{\mathbf{y}}{\|\mathbf{y}\|_1} \Big\|_1 - \min\Big(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}\Big)_+ = \Big\| \frac{\mathbf{y}}{\|\mathbf{y}\|_1} - \min\Big(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}\Big) \Big\|_1 = m,$$

which implies $\lambda = m = \| \frac{\mathbf{y}}{\|\mathbf{y}\|_1} - \min(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}) \|_1$, concluding the proof.

5 Proof of Proposition 6

Let $(\mathbf{y}, \hat{\mathbf{y}})$ be two single-label vectors. In this case, $\frac{\mathbf{y}}{\|\mathbf{y}\|_1} = \mathbf{y}$ and $\frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1} = \hat{\mathbf{y}}$, hence

$$\pi(\mathbf{y}, \hat{\mathbf{y}}) = \mathrm{diag}\big(\min(\mathbf{y}, \hat{\mathbf{y}})\big) + \frac{\big(\mathbf{y} - \min(\mathbf{y}, \hat{\mathbf{y}})\big) \otimes \big(\hat{\mathbf{y}} - \min(\mathbf{y}, \hat{\mathbf{y}})\big)}{\|\mathbf{y} - \min(\mathbf{y}, \hat{\mathbf{y}})\|_1}.$$

If $\mathbf{y} = \hat{\mathbf{y}}$ with label and prediction i, then $\mathbf{y} = \hat{\mathbf{y}} = e_i$, where e_i denotes the canonical basis vector. Then

$$\min(\mathbf{y}, \hat{\mathbf{y}}) = e_i, \qquad \mathbf{y} - \min(\mathbf{y}, \hat{\mathbf{y}}) = 0,$$

and therefore

$$\pi(\mathbf{y}, \hat{\mathbf{y}}) = \mathrm{diag}(e_i) = E_{ii},$$

which corresponds to the elementary matrix of the single-label confusion matrix.

If $\mathbf{y} \neq \hat{\mathbf{y}}$, assume $\mathbf{y} = e_i$ and $\hat{\mathbf{y}} = e_j$ with $i \neq j$. Then

$$\min(\mathbf{y}, \hat{\mathbf{y}}) = 0, \qquad \mathbf{y} - \min(\mathbf{y}, \hat{\mathbf{y}}) = e_i, \qquad \hat{\mathbf{y}} - \min(\mathbf{y}, \hat{\mathbf{y}}) = e_j,$$

and $\|\mathbf{y} - \min(\mathbf{y}, \hat{\mathbf{y}})\|_1 = 1$. Hence,

$$\pi(\mathbf{y}, \hat{\mathbf{y}}) = e_i \otimes e_j = E_{ij}.$$

In both cases, $\pi(\mathbf{y}, \hat{\mathbf{y}})$ coincides with the elementary matrix of the standard single-label confusion matrix, which concludes the proof.

6 Proof of proposition 7

Let $(\mathbf{y}, \hat{\mathbf{y}})$ be two multi-label vectors. Let Y and $\hat{Y}$ be the sets of present classes in $\mathbf{y}$ and $\hat{\mathbf{y}}$, respectively. The four formulas presented in [64] are presented in set notation; we translate them in vector form:

(i) In the case where $Y = \hat{Y}$, contribution formula is:

$$\mathrm{diag}(\mathbf{y})$$

(ii) In the case where $Y \subsetneqq \hat{Y}$, contribution formula is:

$$\frac{\mathbf{y} \otimes \Big(\hat{\mathbf{y}} - \min(\mathbf{y}, \hat{\mathbf{y}})\Big)}{\|\hat{\mathbf{y}}\|_1} + \mathrm{diag}(\mathbf{y}) \frac{\|\mathbf{y}\|_1}{\|\hat{\mathbf{y}}\|_1}$$

(iii) In the case where $\hat{Y} \subsetneqq Y$, contribution formula is:

$$\frac{\Big(\mathbf{y} - \min(\mathbf{y}, \hat{\mathbf{y}})\Big) \otimes \hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1} + \mathrm{diag}(\hat{\mathbf{y}})$$

(iv) In none of the previous cases, the contribution formula is:

$$\frac{\Big(\mathbf{y} - \min(\mathbf{y}, \hat{\mathbf{y}})\Big) \otimes \Big(\hat{\mathbf{y}} - \min(\mathbf{y}, \hat{\mathbf{y}})\Big)}{\|\hat{\mathbf{y}} - \min(\mathbf{y}, \hat{\mathbf{y}})\|_1} + \mathrm{diag}\Big(\min(\mathbf{y}, \hat{\mathbf{y}})\Big)$$

It is straightforward to verify that f is homogeneous, in the sense that

$$f(\alpha \mathbf{u}, \alpha \hat{\mathbf{u}}) = \alpha f(\mathbf{u}, \hat{\mathbf{u}})$$

for any $\alpha \geq 0$. Therefore,

$$\begin{aligned}
&\|\mathbf{y}\|_1 \pi(\mathbf{y}, \hat{\mathbf{y}}) \\
&\quad = f(\mathbf{y}, \hat{\mathbf{y}}\frac{\|\mathbf{y}\|_1}{\|\hat{\mathbf{y}}\|_1}) \\
&\quad = \operatorname{diag}\Big(\min\Big(\mathbf{y}, \hat{\mathbf{y}}\frac{\|\mathbf{y}\|_1}{\|\hat{\mathbf{y}}\|_1}\Big)\Big) + \frac{\Big(\mathbf{y} - \min\Big(\mathbf{y}, \hat{\mathbf{y}}\frac{\|\mathbf{y}\|_1}{\|\hat{\mathbf{y}}\|_1}\Big)\Big) \otimes \Big(\hat{\mathbf{y}}\frac{\|\mathbf{y}\|_1}{\|\hat{\mathbf{y}}\|_1} - \min\Big(\mathbf{y}, \hat{\mathbf{y}}\frac{\|\mathbf{y}\|_1}{\|\hat{\mathbf{y}}\|_1}\Big)\Big)}{\Big\|\mathbf{y} - \min\Big(\mathbf{y}, \hat{\mathbf{y}}\frac{\|\mathbf{y}\|_1}{\|\hat{\mathbf{y}}\|_1}\Big)\Big\|_1}
\end{aligned}$$

We now compare the proposed expression with the four contribution formulas of [64].

6.1 Case $Y = \hat{Y}$

In this case,

$$\hat{\mathbf{y}}\frac{\|\mathbf{y}\|_1}{\|\hat{\mathbf{y}}\|_1} = \mathbf{y},$$

so that

$$\min\left(\mathbf{y}, \hat{\mathbf{y}}\frac{\|\mathbf{y}\|_1}{\|\hat{\mathbf{y}}\|_1}\right) = \mathbf{y}.$$

Hence the outer-product term vanishes, and

$$\|\mathbf{y}\|_1 \pi(\mathbf{y}, \hat{\mathbf{y}}) = \operatorname{diag}(\mathbf{y}),$$

which is exactly the contribution formula in case (i).

6.2 Case $Y \subsetneqq \hat{Y}$ and $\hat{Y} \subsetneqq Y$

Since $\min(\mathbf{y}, \hat{\mathbf{y}}) = \mathbf{y}$, one has

$$\min\left(\mathbf{y}, \hat{\mathbf{y}}\frac{\|\mathbf{y}\|_1}{\|\hat{\mathbf{y}}\|_1}\right) = \mathbf{y}\frac{\|\mathbf{y}\|_1}{\|\hat{\mathbf{y}}\|_1}.$$

Therefore,

$$\operatorname{diag}\left(\min\left(\mathbf{y}, \hat{\mathbf{y}}\frac{\|\mathbf{y}\|_1}{\|\hat{\mathbf{y}}\|_1}\right)\right) = \operatorname{diag}(\mathbf{y})\frac{\|\mathbf{y}\|_1}{\|\hat{\mathbf{y}}\|_1},$$

and the second term simplifies to

$$\frac{\mathbf{y} \otimes (\hat{\mathbf{y}} - \mathbf{y})}{\|\hat{\mathbf{y}}\|_1}.$$

Thus,

$$\|\mathbf{y}\|_1 \pi(\mathbf{y}, \hat{\mathbf{y}}) = \operatorname{diag}(\mathbf{y})\frac{\|\mathbf{y}\|_1}{\|\hat{\mathbf{y}}\|_1} + \frac{\mathbf{y} \otimes (\hat{\mathbf{y}} - \mathbf{y})}{\|\hat{\mathbf{y}}\|_1},$$

which is exactly the contribution formula in case (ii).

The case $\hat{Y} \subsetneqq Y$ is similar to case (ii).

6.3 $Y \not\subset \hat{Y}$ and $\hat{Y} \not\subset Y$

In this case, the obtained expression does *not* coincide with the contribution formula proposed in [64]. Therefore, cases (i)–(iii) are recovered exactly, whereas case (iv) leads to a different formula.

7 Proof of Proposition 8

Silván-Cárdenas and Wang [100] introduce a soft-label confusion matrix defined on the simplex. Let $(\mathbf{y}, \hat{\mathbf{y}})$ be two points in the simplex of dimension C. They define the *MIN-LEAST* and *MIN-MIN* operators to construct the sub-pixel confusion matrix.

The diagonal entries of SCM and TCM coincide. The off-diagonal entries (i, j) are given as intervals:

$$I(i,j) := \Bigg[\max\Big(0, \hat{\mathbf{y}}_j - \min(\mathbf{y}_j, \hat{\mathbf{y}}_j) - \sum_{k \neq i} \big(\mathbf{y}_k - \min(\mathbf{y}_k, \hat{\mathbf{y}}_k)\big)\Big), \\ \min\Big(\mathbf{y}_i - \min(\mathbf{y}_i, \hat{\mathbf{y}}_i), \hat{\mathbf{y}}_j - \min(\mathbf{y}_j, \hat{\mathbf{y}}_j)\Big)\Bigg].$$

We first show that, for all $\pi \in T^{\mathrm{opt}}(\mathbf{y}, \hat{\mathbf{y}})$ and all $i \neq j$, one has $\pi_{ij} \in I(i, j)$. We then prove that the bounds are tight.

7.1 Interval Bounds for Optimal Plan Entries

Let $\pi \in T^{\mathrm{opt}}(\mathbf{y}, \hat{\mathbf{y}})$ and $i \neq j$.

Upper bound. By admissibility,

$$\pi_{ij} \leq \sum_{k \neq i} \pi_{ik} = \mathbf{y}_i - \min(\mathbf{y}_i, \hat{\mathbf{y}}_i), \qquad \pi_{ij} \leq \sum_{k \neq j} \pi_{kj} = \hat{\mathbf{y}}_j - \min(\mathbf{y}_j, \hat{\mathbf{y}}_j).$$

Hence,

$$\pi_{ij} \leq \min\Big(\mathbf{y}_i - \min(\mathbf{y}_i, \hat{\mathbf{y}}_i), \hat{\mathbf{y}}_j - \min(\mathbf{y}_j, \hat{\mathbf{y}}_j)\Big),$$

which proves the upper bound.

Lower bound. We prove the lower bound by contradiction. Assume

$$\pi_{ij} < \max\Big(0, \hat{\mathbf{y}}_j - \min(\mathbf{y}_j, \hat{\mathbf{y}}_j) - \sum_{k \neq i} \big(\mathbf{y}_k - \min(\mathbf{y}_k, \hat{\mathbf{y}}_k)\big)\Big).$$

Then,

$$\begin{aligned}\sum_{k \neq j} \pi_{kj} &= \pi_{ij} + \sum_{k \neq i,j} \pi_{kj} \\ &\leq \pi_{ij} + \sum_{k \neq i,j} \big(\mathbf{y}_k - \min(\mathbf{y}_k, \hat{\mathbf{y}}_k)\big) \\ &< \hat{\mathbf{y}}_j - \min(\mathbf{y}_j, \hat{\mathbf{y}}_j),\end{aligned}$$

which contradicts the column constraint

$$\sum_{k \neq j} \pi_{kj} = \hat{\mathbf{y}}_j - \min(\mathbf{y}_j, \hat{\mathbf{y}}_j).$$

Thus, the lower bound holds.

We conclude that, for all $\pi \in T^{\mathrm{opt}}(\mathbf{y}, \hat{\mathbf{y}})$ and all $i \neq j$, one has $\pi_{ij} \in I(i, j)$.

7.2 Existence of Elements Attaining the Bounds

Let $i \neq j$ and

$$r_k := \mathbf{y}_k - \min(\mathbf{y}_k, \hat{\mathbf{y}}_k), \qquad c_k := \hat{\mathbf{y}}_k - \min(\mathbf{y}_k, \hat{\mathbf{y}}_k).$$

Then, for any $\pi \in T^{\mathrm{opt}}(\mathbf{y}, \hat{\mathbf{y}})$, the off-diagonal entries form a transportation problem with row sums (r_k) and column sums (c_k). We now describe constructions attaining the upper and lower bounds.

Upper bound. We maximize the mass at (i, j) by assigning $\min(r_i, c_j)$ to this entry. The remaining mass can then be distributed to satisfy the residual row and column sums, yielding a feasible plan. Hence, the upper bound is attained.

Lower bound. We minimize the mass at (i, j) by assigning the excess mass in class j (quantified by c_j) first to underestimated classes $k \neq i$, and then to class i only if necessary.

If the total underestimation of classes other than i, given by $\sum_{k \neq i} r_k$, is greater than or equal to c_j, then these classes can fully absorb the excess of j. Hence, no mass needs to be assigned from i to j, and

$$\pi_{ij} = 0 = \max\Big(0,\ c_j - \sum_{k \neq i} r_k\Big).$$

If the total underestimation of classes other than i is strictly smaller than c_j, then the remaining excess must be assigned to class i. Therefore,

$$\pi_{ij} = c_j - \sum_{k \neq i} r_k = \max\Big(0,\ c_j - \sum_{k \neq i} r_k\Big).$$

Chapter C
A Local Sampling Strategy

1	Proofs	**115**
1.1	Proof of Proposition 10	115
1.2	Proof of Proposition 11	116
1.3	Proof of Proposition 12	116
2	Model Architectures	**116**

1 Proofs

1.1 Proof of Proposition 10

For each honest worker i,

$$\begin{aligned} g_i - g^\star &= \sum_{c=1}^{C} q_{i,c} g_i^{(c)} - \sum_{c=1}^{C} q_c g^{(c)} \\ &= \sum_{c=1}^{C} q_{i,c}\left(g_i^{(c)} - g^{(c)}\right) + \sum_{c=1}^{C} (q_{i,c} - q_c) g^{(c)}. \end{aligned}$$

Define

$$A_i = \sum_{c=1}^{C} q_{i,c}\left(g_i^{(c)} - g^{(c)}\right), \qquad B_i = \sum_{c=1}^{C} (q_{i,c} - q_c) g^{(c)}.$$

Then $g_i - g^\star = A_i + B_i$, and therefore

$$\|g_i - g^\star\|^2 = \|A_i\|^2 + \|B_i\|^2 + 2\langle A_i, B_i\rangle.$$

We now show that the cross term has zero expectation. Since $\mathbf{q}_i$ depends only on the class counts and the class counts are independent of the observations, the class-wise estimators are conditionally unbiased:

$$\mathbb{E}[g_i^{(c)} \mid \mathbf{q}_i] = g^{(c)}.$$

It follows that

$$\mathbb{E}[g_i^{(c)} - g^{(c)} \mid \mathbf{q}_i] = 0.$$

Hence

$$\mathbb{E}[A_i \mid \mathbf{q}_i] = \sum_{c=1}^{C} q_{i,c} \mathbb{E}[g_i^{(c)} - g^{(c)} \mid \mathbf{q}_i] = 0.$$

Moreover, B_i is a function of $\mathbf{q}_i$. Therefore,

$$\begin{aligned} \mathbb{E}[\langle A_i, B_i\rangle] &= \mathbb{E}\left[\mathbb{E}[\langle A_i, B_i\rangle \mid \mathbf{q}_i]\right] \\ &= \mathbb{E}\left[\langle \mathbb{E}[A_i \mid \mathbf{q}_i], B_i\rangle\right] \\ &= 0. \end{aligned}$$

Since honest workers are i.i.d. and the assumed second-moment conditions ensure integrability, the strong law of large numbers gives

$$\frac{1}{|\mathcal{H}|}\sum_{i\in\mathcal{H}}\|g_i - g^\star\|^2 \xrightarrow[|\mathcal{H}|\to\infty]{\text{a.s.}} \mathbb{E}[\|A_i\|^2] + \mathbb{E}[\|B_i\|^2].$$

Substituting the definitions of A_i and B_i completes the proof.

1.2 Proof of Proposition 11

For each honest worker i,

$$g_{\mathrm{NS}\,i} - g^\star_{\mathrm{NS}} = \sum_{c=1}^{C} p_c \left(g^{(c)}_{\mathrm{NS}\,i} - g^{(c)}\right).$$

Taking squared norms and applying the strong law of large numbers gives the result.

1.3 Proof of Proposition 12

By definition,

$$g^\star_{\mathrm{NS}} = \sum_{c=1}^{C} p_c g^{(c)}, \qquad g^\star = \sum_{c=1}^{C} q_c g^{(c)}.$$

If $\mathbf{p} = \mathbf{q}$, then $p_c = q_c$ for every class c, and therefore $g^\star_{\mathrm{NS}} = g^\star$.

2 Model Architectures

We adopt the following abreviations : L(#outputs) denotes a fully connected linear layer; R is a ReLU activation; T is a Tanh activation; C(#channels) represents a 2D convolutional layer with kernel size 5, padding 0, and stride 1; M denotes 2D max-pooling with kernel size 2; B stands for batch normalization; and D represents dropout with a fixed probability of 0.25.

The model architectures are as follows:

- MNIST, Fashion MNIST, Kuzushiji-MNIST (CNN from [4]): C(20)-R-M–C(20)-R-M–L(500)-R–L(10)
- CIFAR-10 (CNN from [4]):(3,32×32)-C(64)-R-B-C(64)-R-B-M-D-C(128)-R-B-C(128)-R-B-M-D-L(128)-R-D-L(10)

Chapter D
Inside the Convex Hull of Gradients

1	Additionnal Experiments	**117**
2	Variance Decomposition of the Gradient Estimator	**117**
3	Theoretical Analysis	**119**

1 Additionnal Experiments

This section reports additional results in Tables D.1, D.2, D.3, D.4, D.5, and D.6.

2 Variance Decomposition of the Gradient Estimator

Let $P \in \mathbb{R}^{h' \times C}$ and $U \in \mathbb{R}^{h' \times D}$ denote the label distribution and update matrices of the honest workers in A, respectively. Similarly, let $Q \in \mathbb{R}^{f' \times C}$, $V \in \mathbb{R}^{f' \times D}$ denote the corresponding matrices for the Byzantine workers. Let $\mathcal{U} \in \mathbb{R}^{C \times D}$ denote the expected class-wise gradients over the global dataset $\mathcal{D}$.

To model the random selection of a subgroup $B \subseteq A$ with s workers, we introduce two diagonal matrices $S = \operatorname{diag}(s_1, \ldots, s_{h'})$ and $T = \operatorname{diag}(t_1, \ldots, t_{f'})$, whose diagonal entries are identically distributed Bernoulli variables. These variables indicate whether a worker is selected in the subgroup. The subgroup size is fixed by the constraint $\operatorname{Tr}(S + T) = s$.

The following label distribution and update matrices represent the workers in subgroup B; rows corresponding to workers not selected in B are set to zero. We denote by $\hat{\mathcal{G}}$ the global class-wise gradient estimator; zero rows have no effect on these estimators:

$$\underbrace{\begin{pmatrix} SP \\ TQ \end{pmatrix},}_{\text{label distributions } h\times C} \qquad \underbrace{\begin{pmatrix} SU \\ TV \end{pmatrix},}_{\text{updates } h\times D} \qquad \underbrace{\hat{\mathcal{G}} = \begin{pmatrix} SP \\ TQ \end{pmatrix}^{\dagger} \begin{pmatrix} SU \\ TV \end{pmatrix},}_{\text{global class gradient estimator } C\times D}$$

For any realizations of S and T, we assume that the label distribution matrix has full column rank.

Then, it holds

$$\begin{aligned}
\hat{\mathcal{G}} &= \begin{pmatrix} SP \\ TQ \end{pmatrix}^{\dagger} \begin{pmatrix} SU \\ TV \end{pmatrix} \\
&= (\begin{pmatrix} SP \\ TQ \end{pmatrix}^{\top} \begin{pmatrix} SP \\ TQ \end{pmatrix})^{-1} \begin{pmatrix} SP \\ TQ \end{pmatrix}^{\top} \begin{pmatrix} SU \\ TV \end{pmatrix} \\
&= \left(P^{\top}SP + Q^{\top}TQ\right)^{-1}\left(P^{\top}SU + Q^{\top}TV\right) \\
&= \underbrace{\left(P^{\top}SP + Q^{\top}TQ\right)^{-1}P^{\top}S}_{\widetilde{P}}\,U + \underbrace{\left(P^{\top}SP + Q^{\top}TQ\right)^{-1}Q^{\top}T}_{\widetilde{Q}}\,V \\
&= \widetilde{P}U + \widetilde{Q}V,
\end{aligned}$$

Considering the Frobenius variance of $\hat{\mathcal{G}}$, Proposition 15 yields

$$\begin{aligned}
&\sigma^2(\hat{\mathcal{G}}) \\
&= \mathbb{E}\left[\left\|\hat{\mathcal{G}} - \mathbb{E}\left[\hat{\mathcal{G}}\right]\right\|_F^2\right] \\
&= \mathbb{E}\left[\left\|\widetilde{P}(U - \mathbb{E}[U])\right\|_F^2\right] + \mathbb{E}\left[\left\|\widetilde{Q}(V - \mathbb{E}[V])\right\|_F^2\right] + \mathbb{E}\left[\left\|(\widetilde{Q} - \mathbb{E}[\widetilde{Q}])(\mathbb{E}[V] - Q\mathcal{U})\right\|_F^2\right]
\end{aligned}$$

We assume that Byzantine updates are convex combinations of C vectors consistent with their broadcast label distributions. Let $\mathcal{V} \in \mathbb{R}^{C \times D}$ denote the Byzantine analogue of the class-wise gradient matrix $\mathcal{U}$. Consequently, $\mathbb{E}[V] = Q\mathcal{V}$.

Let $\hat{\mathcal{U}}$ denote the class-wise gradient estimator constructed from the honest workers selected by S, and $\hat{\mathcal{V}}$ the corresponding estimator constructed from the Byzantine workers selected by T, defined as

$$\hat{\mathcal{U}} = (SP)^{\dagger}SU, \quad \hat{\mathcal{V}} = (TQ)^{\dagger}TV$$

Let W be the matrix defined as $W = \widetilde{P}P$.

Then, Proposition 18 gives

$$\begin{aligned}
&\sigma^2(\hat{\mathcal{G}}) \\
&= \mathbb{E}\left[\left\|W(\hat{\mathcal{U}} - \mathcal{U})\right\|_F^2\right] + \mathbb{E}\left[\left\|(I - W)(\hat{\mathcal{V}} - \mathcal{V})\right\|_F^2\right] + \mathbb{E}\left[\|(W - \mathbb{E}[W])(\mathcal{V} - \mathcal{U})\|_F^2\right],
\end{aligned}$$

with W equals to

$$\begin{aligned}
W &= (X + Y)^{-1}X, \\
I - W &= (X + Y)^{-1}Y, \\
\text{where} \quad X &= P^{\top}SP, \quad Y = Q^{\top}TQ.
\end{aligned}$$

The matrices W and $I - W$ measures the share of total information $X + Y$ contributed by honest workers, while $I - W = (X + Y)^{-1}Y$ measures the contribution of Byzantine workers. Additionally, W and $I - W$ are diagonalisable and their singular values lie in the interval $[0, 1]$.

We now consider the following simplification, which does not hold in practice but helps clarify the equations: (i) We assume that Byzantine label distributions perfectly mimic the honest ones, i.e., for any realizations of S and T we have $Y = \lambda X$. This implies, $\lambda = \mathrm{Tr}(Y)/\mathrm{Tr}(X)$, and

$$W = (X + Y)^{-1}X = \frac{1}{1 + \lambda}I = \frac{\mathrm{Tr}(X)}{s}\,I, \qquad I - W = \frac{\lambda}{1 + \lambda}I = \frac{\mathrm{Tr}(Y)}{s}\,I.$$

Assume moreover that (ii) $\mathrm{Tr}(X)$ is independent of $\hat{\mathcal{U}}$.

Substituting this into the previous expression for $\sigma^2(\hat{\mathcal{G}})$, and neglecting the second term, which corresponds to the variance of the Byzantine estimator and can be made zero by an appropriate Byzantine strategy, we obtain

$$\begin{aligned}\sigma^2(\hat{\mathcal{G}}) &= \mathbb{E}\left[\left(\frac{\mathrm{Tr}(X)}{s}\right)^2\right]\mathbb{E}\left[\|\hat{\mathcal{U}}-\mathcal{U}\|_F^2\right] + \mathbb{E}\left[\left(\frac{\mathrm{Tr}(X)}{s} - \mathbb{E}\left[\frac{\mathrm{Tr}(X)}{s}\right]\right)^2\right]\|\mathcal{V}-\mathcal{U}\|_F^2 \\ &= \frac{\mathbb{E}[\mathrm{Tr}(X)^2]}{s^2}\mathbb{E}\left[\|\hat{\mathcal{U}}-\mathcal{U}\|_F^2\right] + \frac{\mathrm{Var}(\mathrm{Tr}(X))}{s^2}\|\mathcal{V}-\mathcal{U}\|_F^2.\end{aligned}$$

In practice, the mismatch term $\|\mathcal{V}-\mathcal{U}\|_F^2$ is often much larger than the honest estimation error $\mathbb{E}[\|\hat{\mathcal{U}}-\mathcal{U}\|_F^2]$. Under this assumption, the behavior of $\sigma^2(\hat{\mathcal{G}})$ is primarily driven by the second term. Consequently, increasing h' tends to reduce the variance whenever it decreases $\mathrm{Var}(\mathrm{Tr}(X))$. In this regime, the worker score strategy is theoretically grounded by Proposition 19 in Appendix.

3 Theoretical Analysis

Proposition 13.

$$(\widetilde{P}-\mathbb{E}[\widetilde{P}])\mathbb{E}[U] = -\big(\widetilde{Q}-\mathbb{E}[\widetilde{Q}]\big)Q\mathcal{U}$$

Proof. It holds that $\mathbb{E}[U] = P\mathcal{U}$. Since P is deterministic, we obtain

$$\begin{aligned}(\widetilde{P}-\mathbb{E}[\widetilde{P}])\mathbb{E}[U] &= (\widetilde{P}-\mathbb{E}[\widetilde{P}])P\mathcal{U} \\ &= (\widetilde{P}P-\mathbb{E}[\widetilde{P}P])\mathcal{U}\end{aligned}$$

Observing that $\widetilde{P}P = I - \widetilde{Q}Q$, it follows that

$$\begin{aligned}(\widetilde{P}P-\mathbb{E}[\widetilde{P}P])\mathcal{U} &= \big(I-\widetilde{Q}Q-\mathbb{E}[I-\widetilde{Q}Q]\big)\mathcal{U} \\ &= -\big(\widetilde{Q}-\mathbb{E}[\widetilde{Q}]\big)Q\mathcal{U}\end{aligned}$$

□ □

Proposition 14.

$$\hat{\mathcal{G}}-\mathbb{E}[\hat{\mathcal{G}}] = \widetilde{P}(U-\mathbb{E}[U]) + \widetilde{Q}(V-\mathbb{E}[V]) + (\widetilde{Q}-\mathbb{E}[\widetilde{Q}])(\mathbb{E}[V]-Q\mathcal{U})$$

Proof. Since S and T are independent of U and V, it follows that $\widetilde{P}$ and $\widetilde{Q}$ are also independent of U and V. It holds that

$$\mathbb{E}[\hat{\mathcal{G}}] = \mathbb{E}[\widetilde{P}U+\widetilde{Q}V] = \mathbb{E}[\widetilde{P}]\mathbb{E}[U] + \mathbb{E}[\widetilde{Q}]\mathbb{E}[V]$$

Injecting previous result in $\hat{\mathcal{G}}-\mathbb{E}[\hat{\mathcal{G}}]$, it holds that

$$\begin{aligned}\hat{\mathcal{G}}-\mathbb{E}[\hat{\mathcal{G}}] &= \widetilde{P}U+\widetilde{Q}V-\mathbb{E}[\widetilde{P}]\mathbb{E}[U]-\mathbb{E}[\widetilde{Q}]\mathbb{E}[V] \\ &= \widetilde{P}(U-\mathbb{E}[U]) + \widetilde{Q}(V-\mathbb{E}[V]) + (\widetilde{P}-\mathbb{E}[\widetilde{P}])\mathbb{E}[U] + (\widetilde{Q}-\mathbb{E}[\widetilde{Q}])\mathbb{E}[V]\end{aligned}$$

According to Proposition 13, it holds

$$(\widetilde{P}-\mathbb{E}[\widetilde{P}])\mathbb{E}[U] = -\big(\widetilde{Q}-\mathbb{E}[\widetilde{Q}]\big)Q\mathcal{U},$$

which leads to

$$\hat{\mathcal{G}}-\mathbb{E}[\hat{\mathcal{G}}] = \widetilde{P}(U-\mathbb{E}[U]) + \widetilde{Q}(V-\mathbb{E}[V]) + (\widetilde{Q}-\mathbb{E}[\widetilde{Q}])(\mathbb{E}[V]-Q\mathcal{U})$$

□ □

Proposition 15.

$$\sigma^2(\hat{\mathcal{G}}) =$$
$$\mathbb{E}\left[\left\|\widetilde{P}(U-\mathbb{E}[U])\right\|_F^2\right]+\mathbb{E}\left[\left\|\widetilde{Q}(V-\mathbb{E}[V])\right\|_F^2\right]+\mathbb{E}\left[\left\|(\widetilde{Q}-\mathbb{E}[\widetilde{Q}])(\mathbb{E}[V]-Q\mathcal{U})\right\|_F^2\right]$$

Proof. According to Proposition 14, it holds

$$\sigma^2(\hat{\mathcal{G}})-\mathbb{E}[\sigma^2(\hat{\mathcal{G}})]=\widetilde{P}(U-\mathbb{E}[U])+\widetilde{Q}(V-\mathbb{E}[V])+(\widetilde{Q}-\mathbb{E}[\widetilde{Q}])(\mathbb{E}[V]-Q\mathcal{U}),$$

using this, we obtain

$$\begin{aligned}
&\mathbb{E}\left[\left\|\hat{\mathcal{G}}-\mathbb{E}\left[\hat{\mathcal{G}}\right]\right\|_F^2\right]\\
&=\mathbb{E}\left[\left\|\widetilde{P}(U-\mathbb{E}[U])+\widetilde{Q}(V-\mathbb{E}[V])+(\widetilde{Q}-\mathbb{E}[\widetilde{Q}])(\mathbb{E}[V]-Q\mathcal{U})\right\|_F^2\right]\\
&=\mathbb{E}\left[\mathbb{E}\left[\left\|\widetilde{P}(U-\mathbb{E}[U])+\widetilde{Q}(V-\mathbb{E}[V])+(\widetilde{Q}-\mathbb{E}[\widetilde{Q}])(\mathbb{E}[V]-Q\mathcal{U})\right\|_F^2 \,\middle|\, S,T\right]\right],
\end{aligned}$$

where the last equality follows from the law of total expectation. Inside the conditional expectation, $\widetilde{P}$ and $\widetilde{Q}$ are deterministic. It holds that

$$\begin{aligned}
&\mathbb{E}\left[\left\|\widetilde{P}(U-\mathbb{E}[U])+\widetilde{Q}(V-\mathbb{E}[V])+(\widetilde{Q}-\mathbb{E}[\widetilde{Q}])(\mathbb{E}[V]-Q\mathcal{U})\right\|_F^2 \,\middle|\, S,T\right]\\
&\quad=\mathbb{E}\left[\left\|\widetilde{P}(U-\mathbb{E}[U])\right\|_F^2 \,\middle|\, S,T\right]+\mathbb{E}\left[\left\|\widetilde{Q}(V-\mathbb{E}[V])\right\|_F^2 \,\middle|\, S,T\right]\\
&\qquad+\left\|(\widetilde{Q}-\mathbb{E}[\widetilde{Q}])(\mathbb{E}[V]-Q\mathcal{U})\right\|_F^2,
\end{aligned}$$

where we used that, conditional on (S,T), the random matrices $\widetilde{P}(U-\mathbb{E}[U])$ and $\widetilde{Q}(V-\mathbb{E}[V])$ are centered and independent, and are also independent of $(\widetilde{Q}-\mathbb{E}[\widetilde{Q}])(\mathbb{E}[V]-Q\mathcal{U})$, which is deterministic given (S,T). Therefore,

$$\begin{aligned}
&\mathbb{E}\left[\left\|\hat{\mathcal{G}}-\mathbb{E}\left[\hat{\mathcal{G}}\right]\right\|_F^2\right]\\
&=\mathbb{E}\left[\left\|\widetilde{P}(U-\mathbb{E}[U])\right\|_F^2\right]+\mathbb{E}\left[\left\|\widetilde{Q}(V-\mathbb{E}[V])\right\|_F^2\right]+\mathbb{E}\left[\left\|(\widetilde{Q}-\mathbb{E}[\widetilde{Q}])(\mathbb{E}[V]-Q\mathcal{U})\right\|_F^2\right]
\end{aligned}$$

□ □

Proposition 16.
$$\widetilde{P}P(SP)^\dagger=\widetilde{P},\quad \widetilde{Q}Q(TQ)^\dagger=\widetilde{Q}$$

Proof.

$$\begin{aligned}
\widetilde{P}P(SP)^\dagger&=\left(P^\top SP+Q^\top TQ\right)^{-1}P^\top SP(SP)^\dagger\\
&=\left(P^\top SP+Q^\top TQ\right)^{-1}(SP)^\top SP(SP)^\dagger
\end{aligned}$$

It holds $M^\top=M^\top MM^\dagger$ for any matrix M. Applying this with $M=SP$ gives

$$\begin{aligned}
\widetilde{P}P(SP)^\dagger&=\left(P^\top SP+Q^\top TQ\right)^{-1}(SP)^\top\\
&=\left(P^\top SP+Q^\top TQ\right)^{-1}P^\top S\\
&=\widetilde{P}
\end{aligned}$$

Similarly,

$$\begin{aligned}\widetilde{Q}Q(TQ)^{\dagger} &= \left(P^{\top}SP + Q^{\top}TQ\right)^{-1}Q^{\top}TQ(TQ)^{\dagger}\\ &= \left(P^{\top}SP + Q^{\top}TQ\right)^{-1}(TQ)^{\top}TQ(TQ)^{\dagger}\\ &= \left(P^{\top}SP + Q^{\top}TQ\right)^{-1}(TQ)^{\top}\\ &= \left(P^{\top}SP + Q^{\top}TQ\right)^{-1}Q^{\top}T\\ &= \widetilde{Q}\end{aligned}$$

□ □

Proposition 17.
$$\widetilde{P}(U - \mathbb{E}[U]) = W(\hat{\mathcal{U}} - \mathcal{U}).$$

Proof. We have $\mathbb{E}[U] = P\mathcal{U}$. Hence, $\widetilde{P}\mathbb{E}[U] = \widetilde{P}P\mathcal{U} = W\mathcal{U}$.

By Proposition 16, $\widetilde{P} = W(SP)^{+}$. Moreover, since $\widetilde{P} = \widetilde{P}S$, we obtain

$$\widetilde{P}U = \widetilde{P}SU = W(SP)^{+}SU = W\hat{\mathcal{U}}.$$

Combining the two identities,

$$\widetilde{P}(U - \mathbb{E}[U]) = \widetilde{P}U - \widetilde{P}\mathbb{E}[U] = W\hat{\mathcal{U}} - W\mathcal{U} = W(\hat{\mathcal{U}} - \mathcal{U}).$$

□ □

Proposition 18. *Under Byzantine convex combination assumption,*

$$\begin{aligned}&\sigma^2[\hat{\mathcal{G}}]\\ &= \mathbb{E}\left[\left\|W(\hat{\mathcal{U}} - \mathcal{U})\right\|_F^2\right] + \mathbb{E}\left[\left\|(I - W)(\hat{\mathcal{V}} - \mathcal{V})\right\|_F^2\right] + \mathbb{E}\left[\|(W - \mathbb{E}[W])(\mathcal{V} - \mathcal{U})\|_F^2\right]\end{aligned}$$

Proof. Under Byzantine convex combination assumption, $\mathbb{E}[V] = Q\mathcal{V}$.

Moreover, applying the reasoning of Proposition 17 and using the identity $\widetilde{Q}Q = (I - W)$, we obtain $\widetilde{Q}(V - \mathbb{E}[V]) = (I - W)(\hat{\mathcal{V}} - \mathcal{V})$.

Proposition 15 gives

$$\begin{aligned}&\mathbb{E}\left[\left\|\widehat{\mathcal{U}}(\mathfrak{g}) - \mathbb{E}\left[\widehat{\mathcal{U}}(\mathfrak{g})\right]\right\|_F^2\right]\\ &= \mathbb{E}\left[\left\|\widetilde{P}(U - \mathbb{E}[U])\right\|_F^2\right] + \mathbb{E}\left[\left\|\widetilde{Q}(V - \mathbb{E}[V])\right\|_F^2\right] + \mathbb{E}\left[\left\|(\widetilde{Q} - \mathbb{E}[\widetilde{Q}])(\mathbb{E}[V] - Q\mathcal{U})\right\|_F^2\right]\end{aligned}$$

Proposition 17 gives $\widetilde{P}(U - \mathbb{E}[U]) = W(\hat{\mathcal{U}} - \mathcal{U})$.

Combining all these results, we get:

$$\begin{aligned}&\sigma^2[\widehat{\mathcal{U}}(\mathfrak{g})]\\ &= \mathbb{E}\left[\left\|W(\hat{\mathcal{U}} - \mathcal{U})\right\|_F^2\right] + \mathbb{E}\left[\left\|(I - W)(\hat{\mathcal{V}} - \mathcal{V})\right\|_F^2\right] + \mathbb{E}\left[\|(W - \mathbb{E}[W])(\mathcal{V} - \mathcal{U})\|_F^2\right]\end{aligned}$$

□ □

Proposition 19. *Assume that each candidate is either good or bad. Let there be h good candidates and f bad candidates with $f < N/3$, $N = h + f$. Teams are formed by sampling h candidates. Assume that the time achieved by a team depends only on the proportion of good rowers in the boat.*

Then, for any good candidate i and bad candidate j, $\mathrm{Score}(i) > \mathrm{Score}(j)$.

Proof. Fix a good candidate i and a bad candidate j. Couple the random teams used to define Score(i) and Score(j) as follows: sample the same set of $h-1$ other candidates in both cases, excluding i and j. Then compare the two teams

$$T_i = i \cup A \qquad \text{and} \qquad T_j = j \cup A,$$

where A is any subset of size $h-1$ of the remaining $N-2$ candidates.

The two teams differ only in that T_i contains the good candidate i whereas T_j contains the bad candidate j. Hence T_i has exactly one more good candidate than T_j. Since the team time depends only on the proportion of good candidates, and improves when this proportion increases, the team containing i always achieves a strictly better time than the team containing j.

Therefore, for every realization of A, the score contribution of i is strictly larger than that of j. Taking expectation over A yields

$$\text{Score}(i) > \text{Score}(j).$$

This concludes the proof. □ □

Proposition 20. *Let $\varepsilon_i = p_i\mathcal{U}_i - p_i\mathcal{U}$ and*

$$\varepsilon = \begin{pmatrix} \varepsilon_1 \\ \vdots \\ \varepsilon_h \end{pmatrix} \in \mathbb{R}^{h\times D}.$$

Then $(\varepsilon_i)_{i=1}^h$ are independent random vectors and, for all $i = 1, \dots, h$,

$$\begin{aligned}
&\mathbb{E}[\varepsilon_i] = 0,\\
&\mathbb{V}(\varepsilon_{id}) = \frac{1}{s^2}\sum_{c=1}^{C} |\mathcal{D}_i^c|\, \mathbb{V}[\nabla\ell(\Phi(x), c)_d]\,,
\end{aligned}$$

where $|\mathcal{D}_i| = s$ is the number of samples used to compute the local update (typically the batch size), and is the same for all workers.

In addition, if the variance is independent of the class, i.e., $\mathbb{V}[\nabla\ell(\Phi(x), c)_d] = \sigma_d^2$ for all c, then

$$\mathbb{V}(\varepsilon_{id}) = \frac{\sigma_d^2}{s}.$$

In particular, the noise variance is the same across workers, which corresponds to a homoscedasticity assumption.

Proof. The null expectation follows is straightforward. Moreover, since the observations are independent and each error term is computed from a disjoint set of observations, the vectors ε_i are independent. We now compute the variance.

For notational simplicity, denote the column vector $(\mathcal{U}_i)_{:d}$ by q. Note that $\mathbb{E}(q) = \mathbb{E}((\mathcal{U}_i)_{:d}) = \mathcal{U}_{:d}$. We then have

$$\begin{aligned}
\mathbb{V}(\varepsilon_{id}) = \mathbb{E}[\varepsilon_{id}^2] &= \mathbb{E}\big[(p_i q - p_i\mathbb{E}(q))^2\big]\\
&= \mathbb{E}\big[(p_i(q - \mathbb{E}(q)))^2\big]\\
&= \mathbb{E}\big[p_i(q-\mathbb{E}(q))(q-\mathbb{E}(q))^\top p_i^\top\big]\\
&= p_i\mathbb{E}\big[(q-\mathbb{E}(q))(q-\mathbb{E}(q))^\top\big]p_i^\top\\
&= p_i\text{Cov}(q)p_i^\top.
\end{aligned}$$

We now study Cov(q). Note that components of q are independent. As a result,

$$\begin{aligned}
\text{Cov}(q) &= \text{diag}\Big(\mathbb{V}[q_1], \dots, \mathbb{V}[q_C]\Big)\\
&= \text{diag}\Big(\mathbb{V}[(u_i^1)_d], \dots, \mathbb{V}[(u_i^C)_d]\Big)
\end{aligned}$$

Let's now study the component variance:

$$
\begin{aligned}
\mathbb{V}[(u_i^c)_d] &= \mathbb{V}\left[\frac{1}{|\mathcal{D}_i^c|}\sum_{(x,y)\in\mathcal{D}_i^c}\nabla\ell(\Phi(x),c)_d\right] \\
&= \frac{1}{|\mathcal{D}_i^c|^2}\mathbb{V}\left[\sum_{(x,y)\in\mathcal{D}_i^c}\nabla\ell(\Phi(x),c)_d\right] \\
&= \frac{1}{|\mathcal{D}_i^c|^2}\sum_{(x,y)\in\mathcal{D}_i^c}\mathbb{V}[\nabla\ell(\Phi(x),c)_d] \\
&= \frac{1}{|\mathcal{D}_i^c|}\mathbb{V}[\nabla\ell(\Phi(x),c)_d]
\end{aligned}
$$

Finally, compiling the previous results, we obtain

$$
\begin{aligned}
\mathbb{V}(\varepsilon_{id}) &= p_i \operatorname{Cov}(q)\, p_i^\top \\
&= \sum_{c=1}^{C}(p_i)_c \operatorname{Cov}(q)_{cc}(p_i)_c \\
&= \sum_{c=1}^{C}(p_i)_c^2\,\frac{1}{|\mathcal{D}_i^c|}\,\mathbb{V}[\nabla\ell(\Phi(x),c)_d]\,.
\end{aligned}
$$

Using $(p_i)_c = |\mathcal{D}_i^c|/|\mathcal{D}_i|$, we further get

$$
\begin{aligned}
\mathbb{V}(\varepsilon_{id}) &= \sum_{c=1}^{C}\frac{|\mathcal{D}_i^c|^2}{|\mathcal{D}_i|^2}\,\frac{1}{|\mathcal{D}_i^c|}\,\mathbb{V}[\nabla\ell(\Phi(x),c)_d] \\
&= \frac{1}{|\mathcal{D}_i|^2}\sum_{c=1}^{C}|\mathcal{D}_i^c|\,\mathbb{V}[\nabla\ell(\Phi(x),c)_d]\,.
\end{aligned}
$$

All worker updates are based on the same number of samples, i.e., $|\mathcal{D}_i| = s$. Additionally, assuming that the variance is independent of the class, i.e., $\mathbb{V}[\nabla\ell(\Phi(x),c)_d] = \sigma_d^2$ for all c, it holds

$$
\mathbb{E}\big[\varepsilon_{id}^2\big] = \frac{1}{|\mathcal{D}_i|^2}\sum_{c=1}^{C}|\mathcal{D}_i^c|\,\sigma_d^2 = \frac{\sigma_d^2}{|\mathcal{D}_i|} = \frac{\sigma_d^2}{s},
$$

ending the proof. □ □

Table D.1: Mean Accuracy (%) comparison over last 200 training steps between robust aggregators (BKT + CWMed, BKT + CwTM, BKT + RFA, BKT + GAS, HullGuard + GAS) on CIFAR-10 for different concentration parameter $\alpha \in \{10, 1, 0.1\}$ under several Byzantine attacks (ALIE, FOE, Mimic, MinSum, LF) and different numbers of workers $f \in \{8, 14, 20, 26\}$. The Gain row reports the accuracy improvement of HullGuard paired with GAS robust aggregator over the best competing method for each configuration. A negative gain indicates that HullGuard is being outperformed by another method.

		ALIE				FOE				Mimic				MinSum				LF			
		8	14	20	26	8	14	20	26	8	14	20	26	8	14	20	26	8	14	20	26
10	CWMed	80.01	64.81	24.43	17.48	78.77	77.08	48.94	29.59	78.37	76.74	62.04	52.41	81.03	81.13	81.04	80.20	77.09	75.72	69.98	61.98
	CwTM	77.69	58.69	22.06	17.71	80.21	78.66	64.38	30.33	80.25	78.15	66.16	52.81	81.29	81.29	81.07	80.67	79.49	77.37	71.26	66.63
	RFA	77.68	61.67	23.32	18.97	80.03	78.16	64.14	3.60	80.03	77.95	55.90	48.17	81.51	81.35	81.39	80.40	79.71	78.71	78.34	75.36
	GAS	75.47	45.12	27.31	18.70	80.27	78.21	76.82	67.05	78.26	74.03	71.61	56.73	80.53	81.28	80.92	80.56	80.58	80.25	80.30	80.36
	HullGas	80.01	80.33	80.20	79.89	80.12	80.09	80.20	79.87	80.09	80.42	80.24	80.15	80.20	80.62	80.52	80.46	80.25	80.38	80.38	80.38
	Gain	0.00	15.52	**52.89**	**60.92**	-0.15	1.43	3.38	**12.82**	-0.16	2.27	8.63	**23.42**	-1.31	-0.73	-0.87	-0.21	-0.33	0.13	0.08	0.02
1	CWMed	79.24	52.87	18.10	13.03	76.41	73.93	34.44	16.95	75.84	74.20	53.87	42.03	79.18	79.84	79.78	78.80	75.16	72.99	65.10	56.11
	CwTM	76.65	42.06	23.58	12.98	79.17	77.58	57.85	25.86	78.17	76.18	60.23	43.96	79.81	79.42	79.20	79.19	78.04	75.92	69.05	56.77
	RFA	76.08	50.00	25.38	14.34	78.68	77.17	53.68	16.89	77.51	76.30	52.16	38.91	79.89	79.91	79.19	79.19	77.70	76.40	75.66	66.06
	GAS	74.73	31.25	16.49	12.58	78.14	77.06	73.94	56.96	76.73	71.39	67.48	50.09	79.73	80.05	79.24	78.74	78.81	78.40	78.79	77.58
	HullGas	78.40	79.05	78.60	78.55	79.01	78.68	78.94	78.11	78.49	78.99	78.39	77.88	79.15	78.70	78.65	78.16	78.63	79.10	79.23	78.34
	Gain	-0.84	**26.18**	**53.22**	**64.21**	-0.16	1.10	5.00	**21.15**	0.32	2.69	**10.91**	**27.79**	-0.74	-1.35	-1.13	-1.03	-0.18	0.70	0.44	0.76
0.1	CWMed	64.38	29.47	11.08	7.87	60.18	50.06	28.52	10.75	61.97	53.87	29.50	25.47	67.39	68.01	65.66	67.22	57.58	52.71	36.75	38.78
	CwTM	60.29	26.04	12.71	16.68	63.86	53.55	32.05	28.49	63.93	58.12	42.36	27.10	66.82	67.84	64.82	66.33	60.01	55.33	38.01	35.98
	RFA	61.19	34.11	18.72	4.30	64.93	62.47	25.98	7.20	64.46	62.64	37.96	26.92	66.97	67.54	64.97	64.94	61.28	62.18	55.71	51.46
	GAS	58.88	15.70	15.04	4.30	65.01	64.03	50.63	22.53	64.94	58.40	50.12	28.00	66.83	68.34	65.54	66.74	64.09	62.88	57.31	59.89
	HullGas	63.71	63.31	61.79	65.94	65.32	65.10	64.11	66.64	65.07	66.06	63.47	66.23	66.25	65.61	65.21	66.12	63.98	65.56	64.40	64.49
	Gain	-0.67	**29.20**	**43.07**	**49.26**	0.31	1.07	**13.48**	**38.15**	0.13	3.42	**13.35**	**38.23**	-1.14	-2.73	-0.45	-1.10	-0.11	2.68	7.09	4.60

Table D.2: Mean Accuracy (%) comparison over last 200 training steps between robust aggregators (BKT + CWMed, BKT + CwTM, BKT + RFA, BKT + GAS, HullGuard with byzantine distributions Dirac (D), Uniform (U)) on CIFAR-10 for different concentration parameter $\alpha \in \{10, 1, 0.1\}$ under several Byzantine attacks (ALIE, FOE, Mimic, MinSum, LF) and different numbers of workers (8, 14, 20, 26). The Gain row reports the accuracy improvement of HullGuard over the best competing method for each configuration. A negative gain indicates that HullGuard is being outperformed by another method.

		ALIE				FOE				Mimic				MinSum				LF			
		8	14	20	26	8	14	20	26	8	14	20	26	8	14	20	26	8	14	20	26
10	CWMed	80.01	64.81	24.43	17.48	78.77	77.08	48.94	29.59	78.37	76.74	62.04	52.41	81.03	81.13	81.04	80.20	77.09	75.72	69.98	61.98
	CwTM	77.69	58.69	22.06	17.71	80.21	78.66	64.38	30.33	80.25	78.15	66.16	52.81	81.29	81.29	81.07	80.67	79.49	77.37	71.26	66.63
	RFA	77.68	61.67	23.32	18.97	80.03	78.16	64.14	3.60	80.03	77.95	55.90	48.17	81.51	81.35	81.39	80.40	79.71	78.71	78.34	75.36
	GAS	75.47	45.12	27.31	18.70	80.27	78.21	76.82	67.05	78.26	74.03	71.61	56.73	80.53	81.28	80.92	80.56	80.58	80.25	80.30	80.36
	HullGuard + D	80.32	78.45	77.90	79.47	80.26	77.96	77.52	80.40	80.49	78.06	77.24	80.63	80.65	78.71	77.65	80.82	80.39	77.98	77.38	80.05
	Gain	0.31	**13.64**	**50.59**	**60.50**	-0.01	-0.70	0.70	**13.35**	0.24	-0.09	**5.63**	**23.90**	-0.86	-2.64	-3.74	0.15	-0.19	-2.27	-2.92	-0.31
	HullGuard + U	80.35	78.58	77.97	79.36	80.26	77.96	77.52	80.40	80.49	78.06	77.24	80.63	80.65	78.71	77.65	80.82	80.39	77.98	77.38	80.05
	Gain	0.34	**13.77**	**50.66**	**60.39**	-0.01	-0.70	0.70	**13.35**	0.24	-0.09	**5.63**	**23.90**	-0.86	-2.64	-3.74	0.15	-0.19	-2.27	-2.92	-0.31
1	CWMed	79.24	52.87	18.10	13.03	76.41	73.93	34.44	16.95	75.84	74.20	53.87	42.03	79.18	79.84	79.78	78.80	75.16	72.99	65.10	56.11
	CwTM	76.65	42.06	23.58	12.98	79.17	77.58	57.85	25.86	78.17	76.18	60.23	43.96	79.81	79.42	79.20	79.19	78.04	75.92	69.05	56.77
	RFA	76.08	50.00	25.38	14.34	78.68	77.17	53.68	16.89	77.51	76.30	52.16	38.91	79.89	79.91	79.19	79.19	77.70	76.40	75.66	66.06
	GAS	74.73	31.25	16.49	12.58	78.14	77.06	73.94	56.96	76.73	71.39	67.48	50.09	79.73	80.05	79.24	78.74	78.81	78.40	78.79	77.58
	HullGuard + D	79.04	77.03	75.52	78.05	78.79	75.36	74.34	79.02	78.91	75.61	74.16	78.54	79.47	76.52	75.11	78.45	79.16	75.53	74.68	79.03
	Gain	-0.20	**24.16**	**50.14**	**63.71**	-0.38	-2.22	0.40	**22.06**	0.74	-0.69	**6.68**	**28.45**	-0.42	-3.53	-4.67	-0.74	0.35	-2.87	-4.11	1.45
	HullGuard + U	79.15	76.25	74.94	77.61	78.79	75.36	74.34	79.02	78.91	75.61	74.16	78.54	79.47	76.52	75.11	78.45	79.16	75.53	74.68	79.03
	Gain	-0.09	**23.38**	**49.56**	**63.27**	-0.38	-2.22	0.40	**22.06**	0.74	-0.69	**6.68**	**28.45**	-0.42	-3.53	-4.67	-0.74	0.35	-2.87	-4.11	1.45
0.1	CWMed	64.38	29.47	11.08	7.87	60.18	50.06	28.52	10.75	61.97	53.87	29.50	25.47	67.39	68.01	65.66	67.22	57.58	52.71	36.75	38.78
	CwTM	60.29	26.04	12.71	16.68	63.86	53.55	32.05	28.49	63.93	58.12	42.36	27.10	66.82	67.84	64.82	66.33	60.01	55.33	38.01	35.98
	RFA	61.19	34.11	18.72	4.30	64.93	62.47	25.98	7.20	64.46	62.64	37.96	26.92	66.97	67.54	64.97	64.94	61.28	62.18	55.71	51.46
	GAS	58.88	15.70	15.04	4.30	65.01	64.03	50.63	22.53	64.94	58.40	50.12	28.00	66.83	68.34	65.54	66.74	64.09	62.88	57.31	59.89
	HullGuard + D	63.00	65.49	63.41	54.75	65.07	61.03	56.21	65.36	63.82	64.89	57.22	64.86	66.50	67.51	63.41	66.89	60.85	59.74	54.18	62.35
	Gain	-1.38	**31.38**	**44.69**	**38.07**	0.06	-3.00	**5.58**	**36.87**	-1.12	2.25	**7.10**	**36.86**	-0.89	-0.83	-2.25	-0.33	-3.24	-3.14	-3.13	2.46
	HullGuard + U	64.20	65.80	62.04	55.36	65.07	61.03	56.21	65.36	63.82	64.89	57.22	64.86	66.50	67.51	63.41	66.89	60.85	59.74	54.18	62.35
	Gain	-0.18	**31.69**	**43.32**	**38.68**	0.06	-3.00	**5.58**	**36.87**	-1.12	2.25	**7.10**	**36.86**	-0.89	-0.83	-2.25	-0.33	-3.24	-3.14	-3.13	2.46

Table D.3: Mean Accuracy (%) comparison over last 200 training steps between robust aggregators (BKT + CWMed, BKT + CwTM, BKT + RFA, BKT + GAS, HullGuard with byzantine distributions Random (R), Dirac (D), Uniform (U)) on MNIST for different concentration parameter $\alpha \in \{10, 1, 0.1\}$ under several Byzantine attacks (ALIE, FOE, Mimic, MinSum, LF) and different numbers of workers (8, 14, 20, 26). The Gain row reports the accuracy improvement of HullGuard over the best competing method for each configuration. A negative gain indicates that HullGuard is being outperformed by another method.

		ALIE				FOE				Mimic				MinSum				LF			
		8	14	20	26	8	14	20	26	8	14	20	26	8	14	20	26	8	14	20	26
	CWMed	98.74	98.53	96.98	93.72	98.69	98.32	96.17	95.35	98.71	98.50	97.60	97.10	99.01	99.04	98.79	99.03	98.41	97.91	97.40	95.09
	CwTM	98.68	98.48	96.90	91.95	98.66	98.31	96.44	94.44	98.72	98.48	97.78	96.98	98.96	99.04	98.96	99.04	98.45	98.09	97.06	95.33
10	RFA	98.75	98.54	97.19	91.57	98.72	98.56	90.49	75.45	98.76	98.61	98.04	97.45	99.02	99.04	98.98	99.08	98.86	98.65	98.77	98.69
	GAS	98.65	98.38	97.76	89.90	98.67	98.36	98.10	96.53	98.64	98.42	98.36	97.27	98.92	99.11	99.06	99.10	98.78	98.81	98.77	98.77
	HullGuard + R	98.69	98.77	98.74	98.76	98.72	98.76	98.68	98.83	98.76	98.77	98.70	98.80	98.78	98.75	98.75	98.86	98.70	98.83	98.68	98.82
	Gain	-0.06	0.23	0.98	**5.04**	0.00	0.20	0.58	2.30	0.00	0.16	0.34	1.35	-0.24	-0.36	-0.31	-0.24	-0.16	0.02	-0.09	0.05
	HullGuard + D	98.79	98.86	98.68	98.75	98.72	98.76	98.68	98.83	98.76	98.77	98.70	98.80	98.78	98.75	98.75	98.86	98.70	98.83	98.68	98.82
	Gain	0.04	0.32	0.92	**5.03**	0.00	0.20	0.58	2.30	0.00	0.16	0.34	1.35	-0.24	-0.36	-0.31	-0.24	-0.16	0.02	-0.09	0.05
	HullGuard + U	98.69	98.77	98.74	98.76	98.72	98.76	98.68	98.83	98.76	98.77	98.70	98.80	98.78	98.75	98.75	98.86	98.70	98.83	98.68	98.82
	Gain	-0.06	0.23	0.98	**5.04**	0.00	0.20	0.58	2.30	0.00	0.16	0.34	1.35	-0.24	-0.36	-0.31	-0.24	-0.16	0.02	-0.09	0.05
	CWMed	98.68	98.43	94.02	84.86	98.63	98.29	93.81	87.95	98.69	98.38	97.30	95.30	99.10	99.06	98.85	99.00	98.64	98.08	96.81	90.93
	CwTM	98.58	98.27	95.37	74.34	98.66	98.26	94.60	89.35	98.62	98.39	97.81	96.00	99.07	98.98	99.11	98.98	98.70	98.25	97.13	92.27
1	RFA	98.73	98.40	95.91	75.08	98.71	98.43	75.20	16.51	98.63	98.64	97.89	96.35	99.08	99.09	99.11	98.93	98.82	98.68	98.64	98.54
	GAS	98.65	98.09	96.54	63.32	98.60	98.07	97.30	87.58	98.51	98.17	98.32	96.54	99.00	98.96	99.03	98.37	98.82	98.50	98.77	98.78
	HullGuard + R	98.80	98.71	98.65	98.63	98.71	98.69	98.58	98.70	98.71	98.78	98.75	98.75	98.82	98.83	98.84	98.86	98.75	98.69	98.67	98.68
	Gain	0.07	0.28	2.11	**13.77**	0.00	0.26	1.28	**9.35**	0.02	0.14	0.43	2.21	-0.28	-0.26	-0.27	-0.14	-0.07	0.01	-0.10	-0.10
	HullGuard + D	98.70	98.66	98.78	98.66	98.71	98.69	98.58	98.70	98.71	98.78	98.75	98.75	98.82	98.83	98.84	98.86	98.75	98.69	98.67	98.68
	Gain	-0.03	0.23	2.24	**13.80**	0.00	0.26	1.28	**9.35**	0.02	0.14	0.43	2.21	-0.28	-0.26	-0.27	-0.14	-0.07	0.01	-0.10	-0.10
	HullGuard + U	98.80	98.71	98.65	98.63	98.71	98.69	98.58	98.70	98.71	98.78	98.75	98.75	98.82	98.83	98.84	98.86	98.75	98.69	98.67	98.68
	Gain	0.07	0.28	2.11	**13.77**	0.00	0.26	1.28	**9.35**	0.02	0.14	0.43	2.21	-0.28	-0.26	-0.27	-0.14	-0.07	0.01	-0.10	-0.10
	CWMed	97.85	92.21	44.80	9.34	96.71	85.36	54.08	27.68	97.25	89.20	62.27	39.76	99.08	99.15	98.67	97.86	97.07	89.16	53.11	45.52
	CwTM	98.00	92.60	30.47	11.37	97.46	88.38	52.56	28.99	97.43	91.69	59.70	52.91	98.95	99.07	98.95	98.75	97.43	93.46	50.55	48.89
0.1	RFA	98.57	98.05	20.95	8.76	98.69	98.12	18.99	4.66	98.62	98.43	94.89	83.34	99.08	99.02	98.98	98.01	98.74	98.59	98.40	97.08
	GAS	98.51	96.61	74.34	8.76	98.21	87.63	80.72	58.06	98.08	96.60	86.73	87.73	98.79	98.97	98.96	98.32	98.51	98.35	98.57	98.56
	HullGuard + R	98.52	98.59	98.49	97.96	98.33	98.10	98.32	98.56	98.63	98.23	98.26	98.37	98.86	98.85	98.84	98.92	98.47	98.18	98.04	98.18
	Gain	-0.05	0.54	**24.15**	**86.59**	-0.36	-0.02	**17.60**	**40.50**	0.01	-0.20	3.37	10.64	-0.22	-0.30	-0.14	0.17	-0.27	-0.41	-0.53	-0.38
	HullGuard + D	98.37	98.55	98.45	98.04	98.33	98.10	98.32	98.56	98.63	98.23	98.26	98.37	98.86	98.85	98.84	98.92	98.47	98.18	98.04	98.18
	Gain	-0.20	0.50	**24.11**	**86.67**	-0.36	-0.02	**17.60**	**40.50**	0.01	-0.20	3.37	10.64	-0.22	-0.30	-0.14	0.17	-0.27	-0.41	-0.53	-0.38
	HullGuard + U	98.52	98.59	98.49	97.96	98.33	98.10	98.32	98.56	98.63	98.23	98.26	98.37	98.86	98.85	98.84	98.92	98.47	98.18	98.04	98.18
	Gain	-0.05	0.54	**24.15**	**86.59**	-0.36	-0.02	**17.60**	**40.50**	0.01	-0.20	3.37	10.64	-0.22	-0.30	-0.14	0.17	-0.27	-0.41	-0.53	-0.38

Table D.4: Mean Accuracy (%) comparison over last 200 training steps between robust aggregators (BKT + CWMed, BKT + CwTM, BKT + RFA, BKT + GAS, HullGuard with byzantine distributions Random (R), Dirac (D), Uniform (U)) on Fashion-MNIST for different concentration parameter $\alpha \in \{10, 1, 0.1\}$ under several Byzantine attacks (ALIE, FOE, Mimic, MinSum, LF) and different numbers of workers (8, 14, 20, 26). The Gain row reports the accuracy improvement of HullGuard over the best competing method for each configuration. A negative gain indicates that HullGuard is being outperformed by another method.

		ALIE				FOE				Mimic				MinSum				LF			
		8	14	20	26	8	14	20	26	8	14	20	26	8	14	20	26	8	14	20	26
	CWMed	86.74	85.79	77.74	67.27	86.13	85.39	78.27	74.68	86.69	85.59	82.74	80.46	88.43	88.75	88.58	88.89	85.36	83.12	80.71	72.55
	CwTM	86.78	85.36	77.42	67.19	86.44	85.58	79.19	75.33	86.85	85.61	82.87	80.27	88.40	88.91	88.59	88.48	86.01	83.89	80.30	74.04
10	RFA	86.93	85.64	78.37	68.64	86.44	85.30	68.44	57.40	86.73	86.01	82.81	79.93	88.62	88.87	88.89	88.36	86.78	86.47	86.50	86.20
	GAS	86.17	83.80	80.52	68.20	85.61	84.70	83.60	76.41	85.79	84.62	83.60	80.18	88.27	88.56	88.88	87.76	86.32	86.51	87.05	87.10
	HullGuard + R	86.82	86.72	86.20	86.63	86.80	86.34	85.99	86.63	86.75	86.22	86.07	87.02	87.05	86.60	86.29	87.24	86.98	86.40	86.26	86.67
	Gain	-0.11	0.93	**5.68**	**17.99**	0.36	0.76	2.39	**10.22**	-0.10	0.21	2.47	**6.56**	-1.57	-2.31	-2.60	-1.65	0.20	-0.11	-0.79	-0.43
	HullGuard + D	86.73	86.52	86.01	86.71	86.80	86.34	85.99	86.63	86.75	86.22	86.07	87.02	87.05	86.60	86.29	87.24	86.98	86.40	86.26	86.67
	Gain	-0.20	0.73	b	**18.07**	0.36	0.76	2.39	**10.22**	-0.10	0.21	2.47	**6.56**	-1.57	-2.31	-2.60	-1.65	0.20	-0.11	-0.79	-0.43
	HullGuard + U	86.82	86.72	86.20	86.63	86.80	86.34	85.99	86.63	86.75	86.22	86.07	87.02	87.05	86.60	86.29	87.24	86.98	86.40	86.26	86.67
	Gain	-0.11	0.93	**5.68**	**17.99**	0.36	0.76	2.39	10.22	-0.10	0.21	2.47	6.56	-1.57	-2.31	-2.60	-1.65	0.20	-0.11	-0.79	-0.43
	CWMed	85.48	82.39	64.55	52.87	84.09	82.38	70.56	57.29	85.15	85.01	79.91	76.29	88.76	88.73	87.84	87.30	83.42	82.39	70.14	56.97
	CwTM	84.87	81.13	70.45	52.58	85.02	82.58	71.28	60.34	85.79	85.08	80.72	77.87	89.05	88.80	86.87	87.63	84.60	83.29	72.42	58.29
1	RFA	86.31	83.64	71.44	47.99	85.95	84.12	55.02	32.98	86.33	85.83	80.58	77.12	89.15	88.79	88.69	87.82	86.24	86.04	86.17	83.02
	GAS	85.22	81.07	73.19	25.32	84.14	82.86	82.24	66.62	85.29	83.64	81.77	78.06	87.75	88.24	87.31	87.76	85.29	85.70	86.93	6.70
	HullGuard + R	85.82	85.10	85.49	84.61	85.86	85.07	84.69	85.34	85.83	84.90	84.50	85.11	86.33	86.23	85.60	86.63	85.75	84.50	84.94	85.16
	Gain	-0.49	1.46	**12.30**	**31.74**	-0.09	0.95	2.45	**18.72**	-0.50	-0.93	2.73	**7.05**	-2.82	-2.57	-3.09	-1.19	-0.49	-1.54	-1.99	2.14
	HullGuard + D	85.72	84.85	85.27	85.10	85.86	85.07	84.69	85.34	85.83	84.90	84.50	85.11	86.33	86.23	85.60	86.63	85.75	84.50	84.94	85.16
	Gain	-0.59	1.21	**12.08**	**32.23**	-0.09	0.95	2.45	**18.72**	-0.50	-0.93	2.73	**7.05**	-2.82	-2.57	-3.09	-1.19	-0.49	-1.54	-1.99	2.14
	HullGuard + U	85.82	85.10	85.49	84.61	85.86	85.07	84.69	85.34	85.83	84.90	84.50	85.11	86.33	86.23	85.60	86.63	85.75	84.50	84.94	85.16
	Gain	-0.49	1.46	**12.30**	31.74	-0.09	0.95	2.45	18.72	-0.50	-0.93	2.73	7.05	-2.82	-2.57	-3.09	-1.19	-0.49	-1.54	-1.99	2.14
	CWMed	79.48	67.62	33.43	11.37	74.80	70.70	41.86	17.54	78.08	70.93	52.70	35.25	89.69	90.11	88.87	18.01	79.79	70.67	56.59	47.07
	CwTM	80.11	64.21	36.92	11.37	76.89	71.63	30.20	19.28	79.31	69.49	48.51	39.55	89.82	90.02	89.70	88.90	78.50	71.96	58.08	54.44
0.1	RFA	84.89	79.52	28.49	8.79	85.37	83.05	16.35	9.16	85.75	84.38	72.93	56.46	90.23	90.36	88.55	88.31	86.18	86.73	85.04	77.86
	GAS	81.48	72.51	15.27	8.72	82.00	79.94	73.16	45.16	80.62	77.05	74.16	60.98	87.97	88.01	87.33	18.01	83.60	83.62	86.51	85.85
	HullGuard + R	83.12	84.32	83.93	81.97	83.70	82.06	83.25	85.06	83.48	83.37	83.76	83.20	87.16	87.34	86.71	89.95	82.54	80.66	82.65	82.72
	Gain	-1.77	4.80	**47.01**	**70.60**	-1.67	-0.99	**10.09**	**39.90**	-2.27	-1.01	**9.60**	**22.22**	-3.07	-3.02	-2.99	1.05	-3.64	-6.07	-3.86	-3.13
	HullGuard + D	82.09	84.07	84.19	81.55	83.70	82.06	83.25	85.06	83.48	83.37	83.76	83.20	87.16	87.34	86.71	89.95	82.54	80.66	82.65	82.72
	Gain	-2.80	4.55	**47.27**	**70.18**	-1.67	-0.99	**10.09**	**39.90**	-2.27	-1.01	**9.60**	**22.22**	-3.07	-3.02	-2.99	1.05	-3.64	-6.07	-3.86	-3.13
	HullGuard + U	83.12	84.32	83.93	81.97	83.70	82.06	83.25	85.06	83.48	83.37	83.76	83.20	87.16	87.34	86.71	89.95	82.54	80.66	82.65	82.72
	Gain	-1.77	4.80	**47.01**	**70.60**	-1.67	-0.99	**10.09**	**39.90**	-2.27	-1.01	**9.60**	**22.22**	-3.07	-3.02	-2.99	1.05	-3.64	-6.07	-3.86	-3.13

Table D.5: Mean Accuracy (%) comparison over last 200 training steps between robust aggregators (BKT + CWMed, BKT + CwTM, BKT + RFA, BKT + GAS, HullGuard with byzantine distributions Random (R), Dirac (D), Uniform (U)) on EuroSAT for different concentration parameter $\alpha \in \{10, 1, 0.1\}$ under several Byzantine attacks (ALIE, FOE, Mimic, MinSum, LF) and different numbers of workers (8, 14, 20, 26). The Gain row reports the accuracy improvement of HullGuard over the best competing method for each configuration. A negative gain indicates that HullGuard is being outperformed by another method.

α	Method	ALIE 8	ALIE 14	ALIE 20	ALIE 26	FOE 8	FOE 14	FOE 20	FOE 26	Mimic 8	Mimic 14	Mimic 20	Mimic 26	MinSum 8	MinSum 14	MinSum 20	MinSum 26	LF 8	LF 14	LF 20	LF 26
	CWMed	87.68	81.64	42.79	27.40	88.05	87.59	69.50	59.99	83.13	81.21	70.46	35.71	88.58	84.83	80.23	61.23	85.99	83.89	82.49	75.87
	CwTM	87.42	80.82	60.22	25.67	88.57	87.53	73.33	59.07	86.19	83.41	72.19	37.02	85.99	85.63	80.03	44.44	87.19	86.62	82.97	76.17
10	RFA	88.13	82.41	64.61	29.55	88.79	87.17	71.70	23.65	86.82	83.60	70.80	36.58	86.27	84.50	81.13	30.98	88.18	88.08	88.33	87.96
	GAS	84.84	75.70	67.08	27.30	84.58	84.07	85.51	77.11	83.45	75.53	71.70	31.18	85.05	78.01	75.59	10.94	86.26	82.09	87.65	85.04
	HullGuard + R	87.04	86.09	83.34	86.04	87.97	85.41	83.36	87.06	87.88	85.05	83.42	86.48	87.17	85.85	84.57	84.13	87.39	85.88	83.93	87.16
	Gain	-1.09	3.68	**16.26**	**56.49**	-0.82	-2.18	-2.15	**9.95**	1.06	1.45	**11.23**	**49.46**	-1.41	0.22	3.44	**22.90**	-0.79	-2.20	-4.40	-0.80
	HullGuard + D	87.04	86.09	83.34	86.04	87.97	85.41	83.36	87.06	87.88	85.05	83.42	86.48	87.17	85.85	84.57	84.13	87.39	85.88	83.93	87.16
	Gain	-1.09	3.68	**16.26**	**56.49**	-0.82	-2.18	-2.15	**9.95**	1.06	1.45	**11.23**	**49.46**	-1.41	0.22	3.44	**22.90**	-0.79	-2.20	-4.40	-0.80
	HullGuard + U	87.04	86.09	83.34	86.04	87.97	85.41	83.36	87.06	87.88	85.05	83.42	86.48	87.17	85.85	84.57	84.13	87.39	85.88	83.93	87.16
	Gain	-1.09	3.68	**16.26**	**56.49**	-0.82	-2.18	-2.15	**9.95**	1.06	1.45	**11.23**	**49.46**	-1.41	0.22	3.44	**22.90**	-0.79	-2.20	-4.40	-0.80
	CWMed	84.83	75.57	25.14	10.24	82.61	78.37	57.08	28.37	80.30	77.71	57.58	32.09	83.66	80.22	11.96	11.24	81.58	79.19	73.39	62.92
	CwTM	84.28	71.68	33.18	18.98	84.89	81.08	69.88	42.31	82.38	80.55	62.38	43.45	84.29	82.71	36.45	11.24	81.94	80.28	74.91	65.62
1	RFA	85.53	71.21	44.90	17.80	86.09	83.79	56.93	33.88	83.00	81.12	64.11	29.77	82.98	79.84	27.54	11.24	84.19	84.60	81.19	76.53
	GAS	79.39	62.60	53.47	9.96	78.52	78.44	84.07	70.36	77.58	70.92	69.62	33.45	78.64	21.82	11.96	11.36	78.47	76.77	83.52	37.06
	HullGuard + R	84.39	79.85	82.49	81.52	82.21	79.50	79.21	83.92	82.95	79.46	79.44	83.34	85.15	81.76	80.16	82.46	82.18	79.54	76.58	81.06
	Gain	-1.14	4.28	**29.02**	**62.54**	-3.88	-4.29	-4.86	**13.56**	-0.05	-1.66	**9.82**	**39.89**	0.86	-0.95	**43.71**	**71.10**	-2.01	-5.06	-6.94	4.53
	HullGuard + D	84.39	79.85	82.49	81.52	82.21	79.50	79.21	83.92	82.95	79.46	79.44	83.34	85.15	81.76	80.16	82.46	82.18	79.54	76.58	81.06
	Gain	-1.14	4.28	**29.02**	**62.54**	-3.88	-4.29	-4.86	**13.56**	-0.05	-1.66	**9.82**	**39.89**	0.86	-0.95	**43.71**	**71.10**	-2.01	-5.06	-6.94	4.53
	HullGuard + U	84.39	79.85	82.49	81.52	82.21	79.50	79.21	83.92	82.95	79.46	79.44	83.34	85.15	81.76	80.16	82.46	82.18	79.54	76.58	81.06
	Gain	-1.14	4.28	**29.02**	**62.54**	-3.88	-4.29	-4.86	**13.56**	-0.05	-1.66	**9.82**	**39.89**	0.86	-0.95	**43.71**	**71.10**	-2.01	-5.06	-6.94	4.53
	CWMed	60.65	28.80	10.68	15.33	68.52	58.76	31.89	22.90	59.11	58.12	25.41	14.57	72.13	73.24	64.59	15.33	66.92	46.91	35.05	36.38
	CwTM	62.67	26.77	14.79	13.33	60.53	58.62	41.35	20.27	61.55	59.56	27.31	22.01	68.69	66.10	41.50	15.01	61.38	44.37	29.38	32.02
0.1	RFA	60.04	26.95	10.26	4.81	70.33	70.70	40.68	11.58	73.16	73.70	31.85	17.30	74.40	73.00	16.45	15.33	67.76	47.12	46.06	16.26
	GAS	58.20	22.28	16.02	2.21	59.59	64.45	48.87	30.20	62.01	43.06	25.78	14.19	62.44	14.33	16.45	15.33	64.35	62.24	63.43	31.65
	HullGuard + R	67.58	72.64	71.22	52.91	68.54	70.12	64.82	66.26	65.66	67.32	60.94	67.14	69.05	70.49	72.43	70.54	63.20	64.57	56.78	70.04
	Gain	4.91	**43.84**	**55.20**	**37.58**	-1.79	-0.58	**15.95**	**36.06**	-7.50	-6.38	**29.09**	**45.13**	-5.35	-2.75	**7.84**	**55.21**	-4.56	2.33	-6.65	**33.66**
	HullGuard + D	67.58	72.64	71.22	52.91	68.54	70.12	64.82	66.26	65.66	67.32	60.94	67.14	69.05	70.49	72.43	70.54	63.20	64.57	56.78	70.04
	Gain	4.91	**43.84**	**55.20**	**37.58**	-1.79	-0.58	**15.95**	**36.06**	-7.50	-6.38	**29.09**	**45.13**	-5.35	-2.75	**7.84**	**55.21**	-4.56	2.33	-6.65	**33.66**
	HullGuard + U	67.58	72.64	71.22	52.91	68.54	70.12	64.82	66.26	65.66	67.32	60.94	67.14	69.05	70.49	72.43	70.54	63.20	64.57	56.78	70.04
	Gain	4.91	**43.84**	**55.20**	**37.58**	-1.79	-0.58	**15.95**	**36.06**	-7.50	-6.38	**29.09**	**45.13**	-5.35	-2.75	**7.84**	**55.21**	-4.56	2.33	-6.65	**33.66**

Table D.6: Mean training accuracy comparison over the last 200 steps for aggregators NNM + CWMed, NNM + CwTM, NNM + RFA, NNM + GAS and HullGuard, for concentration parameters 10, 1, 0.1, using dataset CIFAR-10, against ALIE attack with byzantine number 8, 14, 20, 29. Settings for HullGuard are the ones described in section 4.1.

				ALIE			
				8	14	22	26
CIFAR10	10	CWMed	NNM	77.78	59.87	19.66	80.36
		CwTM	NNM	77.74	60.29	26.38	80.09
		RFA	NNM	78.28	59.05	25.47	73.90
		GAS	NNM	77.72	59.76	23.99	80.71
		HullGuard		80.35	78.58	77.97	79.36
	1	CWMed	NNM	76.78	50.82	15.56	28.05
		CwTM	NNM	76.14	52.13	20.11	21.96
		RFA	NNM	76.64	51.68	18.20	18.92
		GAS	NNM	76.71	50.67	19.30	76.89
		HullGuard		79.15	76.25	74.94	77.61
	0.1	CWMed	NNM	61.31	31.75	9.44	25.26
		CwTM	NNM	61.13	30.15	12.05	27.50
		RFA	NNM	61.64	31.89	11.90	24.45
		GAS	NNM	60.89	30.97	13.27	47.23
		HullGuard		64.20	65.80	62.04	55.36

Bibliography

[1] Gagan Aggarwal, Nina Mishra, and Benny Pinkas. "Secure Computation of the k-th Ranked Element". In: *EUROCRYPT*. 2004.

[2] Umang Aggarwal, Adrian Popescu, and Céline Hudelot. "Minority Class Oriented Active Learning for Imbalanced Datasets". In: *2020 25th International Conference on Pattern Recognition (ICPR)*. IEEE, 2021, pp. 9920–9927.

[3] Zeyuan Allen-Zhu et al. "Byzantine-Resilient Non-Convex Stochastic Gradient Descent". 2020. arXiv: `2012.14368`.

[4] Youssef Allouah et al. "Fixing by Mixing: A Recipe for Optimal Byzantine Ml under Heterogeneity". In: *International Conference on Artificial Intelligence and Statistics*. PMLR, 2023, pp. 1232–1300.

[5] Fahmy Amin and M Mahmoud. "Confusion matrix in binary classification problems: A step-by-step tutorial". In: *Journal of Engineering Research* 6.5 (2022).

[6] Jayanth R Banavar, Amos Maritan, and Igor Volkov. "Applications of the Principle of Maximum Entropy: From Physics to Ecology". In: *Journal of Physics: Condensed Matter* 22.6 (2010), p. 063101.

[7] Wenxuan Bao, Jun Wu, and Jingrui He. "BOBA: Byzantine-robust Federated Learning with Label Skewness". In: *International Conference on Artificial Intelligence and Statistics*. PMLR, 2024, pp. 892–900.

[8] Gilad Baruch, Moran Baruch, and Yoav Goldberg. "A Little Is Enough: Circumventing Defenses for Distributed Learning". In: *Advances in Neural Information Processing Systems* 32 (2019).

[9] Sara Beery et al. "Synthetic Examples Improve Generalization for Rare Classes". In: *Proceedings of the Ieee/Cvf Winter Conference on Applications of Computer Vision*. 2020, pp. 863–873.

[10] Elisabetta Binaghi et al. "A Fuzzy Set-Based Accuracy Assessment of Soft Classification". In: *Pattern recognition letters* 20.9 (1999), pp. 935–948.

[11] Peva Blanchard et al. "Machine Learning with Adversaries: Byzantine Tolerant Gradient Descent". In: *Advances in neural information processing systems* 30 (2017).

[12] Stephen P Boyd and Lieven Vandenberghe. *Convex Optimization*. Cambridge university press, 2004.

[13] Mateusz Buda, Atsuto Maki, and Maciej A Mazurowski. "A Systematic Study of the Class Imbalance Problem in Convolutional Neural Networks". In: *Neural networks* 106 (2018), pp. 249–259.

[14] Xiaoyu Cao et al. "Fltrust: Byzantine-robust Federated Learning via Trust Bootstrapping". 2020. arXiv: `2012.13995`.

[15] Ekram Chamseddine et al. "Handling Class Imbalance in COVID-19 Chest X-ray Images Classification: Using SMOTE and Weighted Loss". In: *Applied Soft Computing* 129 (2022), p. 109588.

[16] Zelei Cheng et al. "Soft-label integration for robust toxicity classification". In: *Advances in Neural Information Processing Systems* 37 (2024), pp. 94776–94807.

[17] Tarin Clanuwat et al. *Deep Learning for Classical Japanese Literature*. 2018. arXiv: 1812.01718 [cs.CV].

[18] Adam Coates, Andrew Ng, and Honglak Lee. "An Analysis of Single-Layer Networks in Unsupervised Feature Learning". In: *Proceedings of the Fourteenth International Conference on Artificial Intelligence and Statistics*. JMLR Workshop and Conference Proceedings, 2011, pp. 215–223.

[19] Katherine M. Collins, Umang Bhatt, and Adrian Weller. "Eliciting and Learning with Soft Labels from Every Annotator". In: *Proceedings of the AAAI Conference on Human Computation and Crowdsourcing* 10.1 (Oct. 2022), pp. 40–52. ISSN: 2769-1349, 2769-1330. DOI: 10.1609/hcomp.v10i1.21986. URL: https://ojs.aaai.org/index.php/HCOMP/article/view/21986 (visited on 07/13/2023).

[20] Russell G Congalton. "A Review of Assessing the Accuracy of Classifications of Remotely Sensed Data". In: *Remote sensing of environment* 37.1 (1991), pp. 35–46.

[21] Russell G Congalton. "Accuracy Assessment and Validation of Remotely Sensed and Other Spatial Information". In: *International journal of wildland fire* 10.4 (2001), pp. 321–328.

[22] Nelson Filipe Costa and Leila Kosseim. "Exploring soft-label training for implicit discourse relation recognition". In: *Proceedings of the 5th Workshop on Computational Approaches to Discourse (CODI 2024)*. 2024, pp. 120–126.

[23] Solemane Coulibaly et al. "Deep Convolution Neural Network Sharing for the Multi-Label Images Classification". In: *Machine learning with applications* 10 (2022), p. 100422.

[24] Deepesh Data and Suhas Diggavi. "Byzantine-Resilient High-Dimensional SGD with Local Iterations on Heterogeneous Data". In: *International Conference on Machine Learning*. PMLR, 2021, pp. 2478–2488.

[25] S Deepak and PM Ameer. "Brain Tumor Categorization from Imbalanced MRI Dataset Using Weighted Loss and Deep Feature Fusion". In: *Neurocomputing* 520 (2023), pp. 94–102.

[26] W Edwards Deming and Frederick F Stephan. "On a Least Squares Adjustment of a Sampled Frequency Table When the Expected Marginal Totals Are Known". In: *The Annals of Mathematical Statistics* 11.4 (1940), pp. 427–444.

[27] Petros Drineas et al. "Faster least squares approximation". In: *Numerische mathematik* 117.2 (2011), pp. 219–249.

[28] El Mahdi El Mhamdi, Rachid Guerraoui, and Sébastien Louis Alexandre Rouault. "Distributed Momentum for Byzantine-Resilient Stochastic Gradient Descent". In: *9th International Conference on Learning Representations (ICLR)*. 2021.

[29] Dominik Maria Endres and Johannes E Schindelin. "A new metric for probability distributions". In: *IEEE Transactions on Information theory* 49.7 (2003), pp. 1858–1860.

[30] Johan Erbani et al. "Benchmarking Confusion Matrices Beyond Single-Label Classification". 2026. arXiv: 2506.09824.

[31] Johan Erbani et al. "Confusion Matrices: A Unified Theory". In: *IEEE Access* (2024).

[32] Johan Erbani et al. "Inside the Convex Hull of Gradients: Byzantine-Robust Federated Learning under Label Skew". 2026. arXiv: 2506.09824.

[33] Johan Erbani et al. "NorthStar: Local Sampling for Byzantine-Robust Federated Learning under Label Skew". 2026. arXiv: 2506.09824.

[34] Johan Erbani et al. "On the Normalization of Confusion Matrices: Methods and Geometric Interpretations". In: *The 29th International Conference on Artificial Intelligence and Statistics*. 2026. URL: https://openreview.net/forum?id=DH3fFrJDs4.

[35] Johan Erbani et al. "Weighted Loss Methods for Robust Federated Learning under Data Heterogeneity". 2025. arXiv: 2506.09824.

[36] Philipp M Faller et al. "Self-compatibility: Evaluating causal discovery without ground truth". In: *International Conference on Artificial Intelligence and Statistics*. PMLR. 2024, pp. 4132–4140.

[37] Minghong Fang et al. "Do We Really Need to Design New Byzantine-robust Aggregation Rules?" 2025. arXiv: 2501.17381.

[38] Minghong Fang et al. "Local Model Poisoning Attacks to {Byzantine-Robust} Federated Learning". In: *29th USENIX Security Symposium (USENIX Security 20)*. 2020, pp. 1605–1622.

[39] Sadegh Farhadkhani et al. "Byzantine Machine Learning Made Easy by Resilient Averaging of Momentums". In: *International Conference on Machine Learning*. PMLR, 2022, pp. 6246–6283.

[40] Muhammad Hasan Ferdous, Emam Hossain, and Md Osman Gani. "Timegraph: Synthetic benchmark datasets for robust time-series causal discovery". In: *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining V. 2*. 2025, pp. 5425–5435.

[41] K Ruwani M Fernando and Chris P Tsokos. "Dynamically Weighted Balanced Loss: Class Imbalanced Learning and Confidence Calibration of Deep Neural Networks". In: *IEEE Transactions on Neural Networks and Learning Systems* 33.7 (2021), pp. 2940–2951.

[42] Kushankur Ghosh et al. "The Class Imbalance Problem in Deep Learning". In: *Machine Learning* 113.7 (2024), pp. 4845–4901.

[43] Jochen Görtler et al. "Neo: Generalizing Confusion Matrix Visualization to Hierarchical and Multi-Output Labels". In: *CHI Conference on Human Factors in Computing Systems*. CHI '22. ACM, Apr. 2022. DOI: 10.1145/3491102.3501823. URL: http://dx.doi.org/10.1145/3491102.3501823.

[44] Rachid Guerraoui, Nirupam Gupta, and Rafael Pinot. "Byzantine Machine Learning: A Primer". In: *ACM Computing Surveys* 56.7 (2024), pp. 1–39.

[45] Rachid Guerraoui, Nirupam Gupta, and Rafael Pinot. "Fundamentals of Robust Machine Learning". In: *Robust Machine Learning: Distributed Methods for Safe AI*. Springer, 2024, pp. 55–92.

[46] Rachid Guerraoui, Sébastien Rouault, et al. "The Hidden Vulnerability of Distributed Learning in Byzantium". In: *International Conference on Machine Learning*. PMLR, 2018, pp. 3521–3530.

[47] Silviu Guiasu and Abe Shenitzer. "The Principle of Maximum Entropy". In: *The mathematical intelligencer* 7 (1985), pp. 42–48.

[48] Nirupam Gupta, Shuo Liu, and Nitin Vaidya. "Byzantine Fault-Tolerant Distributed Machine Learning with Norm-Based Comparative Gradient Elimination". In: *2021 51st Annual IEEE/IFIP International Conference on Dependable Systems and Networks Workshops (DSN-W)*. IEEE, 2021, pp. 175–181.

[49] Perry J Hardin and J Matthew Shumway. "Statistical Significance and Normalized Confusion Matrices". In: *Photogrammetric engineering and remote sensing* 63.6 (1997), pp. 735–739.

[50] Patrick Helber et al. "Eurosat: A Novel Dataset and Deep Learning Benchmark for Land Use and Land Cover Classification". In: *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing* (2019).

[51] Mohammadreza Heydarian, Thomas E Doyle, and Reza Samavi. "MLCM: Multi-label Confusion Matrix". In: *IEEE Access* 10 (2022), pp. 19083–19095.

[52] Tzu-Ming Harry Hsu, Hang Qi, and Matthew Brown. “Measuring the Effects of Non-Identical Data Distribution for Federated Visual Classification”. 2019. arXiv: 1909.06335.

[53] Martin Idel. “A Review of Matrix Scaling and Sinkhorn’s Normal Form for Matrices and Positive Maps”. 2016. arXiv: 1609.06349.

[54] C Terrance Ireland and Solomon Kullback. “Contingency Tables with given Marginals”. In: *Biometrika* 55.1 (1968), pp. 179–188.

[55] Justin Johnson et al. “Clevr: A diagnostic dataset for compositional language and elementary visual reasoning”. In: *Proceedings of the IEEE conference on computer vision and pattern recognition*. 2017, pp. 2901–2910.

[56] Sai Praneeth Karimireddy, Lie He, and Martin Jaggi. “Byzantine-Robust Learning on Heterogeneous Datasets via Bucketing”. 2020. arXiv: 2006.09365.

[57] Sai Praneeth Karimireddy, Lie He, and Martin Jaggi. “Learning from History for Byzantine Robust Optimization”. In: *International Conference on Machine Learning*. PMLR, 2021, pp. 5311–5319.

[58] Ryosuke Kawamura et al. “MIDAS: Mixing Ambiguous Data with Soft Labels for Dynamic Facial Expression Recognition”. In: *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*. 2024, pp. 6552–6562.

[59] Hyunjik Kim and Andriy Mnih. “Disentangling by Factorising”. In: *International Conference on Machine Learning*. 2018.

[60] Alex Krizhevsky, Vinod Nair, Geoffrey Hinton, et al. “The CIFAR-10 Dataset”. In: *online: http://www. cs. toronto. edu/kriz/cifar. html* 55.5 (2014), p. 2.

[61] Alex Krizhevsky, Ilya Sutskever, and Geoffrey E Hinton. “Imagenet Classification with Deep Convolutional Neural Networks”. In: *Advances in neural information processing systems* 25 (2012).

[62] Damir Krstinić, Ljiljana Šerić, and Ivan Slapničar. “Comments on” MLCM: Multi-Label Confusion Matrix””. In: *IEEE Access* (2023).

[63] Damir Krstinić et al. “Multi-Label Classifier Performance Evaluation with Confusion Matrix”. In: *Computer Science & Information Technology* 1 (2020).

[64] Damir Krstinić et al. “Multi-Label Confusion Tensor”. In: *IEEE access* (2024).

[65] Sven Kurras. “Symmetric Iterative Proportional Fitting”. In: *Artificial Intelligence and Statistics*. PMLR, 2015, pp. 526–534.

[66] Jack Lanchantin et al. “General Multi-Label Image Classification with Transformers”. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2021, pp. 16478–16488.

[67] AJ Larner. *The 2x2 matrix: contingency, confusion and the metrics of binary classification*. Springer Nature, 2024.

[68] Andrew R Lawrence et al. “Data generating process to evaluate causal discovery techniques for time series data”. In: *arXiv preprint arXiv:2104.08043* (2021).

[69] Yann LeCun, Corinna Cortes, and CJ Burges. “MNIST handwritten digit database”. In: *ATT Labs [Online]. Available: http://yann.lecun.com/exdb/mnist* 2 (2010).

[70] Yann LeCun et al. “Gradient-Based Learning Applied to Document Recognition”. In: *Proceedings of the IEEE* 86.11 (1998), pp. 2278–2324.

[71] Joffrey L Leevy et al. “A Survey on Addressing High-Class Imbalance in Big Data”. In: *Journal of Big Data* 5.1 (2018), pp. 1–30.

[72] Guang Li et al. “Compressed Gastric Image Generation Based on Soft-Label Dataset Distillation for Medical Data Sharing”. In: *Computer Methods and Programs in Biomedicine* 227 (2022), p. 107189.

[73] Huiting Liu et al. “Multi-Label Text Classification via Joint Learning from Label Embedding and Label Correlation”. In: *Neurocomputing* 460 (2021), pp. 385–398.

[74] Yang Liu et al. "Synthetic benchmarks for scientific research in explainable machine learning". In: *arXiv preprint arXiv:2106.12543* (2021).

[75] Yuchen Liu et al. "Byzantine-Robust Learning on Heterogeneous Data via Gradient Splitting". In: *International Conference on Machine Learning*. PMLR, 2023, pp. 21404–21425.

[76] Wanqiu Long, N Siddharth, and Bonnie Webber. "Multi-label classification for implicit discourse relation recognition". In: *Findings of the Association for Computational Linguistics: ACL 2024*. 2024, pp. 8437–8451.

[77] Tohar Lukov et al. "Teaching with Soft Label Smoothing for Mitigating Noisy Labels in Facial Expressions". In: *European Conference on Computer Vision*. Springer, 2022, pp. 648–665.

[78] Anqi Mao, Mehryar Mohri, and Yutao Zhong. "Multi-label learning with stronger consistency guarantees". In: *Advances in neural information processing systems* 37 (2024), pp. 2378–2406.

[79] Loic Matthey et al. *dSprites: Disentanglement testing Sprites dataset*. 2017.

[80] Hiren Mewada et al. "Quantum convolutional neural network for bone fracture classification from X-Ray images". In: *Procedia Computer Science* 256 (2025), pp. 1143–1150.

[81] Miquel Miró-Nicolau, Antoni Jaume-i-Capó, and Gabriel Moyà-Alcover. "Assessing fidelity in xai post-hoc techniques: A comparative study with ground truth explanations datasets". In: *Artificial Intelligence* 335 (2024), p. 104179.

[82] Petru Mironescu. *Mesure, Intégration, Eléments d'Analyse Fonctionnelle*. 2023.

[83] Joris M Mooij et al. "Distinguishing cause from effect using observational data: methods and benchmarks". In: *Journal of Machine Learning Research* 17.32 (2016), pp. 1–102.

[84] Srikanth Nagineni and Rajesh M Hegde. "On Line Client-Wise Cohort Set Selection for Speaker Verification Using Iterative Normalization of Confusion Matrices". In: *2010 18th European Signal Processing Conference*. IEEE, 2010, pp. 576–580.

[85] Nathan Nowakowski et al. "GrootWatch at EXIST 2025: Automatic Sexism Detection on Social Networks-Classification of Tweets and Memes". In: *Lecture Notes in Computer Science* (2025).

[86] Sandeep Kumar Panda and Madhukar Dwivedi. "Minimizing food wastage using machine learning: a novel approach". In: *Smart Intelligent Computing and Applications: Proceedings of the Third International Conference on Smart Computing and Informatics, Volume 1*. Springer. 2019, pp. 465–473.

[87] Ju-Won Park et al. "I/O-signature-based feature analysis and classification of high-performance computing applications". In: *Cluster Computing* 27.3 (2024), pp. 3219–3231.

[88] Ana Maria Peco Chacon and Fausto Pedro García Márquez. "Support vector machine and K-fold Cross-validation to detect false alarms in wind turbines". In: *Sustainability: Cases and Studies in Using Operations Research and Management Science Methods*. Springer, 2023, pp. 81–97.

[89] F. Pedregosa et al. "Scikit-Learn: Machine Learning in Python". In: *Journal of Machine Learning Research* 12 (2011), pp. 2825–2830.

[90] Jie Peng, Weiyu Li, and Qing Ling. "Mean Aggregator Is More Robust than Robust Aggregators under Label Poisoning Attacks". In: *Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence*. 2024, pp. 4797–4805.

[91] Krishna Pillutla, Sham M Kakade, and Zaid Harchaoui. "Robust Aggregation for Federated Learning". In: *IEEE Transactions on Signal Processing* 70 (2022), pp. 1142–1154.

[92] Laura Plaza et al. “Overview of exist 2025: Learning with disagreement for sexism identification and characterization in tweets, memes, and tiktok videos”. In: *International Conference of the Cross-Language Evaluation Forum for European Languages*. Springer. 2025, pp. 266–289.

[93] Robert Gilmore Pontius Jr and Mang Lung Cheuk. “A Generalized Cross-Tabulation Matrix to Compare Soft-Classified Maps at Multiple Resolutions”. In: *International Journal of Geographical Information Science* 20.1 (2006), pp. 1–30.

[94] Jayanth Regatti, Hao Chen, and Abhishek Gupta. “Byzantine Resilience with Reputation Scores”. In: *2022 58th Annual Allerton Conference on Communication, Control, and Computing (Allerton)*. IEEE, 2022, pp. 1–8.

[95] Claude Sammut and Geoffrey I Webb. *Encyclopedia of Machine Learning*. Springer Science & Business Media, 2011.

[96] Aaron Sander et al. “Large-scale stochastic simulation of open quantum systems”. In: *Nature Communications* (2025).

[97] Robert I Saye. “High-Order Quadrature Methods for Implicitly Defined Surfaces and Volumes in Hyperrectangles”. In: *SIAM Journal on Scientific Computing* 37.2 (2015), A993–A1019.

[98] David W Scott. *Multivariate Density Estimation: Theory, Practice, and Visualization*. John Wiley & Sons, 2015.

[99] Virat Shejwalkar and Amir Houmansadr. “Manipulating the byzantine: Optimizing model poisoning attacks and defenses for federated learning”. In: *Ndss*. 2021.

[100] José Luis Silván-Cárdenas and Lizhu Wang. “Sub-Pixel Confusion–Uncertainty Matrix for Assessing Soft Classifications”. In: *Remote Sensing of Environment* 112.3 (2008), pp. 1081–1095.

[101] Terence Tao. *An introduction to measure theory*. Vol. 126. American Mathematical Soc., 2011.

[102] Adane Nega Tarekegn, Mohib Ullah, and Faouzi Alaya Cheikh. “Deep learning for multi-label learning: A comprehensive survey”. In: *arXiv preprint arXiv:2401.16549* (2024).

[103] Mohammed Temraz and Mark T Keane. “Solving the Class Imbalance Problem Using a Counterfactual Method for Data Augmentation”. In: *Machine Learning with Applications* 9 (2022), p. 100375.

[104] Nitin Thaper et al. “Dynamic Multidimensional Histograms”. In: *Proceedings of the 2002 ACM SIGMOD International Conference on Management of Data*. 2002, pp. 428–439.

[105] Jilun Tian et al. “A Novel Data Augmentation Approach to Fault Diagnosis with Class-Imbalance Problem”. In: *Reliability Engineering & System Safety* 243 (2024), p. 109832.

[106] Surya T Tokdar and Robert E Kass. “Importance Sampling: A Review”. In: *Wiley Interdisciplinary Reviews: Computational Statistics* 2.1 (2010), pp. 54–60.

[107] Alexandra N Uma et al. “Learning from Disagreement: A Survey”. In: *Journal of Artificial Intelligence Research* 72 (2021), pp. 1385–1470.

[108] Cédric Villani. *Topics in Optimal Transportation*. Vol. 58. American Mathematical Soc., 2021.

[109] Sjoerd de Vries and Dirk Thierens. “Learning with confidence: training better classifiers from soft labels”. In: *Machine Learning* 114.11 (2025), p. 238.

[110] Ching-Wei Wang et al. “A Soft Label Deep Learning to Assist Breast Cancer Target Therapy and Thyroid Cancer Diagnosis”. In: *Cancers* 14.21 (2022), p. 5312.

[111] Daokang Wang et al. "Soft-label for multi-domain fake news detection". In: *IEEe Access* 11 (2023), pp. 98596–98606.

[112] Na Wu et al. "Deep Convolution Neural Network with Weighted Loss to Detect Rice Seeds Vigor Based on Hyperspectral Imaging under the Sample-Imbalanced Condition". In: *Computers and Electronics in Agriculture* 196 (2022), p. 106850.

[113] Han Xiao, Kashif Rasul, and Roland Vollgraf. "Fashion-MNIST: a Novel Image Dataset for Benchmarking Machine Learning Algorithms". In: *CoRR* abs/1708.07747 (2017). arXiv: 1708.07747. URL: http://arxiv.org/abs/1708.07747.

[114] Han Xiao, Kashif Rasul, and Roland Vollgraf. "Fashion-Mnist: A Novel Image Dataset for Benchmarking Machine Learning Algorithms". 2017. arXiv: 1708.07747.

[115] Cong Xie, Oluwasanmi Koyejo, and Indranil Gupta. "Fall of Empires: Breaking Byzantine-Tolerant Sgd by Inner Product Manipulation". In: *Uncertainty in Artificial Intelligence*. PMLR, 2020, pp. 261–270.

[116] Cong Xie, Sanmi Koyejo, and Indranil Gupta. "Zeno: Distributed Stochastic Gradient Descent with Suspicion-Based Fault-Tolerance". In: *International Conference on Machine Learning*. PMLR, 2019, pp. 6893–6901.

[117] Yueqi Xie, Minghong Fang, and Neil Zhenqiang Gong. "Fedredefense: Defending against model poisoning attacks for federated learning using model update reconstruction error". In: International Conference on Machine Learning. 2024.

[118] Dong Yin et al. "Byzantine-Robust Distributed Learning: Towards Optimal Statistical Rates". In: *International Conference on Machine Learning*. Pmlr, 2018, pp. 5650–5659.

[119] Vithya Yogarajan et al. "Transformers for Multi-Label Classification of Medical Text: An Empirical Comparison". In: *International Conference on Artificial Intelligence in Medicine*. Springer, 2021, pp. 114–123.

[120] Jichang Zhang, Yuanjie Zheng, and Yunfeng Shi. "A Soft Label Method for Medical Image Segmentation with Multirater Annotations". In: *Computational Intelligence and Neuroscience* 2023.1 (2023), p. 1883597.

[121] Jie Zhang et al. "Federated Learning with Label Distribution Skew via Logits Calibration". In: *International Conference on Machine Learning*. PMLR, 2022, pp. 26311–26329.

[122] Ying Zhou, Hui Xue, and Xin Geng. "Emotion Distribution Recognition from Facial Expressions". In: *Proceedings of the 23rd ACM International Conference on Multimedia*. 2015, pp. 1247–1250.

[123] Enwei Zhu, Yiyang Liu, and Jinpeng Li. "Deep Span Representations for Named Entity Recognition". 2022. arXiv: 2210.04182.

FOLIO ADMINISTRATIF

THESE DE L'INSA LYON, MEMBRE DE L'UNIVERSITE DE LYON

NOM : **ERBANI** **DATE de SOUTENANCE** : **17/07/2026**

Prénoms : **Johan**

TITRE : **Understanding Machine Learning Vectors: From Confusion Matrices to Byzantine-Robust Federated Learning**

NATURE : **Doctorat** **Numéro d'ordre** :

École Doctorale : **InfoMaths**

Spécialité : **Informatique**

RÉSUMÉ :
Ce travail étudie la classification dans des cadres complexes à partir de vecteurs issus de l'apprentissage automatique, en particulier les sorties de modèles et les gradients. Il se divise en deux parties.
La première partie porte sur les matrices de confusion. Ces matrices permettent d'analyser les erreurs d'un modèle au niveau des classes, contrairement aux métriques globales comme l'accuracy. Cependant, lorsque les distributions d'étiquettes sont déséquilibrées, leurs entrées peuvent être biaisées et masquer les similarités réelles entre classes. De plus, les cadres multi-labels et à soft-labels capturent mieux la complexité des données réelles, mais la généralisation de la matrice de confusion à ces contextes reste non triviale.
Nous proposons plusieurs contributions dans cette direction. Nous introduisons une nouvelle normalisation de la matrice de confusion pour mieux faire apparaître les similarités inter-classes. Nous établissons aussi des liens entre normalisation, importance sampling et représentations latentes des classes. Enfin, nous proposons une généralisation fondée sur le transport optimal, adaptée aux cadres mono-label, multi-labels et à soft-labels, ainsi qu'un cadre expérimental pour évaluer ces extensions.
La seconde partie porte sur l'apprentissage fédéré robuste. Dans ce cadre, plusieurs clients entraînent un modèle commun sans partager leurs données. À chaque round, chaque client calcule une mise à jour locale, par exemple un gradient, que le serveur agrège. Certains clients, qualifiés de byzantins, peuvent être défaillants ou malveillants. Les méthodes robustes cherchent souvent à écarter les gradients atypiques. Cependant, lorsque les données sont hétérogènes, notamment en présence de déséquilibre de labels entre clients, les gradients honnêtes peuvent eux aussi être très différents. Cela rend les clients honnêtes plus difficiles à distinguer des clients byzantins.
Nous proposons deux contributions dans ce cadre. La première réduit l'hétérogénéité des gradients honnêtes grâce à une stratégie d'échantillonnage local. Elle agit côté client, dès la construction des mises à jour, avant leur envoi au serveur. La seconde est un filtre byzantin côté serveur. Ce filtre s'appuie sur l'observation que les gradients locaux peuvent être vus comme des combinaisons convexes bruitées des gradients globaux propres à chaque classe. Il identifie alors les mises à jour suspectes à partir de leur cohérence avec cette structure géométrique.

MOTS-CLÉS : **Matrice de confusion, Normalisation, Multi-label, Federated Learning, Byzantine Robustness, Label Skew**

Laboratoire(s) de recherche : **Laboratoire d'InfoRmatique en Image et Systèmes d'information (LIRIS)**

Directeur de thèse : **Egyed-Zsigmond Elöd**

Président du Jury :

Composition du Jury :

**Gianini Gabriel, Tommasi Marc, Gouy-Pailler Cédric, Réveillère Laurent,
Egyed-Zsigmond Elöd, Ben Mokhtar Sonia, Nurbakova Diana**